

SEED AND SEED TECHNOLOGY-DEFINITION-IMPORTANCE-CHARACTERS OF GOOD QUALITY SEED

Seed

Seed is a basic input in agriculture upon which other inputs are applied. A good vigorous Seed utilizes all the resources and realized as a reasonable output to the grower. It is wealth to the farmer, it is the yesterday's harvest and tomorrow's hope. Good seed in good soil realize good yield.

Importance of seed

Seed is the vital input in crop production because through seed only the investment made on other inputs like pesticide, fertilizer, irrigation and crop maintenance can be realized.

The seed required for raising the crop is quite small and its cost is also less compare to other inputs, but the greater income farmer gets depends upon the quality of the small quantity of seed he uses. In addition to above seed is the basic for the following event of agriculture.

1. A carrier of new technologies
2. A basic tool for secured food supply
3. The principal means to secure crop yields in less favourable production areas
4. A medium for rapid rehabilitation of agriculture in cases of natural disaster

Seed is known as (Definitions)

- ✓ A ripened ovule or a fertilized matured ovule containing embryo which has developed after fertilisation.
- ✓ The dry dispersal unit or matured ovule developed after fertilization
- ✓ Any part (or) organ of plant which has the capability to regenerate into a new plant
- ✓ A propagule responsible for maintaining the intrinsic (or) genetic qualities of the variety / hybrid.
- ✓ An 'embryo', a living organism embeded in the supporting (or) the food storage tissue and a protective coat.
- ✓ Any propagative material.
- ✓ Miniature plant.
- ✓ Dormant plant
- ✓ Link between two generations
- ✓ Carrier of service material.
- ✓ Generative part of a plant that develop into a new plant.

However, the widely accepted definition for a “seed is matured ovule that consisting of an embryonic plant together with a store of food, all surrounded by a protective coat”.

(OR)

Seed is defined as a ‘matured ovule’ or a reproductive unit formed from fertilized ovule, consisting of an embryo, reserve food and a protective cover.

As per Seed Act (1966) seed includes

- ✓ Seed of food crops including edible oil seeds and seeds of fruits & vegetables.
- ✓ Cotton seeds
- ✓ Seeds of cattle fodder
- ✓ Jute seeds
- ✓ Seedlings, tubers, bulbs, rhizomes, roots, cuttings, all types of grafts and other vegetatively propagated material for food crops (or) cattle fodder.

SEED STRUCTURE

The structure of a seed includes the following

- (1) **Testa:** which is really the two dried integuments, the testa (outer integument) and the tegmen may sometimes get so much fused together that they cannot be separated
- (2) **Perisperm:** derived from the **nucellus**
- (3) **Endosperm:** produced as a result of fusion between one male generative nucleus and the two polar nuclei to form the triploid endosperm nucleus. The perisperm and endosperm are the nutritive tissue.
- (4) **Embryo:** the result of fertilization of the oosphere (ovum) by a male nucleus. It is the rudimentary plant or miniature of the plant which is the most important part of the seed.
- (5) **Cotyledon** is the part of the embryo which is the special type of leaf called seed leaf.

The single cotyledon of the monocot is known as “scutellum”

Did you know?

- Orchid seeds lack endosperm.
- Monocots have single endosperm eg. rice, wheat, sorghum
- Dicots have two cotyledons
- eg. greengram, blackgram, chickpea, pigeon pea

Did you know?

Cotyledons store food for developing seedling. Cotyledons are used by humans for most of their nutrition needs like carbohydrate, fat and protein.

Angiosperms are divided into the Monocotyledons and Dicotyledons according to the number of cotyledons. Polycotyledonous types are known among Gymnosperms. Some angiospermic seeds do not show a proper differentiation of cotyledons but other characteristics put them along with either monocotyledons or dicotyledons. Both, dicotyledonous and monocotyledonous plants may be albuminous (endospermic) or exalbuminous (exendospermic) according as whether the endosperm is present or not (Fig.1).

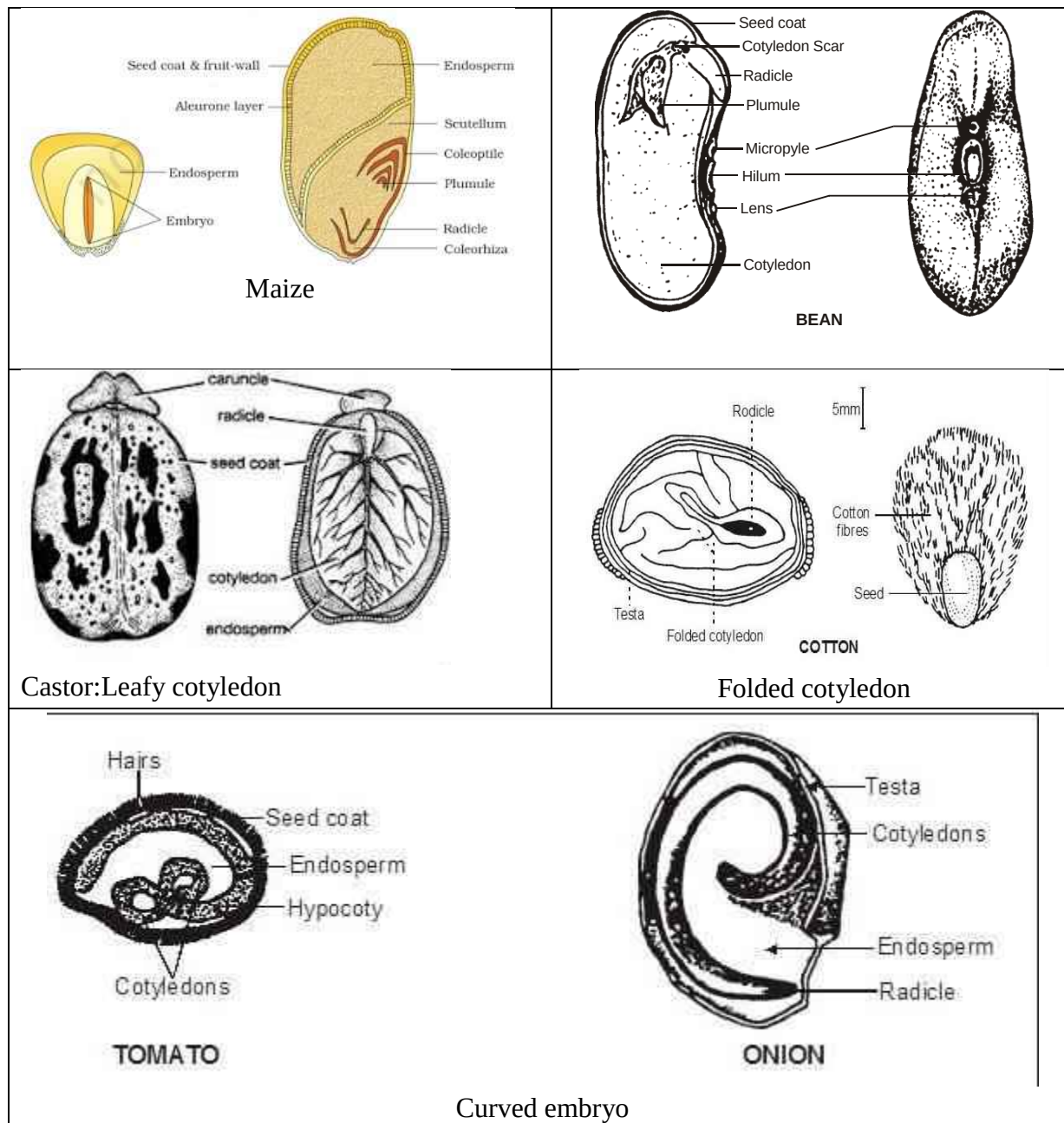


Fig. 1: Structure of Monocotyledons and Dicotyledons seeds

Classification of seed based on structure

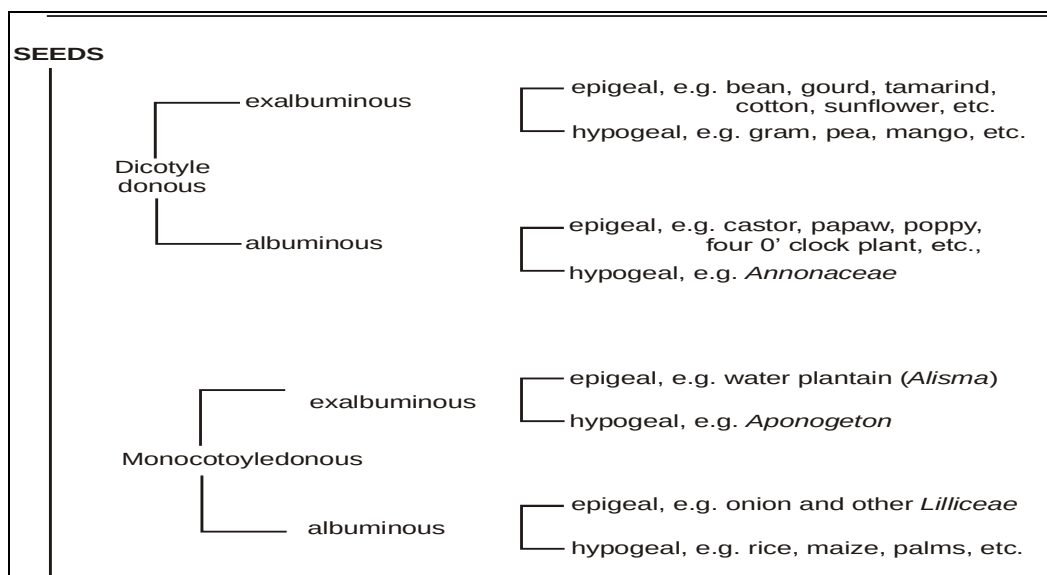
Based on the storage organ present in seed to nourish the growing embryo, the kind of seeds can be broadly classified into two viz., monocot and dicot seeds

Monocot seeds/ Monocotyledons

- The seeds of monocots are single seeded, embedded in a fruit called “caryopsis, utricle”etc. The seed coat and fruit coat are fused as pericarp.
- The embryo is minute and confined to one end of the seed and is viable with $2n$ number of chromosome. **“The storage tissue is known as endosperm and is non-viable”** with $3n$ number of chromosome and it occupies the major portion of the seed.
- The embryo consists of **“radicle and plumule which are enclosed in coleorhiza and coleoptile, respectively”**.
- The single cotyledon of the monocot is known as **“scutellum”**. It is situated in between the radicle and plumule.
- **Aleurone layer is a protein body** which remains at the periphery of the endosperm. It **helps in absorption of water and releases enzymes** which help in **germination**.
- Monocot seeds are known as **albuminous or endospermous seeds**.

Dicot seeds/ Dicotyledons

- The dicot seeds consist of a **primary axis and living storage tissue, the cotyledons which combinely from the embryo of the seed**.
- This **embryo is enclosed in the seed coat**.
- Mostly dicots **do not have endosperm** which is fully utilized for the development of embryo. Sometimes they have well developed endosperm (Castor, Fenugreek)
- Mostly they are **ex-endospermous or ex- albuminous** in nature (legumes).
- In some cases perisperm which is the remaining, unfertilized portion of nucellus is also seen in dicot seeds (eg.) Pepper.



Internal structures of seed:

1.	Embryo	It is a miniature plant consisting of plumule, radicle and cotyledon (embryonic axis + cotyledon). Plumule + radicle without cotyledon is known as primary axis.
2.	Endosperm	Storage tissue of the seeds which are formed by fusion of pollen cell or sperm nuclei with polar nuclei. It is triploid in nature and non-viable.
3.	Cotyledon	It is the storage tissue of dicots and is a part of the embryo. It is viable.
4.	Scutellum	It is the cotyledon of monocot seeds. It is viable.
5.	Coleoptyle	It is the covering of miniature plumule the shoot portion in monocots.
6.	Coleorhiza	It is the covering of radicle, the root portion in monocots.
7.	Plumule	It is a miniature part of shoot region seen both in dicots and monocots
8.	Radicle	It is the miniature part of root region.
9.	Hypocotyl	Part of seedling which is below the cotyledon and above the radicle.
10.	Perisperm	It is an unutilized part of the nucellar region which is normally visible as a papery growth outside the endosperm and inside the seed coat.
11.	Epicotyl	Part of seedling above the cotyledon and below the primary leaf of seed.

External structures of seed

1.	Seed coat	It is the outer covering of seed which gives protection. It is developed from the two integuments of ovule. The outer layer of seed coat which is smooth and rough is known as testa and it is formed from outer integument. The inner layer of seed coat is called as tegmen and is formed from inner integument.
2.	Pericarp	The body of a fruit developed from the ovary wall and enclosing the seed
3.	Raphe	The area between the micropyle and chalaza. The raphe will be visible on the seed coat of some species.
4.	Micropyle	The point where the integuments meet at nucellar apex
5.	Hilum	It is the scar left on the seed where it was formerly attached to the funicle or placenta
6.	Caruncle	White spongy out growth of the micropyle seen near the hilum of seed. Eg. castor, tapioca

Facts:

- ☞ **Leafy cotyledon** present in castor
- ☞ **Curved embryo** present in Tomato, brinjal and chilli
- ☞ **Folded cotyledon** present in Bhendi and cotton

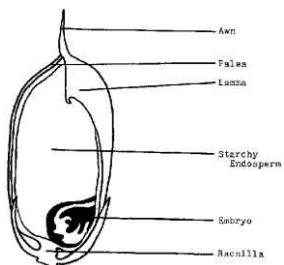
Appendages:

External attachments on the seed that favours dispersal of seeds or in identification of genotypes.

APPENDAGES OF SEED

AWN:

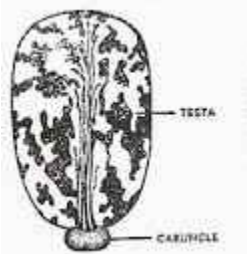
Awn- Thread like projection at the tip of seed is known as Awn



ARIL:

Aril- Coloured fleshy mass present on the outer side of the seed is known as Aril



Paddy	Nutmeg
WINGS: Wings: It is the papery structure attached to the side of the seed coat either to a specific side or to all sides is known as Wings.	CARUNCLE: Caruncle: An elaiosome (Fleshy structure attached to the seed) is known as caruncle
 Samara Moringa	 Castor

Seed technology: is the science dealing with the methods of improving physical and genetical characteristics of **seed**. It involves such activities as variety development, evolution and release of varieties, **seed** production, **seed** processing, **seed** certification and storage.

Differences between seed and grain

SEED	GRAIN
It should be a viable one	Need not be a viable one
It should have maximum genetic & physical purity	Not so
Should satisfy minimum seed certification standards	No such requirements
It should be completely treated with pesticide /fungicide to protect seed against storage pests and fungi	It should never be treated with any chemicals, since used for consumption
Respiration rate and other physiological and biological processes should be kept at low level during storage	No such specifications
Should be compulsorily certified / truthful labeled	No such condition in grain production
Should never be converted into grain unless warranted	Can be converted as seed provided the situation warrants
It should satisfy all the quality norms	Not considered

CHARACTERS OF GOOD QUALITY SEED:

1. Seed should be viable with high germination % and vigour
2. Genetically pure / true to type to mother plant

Nucleus/ breeder seed	:	100 %
Foundation seed	:	99.5 %
Certified seed	:	99 %

3. Physically pure ie., free from inseparable other crop seeds, weed seeds, designated diseased seeds and inert matter

a. Inseparable other crop seeds

These are plants or different crops which have seed similar to seed crop

Crop	Inseparable crop plants
Wheat	Barely, oats, gram
Barely	Oats, gram, wheat
Oats	Barely, gram, wheat

b. Objectionable weed seeds

These are weeds

1. Whose seeds are difficult to separate once mixed
2. Which are poisonous
3. Which have smothering effect on the main crop
4. Which are difficult to eradicate once established

S.No	Crop	Common name of the weed	Botanical name
1	Paddy	Wild rice	<i>Oryza sativa var fatua</i>
2	Wheat	Wild morning glory	<i>Convolvulus arvensis</i>
3	Sunflower	Wild sunflower	<i>Helianthus spp</i>
4	Bhendi	Wild okra	<i>Abelmaschus spp</i>
5	Rape and mustard	Mexican prickly poppy	<i>Argemone mexicana</i>
6	Lucerne	Dodder	<i>Cuscuta spp</i>

c. Designated diseases

S.No	Crop	Designated diseases	Botanical name
1	Wheat	Loose smut	<i>Ustilago tritici</i>
2	Sorghum	Grain smut	<i>Sphacelotheca sorghii</i>
3	Cumbu	Grain smut Ergot	<i>Tolyposporium penicillariae</i> <i>Claviceps microcephala</i>
4	Cowpea	Anthraxnose	<i>Colletotricum sp</i>
5	Sesame	Leaf spot	<i>Alternaria sp</i>

6	Rape and mustard	Alternaria blight	<i>Alternaria sp</i>
7	Brinjal	Little leaf	<i>Datura virus</i>
8	Chillies	Leaf blight	<i>Alternaria solani</i>
9	Bhendi	Yellow vein mosaic	
10	Tomato	Early blight Leaf spot	<i>Alternaria solani</i> <i>Xanthomonas sp</i>
11	Potato	Brown rot	<i>Pseudomonas solanacearum</i>
12	Cucurbits	Mosaic	<i>Cucumis virus</i>

d. Free from inert matter, sclerotia, nematode galls etc.,

e. Higher physical purity percentage

Name of the crop	:	Purity %
Carrot	:	95
Groundnut	:	96
Sesame, soybean and Jute	:	97
All crops (most)	:	98
Bhendi	:	99

4. Have optimum moisture content for storage

- long term storage - 8 % & below
- short term storage -10-13%

Significance of a good quality seed

- Ensure increase in yield by at least 10 – 12 %
- Ensures genetic and physical purity of the crops.
- Gives desired plant population.
- Able to withstand the adverse conditions.
- Free from pest and diseases.
- Seedlings will be more vigorous and fast growing
- Can resist pest and disease incidence to certain extent.
- Ensures uniform growth and maturity
- Efficient absorption of nutrients results in higher yield.
- Respond well to added fertilizer and other inputs.

Advantage of seed propagation

- Rapid multiplication
- Assured and timely supply
- Easy in handling and utilisation
- Reasonable price

- More crop coverage

Seed Production

Systemized crop production is known as seed production. In seed production adequate care is given from the purchase of seeds upto harvest adopting proper seed and crop management techniques. The benefits of seed production are,

- ❖ Higher income
- ❖ Higher quality seed for next sowing

Difference between seed and crop production

Seed production	Crop production
Basic seed should be from an authentic source	Any seed material can be used
Seed plot should be selected carefully for better performance, as per edaphic and environmental requirement	Can be grown in any area
Needs isolation from other varieties	Isolation is not necessary
Needs technical skill for maintenance of quality	Special technical skill is not required
Maintenance of genetic purity is important	Genetic purity is not required
Roguing is compulsorily practiced	Roguing is not practiced
Harvesting should be done at physiological/ harvestable maturity	Harvested at field maturity
Resultant seed should be vigorous and viable	Question of viability does not arise
Importance is given to seed quality rather than the yield	Importance is given more to yield

There are two types (major) of seed production *ie.*, varietal and hybrid. Seed production based on the type of seed used for multiplication.

Difference between varietal and hybrid seed production

Varietal seed production	Hybrid seed production
It is single parent multiplication	It needs two to many parents
Isolation distance requirement is less	Isolation distance requirement is more
Production is by open pollination	Production is by managed control pollination (Female)
Seed can be used continuously for 3/4/5 generations	Seed has to be changed every time
Production technique is uniform (multiplication)	Technique differ with crop
Production care is less	Production care is more
Yield will be lower	Yield will be higher
Profit is less	Profit is higher

SEED DEVELOPMENT AND MATURATION

Angiosperms are true flowering plants in which seeds are enclosed inside the fruits, because the seed develops from an ovule situated inside the ovary of a flower, usually after fertilization. The ovules that develop into seeds are totally enclosed within the ovary.

The different phases of growth, development and maturation in seeds may be summarized briefly as i) the formation of stamens and pistils in separate buds or in the same bud in species having perfect flowers, ii) the blossoming or attainment of sexual maturity of these organs, iii) pollination, consisting of the distribution of pollen from the stamen to the pistil of the same or another flower and germination of pollen grain, iv) fertilization or union of pollen tube and embryo sac, v) growth and differentiation of the fertilized egg into an embryo; in angiosperms there is the added formation of endosperm, and vi) maturity of the seed accompanied by more or less profound chemical changes.

The reproductive organ of the plant is called as “flower”, which consists of calyx (all sepals), corolla (petals), perianth (calyx and corolla together), androecium (all stamens – male part and gynoecium (the female part). Pollen is produced in the anther of the stamen; the stamens are specialized microsporophylls. The pollen is enclosed within the anthers and released at maturity. The pollinating agents may be wind, insects, bats or birds. The ovules are produced and enclosed in the pistil, which may be formed by one or more carpels (gynoecium).

Seed formation

A true seed is defined as a fertilized mature ovule consisting of embryo, stored food material and protective coats. The important events involved in seed development and maturation include

1. Pollination
2. Fertilization
3. Development of the fertilized ovule by
 - a. Cell division
 - b. Accumulation of reserve food material and
 - c. Dehydration.

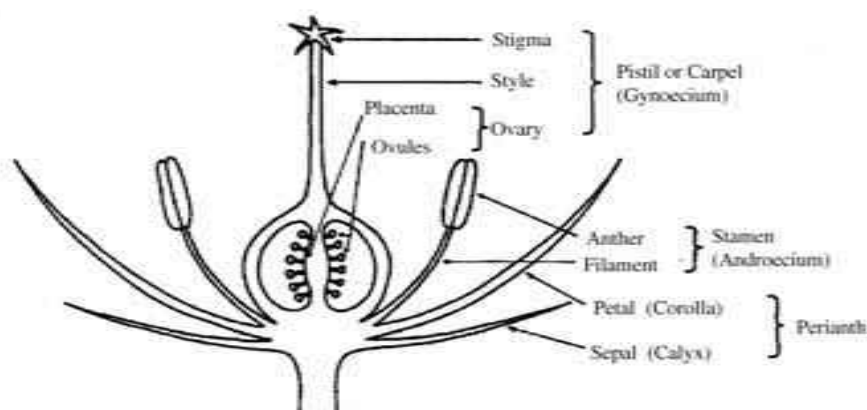
Pollination

Floral biology

Flower is a reproductive organ bearing pistil, stamen and usually sepals and petals. Male part of a flower is **androecium** consisting of anther sac, anthers and pollen grains. Female part is **gynoecium** consisting of ovary, style and stigma.

The flower is said to be **perfect**, when they contain both **male and female** parts. A flower with both **functional male and female** is called as **bisexual** or **hermaphrodite**. Sometimes male or female **mature** slightly at **different times**.

This nature is called **dichogamy** which favours **cross pollination**. If **male** matures first it is called as **protandry**, if **female – protogyny**.



Imperfect flowers have either male (staminate flower) or female (pistillate flower) part. Such flowers are called as **unisexual flowers**. When both type of flowers occur in same plant – **monoecious**, if they occur in different plants – **dioceious**.

Seeds usually developed from the fertilized flower for which pollination (transfer of pollen from anthers to stigma) is pre requisite. For successful fertilization viable pollen and receptive stigma are two pre requisites. The mature anthers dehisce and release pollen grains (haploid microspores). When pollen grains are transferred from an anther to the stigma of the same flower the process is called **self-pollination** or autogamy. If they are transferred to the stigma of another flower, **cross-pollination** or allogamy is said to have occurred.

Self-pollination occurs in those plants where bisexual flowers achieve anther dehiscence and stigma receptivity simultaneously. The majority of angiosperms bear chasmogamous flowers i.e., flowers do not open before pollination. In some plants, flowers do not open at all such

flowers is called cleistogamous, and this is the most efficient floral adaptation for promoting self-pollination.

Cross-pollination is ensured in plants which bear unisexual flowers. In bisexual flowers also self-pollination may be prevented by self-sterility, dichogamy (maturation of male and female organs at different times), herkogamy (where the structure of male and female sex organs proves a barrier to self pollination) and heterostyly (where flowers are of different types depending on the length of the style and stigma and pollination occurs only between 2 dissimilar types).

The important self-pollinated crops are wheat, rice, barely, mungbean and cowpea and cross pollinated are maize, rye, forage legumes and vegetables like carrot, cauliflower and onion. There is yet another category of crops called often cross pollinated crops such as cotton and pigeon pea where there may be 10-40% cross pollination.

Fertilization

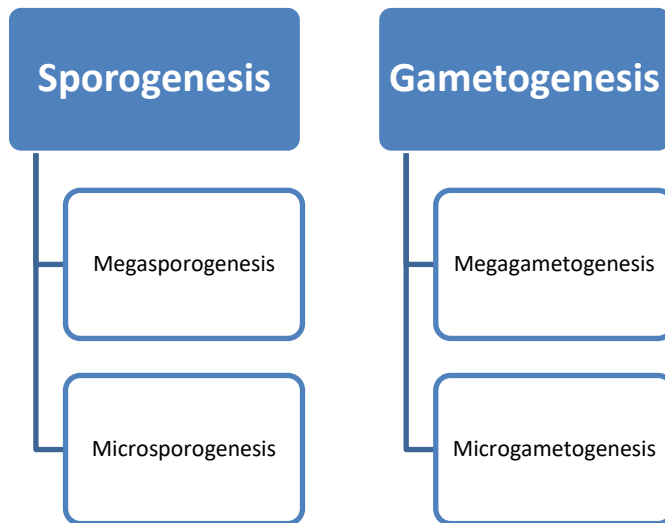
Ovules are developed in the female gametophyte and pollen in the male gametophyte. Fertilization always takes place in female gametophyte, therefore pollen must transferred from male to female by pollen vector which may be abiotic including wind (anemophily) and water (hydrophily) or biotic including insects (entomophily) and bats (cheiropterophily).

Apomixis

Development of seed without fertilization i.e. the seed formation occurs without sexual fusion, the process is known as Apomixis. This occurs by several mechanism, however, all apomitic seed have genetic material only from the female plant. Apomixis may or may not require pollination and pollen tube germination to initiate seed formation, however sexual union never occurs.

Parthenocarpy: Development of fruit without fertilization

Seed Formation: The seed formation takes place by the following steps:

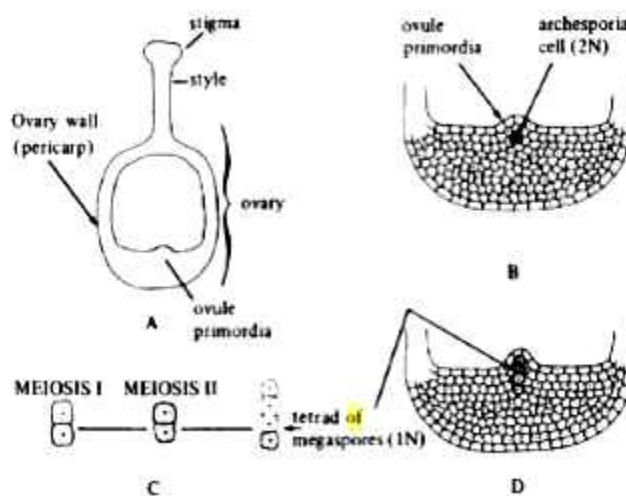


Sporogenesis

- ✓ Production of **microspores and megaspores** is known as “sporogenesis”.
- ✓ In **ovules**, the megaspores are formed through “megasporogenesis”.
- ✓ In **anthers**, microspores are formed through “microsporogenesis”

Megasporogenesis

The seeds of angiosperms originate from the meristamatic tissue of the ovary wall called ‘*ovule primordia*’. Within the ovule primordia, one cell known as the archesporial cell, increases in size, its nucleus becomes larger, its cytoplasm becomes dense and undergoes mitotic cell division to form ‘megaspore mother cell and parietal cell’. The parietal cell soon degenerates and megaspore mother cell (2N) undergoes two step cell division (meiosis) to form four megaspores (1N). Normally, only one megaspore is functional while the other degenerates.



Megasporogenesis: (A) Location of ovule development (B) cutaway section of the lower region of the ovary wall (pericarp), showing origin of the archesporial cell, note that it is larger than surrounding cells, having a larger nucleus and denser cytoplasm (C) cell division during megasporogenesis, (D) cutaway section of lower part of the ovary, showing location of the four megaspores, three of which normally degenerate

Microsporogenesis vs Megasporogenesis

Microsporogenesis	Megasporogenesis
The sporophytic cells in the pollen sacs of anther which undergo meiotic division to form haploid i.e., microspores are called microspore mother cell (MMC) or pollen mother cell (PMC) and the process is called “ microsporogenesis ”.	A single sporophytic cell inside the ovule, which undergo meiotic division to form haploid megaspore, is called megaspore mother cell (MMC) and the process is called “ megasporogenesis ”.
Each PMC produce four microspores and each microspore after thickening of the wall transforms into pollen grain.	Each MMC produces four megaspores out of which three degenerate resulting in a single functional megaspore.

Gametogenesis

The production of male and female gametes in the microspores and megaspores is known as “**gametogenesis**”.

Microgametogenesis

This is nothing but the production of “**male gametes or sperm**”.

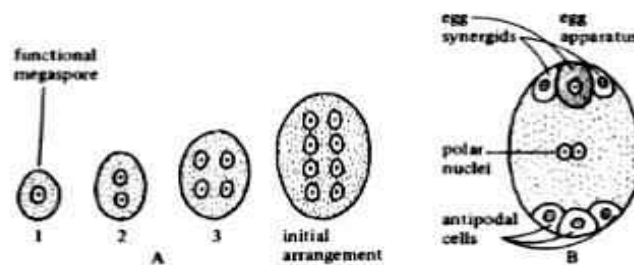
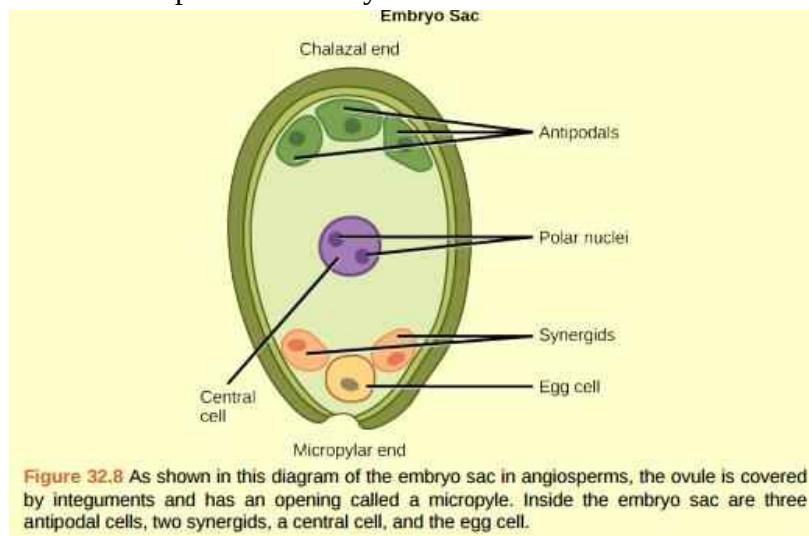
On maturation of the pollen,

- The microspore nucleus divides mitotically to produce a generative and a vegetative or tube nucleus.
- The pollen is generally released in this binucleate stage.
- The reach of pollen over the stigma is called pollination.
- After the pollination, the pollen germinates.
- The pollen tube enters the stigma and travels down the style.
- The generative nucleus at this phase undergoes another mitotic division to produce two male gametes or sperm nuclei.
- The pollen along with the pollen tube possessing a pair of sperm nuclei is called microgametophyte.
- The pollen tube enters the embryo sac through micropyle and discharges the two sperm nuclei.

Megagametogenesis

- ✓ The nucleus of the functional megaspore undergoes three mitotic divisions to produce eight or more nuclei.
- ✓ The exact number of nuclei and their arrangement varies from one species to another.
- ✓ The megaspore nucleus divides thrice to produce eight nuclei.

- ✓ Three of these nuclei move to one pole and produce a central egg cell and two synergid cells on either side.
- ✓ Another three nuclei migrate to the opposite pole to develop into three antipodal cells.
- ✓ The two nuclei remaining in the center, the polar nuclei, fuse to form the secondary nucleus.
- ✓ The megaspore thus develops into a mature female gametophyte called megagametophyte or embryo sac.
- ✓ The development of embryo sac from a megaspore is known as “**megagametogenesis**”.
- ✓ The embryo sac generally contains one egg cell, two synergids with the apparent function of guiding the sperm nucleus towards the egg cell and three antipodals which forms the prothalamus cells and one diploid secondary nucleus.

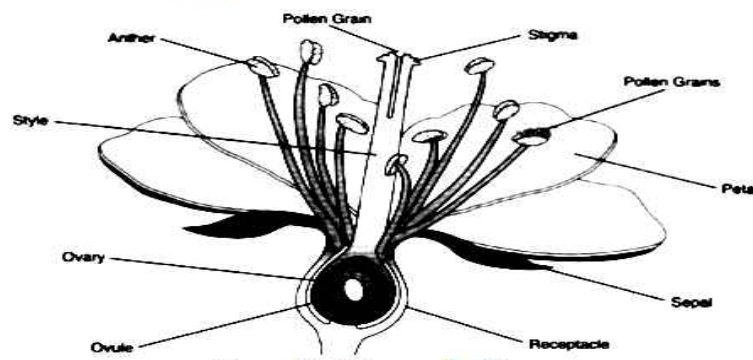


Megagametogenesis: A. Three normal mitotic nuclear divisions leading to one large cell enclosing eight nuclei. Later, cell walls enclose the nuclei and the entire structure

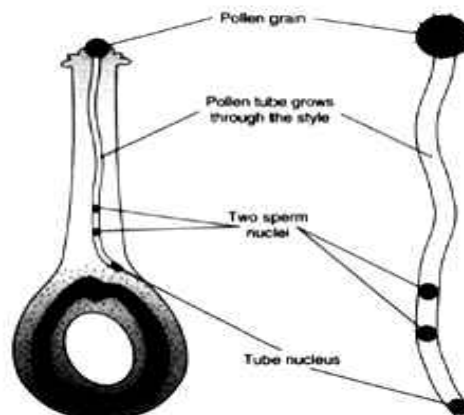
Pollination and fertilization

The arrival of pollen on the stigma of the female flower marks the pollination in angiosperm. Upon landing on the stigma, the pollen nucleus divides into two (sperm nucleus and tube nucleus). The tube nucleus forms the pollen tube that grows and enters into stigma. After

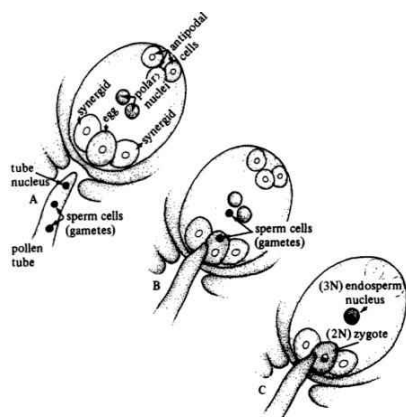
entering the ovule the pollen tube penetrates the nucellus and reaches the embryo sac, penetrates the embryo sac and reaches the egg apparatus. The tip of the pollen tube then swells up and bursts. The two male gametes are then liberated and the tube nucleus disintegrates. One of the two male gametes fuses with the female nucleus or egg cell, also known as oosphere to form the oospore which later becomes the embryo. The other male gamete pushes further into the embryo sac and fuses with the secondary nucleus formed from the two polar nuclei. The fusion of a male gamete with the secondary nucleus is known as triple fusion and forms the primary endosperm nucleus. The endosperm nucleus gives rise to the endosperm. The whole process is known as **double fertilization**.



Structure of a complete angiosperm flower



Male gametophyte in the pistil of a flower (Left side) Male gametophyte alone



Seed development and maturation

i) Embryogeny and storage tissue formation

From the moment of fertilization, certain morphological, physiological and functional changes are initiated in the ovule and ovary, which culminate in the formation of a mature seed capable of reproducing another plant. Subsequent to the fertilization, the embryo and storage tissues are formed.

The **embryo** is the new genetic combination, which is formed from the zygote by cell division and differentiation. The embryo consists of radicle, cotyledons, plumule and hypocotyl.

The **endosperm** is a nutritive tissue surrounding the embryo. It commonly grows more rapidly than the embryo immediately after fertilization thus covers the embryo.

Did you know?

The endosperm is **triploid** while seed coat, fruit coat,

Did you know?

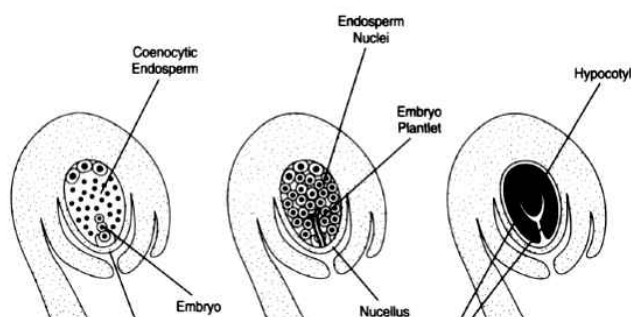
The scutellum is the cotyledon in monocots. Being single or mono it lends it name to monocots.

In monocot seeds, cell walls form about two days after fertilization, beginning at the periphery of the endosperm, which later becomes aleurone layer. During the early endosperm growth, the proembryo also begins to grow and differentiate. However, the growth of endosperm is faster and occupies most part of the seed, while the embryo is restricted to a small region at the periphery (eg., rice, maize, wheat *etc.*).

In dicot seeds, normally the endosperm reserves are depleted and occluded by the developing embryo and reorganized in the cotyledons, which act as the source of stored reserves (eg., black gram, ground nut *etc.*). In certain seeds endosperm is retained as a permanent storage tissue in the endospermic dicot seeds eg., coriander, castor.

The nucellus of the ovule may persist in some genera as a thin layer called **perisperm**. In dicotyledonous seeds, embryo absorbs the nutrients and the endosperm disappears at the maturity of the seed. In the monocotyledons seeds such as rice, maize etc., the endosperm develops rapidly and pushes the embryo to the periphery.

The secondary growth in the periphery of the ovule is called collars or **integuments**, which ultimately fuse to become the **testa** (seed coat) of the mature ovule. The developing ovule is attached to the placenta of the ovary by the **funiculus** which forms the **hilum** when the seed detaches at maturity. The point where the integuments meet is the **micropyle** and the region of integumental origin and attachment, usually opposite to the micropyle, is the **chalaza**. Often the integuments and the nucellus are clearly differentiated only in the region of the micropyle.



New sporophyte has begun to grow in the embryo sac (Left). Cell membranes have been formed around the endosperm nuclei (Centre). The embryo plantlet has

ii) ***Deposition of reserves in storage tissues***

Once the storage tissues are formed, from the early seed development stage up to maturity major reserves such as carbohydrates, fat and protein and the minor reserve viz., phytin are synthesized in mother plants and transported to the storage organs.

Starch synthesis

Fat synthesis

Protein synthesis

Phytin deposition

Hormones in developing seed

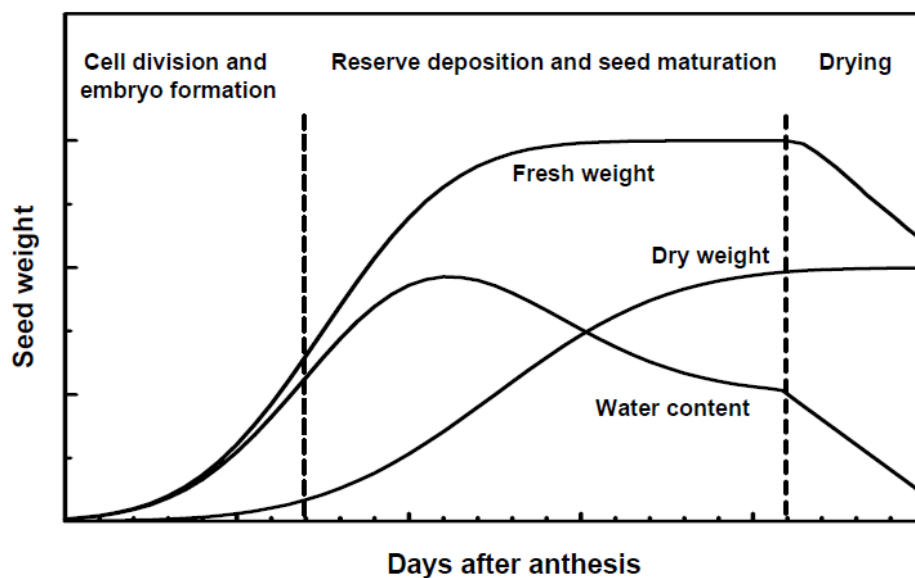
During seed development along with accumulation of major storage reserves, significant changes are also found to occur in the contents of growth regulators or hormones such as auxin, cytokinins, gibberellins and ABA.

Seed maturity

Maturity is defined as the point of maximum dry weight of seed. This point is reached when the seeds are relatively high in moisture content of (35 to 45%). At this stage, the viability and vigour of seeds are also higher. The sooner the seeds are harvested after maturity, the higher the quality. Maturity stage is differentiated into physiological maturity and harvestable maturity.

Physiological maturity: This is the stage at which the maximum dry matter accumulates. The moisture content of the seed will be 25 % to 50%. The seed registers the highest seed germinability and vigour index at this stage. The seeds have to be effectively dried to safe moisture contents.

Harvestable maturity : This is the stage at which the seed for the first time dries to a level of 14 % or less. This allows commercial harvesting and the seeds can be stored without supplemental drying.



Generalized pattern of seed maturation

LECTURE 3

GERMINATION - PHASES OF SEED GERMINATION - DORMANCY –

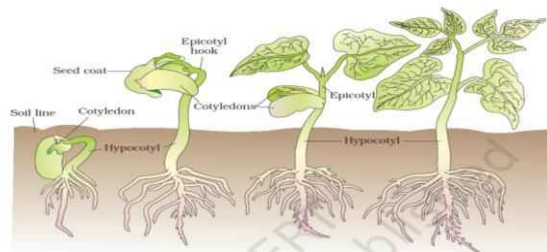
TYPES OF SEED DORMANCY

Seed germination

The process whereby seeds sprout and begin to grow is known as seed germination .

Germination is defined as **the emergence and development from the seed embryo** of those essential structures which for the kind of seed in question, indicate its ability **to produce a normal plant under favorable conditions** .

Seed germination and seedling development in bean



Pre-requisites for seed germination

Water

Water is a basic requirement for germination. It is essential for enzyme activation, breakdown, translocation and use of reserve storage material. Low levels of moisture may trigger initial stages of germination but are not adequate for complete germination. High moisture levels inhibit germination.

Air (Oxygen and Carbon dioxide)

Air is composed of about 20% oxygen, 0.03% carbon dioxide and about 80% nitrogen gas. Oxygen is indispensable for seed germination, carbon dioxide at higher concentrations than 0.03% retards germination, while nitrogen has no influence.

Temperature

The response to temperature depends on a number of factors viz., species, variety, growing region, quality of the seed, etc. As a general rule, seeds of temperate region, require lower temperature than do seeds of tropical region. Wild species have lower temperature requirements than do domesticated plants. High-quality seeds are able to germinate under wider

temperature ranges than low-quality seeds. The optimum temperature for most seeds is between 15 and 30°C. The maximum temperature for most species is between 30 and 40°C. Some species will germinate at temperatures approaching the freezing point.

Light

The greatest promotion of germination occurs in the red area (660 –700 nm) with peak at 660 nm, followed by an inhibition zone in the far-red area above 700 nm.

Morphology of germination

Based on the fate of the cotyledons or storage organs two kinds of seed germination occur.

Epigeal germination

Epigeal germination (*Fig. A*) characteristic of ground nut, black gram and green gram seed. During germination, the cotyledons are raised *above the ground* where they continue to provide nutritive support to the growing points. During root establishment, the hypocotyl begins to elongate in an arch that breaks through the soil, pulling the cotyledons and enclosed plumule through the ground and projecting them into the air. After ward, the cotyledons open, plumule growth continues and the cotyledons wither and fall to the ground.

Hypogeal germination

Hypogeal germination (*Fig. B*) is characteristic of pea seeds, all grasses such bamboos. During germination, the cotyledons or comparable storage organs remain *beneath the soil* while the plumule pushes upward and emerges above the ground, but continue to provide nutritive support to the growing points. The coleoptile, a temporary sheath enclosing the plumule, is associated with hypogeal germination of many species, it provides protection and rigidity to the emerging plumule as it pushes.

Fig.A.Epigeal germination

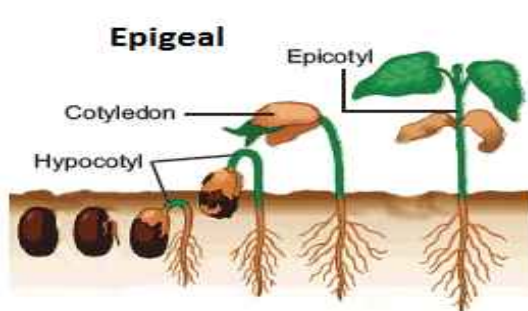
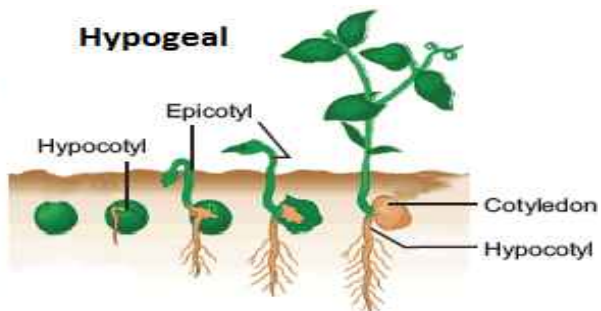


Fig.B.Hypogeal germination



PHASES or STAGES OF SEED GERMINATION

Seed Germination is the sum of events beginning with hydration and culminating with root emergence. The germination process involves several stages which include

- 1) Imbibition of water,
- 2) Respiration
- 3) Activation of enzyme systems and mobilization of stored reserves
- 4) Nucleic acid (RNA, DNA) and protein synthesis
- 5) Emergence of the radicle and growth of the seedling.

1. Seed Imbibition

Seed germination starts with the imbibition of water by dry seed coat. Water absorption has three phases. In

Phase I, shows rapid water absorption. Respiratory activity is resumed within minutes.

Phase II shows stagnation in water absorption. There is increase in enzyme activation and synthesis.

Phase III shows another rapid water uptake. In this stage, enzymes activated in Phase II, acts upon the storage materials such as fats, proteins, carbohydrates (and phosphorous containing compounds) to break them down into low molecular weight substances. These substances are transferred to the growing embryonic axis to support the growth of the seedling until they become autotrophic.

2. Respiration

The onset of respiration *per se* is one of the first activities to resume within minutes of the start of imbibition. As the respiration increases upon imbibitions, the ATP levels also increases. The initial substrates for respiration are soluble sugars (sucrose and oligosaccharides), but storage reserves such as lipids and starch are soon tapped as well. The ATP and NADPH generated via respiration are utilized to initiate synthesis of i) **nucleic acid**, ii) **proteins** and iii) **repair of DNA**.

3) Activation of enzyme systems and mobilization of stored reserves

Seed storage organs such as cotyledons or endosperm contain substantial quantities of major reserves such as carbohydrates, oils and proteins. These high-molecular-weight substances

have to be hydrolysed into low-molecular-weight metabolites so that they can be readily transported to the growing regions (radical and plumule) and utilized for energy-production as well as synthetic events.

Mobilization of food reserves is initiated during the early growth of embryonic axis to enable the emergence of radicle and plumule. However, the major mobilization of reserves commences after the emergence of the radical and plumule, to support the seedling growth of until it becomes photosynthetically active (autotrophic).

Carbohydrate : After imbibition, the embryonic axis secretes gibberellic acid into the aleurone layer which causes synthesis of α -amylase. Starch is hydrolyzed first into maltose in the presence of α -**amylase** and β - **amylase** and then maltose is converted into glucose by **maltase**. The glucose is further converted into soluble sucrose and transported to growing embryonic axis.

Protein: Proteins are hydrolyzed into amino acids by **peptidase** enzyme. The amino acids may either provide energy by oxidation after deamination (removal of amino group) or may be utilized in the synthesis of new proteins.

Lipid: Fats are hydrolyzed into fatty acids and glycerol by lipase enzyme. Fatty acids are further converted into acetyl – CoA by the process, β - oxidation. The acetyl CoA is further converted into sucrose via glyoxylate cycle and is transported to the growing embryonic axis.

Inorganic materials: A number of inorganic materials such as phosphate, calcium, magnesium and potassium are also stored in seeds in the form of phytin. These stored materials are liberated during germination due to the activity of various phosphatases including phytase.

4) Nucleic acid (RNA, DNA) and protein synthesis

i) Protein synthesis

The components of the protein synthesizing complex (rRNA), tRNA and some mRNA) and specific cytoplasmic proteins are stored in a stable form in the dry seed. Protein synthesis resumes within minutes following imbibitions and hydration of the cells.

Initial protein synthesis involves translation of stored mRNAs, as polyribosomes (polysomes) are formed. Over several hours, as these stored messages are released from the complex and used in protein synthesis they are replaced by newly synthesized mRNAs. Many of the proteins which are synthesized during germination are so-called house-keeping or growth maintenance proteins which are required to maintain the metabolic integrity of the cells of the germination seed.

ii) DNA synthesis

On commencement of imbibitions, two discrete phases of DNA synthesis occur in the radicle cells, after imbibition. The first phase, takes place soon after imbibitions. It involves the 'repair of DNA', which was damaged during the seed maturation drying process, rehydration as well as the synthesis of mitochondrial DNA. Part of the repair processes that occur following imbibitions may involve the **new synthesis of telomeric DNA**, using the enzyme telomerase.

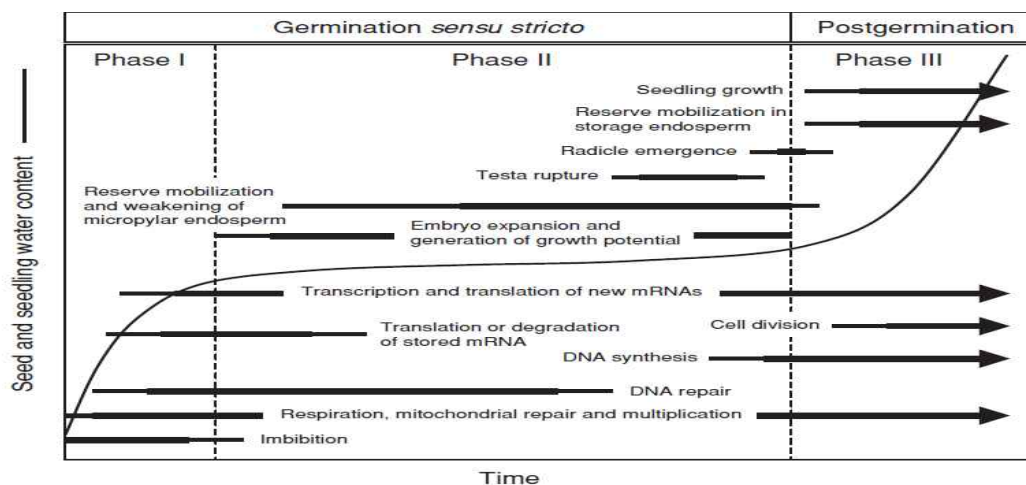
Second phase is associated with synthesis of DNA (replication) during cell division, subsequent to the radical protrusion. Replication of DNA commences usually within the few hours prior to the completion of germination (radical protrusion). The DNA synthesis is a continual process since it is required for nuclear and cell division, associated with radicle and plumule growth and subsequent seedling development. DNA polymerases, necessary for DNA repair and replication, are present in the dry seed,

5) Emergence of the radicle and growth of the seedling.

All the changes described above gradually result in splitting of seed coat and emergence of the growing seedling. The radical comes out first and grows downward, and then plumule comes out and grows upward. Due to the continued growth of this seedling, the plumule comes out of the soil, exposed to light and develops its own photosynthetic apparatus.

Extension of the radicle is a turgor-driven process that requires yielding of walls in those cells of the embryonic root axis that lie between the root cap and the base of the hypocotyls. i) Splitting of seed coat may take place either by imbibition pressure or ii) by internal pressure created by the growing primary root or iii) by hydrolytic enzymes which act on cell wall contents of seed coat and digest it (e.g. cellulose and pectinase).

Events associated with germination and post germination phases.



SEED DORMANCY

Seed dormancy is **the physiological state in which a seed does not germinate, even in the presence of favourable environmental conditions. Classification of seed dormancy.**

Seed dormancy broadly classified into two types followed by

I. Primary dormancy: Dormancy which develops when the seed is attached to the plant and exists when first harvested

II. Secondary dormancy: A condition that develops to prevent germination after the seed has been detached from the plant is exposed to specific unfavourable environmental conditions

I. Primary dormancy

A. Exogenous dormancy

- i) Physical: Impermeability of seed coat or pericarp to water, oxygen or light
- ii) Chemical: Inhibitors in pericarp or seed coat
- iii) Mechanical: Mechanical resistance of pericarp on seed coat to embryo growth.

B. Endogenous dormancy

- i) Morphological : Under developed embryo
- ii) Physiological : Physiological inhibiting mechanism

C. Combined dormancy

- (1) Physical and physiological dormancy: Combination of impermeable seed coat and physiological dormancy in a single seed.
- (2) Morpho - physiological dormancy: Combination of under developed embryo with strong physiological inhibiting mechanism

II. Secondary dormancy

- i) Thermodormancy : Temperature sensitive embryo
- ii) Photodormancy : Light sensitive embryo

I. Primary dormancy

A. Exogenous dormancy

1. Physical dormancy

Primary reason for lack of germination is the **impermeability of seed coat** to water. e.g) Greengram, Blackgram, horsegram, cotton, bhendi, bittergourd, desmanthus, subabul, lucerne, sunn hemp, dainch, *Acacia nilotica* and *Albizia lebbek*

2. Chemical dormancy

Seed coats may contain **germination inhibitors** eg., sunflower (*Helianthus annuus*) and coriander. Light sensitive inhibitors are present in intact seeds of *Betula pubescens* and *Betula pendula*.

3. Mechanical dormancy

It is due to the presence of a hard woody fruit wall, usually endocarp but sometimes mesocarp. It is proposed that the endocarp acts as a mechanical obstacle to a germinating embryo. Long periods of cold stratification are required for breaking of dormancy. Eg. *Pterocarpus* spp. *Terminalia* spp.

B. Endogenous (embryo) Dormancy

1. Morphological dormancy

The embryos are small and immature at the time of seed dispersal. They must develop to full size before the seeds can germinate and this development can be induced or accelerated by treatment. *Fraxinus* spp, requires about 4 months to complete development; In case of *Ginkgo biloba* even fertilization of the ovule is said to take place after the seed falls. *Allium cepa* (Onion) and *Gloriosa superba* seeds also exhibit morphological dormancy

2. Physiological dormancy

Physiological inhibiting mechanism of the embryo that prevents radicle emergence. The presence of inhibitors in the dormant embryos eg., ammonia, hydrogen cyanide, ethylene, essential oils, alkaloids, phenols, unsaturated lactones and unsaturated acids (ABA). Treatments like soaking in water breaks dormancy by decreasing endogenous concentration of inhibitor eg. *Corylus avellana*

C. Double dormancy (or) combined dormancy

In some species, seed dormancy is caused by the combined effect of factors associated with the embryo and seed coat. Seed of *Fraxinus excelsior* exhibited deep dormancy, which was caused by combined effect of restriction of oxygen supply to embryo and a requirement for chilling.

II. Secondary dormancy

1. Thermo dormancy

Seeds of some species do not germinate without pretreatment with appropriate temperatures eg) *Fagus*, *Pinus* and *Eucalyptus* spp.,

2. Photo dormancy

Seeds of some species possess fully developed embryos but do not germinate because they are light sensitive. Seeds germinate only when exposed to appropriate light conditions. It is by a biochemical phytochrome mechanism eg) *Spathodea* and some *Eucalyptus*.

DORMANCY BREAKING TREATMENTS

1. Scarification

Seeds can be scarified by scarification treatments to weaken or soften the seed coat and enhance seed imbibition. This method is more applicable to Malvaceae and Leguminaceae group of seeds.

a) Acid scarification: Treat the seeds with commercial grade (95%) H_2SO_4 @ 100 ml /kg for 2-3 minutes, and wash them in cool running water to remove the traces of acid. The duration of treatment may vary depending on the intensity of hard seed coat.

Eg) Blackgram, Greengram, Rose, Tamarind, Subabul, *Acacia* spp. etc.,

b) Mechanical scarification: Seeds can be rubbed on a sand paper or macerated slightly with a help of mechanical scarifier or by puncturing on seed coat with the help of needle or edge clipped using nail cutter so as to increase the moisture absorption by seeds. For sand scarification, mix the seed with sand in the ratio of 2:1 (sand : seed). Place the mixture in mortar and grind with a pestle for few seconds or minutes. Care should be taken to scarify the seed coat alone but not the embryo. Eg. All gourds, Tamarind etc.,

c) Bio-scarification: Pods of *Acacia* are thick and difficult to extract seeds. They are palatable to goats. Feeding them with the pods will lead to action of digestive juices and microorganisms on the seeds in the digestive system leading to aberration of seed coat and breakdown of seed coat impermeability.

Eg. *Acacia* spp pods are fed to penned sheep or goats and seeds are collected from the droppings.

Insects: Termites are important agents in breaking seed coat dormancy. They feed on the seed helps to remove the exocarp of many tropic species eg. Teak drupes are coated with cow

dung mixture and placed near a termite pit. The termite consume the thick pericarp and expose the seeds.

2. Hot water treatments

It is effective in case of leguminous crop seeds. The seeds should be soaked in boiled water for 1-5 minutes (Hot water). Eg. *Acacia* spp. Hedge lucerne, subabul *etc.*,

Note : Hot water treatment for more than 1 minute is injurious to Bengal gram

3. Dry heat treatment

Exposure of dry seeds to increased temperature by placing in an hot air oven for a specific period of time will enable to overcome dormancy in many cereals. Incubation of seeds in 50°C for 7-14 days removes the dormancy of rice seeds.

4. Scorching

It is a method of subjecting the dormant seeds to heat generated by fire. It is done to improve the seed germination by increasing the seed coat permeability to water. Eg. Teak.

5. Stratification treatment

Morpho-physiological dormancy can be overcome by stratification and imbibing the seeds with hormones like gibberellic acid (or) thiourea to overcome the inhibitory effects of ABA.

i. Cold stratification:

Soaking the seeds in several times their volume of cold water at approximately @ 3-5°C for 48 hrs. Drain the water, and then the seeds are to be mixed in moist sand @ 1: 4 ratio and keep it under constant temperature of 3-5°C for 2-3 days to several months depending on the nature of seed. It helps to overcome the physiological dormancy of Hazel nut, evening prime rose, *Abies* spp., *Eucalyptus* spp. pips, kale, kolerabi and mustard.

ii. Warm stratification:

Soaking the seeds in several times their volume of cold water at approximately @ 3-5°C for 48 hrs. Drain the water, and then the seeds are to be mixed in moist sand @ 1: 4 ratio and keep it under constant temperature of 20 - 25°C or alternating temperature of 20 and 30°C for a period varying from 2 to 6 weeks depending on the species. It helps to overcome the morphological dormancy of oil palm.

6. Leaching of metabolites (Inhibitors)

The seeds can be soaked in water for 3 days. But, once in 12 hours fresh water should be changed to avoid fermentation or seeds can be soaked in running water for a day to leach out the inhibitors such as phenolic compounds. Eg. Coriander (Coumarin), Beet root, Sunflower (HCN).

7. Seed treatment with growth regulators /chemicals

If the dormancy is due to the presence of inhibitors, growth regulators can be applied at the low level.

- GA & Cytokinin and kinetin can be used at concentration of 100-1000 ppm to break dormancy.
- GA_3 and KNO_3 are light substituting chemicals. These can be used for light requiring seeds. Eg. Tomato.
- Thiourea can be used for the seeds require both light and chill. Eg. Thiourea 10^{-2} to 10^{-3} molar for lettuce.
- Plant products like strigol (root exudation from striga parasite host plant) can also be used for breaking the seed dormancy.

Advantage and disadvantages of seed dormancy

A. Advantage of seed dormancy

- Storage of seeds is prolonged
- Seeds can pass through adverse conditions
- Prevents *insitu* germination

B. Disadvantage of seed dormancy

- The view point of weed control, seeds that have longterm viability and sporadic germination are difficult to manage
- Dormancy seeds difficult to establish a good stand
- dormancy seeds do not have uniformity germination and difficult to maintain plant population
- Interferes in seed testing procedures

LECTURE 4

CLASSES OF SEED - GENERATION SYSTEM OF SEED MULTIPLICATION IN SUPPLY CHAIN - SEED REPLACEMENT RATE AND VARIETAL REPLACEMENT - SEED MULTIPLICATION RATIO - SEED RENEWAL PERIOD

Classes of seed-generation system of seed multiplication in supply chain

As per Seed Act (1966) and Rules, seeds are multiplied in definite system which is known as **generation system of seed production**. Production of a particular class of seed from specific class of seed up to certified seed stage is known as **Generation system of seed multiplication**. It involves three to four stages of multiplication known as Nucleus seed, Breeder seed, Foundation seed and Certified seed.

Classes of seed

The four generally recognized classes of seeds are: Nucleus seed, Breeder's seed, foundation seed and certified seed under seed certification system. These classes of seeds are defined as follows:

a) **Nucleus seed:** Initial handful of seed obtained from selected individual plants of a particular variety for the purpose of purifying and maintaining that variety. Nucleus seed is produced by the originating plant breeder. Further multiplication is under his own supervision or supervision by a qualified plant breeder. This class of seed constitutes the basis for all further seed production. It is of 100% genetic purity. There is **no tag** issued for this class of seed.

b) **Breeder seed:** It is the progeny of Nucleus seed and which is used for producing next stage i.e., foundation seed. It is produced by the concerned breeder or sponsoring institute usually by State Agrl. Universities or ICAR Institutes. It is of 100% genetic purity. The quality of breeder seed is assured by the monitoring team constituted by the govt. This class of seed is **exempted from seed certification**. The label/tag issued for breeder seed is **golden yellow** in colour.

c) **Foundation seed:** Foundation seed is the progeny of breeder seed or from foundation seed. It is produced from breeder seed and maintained with specific genetic identity and purity. It is produced on govt. farms or by private seed producers. The quality of foundation seed is certified by certification agency. It has genetic purity of above 99.5%. The certification tag or label issued for foundation seed is **white** in colour

d) **Certified seed:** Certified seed shall be the progeny of foundation seed may be the progeny of certified seed provided this reproduction does not exceed three generations beyond foundation seed stage I to maintain specific genetic identity and purity according to standards prescribed. It has genetic purity of above 99%. The label/tag issued for certified seed is **azur blue** in colour.

Apart from these, one more class of seed which does not require certification but requires compulsory labeling is **TFL seed**.

e) TFL seed (Truthful labeled seeds): If farmers or companies produce and sell their seeds by showing quality parameter through their label **without certification**, those seeds are known as **truthful labeled seeds (TFL seeds)**. Government introduced this system in 1960. Truthfully labelled seed is not field certified to assure genetic purity but the seed standards are not lower than the certified class of seed. The responsibility of ensuring seed quality as per the government norms is by the seed producer. The label / tag issued for TFL seed is **Opal green** in colour.

GENERATION SYSTEM OF SEED MULTIPLICATION

The Generation System of Seed Multiplication is followed to accomplish multiplication of seeds upto required quantities without causing contamination in the different generations of seed multiplication so as to prevent serious reduction in yield potential due to genetic deterioration. It is also important to understand the types of contaminants and the basics of seed deterioration due to different types of contaminants that may occur in the seed production plots.

CLASSES OF SEED /GENERATION SYSTEM OF SEED MULTIPLICATION

AGENCY Produced by	CLASS OF SEED	QUALITY CONTROL SYSTEM	
Concerned breeder (or) Sponsoring institution (or) Breeder/Geneticist himself	NUCLEUS SEED (no specified tag) Stage I ↓	Maintenance breeding	Handful or small quantity of seed, Genetic purity-100 % Nucleus seed have highest purity
Concerned breeder (or) Sponsoring institution (or) Breeder himself	BREEDER SEED (Golden yellow tag) Stage II ↓	Breeder seed monitoring team field inspection to check genetic purity -(Grow out test)	Breeder seed is progeny of Nucleus seed , Genetic purity-100 % Breeder seed is costliest seed

State Department of Agrl. State Seed National seeds Corporation (1963) Cooperative agencies, Central and State Seed Corporations, Private sectors	FOUNDATION SEED Stage III (Colour of foundation seed tag is White tag) ↓	Certification Agency. Field inspection to check and minimum required standards of genetic, physical purity and their quality standards.	Foundation seed is Progeny of Breeder seed , Genetic purity-99.5 %
	CERTIFIED SEED (Azure blue tag) Stage IV ↓	“Farmers commercially used kind seed”	Certified seed is Progeny of Foundation seed , Genetic purity-99 %
Truthful labeled seed (Opel green) ←	← FARMERS		

- ✍ **a. Three - Generation model:** Breeder seed - Foundation seed - Certified seed [**Cross pollinated crops**]
- ✍ **b. Four - Generation model:** Breeder seed - Foundation seed (I) Foundation seed (II) - Certified seed [**Self pollinated crops**]
- ✍ **c. Five - Generation model:** Breeder seed - Foundation seed (I)- Foundation seed (II) - Certified seed (I) - Certified seed (II) [**Often Cross pollinated crops**]
- ☆ Breeder seed is **exemption** from seed certification, so **Seed certification is not required** and seed certification agency **do not certify** breeder seed

Registered seed:

- ☆ **Registered seed** is directly produced from foundation seed and tag colour of Registered seed is **Purple colour**
- ☆ Registered seed class is not produce in INDIA

Seed Multiplication Ratio (SMR)

It is nothing but the “**number of seeds to be produced from a single seed when it is sown and harvested**”. The seed multiplication ratio for different crops are as follows.

Crop	Seed multiplication ratio	Crop	Seed multiplication ratio
Wheat	1:20	Colecrops	1:433
Paddy and Maize	1:80(Variety) 1:100 (Hybrids)	Tomato	1:400
Sorghum, Redgram and Bhendi	1:100	Chilies	1:240
Bajra	1:200	Watermelon	1:100
Brinjal	1:450	Pumpkin	1:160
Gram	1:10	Bitter gourd	1:41
Blackgram, greengram, cowpea, horsegram, mothbean	1:40	Bottle gourd	1:99
Potato	1:4	Ridge gourd	1:83
Ground nut	1:8	Cucumber	1:200
Mustard and rape	1:100	French bean	1:9
Soy bean	1:16	Clusterbean	1:50
Sunflower and cotton	1:50	Peas	1:19
Sesame	1:250	Onion	1:171
Safflower and castor	1:60	Radish	1:100
Linseed	1:50	Carrot	1:83
Jute	1:100	Oats	1:15
Mestha	1:40	Black pepper	1:40
Sun hemp	1:30		
Berseem	1:10		
Lucerne	1:25		

The formula for Seed multiplication ratio (SMR) is

$$\text{“SMR=Seed yield /Seed rate”}$$

SEED RENEWAL PERIOD and SEED REPLACEMENT RATE

Continuous usage of farm saved seeds will lead to varietal deterioration and concomitant yield reduction. Replacement of farm saved seeds with certified seeds is mandatory to minimise varietal deterioration and maintain yield potential of a

particular variety. Therefore, it is mandatory to “renew” the farm saved seeds with certified seeds based on the pollination behaviour of the crop.

The suggested ‘seed renewal period’ is

- i) Once in **Four years** for Self pollinated varieties
- ii) Once in **three years** for Cross pollinated varieties
- iii) Every year for Hybrids
- iv)

SEED REPLACEMENT RATE (SRR) is an important factor that influences the agricultural production of the country. Seed replacement rate (SRR) refers to the ‘the percentage of area sown with certified (excluding farm saved seeds) in a particular season for a particular crop’. The existing Seed Replacement Rate (SRR) may be computed by dividing the quantity of quality seed sown in a particular year by the quantity of seed required for the entire area under a crop.

If X is the quantity of quality seed of a crop variety used in a year and Y is the quantity of seed required to cover the total area under the variety then the seed replacement rate (R) may be computed as

$$R = (X/Y) \times 100$$

VARIETAL REPLACEMENT INDEX

(Brennan and Byerlee ,1991)

The varietal replacement Index (VRI) is a measure of the rate of varietal replacement that takes into consideration the age of varieties and their relative sown area:

$$VRI_t = \sum_i P_{it} (t - y_i), \text{ where;}$$

VRI_t = the weighted average age

P_{it} = the proportion of area sown

If the VRI is larger, the varietal replacement will be slower and only older stock of the variety will be planted. Maintenance of optimum Varietal Replacement Rate is important to ensure optimum yield advantage, varietal longevity with respect to disease susceptibility, and other desired characteristics.

LECTURE 5

VARIETAL DETERIORATION OF CROPS- CAUSES AND MAINTENANCE

CAUSES OF VARIETAL DETERIORATION

Genetic purity (trueness to type) of variety can deteriorate due to several factors during production cycles. The best means to insure genetic purity, therefore, would be to avoid the various factors responsible for genetic deterioration, to the maximum possible extent. The important factors that lead to deterioration were listed by Kadam (1942) as follows:

1. Developmental variations
2. Mechanical mixtures
3. Mutations
4. Natural crossing
5. Minor genetic variations
6. Selective influence of diseases
7. The technique of the plant breeder

Among these natural crossing and selective influence of diseases are perhaps the most important reasons for genetic deterioration of varieties during seed production.

1. Developmental variations

When the seed crops are grown in different environments under different soil and fertility conditions or different climatic conditions, or under different photoperiods or at different elevations for several consecutive generations, it may lead to developmental variations.

2. Mechanical mixtures

This is the most important source of variety deterioration during seed production. mechanical mixtures may often take place at the time of sowing, if more than one variety is sown is sown with same seed drill; through volunteer plants of the same crop in the seed field; or through different varieties grown in adjacent fields. Two varieties growing along side each other in the field are often mixed somehow during harvesting and threshing operations. Further, the gunny bags, seed bins, elevators etc., are also quite often contaminated with seeds of other varieties.

3. Mutations

This is not at all a serious factor of varietal deterioration. In the majority of the cases it is difficult to identify or detect minor mutations. The mutants such as fatuoids in

oats, 'rabbit ear' in peas should be however, be removed by rouging from seed plots to purify the seeds.

4. Natural crossing

In sexually propagated crops natural crossing is another most important source of varietal deterioration. The extent of varietal contamination, however, depends upon the amount of natural cross-fertilization. The deterioration in varieties due to natural crossing occurs due to the following three reasons.

1. Natural crossing with undesirable types
2. Natural crossing with diseased plants
3. Natural crossing with off-type plants

Natural crossing is a serious threat to genetic purity in cross fertilized plant varieties, however, it is not so in the case of self fertilized plants.

5. Minor genetic variations

Minor genetic variations may still exist even in the varieties appearing phenotypically uniform and homogenous at the time of their release. During later production cycle some of these variations may be lost because of selective elimination by the environment. The loss or gain of certain of these may be sufficient to affect yield in one or the other way. Minor genetic variability may be a constant feature in often cross fertilized crop species. Care should be taken during nucleus and breeder seed production to avoid such problems.

6. Selective influence of diseases

The selective influence of diseases in varietal deterioration is also of considerable importance. New crop varieties often become susceptible to new races of diseases in varietal deterioration is also of considerable importance. New crop varieties often become parasites and are out of seed programmes. Similarly the vegetatively propagated stocks deteriorate fast if infected by viral, faungal and bacterial diseases. Therefore, producing seeds under disease free condition is very important.

7. Techniques of plant breeder

In certain instances, serious instabilities may occur in varieties due to cytogenetical irregularities which were not properly assessed in the new varieties, prior to their release.

Premature release of varieties, still segregating for resistance and susceptibility to diseases or other factors may also be important in the deterioration of varieties.

Maintenance of genetic purity during seed production

The important safeguards for maintaining genetic purity during seed production are

1. Control of seed source

Four classes of seeds, namely breeder's, foundation, registered and certified are generally recognized in seed certification. The use of seeds of an appropriate class and from an approved source is necessary for raising a seed crop.

2. Preceding crop requirements

The same kind of crop must not have been grown in the preceding one or two seasons (depending on the crop) unless the crop that was grown was of same variety and was certified for the varietal purity. ‘

3. Isolation Distance

Isolation of seed crops from various sources of contamination listed below is very important

- a) To avoid contamination due to natural crossing with other varieties grown alongside and offtypes present in the seed field
- b) To avoid contamination due to mechanical mixtures at the time of sowing , harvesting , threshing , processing and handling of seeds
- c) To avoid contamination due to seed borne diseases from nearby fields

4. Roguing of seed fields

Off types: Plants differing in their characteristics from those of the seed variety

Rouge: Plants variable in their characteristics from those of the seed variety

Main reasons for off types are i) plants may arise due to presence of some recessive genes in heterozygous conditions at the time of release of varieties. ii) The volunteer plants arising from accidentally planted seed or from seed produced by earlier crops.

Off types individual plants should be rogued out of the seed production fields before pollination occurs. Off types are potent source of genetic contamination. Removal of such plants is called ‘**roguing**’.

5. Seed Certification

The principle objective of seed certification is to maintain and make available crop seeds which are of high seed quality.

To accomplish these purpose, qualified and well experienced personnel of seed certification agencies carry out field inspections at appropriate stages of crop growth. In addition to inspections, seed certification agencies also lay down the field and seed standards to which the seed crop and seed lot respectively must conform to get approval as certified seed.

6. Grow – out tests

Varieties being grown for seed production should periodically be tested for genetic purity by grow-out tests, to make sure that they are being maintained in their true form. Grow – out tests mainly doing for “genetic purity assessment”

LECTURE 6

PRINCIPLES OF SEED PRODUCTION - FACTORS AFFECTING SEED PRODUCTION - PHYSICAL AND GENETIC CONTAMINANTS.

A. Genetic principles in seed production

Pollination behavior of plants

The pollination behavior aids in seed setting and deciding the extent of genetic purity of the seed. The pollination behavior of crops is broadly classified into three as follows.

Self pollinated	Often cross pollinated	Cross pollinated
(0 - 5 % cross pollination)	(6 - 50 % cross pollination)	(> 50 % cross pollination)
Rice, Wheat, Barley, Oats, Blackgram, Greengram, Bengalgram, Soybean, Lablab, Cluster bean, Pea, Groundnut, Tomato, Potato, Jute	Sorghum, Redgram, Sesame, Cotton, Brinjal, Bhendi, Chillies, Limabean, Sweet Pepper, Lucern	Maize, Pearl Millet, Sunflower, Castor, Rape, Mustard, Onion, Cabbage, Cauliflower, Cucurbits, Beet Root, Radish, Turnip, Sugarbeat, Amaranthus,

I. Autogamy

Transfer of pollengrains from the anther to the stigma of same flower is known as autogamy or selfpollination.

There are several mechanisms which promote autogamy.

1. Bisexuality: Presence of male and female organs in the same flower is known as bisexuality.

2. Homogamy: Maturation of anthers and stigma of a flower at the same time is called homogamy.

3. Cleistogamy: When **pollination and fertilization occur in unopened flower bud, it is known as cleistogamy**. Cleistogamy has been reported in some varieties of wheat, barley, oats, Bengal gram and several other grass species.

4. Chasmogamy: **Anthesis or pollination takes place after the opening of flowers it is known as Chasmogamy**. This also promotes selfpollination and is found in crops like wheat, barley, rice and oats.

5. Position of Anthers: In some species, stigma is surrounded by anthers in such a way that selfpollination is ensured. Such situation is found in tomato and brinjal. In some legumes, the stamens and stigma are enclosed by the petals in such a way that self pollination is ensured. Examples are greengram, blackgram, soybean, chickpea and pea.

II. Allogamy

Transfer of pollen grains from the anther of one plant to the stigma of another plant is called **allogamy or cross pollination**.

Mechanisms promotes allogamy

1.Dicliny:It refers to unisexual flowers. This is of two types: viz. i) monoecy and ii) dioecy. When **male and female flowers are separate but present in the same plants, it is known as monoecy**. In some crops, the male and female flowers are present in the same inflorescence such as in mango, castor and banana. In some cases, they are on separate inflorescence as in maize. Other examples are cucurbits, grapes, strawberry, cassava and rubber. When **staminate and pistillate flowers are present on different plants, it is called dioecy**. It includes papaya, date palm, spinach, hemp and asparagus.

2.Dichogamy: It refers to maturation of anthers and stigma of the same flowers at different times. Dichogamy promotes cross pollination even in the hermaphrodite species. Dichogamy is of two types: viz. i) protogyny and ii) protandry. When pistil matures before anthers, it is called protogyny such as in pearl millet. When anthers mature before pistil, it is known as protandry. It is found in maize, sugar beet and several other species.

3.Heterostyly:When styles and filaments in a flower are of different lengths, it is called heterostyly. It promotes cross pollination eg (Brinjal, linseed).

4.Herkogamy: Hinderance to self-pollination due to some physical barriers such as presence of hyaline membrane around the anther is known as herkogamy. Such membrane does not allow the dehiscence of pollen and prevents self-pollination such as in alfalfa.

5.Self incompatibility: The inability of fertile pollens to fertilize the same flower is referred to as self-incompatibility. It prevents self-pollination and promotes cross pollination. Self incompatibility is found in several crop species like Brassica, Radish, Nicotiana and many grass species. It is of two types sporophytic and gametophytic.

6.Male sterility:In some species, the pollen grains are nonfunctional. Such condition is known as male sterility. It prevents self-pollination and promotes cross-pollination. It is of three types: viz. genetic, cytoplasmic and cytoplasmic genetic. It is a useful tool in hybrid seed production.

Pollination behaviour and methods of seed production

a. Seed production in varieties

Raise the crop under isolation and allow the seed to set either by self-pollination in self-pollinated crops through bagging (selfing) or by open pollination in cross pollinated and often cross pollinated crops or by sib-mating in inbreds.

b. Seed production in hybrids

A hybrid is the first generation progeny of a cross between two genetically different identical inbred lines. The individual lines are known as inbreds. Ineffective cross-pollination is

a major constraint in commercial production of hybrid seeds in many crop species. In such crops manual pollination is the way to produce hybrid seeds. Eg. Cotton. In the case of rice, it is highly self-pollinated crop and cross-pollination is a big problem. The increased vigour, uniformity and yield of F1 hybrid have been exploited in most crops where the pollination system allows for convenient and economic cross hybridization. This system is called breeding tool in exploiting the hybrid vigour.

The term **heterosis** in Greek means – *hetero* – different, *osis*- condition was coined by **Skull (1914)**. Heterosis or hybrid vigour refers to the increase or decrease in vigour of off-spring (hybrid) over parents, usually, these two terms are used synonymous. However, *hybrid vigour* refers to increased vigour, while, *heterosis* refers to both increased and decreased vigour. Superiority of the hybrid as ‘hybrid vigour’ and the mechanism by which this superiority is developed is heterosis (Rangaswamy, 1998).

Breeding tools utilized for hybrid seed production

Commercial hybrid seed production demands crossing technique which is easy and also economic to maintain parental lines. Availability of **suitable pollination control system** is one of the key factors determining the successful commercial exploitation of heterosis in any crop species. These techniques are specific to crop floral biology and flowering behaviour. These techniques have their own advantages and disadvantages. Based on the crop behaviour and crossing technique have been adapted for production of hybrid seeds commercially. Only few crossing mechanisms have been adopted for commercial hybrid seed production they are;

Crossing mechanisms

1. Manual modification
2. Self-incompatibility
3. Male sterility

I. MANUAL MODIFICATION

Hybrid seeds are also produced manually by modifying the plant structure by removal of male organ from female plant before anthesis. The male part in the female flower is removed and the pollen grains are collected from the male parent and dusted on the stigma of the emasculated flower. The removal of male part in female plant is called as Emasculation in cotton, detaselling in maize and collection and dusting of pollen from desirable male plant is called as hand pollination or dusting. This system is possible only when the male and female parts of a single flower or plants are separate. The available techniques are,

a. Emasculation (cotton, tomato, bhendi)

This is being adopted in bisexual perfect flowers where the removal of male parts from a flower is known as “Emasculation”. By removing the anther column / or male part from female line, the sterility of female line is created and is dusted with the pollen of desired male parent.

b. Detasseling in maize

This is possible as maize is monoecious and removal of male organ (tassel) is possible before flowering. Here the male sterility is created by manual removal of the tassel and crossed with desired male parent.

II. SELF INCOMPATIBILITY:

Self-incompatibility is a mechanism which avoids self-fertilization through recognition of self pollen in or on stigma on the female pistil. But when pollen from other plant carried by wind or insects are accepted and sets seeds. Self-incompatibility will prevent self-pollination (inbreeding) and promotes crosspollination (out breeding) and creates genetic variability. Self-incompatibility is seen in hermaphrodite and homomorphic flowers. Self-incompatibility is a widespread mechanism in flowering plants that prevents inbreeding and promotes outcrossing.

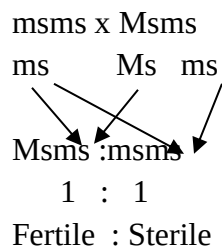
III. MALE STERILITY

In early days occurrence of male sterility was considered as undesirable and had no breeding utility. Later, it became as asset for hybrid seed production. Male sterility was first utilized for hybrid seed production of **sorghum and onion**. Depending on the origin and nature of control, different types of male sterility was reported.

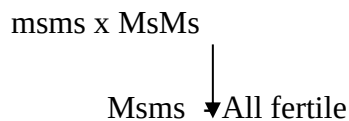
a. Genetic male sterility (GMS)

Also known as nuclear male sterility as it is controlled by nuclear genes. Genetic male sterility is governed by a single recessive gene “msms”. Male sterile lines may be maintained by crossing it with heterozygous (Msms) male fertile plants. Such mating produces 1: 1 male sterile and male fertile plants.

Maintenance of male sterility

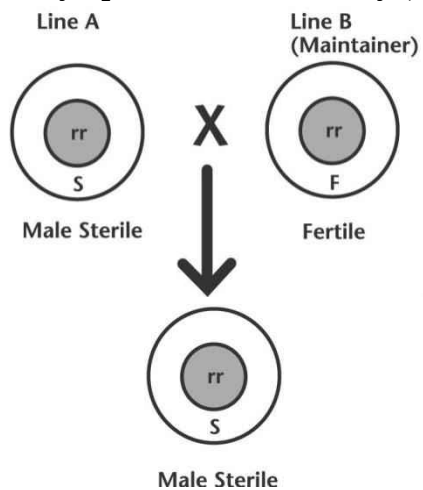


Hybrid seed production



For hybrid seed production, progeny from $msms \times Msms$ crosses are used as female and are interplanted with a homozygous male fertile ($MsMs$) pollinator. The female line would contain both male sterile and male fertile plants, and later male fertile plants must be removed before shedding pollen. Removal of fertile plants is the limiting factor and laborious process. Even though it occurs in rice, barley, cotton and sunflower because of this problem it could not be utilized. It is practiced in redgram hybrid seed production.

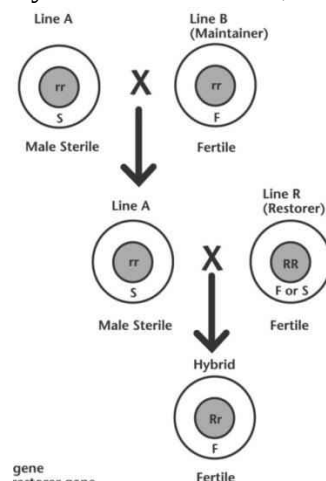
b. Cytoplasmic male sterility (CMS)



This type of male sterility is determined by the cytoplasm and progeny of such male sterile lines would always be male sterile. CMS may be utilized for producing hybrid seeds in certain ornamental sps or in sps where vegetative part is of economic values. But, in those crop plants where seed is the economic part, it is of no use because the hybrid progeny would be male sterile.

c. Cytoplasmic genetic male sterility (CGMS)

It is a result of interactions between male sterility inducing cytoplasm and nuclear fertility restorer gene (s). System consists of A, B and R lines.



A- line - homozygous recessive cytoplasm

B- line - normal cytoplasm

R- line - homozygous dominant nuclear gene (restorer)

The plant would be male sterile in the presence of male sterile cytoplasm if the nuclear genotype were rr. But would be male fertile, if the nucleus were Rr or RR. The fertility restorer gene R is dominant and this gene restores male fertility in male sterile lines hence, it is known as restorer gene.

d. Chemical Hybridising Agents (CHA)

These are the chemicals having gametocidal properties upon spraying on plants at appropriate dosage and crop growth stages, male sterility is created in targeted lines.

Commonly used CHA

<i>Chemicals</i>	<i>Crops</i>
1. Arsenical (Zinc and sodium methyl arsenates)	Rice, rape
2. Dalapon (Sodium 2, 2 –dichloro propionate)	Cotton, pearl millet, Wheat, linseed
3. Ethepon (2-chloromethyl phosphonic acid)	Maize, wheat, barley, oats
4. Sodium alpha–beta dichloroisobutyrate	Sunflower
5. GA ₃ Gibberellic acid	Maize
6. 1-0-methyl threonine	Cotton
7. Maleic Hydroxide(1,2- dichloropyridazine 3, 6-dione)	Sorghum

Commercial hybrids produced in some of the crop species

Inter specific hybrids: Involving parents from entirely different species of a crop.

Cotton – Varalaxmi	Laxmi (G.hirsutum)	x	SB 289 E (G.barbadense)
Cotton – DCH 32 or Jayalaxmi	DS 28 (G.hirsutum)	x	SB (YF) 425 (G.barbadense)

Intraspecific hybrids: Involving parents belong to the same species but between two varieties

Eg., Sorghum – CSH 5, CSH 9, K Tall

Cotton – JKHY 1, Suguna

Types of hybrids

- Single crosshybrid-** (AxB) in case of Bajra, Brinjal, Carrot, Castor, Chillies, Cotton, Cucurbits, Onion, Sorghum, Tobacco, tomato, Water melon, Wheat
- Double crosshybrid-** (AxB) x (CxD) in case of Maize, Sugarbeet
- Three way crosshybrid-** (AxB) x C in case of sweet maize
- Double Top cross hybrid -** (AxB) x OPV in case of maize

e) **Triple crosshybrid** - (AxB)xCx(DxE) x F in case of cabbage

Agronomic principles of seed production

1. Area selection – select places having the following characteristics.

- Free from volunteer plants, weed plants and other crop plants.
- In the proceeding season the same crop should have not been grown.
- Avoid areas where isolation is a problem.
- Having a well distributed rainfall in the cropping period.
- Avoid areas of very high and low rainfall.
- Prevalence of cool and dry weather is preferable.
- Avoid high temperature and relative humidity areas where germination is poor.
- Avoid heavy windy areas.
- Avoid coastal belts and marshy places and heavy windy areas.
- Avoid areas of endemic diseases and pests.
- Select compact areas for hybrid seed production.
- Available of skilled labors especially in hybrid seed production of crops.
- Near the processing operations with transporting facilities and marketing facilities.

2. Selection of land

- Select well leveled field for uniform maturity.
- Select nearest to irrigation sources
- Select fertile fields.
- Avoid weedy fields particularly the presence of abnoxious weeds.
- Avoid problem field like alkaline, saline and sodic soils.
- Avoid sick fields.

3. Selection of season

- Proper season for specific kind of crop.
- Off season is better to avoid isolation problem.

4. Selection of crop and varieties

- The variety should be adapted to the agro climatic conditions of the region.
- The variety should really be a high yielder.
- The variety should be a popular one.
- Select varieties having wide adaptability.
- Select photo and thermal insensitive varieties drought resistant, pest and disease resistant one.

- Should possess other desirable attributes namely, earliness of the crop, grain quality etc.

5. Seeds and sowing

The seed used for raising a seed crop should be of known purity, appropriate class (farmers can produce certified seed but not any class of seed. Breeder Seed can be produced only by the university and Foundation Seed by State Seed Farm / and invariably obtained from an authorized official agency while purchases of seed the following should be carefully examined.

♣ For	B.S	-	N.S	are the basic material for seed production
	F.S	-	B.S	
	C.S	-	F.S	
	Commercial	-	C.S	

- ♣ Other details on the tag – agency, purity, germination.
- ♣ Validity period.
- ♣ Verify that all the bags of the same variety.

6. Preparation of land

The land for the seed crop must be prepared well. Good land preparation helps in improved germination deep ploughing results in destruction of potential weeds. It also aids in water management and good uniform irrigation. Perfect leveling is very important well fertile, avoid sick fields, alkali and sodic soils.

7. Isolation of seed crops

Act of keeping the seed parent from the contamination both genetic and physical is called seed crop isolation. The seed crop has to be isolated from possible genetic physical contaminants from all the four sides either by space, time or barrier.

8. Pre-sowing treatment

Seeds should be appropriately treated. Some of the important pre-sowing seed treatments are,

- ♣ Seed treatment to break dormancy
- ♣ Seed hardening with different chemicals
- ♣ Seed pelleting with macro and micro nutrients
- ♣ Seed protection treatment (fungicide / pesticide)
- ♣ Bio-inoculants (Bio-fertiliser treatment)

9. Seed rate / Transplanting and Line sowing / Transplanting

Seed rate should be based on seed lot viability and vigour. If not there will be lot of gaps in the field indirect sown crop or seedlings may not be sufficient to cover the targeted area to transplant or if high the seed will be wasted. Line sowing is advisable for seed crop adopting correct spacing then only we can achieve required population producing equal opportunity to each plant to develop and mature which is not possible in broadcasted crop.

10. Time of planting

Should be in the proper time.

11. Spacing and sowing

Comparatively wider spacing is required than grain crop.

12. Depth of sowing

Usually double the diameter of the seed.

13. Rogueing spacing

B.S	-	double row planting
F.S	-	30 cm for every 4 feet
C.S	-	30 cm for every 8 feet

14. Border rows

The recommended number of rows of male parental lines to be grown on all the sides of the hybrid seed fields. This may prevent the foreign pollen contamination to some extent and supplement pollen supply to female parents.

15. Planting ratio

It is ratio of number of female rows to male rows in a hybrid seed production plot.

Eg., Rice hybrid seed production 8 : 2 (8 female rows : 2 male rows)

Sorghum hybrid seed production 4 : 2 (4 female rows : 2 male rows)

Number of female rows in any cross combinations will depend upon the pollen production capacity of the pollinator (male line).

16. Live markers

These are the identification plants grown at both the edges of male rows in a hybrid seed production plot to identify the male rows till the harvest of seed crop. Eg. Daincha, sunhemp etc.,

17. Irrigation

Life irrigation and at critical stages depending upon the crop. Irrigate plot by plot and not to plot to plot.

18. Gap filling

Gap filling within 7-10 days beyond means the maturity will be different and harvesting is difficult.

19. Rogueing

Removal of all types of genetic and physical contaminants viz., offtypes, volunteer plants, pollen shedders, partials, diseased plants, etc., from seed production field is called as roguing. It should be at vegetative, pre-flowering, flowering, pre-harvest and harvest stages based on morphological characters.

20. Fertilizers and supplementary pollination

Split application is useful compared to total application. Basal and top dressing of 1 or 2 or 3 times depending upon crop duration. Supplementary pollination and foliar application of fertilizers and growth regulators helps in proper seed set.

21. Weeding

Either manually or by using chemical means.

22. Plant protection measures

Care should be taken for toxicity, residual effect and lethal dose.

23. Harvest – care should be taken for

- The moisture content.
- Physiological maturity.
- Avoid delayed or premature harvest.
- Once over harvest in agricultural crops.
- Staggered harvest in the horticultural crops.
- Harvesting symptoms according to the different crops.

24. Processing

Threshing, drying moisture content is very important to avoid mechanical damage.

25. Seed storage

The type of container, duration of storage, type of seed, chemical composition, seed treatment, storage structures.

FACTORS AFFECTING QUALITY SEED PRODUCTION

Symptom	Crop	Climatic factor
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Blackening of seed due to Black mould	Cumbu, Sorghum	Rainfall or high humidity at maturation
Tip sterility (Low seed set)	Cumbu	Heavy wind at flowering
Off coloured seed	Pulses	Rainfall at maturation
Mottled seed	Peas	High humidity at maturation
Insitu germination	Groundnut	Rainfall at harvest
Pseudo styled flowers	Brinjal	High temperature at flowering
Insitu germination (tomato)	Tomato	Rainfall at harvest
Decayed fruits Soiled fruits	Vegetables (Tomato chillies, brinjal)	Rainfall at maturation
Off coloured seed	Bhendi	Rainfall at maturation

Physical and genetic contaminants

Physical contaminants: Inseparable other crop plants, objectionable weed plants and designated diseased plants.

a. Inseparable other crop seeds

These are plants or different crops which have seed similar to seed crop

Crop	Inseparable crop plants
Wheat	Barely, oats, gram
Barely	Oats, gram, wheat
Oats	Barely, gram, wheat

b. Objectionable weed seeds

These are weeds

1. Whose seeds are difficult to separate once mixed
2. Which are poisonous
3. Which have smothering effect on the main crop
4. Which are difficult to eradicate once established

S.No	Crop	Common name of the weed	Botanical name
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1	Paddy	Wild rice	<i>Oryza sativa var fatua</i>
2	Wheat	Wild morning glory	<i>Convolvulus arvensis</i>
3	Sunflower	Wild sunflower	<i>Helianthus spp</i>
4	Bhendi	Wild okra	<i>Abelmaschus spp</i>
5	Rape and mustard	Mexican prickly poppy	<i>Argemone mexicana</i>
6	Lucerne	Dodder	<i>Cuscuta spp</i>

c. Designated diseases

S.No	Crop	Designated diseases	Botanical name
1	Wheat	Loose smut	<i>Ustilago tritici</i>
2	Sorghum	Grain smut	<i>Sphacelotheca sorghii</i>
3	Cumbu	Grain smut Ergot	<i>Tolyposporium penicillariae</i> <i>Claviceps microcephala</i>
4	Cowpea	Anthrachnose	<i>Colletotricum sp</i>
5	Sesame	Leaf spot	<i>Alternaria sp</i>
6	Rape and mustard	Alternaria blight	<i>Alternaria sp</i>
7	Brinjal	Little leaf	<i>Datura virus</i>
8	Chillies	Leaf blight	<i>Alternaria solani</i>
9	Bhendi	Yellow vein mosaic	
10	Tomato	Early blight Leaf spot	<i>Alternaria solani</i> <i>Xanthomonas sp</i>
11	Potato	Brown rot	<i>Pseudomonas solanacearum</i>
12	Cucurbits	Mosaic	<i>Cucumis virus</i>

Genetical contaminants: Off-types, pollen shedders and shedding tassels.

a. Off type -Plant that differs in morphological characters from the rest of the population of a crop variety. Off type may belong to same species or different species of a given variety. Plants of a different variety are also included under off types.

b. Pollen shedders –presence of 'B' line in 'A' line.

c. Shedding tassel

During detasselling some of the tassel, may remain inside the boot leaf due to **improper removal of tassel**. This tassel may emerge in the subsequent days and may shed pollen and cause genetic contamination. Hence detasselling should be perfect without shedding tassel.

LECTURE 7

SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF RICE (*Oryza sativa*)

Interesting Facts	Rice originated at least 130 million years ago as wild grass. It is suggested that the word ‘rice’ could have been derived from the Tamil word <i>arisi</i> . The two major varieties grown worldwide today are <i>Oryza sativa indica</i> and <i>Oryza sativa Japonica</i> .The domesticated <i>indica</i> subspecies currently appears to be a product of the introgression of favourable alleles from <i>japonica</i> .												
Floral Biology	Rice inflorescence is called as panicle. Panicle emergence in 4-5 days after boot leaf exertion. Flowering proceeds from tip of the branches and completes within 6-8 days. Basipetal nature of anthesis. Time of anthesis is between 7- 10 AM. Receptivity of stigma is for 3 days. Duration of pollen viability is 10 min.												
Pollination	Self pollination (Cross pollination is upto 4 %)												
Methods of seed production	<table><tr><td>Varieties</td><td>Open pollination of plants raised in isolation</td></tr><tr><td>Hybrids</td><td>Cross pollination of male and female plants raised in isolation</td></tr></table>			Varieties	Open pollination of plants raised in isolation	Hybrids	Cross pollination of male and female plants raised in isolation						
Varieties	Open pollination of plants raised in isolation												
Hybrids	Cross pollination of male and female plants raised in isolation												
Land Requirement	The land should be free from volunteer plants (crop of previous season growing in the planting season). The same crop or other varieties of the same crop should not have been grown in the previous season. If it is the same crop (previous), it should have been of the same variety that has been certified. This selection is highly important for maintenance of genetic purity. Adequate irrigation and drainage facilities should be available and the problem soils are not suitable for seed production.												
Isolation	<table><tr><td>Foundation seed</td><td>3 m</td></tr><tr><td>Certified seed</td><td>3m</td></tr></table>			Foundation seed	3 m	Certified seed	3m						
Foundation seed	3 m												
Certified seed	3m												
NURSERY													
Seed rate	<table><tr><th>Varieties</th><th>Seed rate</th></tr><tr><td>Short duration</td><td>60kg ha⁻¹</td></tr><tr><td>Medium duration</td><td>40kg ha⁻¹</td></tr><tr><td>Long duration</td><td>30kg ha⁻¹</td></tr><tr><td>SRI</td><td>7- 8 kg/ha</td></tr></table>	Varieties	Seed rate	Short duration	60kg ha ⁻¹	Medium duration	40kg ha ⁻¹	Long duration	30kg ha ⁻¹	SRI	7- 8 kg/ha		
Varieties	Seed rate												
Short duration	60kg ha ⁻¹												
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Long duration	30kg ha ⁻¹												
SRI	7- 8 kg/ha												

Seed treatment	- I Step : Seed fortification with KCl (1%) for 12 h followed by drying to original seed moisture content - II Step : Seed protection treatment with Bavistin @ 2 g Kg⁻¹ of seed. - III Step : Biofertilizer seed treatment by overnight soaking in a mixture of <i>Pseudomonasfluorescens</i>@ 10 g Kg⁻¹ in 2 l of water, (OR) <i>Azospirillum</i> @ 600 g ha⁻¹ seeds(OR)<i>Phosphobacteria</i> @ 600 g ha⁻¹ seeds, mixed with sufficient quantity of water. - Later, seeds are shade dried for 30 minutes, loosely packed in wet gunny bag and placed in darkness for 24 hours to facilitate sprouting.											
Method of sowing	- Nursery beds are formed in well puddled field. - For planting one hectare, a nursery area of 800 m² is required. - The pre germinated seeds are broad casted uniformly over the bed with thin a film of standing water and allowed them to settle in the soil. - Drain off excess water.											
Transplanting	Age of transplanting <table><tr><th>Varieties</th><th>Age of seedlings</th></tr><tr><td>Short duration</td><td>18-22 days</td></tr><tr><td>Medium duration</td><td>25-30 days</td></tr><tr><td>Long duration</td><td>35-40 days</td></tr></table> Seedlings are to be pulled out without damaging the roots. Transplant two to three seedlings per hill at 2-3 cm depth.				Varieties	Age of seedlings	Short duration	18-22 days	Medium duration	25-30 days	Long duration	35-40 days
Varieties	Age of seedlings											
Short duration	18-22 days											
Medium duration	25-30 days											
Long duration	35-40 days											
	MAIN FIELD											
Land Preparation	Land is puddled repeatedly to obtain fine tilth. Maintain water to a depth of 2.5 cm											
Spacing	- Varieties - Short duration - Medium duration - Long duration	- Spacing (cm) 20 x 10 20 x 15 20 x 20										

Nutrient management	FYM -12.5 tonns ha⁻¹.		
		Dose (Kg ha ⁻¹)	
		N	P₂ O₅
	Short duration	125	40
	Medium & Long duration	150	50
Roguing by Seed producer	The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering.		
	First rouging	Vegetative stage	
	Second rouging	Flowering stage	
	Third rouging	Preharvest stage	
Field Inspection by Seed certification officer	Two inspections		
	Inspection	Growth Stage	Parameters
	I & II	Between flowering to maturity	Off type plants not conforming to varietal purity
Field Standards		Foundation	Certified
	Off-types	0.05	0.20
	Other crop plants	-	-
	Objectionable weed plants (Wild rice)	0.01	0.02
Harvest & Seed Extraction	Harvestable maturity : 90 % of seeds turn golden yellow with 17 -20 % mc. Manual / Mechanical threshing is done when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. Seeds are dried to optimum moisture content ie., 12 %		
Seed Yield	2.5 to 3 mt ha ⁻¹		

HYBRIDRICE

Tool/Hybrid techniques	Cytoplasmic genetic male sterility (CGMS) Three line breeding system A- Male sterile line (CGMS line) (Female Parent) B –Maintainer line (Male Parent) R- Restorer line (Male Parent)
Climate & Soil	Daily mean temperature of 24-30°C & 70-80 % RH. 8-10°C difference between day and night temperature. Cross pollination through wind is the rule therefore, adequate wind flow is a must.
Problems in Hybrid rice seed production	1. Nicking (ensuring synchronization of flowering) Hybrid rice seed production involves two different varieties as male and female parents, hence problem is faced in synchronization of flowering in both parents. 2. Inadequate pollen supply: Flowering period of male (6-8 days) and female (8-10 days) parent differs. Therefore, pollen supply is not available for the entire flowering period of female parent. These problems are addressed through the following Management Technologies 1. Staggered sowing of male and female parent 2. 2-3 staggered sowing of male parent 3. Appropriate planting methods 4. Nutrient management 5. Water management 3. Panicle exertion : Wild abortive based CMS lines have imperfect panicle exertion ie., 10-15 % spikelets remain enclosed within flag leaf . Therefore outcrossing is hindered. <i>Management Technology adopted:</i> 1. Gibberellic acid spray @ 50 Kg ha⁻¹ 4. Outcrossing Rice is a self pollinated crop (0.3-3 % out crossing). But out crossing aided by wind is the rule for hybrid seed production. <i>Management Technology adopted:</i> 1. Supplementary pollination

Method of seed production	• Three steps		
	A line (CMS) multiplication		A x B line
	B and R line multiplication		Open pollination
	Hybrid seed production		A x R line
Steps in rice hybrid seed production			
	Foundation seed		A, B & R line multiplication
	Certified (Hybrid) seed		A x R line
Isolation	Distance isolation		
	Foundation seed		200 m
	Certified seed		100 m
NURSERY			
Nursery preparation	Prepare two separate nurseries for male and female parents. Form separate irrigation channels for male and female parents to avoid genetic contamination.		
Seed rate	Maintain 20 cm distance between female and male block.		
	Female line	20 kg ha ⁻¹	
	Male line	10 kg ha ⁻¹	
Method of sowing	Two separate nurseries for Female (A) and male (R) parents		
	Sparse sowing @ 25 g / m ² . Staggered sowing of female and male line and within male line also.		
Special Operation I	Staggering of female and male line		
	Sowing of male parent is taken up in three consecutive dates, at an interval of 3-4 days to attain synchronization of heading/ flowering.		
MAIN FIELD			
Spacing	Male Parent (R line)		15 x 15 cm
	Female Parent (A line)		15 x 15 cm
	Between Female block and Male block		20 cm

Transplanting	Simultaneous planting of female and male seedlings. Row direction – perpendicular to wind direction. Age of transplanting 20- 25 days 2-3 seedlings per hill at 2-3 cm depth. Planting ratio @ 8:2 (A: R) Border rows – Four rows (Male lines are used for border rows) Advantage of border rows: to supply of additional pollen and to prevent the foreign pollen contamination
Special Operation II	Transplanting of R line Transplanting of R line is completed first. Pull out the seedlings of three different ages, planted in three consecutive dates. Mix the seedlings thoroughly be spreading over one another. Transplant the mixed seedlings in two paired rows @ 15 cm spacing. Leave 145 cm wide block for transplanting A line and continue planting of R line. Transplanting of A line Transplant 8 rows of A line in the 45 cm blocks , in between the 2 rows of R line. Inter and intra row spacing is 15 x 15 cm Leave 20 cm spacing between A line block and R line rows
Special Operation III	Prediction of heading date • It helps to plan for manipulation of flowering, for attaining synchronization. • Observation should be made 30 days prior to flowering , at 3 days interval in both parents. Method • Select the main tiller (longest tiller) and cut at the base where stem and root intercept • Make a longitudinal slit, just above the nodal portion. • Use a magnifying lens and observe the presence of young panicle. • If the panicle is very small <i>ie.</i>, 1st - 3rd stage of development (totally 8 stages), synchronization of flowering will not be attained.
Special Operation IV	Adjustment of flowering in late and early parent Nutrient management • <i>Early Parent</i>: Top dress with Urea @ 35 Kg ha⁻¹ [OR] spray 20 Kg dissolved in 500 litof water at 70 DAS (4th stage of panicle initiation) • <i>Late parent</i> : Spray 2 % DAP • This will help to correct a difference of 4-5 days in flowering between both parents. Water management

	• R lines are more sensitive to water than A line (CMS line). • If R line found to be early in flowering, the field may be drained to maintain a semi dry ecosystem. • If the R line is found to be late in flowering, the field may be irrigated to a higher standing water level . • This will help to correct a difference of 3-4 days in flowering between both parents.												
Special Operation V	Gibberellic acid application:50g/ha Gibberellic acid is known to stimulate the cell elongation. Therefore, it is sprayed over the rice crop to 1. Enhance the panicle exertion. 2. Extend the duration of floret opening. 3. Increase the stigma exertion and stigma receptivity. 4. Improve the growth rate of secondary and tertiary tillers and increase their productivity. Details of application • Optimum dosage is 45-60 g ha⁻¹ • Optimum stage is 5-10 % panicle emergence stage.												
Special Operation VI	Supplementary Pollination Methods i) Rope pulling: Passing of rope across the plant population gently. ii) Rod driving : Shaking of pollen parent with the help of two bamboo sticks. Supplementary pollination should be done at the peak of anthesis ie., when 30 – 40 % of the spikelets are open. It is carried out for 5 times at 30 minutes interval daily during anthesis period, both in the morning (10 - 12 AM) and evening (2 - 4 PM) for 7 - 10 days until no pollen remains on the male parent.												
Rougeing by Seed producer	Three rougeings between ‘heading’ and ‘dough’ stage. All offtypes in both male and female rows. Remove the pollen shedders (maintainer line) and semi sterile plants present in CMS or A line												
Field Inspection by Seed certification officer	• Four inspections <table><tr><th>Inspection</th><th>Growth Stage</th><th>Parameters</th></tr><tr><td>I</td><td>Before flowering</td><td>Isolation, volunteer plants, outcrosses, planting ratio, border rows, etc.,</td></tr><tr><td>II & III</td><td>Flowering</td><td>Isolation, offtypes, pollen shedders etc.,</td></tr><tr><td>IV</td><td>Maturity & before</td><td>Offtypes</td></tr></table>	Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, outcrosses, planting ratio, border rows, etc.,	II & III	Flowering	Isolation, offtypes, pollen shedders etc.,	IV	Maturity & before	Offtypes
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IV	Maturity & before	Offtypes											

	harvest		
Field Standards		Foundation	Certified
	Off-types in seed parent	0.05	0.20
	Off-types in pollinator	0.05	0.20
	Pollen shedding ear heads	0.05	0.10
	Objectionable weed plants (Wild rice)	0.01	0.02
Harvest & Seed Extraction	Harvestable maturity : 90 % of seeds turn golden yellow with 17 -20 % moisture content. Male and female parents should be harvested separately. The male plants are harvested first. Admixture of male and female line should be avoided Seeds from female parent (A line) constitute the <i>hybrid seed</i> .		
Seed Yield	5 to 15 quintals ha ⁻¹ .		
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	10/Kg	20/Kg
	Other distinguishable varieties (maximum)	10/kg	20/kg
	Total weed seeds (maximum)	10/Kg	20/Kg
	Objectionable weed seeds	2/Kg	5/Kg
	Germination (minimum)	80 %	80 %
	Moisture (minimum)	13.0 %	13.0 %

LECTURE 8

SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF MAIZE (Zea mays)

Interesting Facts	• The story of corn began in 1492 when Columbus's men discovered this new grain in Cuba. • Different types of maize exists viz., dent, flint, sweet, flour, pop and waxy corn • First hybrid variety of maize in 1961, marking the dawn of Green Revolution in India (Ganga 1, Ganga 101, Ranjit and Deccan)
Origin & Economic importance	• Originally it was grown in the form of a wild grass called <i>teosinte</i> in Central Mexico. • The original wild form has long been extinct. • Maize is the third largest planted crop after wheat and rice. • It is staple human food, feed for livestock, for fermentation and many industrial uses. It is having abundant starch (65%).
Climate & Soil	• Increase in time to tassel as the diurnal variation increase from 0-17°C. • Maximum rate of maize growth is at 30°C. • Higher the solar radiation, higher will be the photosynthesis in maize.
Floral Biology	• Inflorescence is unisexual and monoecious. • Staminate (male) inflorescence is known as tassel, it is terminal in position • Pistillate (female) inflorescence is known as cob, which is produced in a lateral branch. • The style known as 'silk' is receptive throughout the length and at the tip is usually cleft into two branches. Protandry (male organ mature before female organ) • Maize is protrandous in nature. • Pollen shedding begins three days before the silk emerge from the cob. • Pollen is viable for 12-18 hours. Silk remains receptive for 8-10 days • Anthesis continues upto 2 weeks
Pollination	• Cross pollination
Method of seed production	1. Varieties : Open pollination among plants raised in isolation. 2. Synthetics : The lines that combine well among themselves are mixed and sown. Seed set is by open pollination. 3. Composites : Seed set is open pollination among outstanding lines which were not selected based on combining ability 4. Hybrids : Two true breeding strains are developed and used as male

	and female parent.			
Land Requirement	• The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. • The field should be properly leveled with fine tilth. • Well drained sandy red or black cotton soil is suitable.			
Isolation	• Inbreds, varieties, synthetics, composite			
	Foundation seed	400 m		
	Certified seed	200 m		
Time of sowing	• June-July • September – October • January – February			
Seed rate	• 10 kg ha⁻¹			
Seed Treatment	• Treat the seed with Carbendezim or Thiram at 2g / kg of seed. • One day later, treat the seed with 600 g of Azospirillum using rice gruel • Shade dry for 15 minutes.			
Method of sowing	• Ridges and furrows if preferred. • Dibbling at 1/3 rd of ridge from the bottom. • Depth of seeding is 5 to 6 cm.			
Spacing	• 60X25 cm			
Nutrient management	• 80: 60 : 40 kg NPK ha⁻¹. • FYM -12.5 tonns ha⁻¹.			
	Method of application	(Kg ha ⁻¹)		
		Urea	Super phosphate	Muriate of Potash
	Basal	40	75	40
	Top dressing			
	20 DAS	50	-	-
	40 DAS	10	-	35
	• Micronutrient mixture : 12.5 kg ha⁻¹. • Applied over the soil in furrows (bed system) or at two thirds height of ridges (Ridges& furrows system). • Do not incorporate into the soil.			

Water Management	• Immediately after sowing • Life irrigation three days after sowing, • Subsequently, irrigate once in 10 days. • Critical stages for irrigation – crown root initiation, late tillering, late jointing, flowering and milk stage. • During early stage water stagnation should be avoided.																		
Weed management	I. Spray pre- emergence herbicide Atrazine 50 WP @ 500 g ha⁻¹ (in 900 litres of water) on 3rd day after sowing. II. Hand weeding on the 17th or 18th DAS, if herbicide is not applied.																		
Plant Protection	<table border="1"> <tr> <th colspan="2">Pest</th></tr> <tr> <td>Stem borer</td><td> • Sevin 50 percent W.P. @ 0.2 % (4 g/lit of water) • After 20 days – Lindane @ 1 per cent (OR) Spray Quinalphos 25 EC @ 1 lit/ha </td></tr> <tr> <td>Leaf whorles</td><td> • 20 DAS apply Quinalphos 5 G @ 50 WP 15 kg / ha (OR) Carbaryl 4 G @ 20 kg/ha </td></tr> <tr> <td>Shoot bug and Aphid</td><td> • One week after sowing, Metasystox @ 0.25 % (600 l ha⁻¹) • Subsequent sprays may be done at 10 days interval. </td></tr> <tr> <th colspan="2">Diseases</th></tr> <tr> <td>Downey mildew</td><td> • Mancozeb @ 1 kg ha⁻¹ (OR) Metalaxyl 72 wp @ 1 kg/ha. </td></tr> <tr> <td>Leaf Spot</td><td> • Mancozeb @ 1 kg ha⁻¹ </td></tr> <tr> <td>Leaf blight</td><td> • Apply Zinab @ 2.5 Kg ha⁻¹, 3 to 4 times </td></tr> <tr> <td>Bacterial stalk rot</td><td> • Apply bleaching powder @ 3.3 grams in 10 lit of water at the base of the plants </td></tr> </table>	Pest		Stem borer	• Sevin 50 percent W.P. @ 0.2 % (4 g/lit of water) • After 20 days – Lindane @ 1 per cent (OR) Spray Quinalphos 25 EC @ 1 lit/ha	Leaf whorles	• 20 DAS apply Quinalphos 5 G @ 50 WP 15 kg / ha (OR) Carbaryl 4 G @ 20 kg/ha	Shoot bug and Aphid	• One week after sowing, Metasystox @ 0.25 % (600 l ha⁻¹) • Subsequent sprays may be done at 10 days interval.	Diseases		Downey mildew	• Mancozeb @ 1 kg ha⁻¹ (OR) Metalaxyl 72 wp @ 1 kg/ha.	Leaf Spot	• Mancozeb @ 1 kg ha⁻¹	Leaf blight	• Apply Zinab @ 2.5 Kg ha⁻¹, 3 to 4 times	Bacterial stalk rot	• Apply bleaching powder @ 3.3 grams in 10 lit of water at the base of the plants
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Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Three rougings <table><tr><td>First rouging</td><td>Seedling stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr><tr><td>Third rouging</td><td>Preharvest stage</td></tr></table> - Seedling stage to harvesting stage based on the phenotypic characters. - Off types can be identified through stem colour,plant structure, number of leaves, auricles, nodal colour, tassel colour, sheath colour , grain colour etc.	First rouging	Seedling stage	Second rouging	Flowering stage	Third rouging	Preharvest stage		
First rouging	Seedling stage								
Second rouging	Flowering stage								
Third rouging	Preharvest stage								
Field Inspection by Seed certification officer	The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. <table><tr><td>Inspection</td><td>Growth Stage</td><td>Parameters</td></tr><tr><td>I</td><td>Before flowering</td><td rowspan="2">Isolation, volunteer plants, offtypes etc.,</td></tr><tr><td>II</td><td>During flowering</td></tr></table>	Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, offtypes etc.,	II	During flowering
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Field standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Off-types</td><td>1.0</td><td>1.0</td></tr></table>		Foundation	Certified	Off-types	1.0	1.0		
	Foundation	Certified							
Off-types	1.0	1.0							
Harvest	- Physiological maturity - Darkening of silk or drying of husk into yellow / white colour. - Harvest the crop at Physiological maturity. - Seed moisture content will be around 25 per cent. - Dehusking is done manually or using mechanical dehuskers.								
Special operation (Post harvest inspection)	Cob sorting - It is an important operation to maintain genetic purity. - Sorting is done based on row number, kernel colour, kernel size, pith colour and arrangement of seeds Zenia and Metazenia effects: Zenia: Seed coat colour variation in maize kernal Metazenia: Endosperm colour variation in maize kernal due to contamination of foreign pollen on the female parent.								
Seed Extraction	Shelling Seeds are extracted by shelling at a seed moisture content 15-18 per								

	cent. - Shelling is done manually by beating with pliable stick (OR) with maizesheller. - Shelled seeds are dried to 10-12 percent moisture content.																											
Seed yield	2.5 t ha⁻¹																											
Size Grading	18/ 64 “ round perforated sieve																											
Seed Standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Pure seed (minimum)</td><td>98 %</td><td>98 %</td></tr><tr><td>Inert matter (maximum)</td><td>2.0 %</td><td>2.0 %</td></tr><tr><td>Other crop seed (maximum)</td><td>5/Kg</td><td>10/Kg</td></tr><tr><td>Other distinguishable varieties (maximum)</td><td>10/kg</td><td>20/kg</td></tr><tr><td>Weed seeds (maximum)</td><td>None</td><td>None</td></tr><tr><td>Germination (minimum)</td><td>90 %</td><td>90 %</td></tr><tr><td>Moisture (minimum)</td><td>12.0 %</td><td>12.0 %</td></tr><tr><td>For vapour- proof containers (maximum)</td><td>8.0 %</td><td>8.0 %</td></tr></table>		Foundation	Certified	Pure seed (minimum)	98 %	98 %	Inert matter (maximum)	2.0 %	2.0 %	Other crop seed (maximum)	5/Kg	10/Kg	Other distinguishable varieties (maximum)	10/kg	20/kg	Weed seeds (maximum)	None	None	Germination (minimum)	90 %	90 %	Moisture (minimum)	12.0 %	12.0 %	For vapour- proof containers (maximum)	8.0 %	8.0 %
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Seed storage	Seeds can be stored well upto one year in gunny/cloth bags after seed treatment with thiram @4 kg/ha. In moisture vapour proof container the seeds can be stored upto two years																											

HYBRID MAIZE

Tool	Detasselling : - Physical removal of male inflorescence from female parents, to ensure cross pollination with the male parent, raised next to female parent, in specific planting ratio. - This is enabled by monoecious and protandrous nature of maize
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Steps in Maize hybrid seed production				
		Parental seed production		Hybrid
	Type of hybrid	Breeder seed	Foundation seed	Certified seed
	Single cross	A & B	A & B	A x B
	Three way cross	A, B, & C	A x B & C	(AxB) x C
	Double cross hybrid	A,B,C, & D	(AxB) , (C & D)	(A x B) x (C x D)
Isolation	Depends on the varieties grown near by.			
		Breeder Seed (Inbreds)	Foundation class (Single cross)	Certified class (Hybrid)
	Varieties with same kernel colour & texture	400 m	400 m	200 m
	Varieties with different kernel colour & texture	600 m	600 m	300 m
	Same Inbred/Single cross / Hybrids not conforming with varietal purity standards	400 m	400 m	200 m
	Other Single cross / Hybrids with common male parent conforming with varietal purity standards	-	5 m	5 m
	Other Single cross / Hybrids with common male parent not conforming with varietal purity standards	-	400 m	200 m
	Modification of Isolation distance: - In case the kernel colour and texture of contaminant are similar, <i>Isolation distance</i> can be reduced by increasing the number of border rows as prescribed in the '<i>Indian Minimum Seed Certification Standards</i>'. - For each additional border row (upto 16 rows), isolation distance may be decreased by 12.5 m			

	For each increment in 2 ha (up to 16 ha), isolation distance may be decreased by 5 m.																							
Seed rate	<table><tr><td>Female parent</td><td>12 kg ha⁻¹</td></tr><tr><td>Male parent</td><td>4 kg ha⁻¹</td></tr></table>	Female parent	12 kg ha ⁻¹	Male parent	4 kg ha ⁻¹																			
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Special operation I	Planting ratio & Border rows <table><tr><td></td><td>Planting ratio</td><td>Border rows</td></tr><tr><td>Single cross</td><td>4 : 2</td><td>4</td></tr><tr><td>Three way cross</td><td>6 : 2</td><td>4</td></tr><tr><td>Double cross</td><td>6 : 2</td><td>3</td></tr></table>		Planting ratio	Border rows	Single cross	4 : 2	4	Three way cross	6 : 2	4	Double cross	6 : 2	3											
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Method of sowing	- Ridges and furrows is preferred. - Dibbling at 1/3 rd of ridge from the bottom. - Depth of seeding is 4 to 5 cm.																							
Spacing	Male and Female parent : 60 x 25 cm																							
Nutrient management	100: 75 : 75 kg NPK ha⁻¹. - FYM -12.5 tonns ha⁻¹. <table><tr><td rowspan="2">Method of application</td><td colspan="3">(Kg ha⁻¹)</td></tr><tr><td>Urea</td><td>Super phosphate</td><td>Muriate of Potash</td></tr><tr><td>Basal</td><td>40</td><td>75</td><td>40</td></tr><tr><td colspan="4">Top dressing</td></tr><tr><td>20 DAS</td><td>50</td><td>-</td><td>-</td></tr><tr><td>40 DAS</td><td>10</td><td>-</td><td>35</td></tr></table> - Micronutrient mixture : 12.5 kg ha⁻¹. Foliar application : 2% DAP spray (50 % flowering stage) - Applied over the soil in furrows (bed system) or at two thirds height of ridges (Ridges & furrows system). - Do not incorporate into the soil	Method of application	(Kg ha ⁻¹)			Urea	Super phosphate	Muriate of Potash	Basal	40	75	40	Top dressing				20 DAS	50	-	-	40 DAS	10	-	35
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Special operation II	Detasselling • Removal of tassel (male inflorescence) from the female plant. • It is done in female rows alone • It should be removed when the mature tassel is well out of the boot leaf and before the anthers are shed ie., within 1-2 days after emergence. • It is started from immediately after emergence and continued for 14 days. Procedure • Hold the stem with left hand and pull the tassel with right hand, in one stroke. • Take care that no spikelet is left in the plant. • If detasseling is improper, part of the tassel may remain in the plant and result in shedding of pollen. • Such plants are termed as ‘shedding tassel’ • Presence of ‘shedding tassels’ in female rows will lead to genetic contamination. Caution • Detasseling of immature tasseling may lead to reduced yield and contamination. • After pulling out, the tassel should be dropped to ground there itself and buried in soil, because if the tassel is carried through the field, fall of pollen may lead to contamination • Do not remove the leaves as it will lead to yield reduction.				
Rougeing bySeed producer	• The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. <table border="1" data-bbox="583 1392 1307 1488"> <tr> <td>First rouging</td><td>Vegetative stage (30-40 DAS)</td></tr> <tr> <td>Second rouging</td><td>Flowering stage</td></tr> </table> Shedding tassel • Either full or part of tassel which remain in female plants after detasseling, leads to shedding of pollen which may contaminate the genetic purity of the crop.	First rouging	Vegetative stage (30-40 DAS)	Second rouging	Flowering stage
First rouging	Vegetative stage (30-40 DAS)				
Second rouging	Flowering stage				

Field Inspection by Seed certification officer	Four inspections			
	Inspection	Growth Stage	Parameters	
	I	Before flowering	Isolation, volunteer plants, otypes, planting ratio, border rows, etc.,	
	II, III, & IV	During Flowering	Isolation, off types, shedding tassels etc.,	
Field Standards				
		Breeder (Inbreds)	Foundation (Single crosses)	Certified (Hybrids)
	Off-types	0.20	0.02	0.50
	Shedding Tassels in seed parent in one inspection		0.05	1.00
	Shedding Tassels in seed parent in all three inspections	-	1.00	2.0
	Off type plants in seed parent at final inspection	-	-	0.50
Harvest & Seed Extraction	- Physiological maturity: Darkening of silk and drying up of husk to yellow colour (45 days after flowering). - The male parent is harvested first. - Admixture of male and female line should be avoided - Seeds from female parent constitute the hybrid seed. - Cobs sorting: Remove sheath and check for kernel colour, shank colour, diseased cobs, kernel arrangement etc. - Shelling: It is done when the seed moisture content is 15 %. Mechanical cob shellers or manual rubbing against stone is practiced.			
Seed yield	3-3.5 tonnes/ha			
Seed Grading	18/64” round perforated sieve			

Seed Standards		Breeder (Inbreds)	Foundation (Single crosses)	Certified (Hybrids)
	Pure seed (minimum)	98 %	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %	2.0 %
	Other crop seed (maximum)	5/Kg	5/Kg	10/Kg
	Other distinguishable varieties (maximum)	5/kg	5/kg	10/kg
	Weed seeds (maximum)	None	None	None
	Germination (minimum)	80 %	80 %	90 %
	Moisture (minimum)	12.0 %	12.0 %	12.0 %
	For vapour- proof containers (maximum)	8.0 %	8.0 %	8.0 %

Types of Maize Hybrids

1. Single cross hybrid :

- It involves 2 inbreds (A, B)
- It is a cross between two inbreds, used as male and female parent (A x B).

Eg.,

Hybrid	Male parent	Female parent
Co H 1	UMI 29	UMI 51
Co H 2	UMI 810	UMI 90

2. Three way cross hybrid :

- It involves 3 inbreds (A, B, C)
- It is cross between single cross hybrid (A x B) (female parent) and a inbred C (male parent)

Eg.,

Hybrid	Male parent	Female parent
Deccan hybrid	(CM 104 x CM 105)	(CM 202x Cm 201)
Co H 3	(UMI 101 x UMI 130)	(UMI 90 x UMI 285)

3. Double cross hybrid :

- It involves 4 inbreds (A,B,C,D).
- It is a cross between 2 single crosses (A x B) and (C x D)

Eg.,

Hybrid	Male parent	Female parent
Ganga 5	(CM 202 x CM 111)	CM 500

4. Top cross

- It is a cross between (A x B) and composite
- Eg.,

Hybrid	Male parent	Female parent
Ganga safed Ganga 4		

LECTURE 9

**SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF SORGHUM
AND BAJRA PEARL MILLET (*Pennisetum glaucum*)**

Interesting Facts	- The botanical name of Pearl millet has been changed from time to time: most common names are <i>Pennisetum typhoideum</i>, <i>Panicum spicatum</i> and <i>Pennisetum alopecuroides</i>. - Current name is <i>Pennisetum glaucum</i>
Origin & Economic importance	- Centre of origin of pearl millet is Africa - It is staple food in a short period in the relatively dry tracts of the country. It is the most drought tolerant crop among cereals and millets. - It has wider industrial and household utility. It is used as feed, food and raw material in soft drink industry.
Climate & Soil	- The crop has a wide adaptability to day length and, temperature. It requires low annual (40-50 cm) and dry weather. - Light soils of low inherent fertility with good drainage and mild salinity
Floral Biology	- Inflorescence is a terminal and dense contracted panicle or fusi form panicle or false spike . - Pearl millet spikelets bear 2 types of florets, one is bisexual and other one is staminate. Protogyny(Female organ mature before male organ) - Pearl millet is highly protogynous - The spike emerges after 10 weeks of sowing. - The protrusion of styles start from the tip of the head and gradually proceed downwards. - Exserted stigma remains receptive for 12-24 hours. - Anthers usually emerge after the styles are dry. - The anther dehiscence starts from middle of the spike and proceeds upwards and downwards. - Anthesis takes place in two phases. - The first phases involve bisexual florets (upper floret) and the second phase involves staminate flowers (lower floret), which occur after 2-3 days. - Anthesis occurs throughout the day and night with the peak between

	8.00 PM to 2.00 AM.									
Pollination	• Cross pollination									
Crop duration and Mode of propagation	• 90-100 days • Seed									
Method of seed production	<table><tr><td>Varieties</td><td>Open pollination of plants raised in isolation</td></tr><tr><td>Synthetics</td><td>Open pollination of plants raised in isolation</td></tr><tr><td>Composites</td><td>Open pollination of plants raised in isolation</td></tr><tr><td>Hybrids</td><td>Cross pollination of male and female plants raised in isolation</td></tr></table>		Varieties	Open pollination of plants raised in isolation	Synthetics	Open pollination of plants raised in isolation	Composites	Open pollination of plants raised in isolation	Hybrids	Cross pollination of male and female plants raised in isolation
Varieties	Open pollination of plants raised in isolation									
Synthetics	Open pollination of plants raised in isolation									
Composites	Open pollination of plants raised in isolation									
Hybrids	Cross pollination of male and female plants raised in isolation									
	Synthetics : If more than 5 parental lines are combined ,which are having general combining ability e.g. CO 7, ICMS 7703 Composite: 3-5 inbreds with no general combining ability are mixed and multiplied. WCC 75.(ICRISAT).									
Land Requirement	The land should be free from volunteer plants (crop of previous season growing in the planting season). The same crop or other varieties of the same crop should not have been grown in the previous season. If it is the same crop (previous), it should have been of the same variety that has been certified. This selection is highly important for maintenance of genetic purity. Adequate irrigation and drainage facilities should be available and the problem soils are not suitable for seed production.									
Isolation	<table><tr><td>Foundation seed</td><td>400 m</td></tr><tr><td>Certified seed</td><td>200 m</td></tr></table>		Foundation seed	400 m	Certified seed	200 m				
Foundation seed	400 m									
Certified seed	200 m									
NURSERY										
Time of sowing	• July • October to December									
Seed rate	• 8 kg ha⁻¹									
Seed Treatment	• I Step : Seed fortification with KH₂PO₄ (2%) at 1:0.06 ratio for 16 h									

	followed by drying to original seed moisture content • II Step : Seed protection treatment with Bavistin @ 2 g Kg⁻¹ of seed. • III Step : Biofertilizer treatment with <i>Azospirillum</i>@ 600 g ha⁻¹ + Phosphobacteria @ 600 g ha⁻¹ as slurry treatment Removal of Ergot seeds and Sclerotia • Dissolve 1 kg of common salt in 10 litres of water and add the seeds. • Remove the floating ergot and sclerotia affected seeds. • Wash the seeds in fresh water to remove salt traces and surface dry the seeds .
Method of sowing	• The same variety should not have been cultivated in the previous season to avoid volunteer plants. • For planting one hectare, prepare 7.5 cents m² of nursery area • Raised beds of 20 cm height (2.5 m long x 1.25 m wide) • Sow the seeds at 15 cm spacing and 1 to 2 cm depth • Sprinkle 500 Kg of FYM or compost evenly and cover the seed completely manually.
Nutrient management	• Apply 750 kg FYM or compost and incorporate by ploughing
Water Management	• Allow water to enter the channel and rise up in it. • Turn off the water when the raised bed is wet. • Irrigate properly to ensure that cracks does not develop
Plant protection	• Mix 600 g of Carbofuran 3G with 2 kg of moist sand and apply equally over all the nursery beds to manage shoot fly infestation.
Transplanting	• Nursery is flooded with sufficient water. • Seedlings are to be pulled out without damaging the roots • Age of transplanting is 18 DAS
MAIN FIELD	
Land Preparation	• Form Ridges and furrows at 45 cm spacing Transplanted crop • Seedlings are transplanted on ridges and furrows. Direct sown crop • Dibble the seeds at 1/3 rd (1/2) of ridge from the bottom • Depth of seeding is 3- 5 cm.

	Extra seedlings per hill can be transplanted in the gaps @ 20-25 DAS																															
Spacing	45 x 20 cm																															
Nutrient management	- FYM -12.5 mt ha⁻¹. 70: 35 : 35kg NPK ha⁻¹. <table><tr><td rowspan="2">Method of application</td><td colspan="3">(Kg ha⁻¹)</td></tr><tr><td>N</td><td>P₂ O₅</td><td>K₂O</td></tr><tr><td>Basal</td><td>25</td><td>35</td><td>35</td></tr><tr><td colspan="4">Top dressing</td></tr><tr><td>15 DAS</td><td>50</td><td>-</td><td>-</td></tr><tr><td>30 DAS</td><td>25</td><td>-</td><td>-</td></tr><tr><td colspan="4">Foliar application</td></tr><tr><td>Peak tillering</td><td colspan="3">Diammonium phosphate (1%)</td></tr></table> - Apply 12.5 kg ha⁻¹ of Micronutrient mixture (OR) ZnSo₄@ 25 kg ha⁻¹ by mixing with 50 kg of sand.	Method of application	(Kg ha ⁻¹)			N	P ₂ O ₅	K ₂ O	Basal	25	35	35	Top dressing				15 DAS	50	-	-	30 DAS	25	-	-	Foliar application				Peak tillering	Diammonium phosphate (1%)		
Method of application	(Kg ha ⁻¹)																															
	N	P ₂ O ₅	K ₂ O																													
Basal	25	35	35																													
Top dressing																																
15 DAS	50	-	-																													
30 DAS	25	-	-																													
Foliar application																																
Peak tillering	Diammonium phosphate (1%)																															
Special Operation I	Jerking - The act of pulling out early formed earheads from the first tiller is called jerking - This is done to induce physiological changes in the plant which will induce uniform flowering of all tillers - It is done @ 20-25 days after transplanting [OR] 30-40 days after sowing in direct sown crop																															
Water Management	Direct sown crop - Immediately after sowing - Life irrigation three days after sowing - Subsequently, irrigate once in 10 days. Transplanted crop - Transplantation is done in well irrigated field - Subsequently, irrigate once in 10 days. - Critical stages for irrigation – Primordial initiation stage, vegetative stage, milky and maturation stage.																															
Weed management	I. Spray pre- emergence herbicide Atrazine 50 WP @ 500 g ha ⁻¹ mixed in 1000 lit of water at 3 rd day after sowing. II. Hand weeding on the 30 – 35 DAS																															

Plant Protection	Pest								
	White grub & shoot fly	• Apply Prorate 10 G or Thiodemeton 5 % @ 25-50 kg ha⁻¹ , • During sowing, in seed furrows.							
	Blister beetles	• Spray Carbaryl 50% @ 3 kg ha⁻¹ in 600 l of water.							
	Diseases								
	Sugary disease or Ergot	• Spray Carbendazim @ 500 g (OR) Ziram @ 1000 ml (OR) Mancozeb @ 1 kg ha⁻¹ • During 5 - 10% and 50% flowering stage							
	Rust	• Spray Wettable sulphur WP @ 2.5 kg or Mancozeb @ 1 kg ha⁻¹ • Repeat after 10 days if necessary.							
	Downy Mildew or Green ear	• Seed treatment in hot water at 55°C for 10 min. • Seed treatment Metalaxyl @ 6 g kg⁻¹ • Followed by Metalaxyl @ 500 g ha⁻¹ (or) Ridomil Mz WB @ 2 kg ha⁻¹ (or) Mancozeb 1 kg ha⁻¹							
Rouging by Seed producer	• The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. <table><tr><td>First rouging</td><td>Seedling stage</td></tr><tr><td>Second rouging</td><td>Tillering stage</td></tr><tr><td>Third rouging</td><td>Seed formation stage</td></tr></table> • Rouging is done based on leaf colour, leaf waviness, grain colour, ear head shape, size etc., to maintain genetic purity of the crop.			First rouging	Seedling stage	Second rouging	Tillering stage	Third rouging	Seed formation stage
First rouging	Seedling stage								
Second rouging	Tillering stage								
Third rouging	Seed formation stage								
Field Inspection by Seed certification officer	• Three inspections <table><tr><th>Inspection</th><th>Growth Stage</th><th>Parameters</th></tr><tr><td>I</td><td>Vegetative and flowering stages (30 DAS)</td><td>Isolation, volunteer plants, off types, incidence of downy mildew / green ear, ergot, grain smut, etc.,</td></tr></table>			Inspection	Growth Stage	Parameters	I	Vegetative and flowering stages (30 DAS)	Isolation, volunteer plants, off types, incidence of downy mildew / green ear, ergot, grain smut, etc.,
Inspection	Growth Stage	Parameters							
I	Vegetative and flowering stages (30 DAS)	Isolation, volunteer plants, off types, incidence of downy mildew / green ear, ergot, grain smut, etc.,							

	II	50 % Flowering stage	Isolation, Off types, incidence of downy mildew / green ear, ergot, grain smut, etc.,
	II	Prior to harvest	Off types, incidence of downy mildew / green ear, ergot, grain smut, etc.,
Field Standards			
			Foundation
			Certified
	Off-types		0.05
	Plants affected by downy mildew / green ear (<i>Sclerospora graminicola</i>)		0.10
	Erogottenearheads (<i>Claviceps microcephala</i>)		0.02
Harvest & Seed Extraction	Earheads infected by grain smut (<i>Tolyposporium pencillariae</i>)		0.10
	- Physiological maturity: 90 % of seeds turn straw yellow and the dunken layer (black layer) is formed at the point of attachment of the panicle - Ready for harvest at 85-90 DAS - Harvest the ear heads when 80 % of the ear heads are physiologically matured, where the moisture content will be around 20 %		
	Seed Extraction		
	Dry the ear heads for 2-3 days in the threshing floor.		
	- Manual / Mechanical Threshing is done with pliable sticks when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. - Threshed Seeds are winnowed to remove chaff, dust, empty husk and light grains. - Seeds are dried to optimum moisture content ie., 12 %		
Seed yield	3.5 mt /ha		

Seed Standards		Foundation	Certified	
	Pure seed (minimum)	98 %	98 %	
	Inert matter (maximum)	2.0 %	2.0 %	
	Other crop seed (maximum)	10/Kg	20/Kg	
	Total weed seeds (maximum)	10/Kg	20/Kg	
	Ergot sclerotia, seed entirely or partially modified as sclerotia, broken sclerotia or ergotted seeds (maximum)	0.20 % (by number)	0.40 % (by number)	
	Germination (minimum)	75 %	75 %	
	Moisture (minimum)	12.0 %	12.0 %	
	For vapour- proof containers (maximum)	8.0 %	8.0 %	
Seed Storage period	12 months in moisture pervious			

HYBRID PEARLMILLET

Tool	- Cytoplasmic genetic male sterility (CGMS)- both cytoplasm and nucleus are involved in sterility system - Three line breeding system - A- Male sterile line (CGMS line) (Female Parent) - B –Maintainer line (Male Parent) - R- Restorer line (Male Parent)
Pollination	Cross pollination
Problems in Hybrid rice seed production	Nicking (Ensuring synchronization of flowering) The synchronization problem in pearlmillet seed production is low compared to sorghum and rice due to the following reasons: 1. Tillering habit 2. Low pollen weight and high flying capacity 3. Longer duration of pollen viability and stigma receptivity. However, in case hybrids involve parents with wide variation in flowering synchronization of flowering of both parents may be a problem.

Method of seed production	Three steps	
	A line (CMS) multiplication	A x B line
	B and R line multiplication	Open pollination
	Hybrid seed production	A x R line
Steps in rice seed production	Foundation seed	A, B & R line multiplication
	Certified (Hybrid) seed	A x R line
Isolation		
	Foundation seed	1000 m
	Certified (Hybrid) seed	200 m
NURSERY		
Time of sowing	- July (second fortnight) - October (first fortnight) to December (second fortnight)	
Seed rate		
	Female parent	6 kg ha ⁻¹
	Male parent	2 kg ha ⁻¹
Nursery preparation	- Prepare two separate nurseries for male and female parents - Form separate irrigation channels for male and female parents to avoid contamination.	
Spacing		
	Male Parent (R line)	45 cm solid row
	Female Parent (A line)	45 x 20 cm
	Between Female block and Male block	20 cm
Transplanting	- Simultaneous planting of female and male seedlings. - Row direction – perpendicular to wind direction. - Age of transplanting- is 18 DAS - Two to three seedlings per hill at 2-3 cm depth. <i>Caution : Take care to avoid mixing of male and female seedlings.</i>	
Border rows	- Foundation seed-4rows - Certified seed-8 rows	
Planting ratio	- Foundation seed-4:2 (Female : Male) - Certified seed-6:1 (Female : Male)	

Rougeing by Seed producer	• Vegetative stage • Flowering stage • Preharvest stage			
Field Inspection by Seed certification officer	• Four inspections			
	Inspection	Growth Stage	Parameters	
	I	Before flowering (Before 30 DAS)	Isolation, volunteer plants, outcrosses, planting ratio, border rows, errors in planting , incidence of downy mildew disease etc.,	
	II & III	Flowering	Isolation, pollen shedders offtypes, downy mildew, green ear etc.,	
	IV	Maturity & before harvest	Offtypes, downy mildew / green ear, ergot, grain smut, etc.,	
Field Standards				
		Foundation	Certified	
	Off-types in seed parent	0.05	0.20	
	Off-types in pollinator	0.05	0.20	
	Pollen shedding ear heads	0.05	0.10	
	Inseparable other crop plants (Barley, oats, triticale and gram)	0.010	0.050	
	Seed borne diseases (loose smut (<i>Ustilago tritici</i>))	0.10	0.50	
Seed Standards			Foundation	Certified
	Pure seed (minimum)		98 %	98 %
	Inert matter (maximum)		2.0 %	2.0 %
	Other crop seed (maximum)		10/Kg	20/Kg
	Other distinguishable varieties (maximum)		10/kg	20/kg
	Total weed seeds (maximum)		10/Kg	20/Kg
	Objectionable weed seeds		2/Kg	5/Kg
	Seeds infected by paddy bunt disease (<i>Neovossia horrid</i>)		0.10 % (by number)	0.50 % (by number)
	Germination (minimum)		80 %	80 %
	Moisture (minimum)		13.0 %	13.0 %
	For vapour- proof containers (maximum)		8.0 %	8.0 %

SORGHUM (*Sorghum bicolor*)

Climate & Soil	• Soil should be fertile and neutral in pH. • Coincidence of flower anthesis with rainfall will reduce percentage of seed set. • Prevalence of dry temperature (37°C) during flowering will enhance seed set.											
Floral Biology	• Panicle may be compact or semi compact or loose (lax). • Flower opening starts after 2-4 days of panicle emergence. • Flowering starts from the tip of the panicle and proceeds downwards; it completes within 7 days. • Pollen is viable for 10-20 minutes. • Receptivity of stigma starts two days before opening and remains viable for several days. • Anthesis takes place between 2.00 AM to 8.00 AM.											
Pollination	• Often Cross pollination											
Land Requirement	The land should be free from volunteer plants (crop of previous season growing in the planting season). The same crop or other varieties of the same crop should not have been grown in the previous season. If it is the same crop (previous), it should have been of the same variety that has been certified. This selection is highly important for maintenance of genetic purity. Adequate irrigation and drainage facilities should be available and the problem soils are not suitable for seed production.											
Method of seed production	• Open pollination under isolation											
Isolation	<table><tr><td rowspan="2">Contaminant</td><td colspan="2">Isolation Distance</td></tr><tr><td>Foundation</td><td>Certified</td></tr><tr><td>Other Varieties</td><td>200</td><td>100</td></tr><tr><td>Johnson grass & Forage sorghum</td><td>400</td><td>400</td></tr></table>	Contaminant	Isolation Distance		Foundation	Certified	Other Varieties	200	100	Johnson grass & Forage sorghum	400	400
Contaminant	Isolation Distance											
	Foundation	Certified										
Other Varieties	200	100										
Johnson grass & Forage sorghum	400	400										
Time of sowing	• June - July (Kharif) • November – December (Rabi)											
Seed rate	Irrigated - Direct sown 10.0 kg / ha and transplanted 7.5 kg/ha Rainfed - Direct sown 15.0 kg.											

Seed Treatment	Seeds are soaked in 2% Potassium dihydrogen phosphate (KH_2PO_4) for 10 hrs and then dried back to original moisture content, those hardened seeds used for sowing.																							
Method of sowing	- Ridges and furrows if preferred. - Dibbling at 1/3 rd of ridge from the bottom. - Depth of seeding is 3 to 4 cm.																							
Spacing	45 X 15 cm.																							
Nutrient management	100: 50 : 50 kg NPK ha⁻¹. - FYM -12.5 tonns ha⁻¹. <table><tr><th rowspan="2">Method of application</th><th colspan="3">(Kg ha⁻¹)</th></tr><tr><th>Urea</th><th>Super phosphate</th><th>Muriate of Potash</th></tr><tr><td>Basal</td><td>50</td><td>50</td><td>50</td></tr><tr><td colspan="4">Top dressing</td></tr><tr><td>25 DAS</td><td>25</td><td></td><td></td></tr><tr><td>45 DAS</td><td>25</td><td></td><td></td></tr></table> - ZnSO₄ (25 kg ha⁻¹) and FeSO₄ (50 kg ha⁻¹) for soils deficient in zinc or iron, respectively.	Method of application	(Kg ha ⁻¹)			Urea	Super phosphate	Muriate of Potash	Basal	50	50	50	Top dressing				25 DAS	25			45 DAS	25		
Method of application	(Kg ha ⁻¹)																							
	Urea	Super phosphate	Muriate of Potash																					
Basal	50	50	50																					
Top dressing																								
25 DAS	25																							
45 DAS	25																							
Water Management	- Once in 7-10 days. - Critical stage for irrigation- are primordial initiation, vegetative, milky and maturity . - Water scarcity during these stages will affect seed set and size.																							
Weed Management	I . Apply pre- emergence herbicide Atrazine 50 WP @ 500 g ha⁻¹- in 900 litre of water at 3rd day after sowing. II. Hand weeding on 30-35 DAS																							
Plant Protection	<table><tr><th colspan="2">Pest</th></tr><tr><td>Stem borer</td><td> - Apply Sevin 50 % W.P. @ 0.2 per cent solution (4 g l⁻¹ of water) - After 20 days, apply Lindane granules 1 per cent or - Spray Quinalphos 25 EC @1 l ha⁻¹ on 20 DAS. </td></tr><tr><td>Shoot fly</td><td> - Spray Endosulfan 35 EC @ 500 ml or Methyl </td></tr></table>	Pest		Stem borer	- Apply Sevin 50 % W.P. @ 0.2 per cent solution (4 g l⁻¹ of water) - After 20 days, apply Lindane granules 1 per cent or - Spray Quinalphos 25 EC @1 l ha⁻¹ on 20 DAS.	Shoot fly	Spray Endosulfan 35 EC @ 500 ml or Methyl																	
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Shoot fly	Spray Endosulfan 35 EC @ 500 ml or Methyl																							

		demeton 25 EC @ 500 ml ha ⁻¹					
	Grain midge	• Dust Carbaryl10 D @ 25 kg ha ⁻¹ .					
	Earhead bug	• Dust Phosalone4 D @ 25 kg ha ⁻¹ .					
	Mites	• Spray sulphur W P @ 3.75 kg ha ⁻¹ (or) Dicofol @ 1500 ml ha ⁻¹					
	Disease						
	Grain smut or Covered smut	• Seed treatment with Sulphur D @ 3 to 6 g Kg ⁻¹ or Soak the seed in water for 4 hours and dry back to original moisture content					
	Sugary disease	• Spray Ziram @ 1 kg ha ⁻¹ (or) Mancozeb @ 1 kg ha ⁻¹ at ear head emergence , 50% flowering and after a week, if necessary.					
	Rust	• Spray Mancozeb @ 1 kg ha ⁻¹ ; repeat after 10 days.					
	Charcoal rot	• Seed treatment with <i>Trichodermaviridie</i> @ 4g Kg ⁻¹ .					
Roguing by Seed producer	Downy mildew	• Seed treatment with Metalaxyl @ 4 g Kg ⁻¹ . • Spray Metalaxyl @ 500 g ha ⁻¹ or mancozeb @ 1 kg ha ⁻¹ or , Zineb @ 1 kg ha ⁻¹ or and Ziram @ 1 kg ha ⁻¹ .					
	- The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. <table><tr><td>First rouging</td><td>Before flowering</td></tr><tr><td>Second rouging</td><td>Flowering</td></tr><tr><td>Third rouging</td><td>Maturity stage</td></tr></table> Rogue out the i) off type based on leaf colour, leaf waviness, grain colour, ear head shape, size etc., ii) Diseased plants affected by kernel smut or grain smut and head smut. iii) related plants such as Johnson grass, Sudan grass, forage plants		First rouging	Before flowering	Second rouging	Flowering	Third rouging
First rouging	Before flowering						
Second rouging	Flowering						
Third rouging	Maturity stage						

Field Inspection by Seed certification officer	Three inspections		
	Inspection	Growth Stage	Parameters
	I	Before flowering	Isolation, volunteer plants, etc.,
	II	Flowering	Isolation, offtypes and etc.,
	III	Maturity	Verify true nature of plant etc.,
Field standards			
		Foundation	Certified
	Off-types	0.05	0.10
	Heads infected by kernel smut or grain smut (<i>Sphacelothecasorghi</i>) and Head smut (<i>Sphacelothecareiliana</i>)	0.05	0.10
Harvest & Seed Extraction	- Physiological maturity - black layer at the basal portion of seed at 40-45 days after 50% flowering. - Harvest the crop at physiological maturity - Seed moisture content will be around 20-22 percent - Threshing is done at 15-18 percent moisture content - Later, dry the seeds to 10-12 percent moisture content.		
Seed Yield	2 mt/ha.		
Seed grading	The sorghum seeds can be processed in OSAW cleaner cum grader using 9/64" round perforated metal sieve.		
Seed Standards			
		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	5/Kg	10/Kg
	Total weed seeds (maximum)	5/Kg	10/Kg
	Other distinguishable varieties (maximum)	10/Kg	20/Kg

	Ergot (Claviceps spp) sclerotia, seed entirely or partially modified as sclerotia, broken sclerotia or ergotted seed (maximum)	0.20 % (by number)	0.040 % (by number)
	Germination (minimum)	80 %	80 %
	Moisture (minimum)	13.0 %	13.0 %
	For vapour- proof containers (maximum)	8.0 %	8.0 %

HYBRID SORGHUM

Tool	• Cytoplasmic genetic male sterility (CGMS)- both cytoplasm and nucleus are involved in sterility system • Three line breeding system • A- Male sterile line (CGMS line) (Female Parent) • B –Maintainer line (Male Parent) • R- Restorer line (Male Parent)							
Pollination	• Cross pollination							
Problems in Hybrid rice seed production	• Nicking (Synchronization) Since hybrid rice seed production involves two different varieties as male and female parents, hence problem is faced in the synchronization of flowering of both parents.							
Method of seed production	• Three steps <table><tr><td>A line (CMS) multiplication</td><td>A x B line</td></tr><tr><td>B and R line multiplication</td><td>Open pollination</td></tr><tr><td>Hybrid seed production</td><td>A x R line</td></tr></table>		A line (CMS) multiplication	A x B line	B and R line multiplication	Open pollination	Hybrid seed production	A x R line
A line (CMS) multiplication	A x B line							
B and R line multiplication	Open pollination							
Hybrid seed production	A x R line							
Steps in rice hybrid seed production	<table><tr><td>Foundation seed</td><td>A, B & R line multiplication</td></tr><tr><td>Certified (Hybrid) seed</td><td>A x R line</td></tr></table>		Foundation seed	A, B & R line multiplication	Certified (Hybrid) seed	A x R line		
Foundation seed	A, B & R line multiplication							
Certified (Hybrid) seed	A x R line							

Isolation	Contaminant	Isolation Distance					
		Foundation	Certified				
	Other Varieties	200	100				
	Johnson grass & Forage sorghum	400	400				
Time of sowing	- October to December - Flowering should coincide with dry and cool conditions						
Seed rate	<table><tr><td>Female parent</td><td>7.5 kg ha⁻¹</td></tr><tr><td>Male parent</td><td>5 kg ha⁻¹</td></tr></table>			Female parent	7.5 kg ha ⁻¹	Male parent	5 kg ha ⁻¹
Female parent	7.5 kg ha ⁻¹						
Male parent	5 kg ha ⁻¹						
Special Operation I	Method of sowing - Row direction – perpendicular to wind direction. - Planting ratio @ 5:2 (female: male). Staggered Sowing : The female seed should be sown earlier than male seeds depending upon flowering pattern. - Border rows – four rows						
Spacing	<table><tr><td>A line</td><td>45 x 30 cm</td></tr><tr><td>R line</td><td>45 x 5 cm (solid row)</td></tr></table>			A line	45 x 30 cm	R line	45 x 5 cm (solid row)
A line	45 x 30 cm						
R line	45 x 5 cm (solid row)						
Special Operation II	Adjustment of flowering date in late and early parent <i>Early Parent</i>: Spray 500 ppm MH in one litre of water [OR] CCC @ 300 ppm at 45 DAS - Skip one irrigation . This will help to delay flowering . <i>Late parent</i> : Spray 1% urea solution.						

Roguing by seed producer	- Remove off-types, diseased and pest infested plants. - Three rougings <table><tr><td>First rouging</td><td>Vegetative stage (30-40 DAS)</td></tr><tr><td>Second rouging</td><td>Bud or flowering stage</td></tr><tr><td>Third rouging</td><td>Capsule formation stage</td></tr></table>	First rouging	Vegetative stage (30-40 DAS)	Second rouging	Bud or flowering stage	Third rouging	Capsule formation stage
First rouging	Vegetative stage (30-40 DAS)						
Second rouging	Bud or flowering stage						
Third rouging	Capsule formation stage						
Harvest & Seed Extraction	- Physiological maturity - black layer at the basal portion of seed at 40-45 days after 50% flowering. - Harvest the crop at harvestable maturity. - Seed moisture content will be around 20-22 percent. - Male and female parents should be harvested separately. - The male parent is harvested first. - Admixture of male and female line should be avoided - Seeds from female parent constitute the <i>hybrid seed</i>. Seed Extraction - Seed is extracted by manual or mechanical threshing . - Seed moisture is between 15 and 18 per cent to minimize the mechanical injury. - Dry the seeds 12 % moisture content.						
Seed yield	3000 kg ha⁻¹						
Seed Certification standards							
Field Standards							
Off-types in seed parent	0.05	0.20					
Off-types in pollinator	0.05	0.20					
Pollen shedding ear heads	0.05	0.10					
Objectionable weed plants (Wild rice)	0.01	0.02					
Seed Standards							
	Foundation	Certified					
Pure seed (minimum)	98 %	98 %					
Inert matter (maximum)	2.0 %	2.0 %					
	2.0 %	2.0 %					

Other crop seed (maximum)	10/Kg	20/Kg
Other distinguishable varieties (maximum)	10/kg	20/kg
Total weed seeds (maximum)	10/Kg	20/Kg
Objectionable weed seeds	2/Kg	5/Kg
Seeds infected by paddy bunt disease (<i>Neovossia horrid</i>)	0.10 % (by number)	0.50 % (by number)
Germination (minimum)	80 %	80 %
Moisture (minimum)	13.0 %	13.0 %
For vapour- proof containers (maximum)	8.0 %	8.0 %

Lecture 10-11

SEED PRODUCTION TECHNIQUES IN GREENGRAM AND BLACKGRAM VARIETIES AND RED GRAM VARIETIES AND HYBRIDS REDGRAM (*Cajanus cajan* (L.) Millsp.)

Interesting Facts	Two habit groups: Determinate and Indeterminate	
	Determinate	- Flowering duration is short, - Apical buds of main shoots develop into inflorescence.
	Indeterminate	- Flowering duration is longer - Flowers occur in axillary racemes spread over considerable length of stem
Floral Biology	- Flowers are bisexual, complete and hypogynous. Pollination - Close proximity of anthers and stigma - Simultaneous maturity of anthers and stigma. - Floral biology favours self pollination but out crossing takes place to an extent of 3-65 % - Bees, particularly megachile species are most important pollinators - Anthesis occurs between 8 AM to 5 PM. - Flowers may remain open for 6- 8 hours.	
Pollination	Often cross pollination (Cross pollination is upto 3-40 –%)	

Crop duration		
	Short duration	<150 days
	Medium duration	150 – 180 days
	Long duration	>180 days
Seed Multiplication Stages	Breeder Seed – Foundation Seed – Certified Seed	
Method of seed production	Varieties: Open pollination among plants raised in isolation. Hybrids : Pollination between female and male sterile plants (A line) raised in isolation	
Land Requirement	- The same variety of same seed class should not have been cultivated in the previous 2season to avoid volunteer plants. - Unless the variety grown was the same and approved by a certification agency for varital purity. - The field should be properly leveled. - Well drained, light soil with neutral pH.	
Isolation		
	Foundation seed	200 m
	Certified seed	100 m
Time of sowing	- June – August - September- November - February – March	
Seed rate	Varieties	kg/ha
	Short duration	25
	Long duration	10
Seed treatment	- Soak the seeds for 3 h in 100 ppm ZnSO₄, (10g/100 lit of water) at 1/3 volume before sowing and quickly air dry in shade to their original moisture content. - Slurry treat the seeds with carbendazim 75% WP @ 2 g dissolved in 5 ml of water per kg of seeds and air-dried - Coat the seeds with <i>Rhizobium</i> culture (50 g kg⁻¹ of seed) before sowing	
Methods of sowing	- Sowing – with seed drill or behind country plough. - Depth of seeding - 5 cm.	

Spacing	<table><tr><td>Varieties</td><td>Spacing (cm)</td></tr><tr><td>Short duration</td><td>45 x 30</td></tr><tr><td>Long duration</td><td>90 x 30</td></tr></table>	Varieties	Spacing (cm)	Short duration	45 x 30	Long duration	90 x 30								
Varieties	Spacing (cm)														
Short duration	45 x 30														
Long duration	90 x 30														
Nutrient management	25: 50 : 0 kg NPK ha⁻¹. - FYM -12.5 tonns ha⁻¹. <table><tr><td rowspan="2">Method of application</td><td colspan="2">(Kg ha⁻¹)</td></tr><tr><td>Urea</td><td>Di ammonium Phosphate</td></tr><tr><td>Basal</td><td>25</td><td>50</td></tr><tr><td colspan="3">Foliar application</td></tr><tr><td>55 & 70 DAS</td><td colspan="2">DAP @ 2 per cent</td></tr></table> 25 kg N + 50 kg P₂O₅ in the form of 100 kg DAP/ha. Caution :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .	Method of application	(Kg ha ⁻¹)		Urea	Di ammonium Phosphate	Basal	25	50	Foliar application			55 & 70 DAS	DAP @ 2 per cent	
Method of application	(Kg ha ⁻¹)														
	Urea	Di ammonium Phosphate													
Basal	25	50													
Foliar application															
55 & 70 DAS	DAP @ 2 per cent														
Special Operation I	At first flowering, spray NAA 40 ppm and after a fortnight the second spray is given.														
Water management	- One irrigation at the time of sowing - Life irrigation on third day - Subsequently irrigate once in 7-10 days. - Critical stage for irrigation- flowering stage.														
Weed management	I . Apply pre- emergence herbicide Atrazine @ 500 g ha ⁻¹ - 3 rd day after sowing. II. Hand weeding on 30-35 DAS														
Plant protection	<table><tr><td colspan="2">Pest</td></tr><tr><td>Pod borer</td><td> - Spray Endosulphan @ 200 ml per acre OR - Spray Monocrotophos @ 80 ml per acre </td></tr><tr><td>Blister beetle</td><td> - Manually catch the beetles and destroy </td></tr><tr><td>Termites</td><td>Soil application of - Aldrin (5%) @ 25 kg per acre </td></tr><tr><td colspan="2">Pre-harvest sanitation spray</td></tr><tr><td>Bruchid</td><td> - Done to prevent carry-over of bruchids </td></tr></table>	Pest		Pod borer	- Spray Endosulphan @ 200 ml per acre OR - Spray Monocrotophos @ 80 ml per acre	Blister beetle	Manually catch the beetles and destroy	Termites	Soil application of Aldrin (5%) @ 25 kg per acre	Pre-harvest sanitation spray		Bruchid	Done to prevent carry-over of bruchids		
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Pre-harvest sanitation spray															
Bruchid	Done to prevent carry-over of bruchids														

		- Spray Endosulphon @ 2 ml l⁻¹ 3-5 days prior to harvest.											
	Diseases												
	<i>Downy mildew</i>	Spray wettable sulphur @ 240g per acre											
	<i>Sterility mosaic virus</i>	- Roughing out of infected plants in the early stage of growth OR - Spray monocrotophos @ 500 ml per ha											
Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Three rougings<table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr></table> - Rogue the off-type plants and diseased plants affected by leaf spot and stem canker, yellow mosaic virus and sterility virus from time to time, as required.				First rouging	Vegetative stage	Second rouging	Flowering stage					
First rouging	Vegetative stage												
Second rouging	Flowering stage												
Field Inspection by Seed certification officer	Two inspections<table><tr><th>Inspection</th><th>Growth Stage</th><th>Parameters</th></tr><tr><td>I</td><td>Before flowering</td><td>Isolation, volunteer plants, offtypes, designated diseases, etc.,</td></tr><tr><td>II</td><td>Flowering</td><td>Isolation, offtypes etc.,</td></tr></table>				Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, offtypes, designated diseases, etc.,	II	Flowering	Isolation, offtypes etc.,
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Field Standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Off-types</td><td>0.10</td><td>0.20</td></tr></table>					Foundation	Certified	Off-types	0.10	0.20			
	Foundation	Certified											
Off-types	0.10	0.20											
Harvest & Seed Extraction	- Physiological maturity : 90 % of seeds turn golden yellow with 17 - 20 % mc - Ready for harvest at 85-90 DAS - Harvest the pods at Physiological maturity. 2- 3 pickings may be taken - Threshing is done by beating the plants with sticks when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. - Threshed seeds are winnowed to remove chaff, dust and light												

	grains. • Seeds are dried to optimum moisture content ie., 8-10 per cent.		
Seed yield	• 2000 to 2500 kg/ha.		
Size Grading	10/64 (BSS 5x5)		
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	5/Kg	10/Kg
	Total weed seeds (maximum)	5/Kg	10/Kg
	Other distinguishable varieties (maximum)	10/Kg	20/Kg
	Germination (minimum)	75 %	75 %
	Moisture (minimum)	9.0 %	9.0 %
	For vapour- proof containers (maximum)	7.0 %	7.0 %
Seed storage period	One year		

HYBRID REDGRAM

Genetic Tool	- Genetic Male Sterility - Cytoplasmic genetic male sterility							
Problems in Hybrid Seed Production	Genetic male sterility: Problems are faced in both maintenance of A line and hybrid seed production. Maintenance of A line - Male sterility is governed by single pair (monogenic) of recessive gene (<i>msms</i>) and is being maintained in heterozygous form. - Raising of male sterile line will lead to a population with both fertile (<i>Msms</i>) and sterile (<i>msms</i>) plants in equal proportion (1:1). - Sib-mating of fertile (<i>Msms</i>) and sterile (<i>msms</i>) plants is followed to maintain male sterile line (A Line) - Seeds are collected from male sterile plants alone - Tagging of male sterile plants based on presence of sterile pollen is a tedious step. Hybrid seed production - During hybrid seed production, also A line will segregate into male fertile : male sterile plants in 1:1 proportion. - Male sterile plants are characterized by translucent (white) anthers which do not bear pollen grains. - Identification of male fertile plants and complete rouging in appropriate time is very important to achieve crossing between A line and R line which leads to the production of hybrid seeds.							
Method of seed production	- Two steps <table><tr><td>A line (GMS) multiplication</td><td>Open pollination between male fertile and male sterile plants of A line which are raised in isolation</td></tr><tr><td>R line multiplication</td><td>Open pollination</td></tr><tr><td>Hybrid seed production</td><td>A x R line (with careful rouging of male fertile plants)</td></tr></table> A line multiplication - For the maintenance of male sterile population, male sterile line is raised in isolation. - The population will segregate into both fertile (<i>Ms ms</i>) and sterile		A line (GMS) multiplication	Open pollination between male fertile and male sterile plants of A line which are raised in isolation	R line multiplication	Open pollination	Hybrid seed production	A x R line (with careful rouging of male fertile plants)
A line (GMS) multiplication	Open pollination between male fertile and male sterile plants of A line which are raised in isolation							
R line multiplication	Open pollination							
Hybrid seed production	A x R line (with careful rouging of male fertile plants)							

	(<i>msms</i>) plants in equal proportion (1:1). • At flowering a young bud from each plant is manually opened, and its anthers checked for the presence or absence of pollen grains. • Male sterile plants have to be identified and tagged and the fertile plants have to be retained without tagging. • Natural sib-mating between fertile (<i>Msms</i>) and sterile (<i>msms</i>) plants is allowed through insect pollinators. • The seeds from the tagged male sterile plants are collected and will be used as A line(male sterile lines) for producing hybrid seeds. • In isolation, off types, if any, should be rogued before flowering starts.						
Steps in hybrid seed production	<table border="1"> <tr> <td>Breeder seed</td><td>A line multiplication under isolation</td></tr> <tr> <td>Foundation seed</td><td>R line multiplication under isolation</td></tr> <tr> <td>Certified seed</td><td>A x R line</td></tr> </table>	Breeder seed	A line multiplication under isolation	Foundation seed	R line multiplication under isolation	Certified seed	A x R line
Breeder seed	A line multiplication under isolation						
Foundation seed	R line multiplication under isolation						
Certified seed	A x R line						
Isolation	<table border="1"> <tr> <td>Certified (Hybrid) seed</td><td>100 m</td></tr> <tr> <td>Foundation seed</td><td>200 m</td></tr> </table>	Certified (Hybrid) seed	100 m	Foundation seed	200 m		
Certified (Hybrid) seed	100 m						
Foundation seed	200 m						
Seed rate	Male: 5 kg ha ⁻¹ Female: 30 kg ha ⁻¹						
Spacing	<table border="1"> <tr> <td>Male</td><td>45 x 20 cm</td></tr> <tr> <td>Female</td><td>45 x 10 cm</td></tr> </table>	Male	45 x 20 cm	Female	45 x 10 cm		
Male	45 x 20 cm						
Female	45 x 10 cm						
Special operation I	<table border="1"> <tr> <td>Planting ratio</td><td>6 : 1 (COPH 1) 8 : 2 (COPH2)</td></tr> <tr> <td>Border rows</td><td>2</td></tr> </table> Border rows with sunflower will increase seed yield due to increased activity of bees and corresponding increase in cross pollination between male and female parents of redgram	Planting ratio	6 : 1 (COPH 1) 8 : 2 (COPH2)	Border rows	2		
Planting ratio	6 : 1 (COPH 1) 8 : 2 (COPH2)						
Border rows	2						
Special Operation II	Rouging of Male fertile plants • During the time of flower initiation, ie., one day before flowering, male fertile plants present in the female rows should be identified and removed. • Male fertile plants can be identified based the yellow color of the anthers, whereas the anthers of male sterile plants will be translucent (white) .						

	Roughing should be conducted till flowering is completed.											
Special Operation III	Supplementary Pollination Supplementary pollination can be achieved by adopting the following the techniques <i>Induction of Continuous Flowering</i> : Remove the immature pods set in male plants from time to time to induce continuous flowering - This will be helpful to ensure pollen availability for longer period. <i>Enhancing out crossing</i> : Placing 5-8 bee hives ha⁻¹ and planting sunflower plants in the borders of seed production field will improve bee activity and enhance out crossing											
Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Three rougings <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr></table> - Rogue out the early flowering and late flowering plants.			First rouging	Vegetative stage	Second rouging	Flowering stage					
First rouging	Vegetative stage											
Second rouging	Flowering stage											
Field Inspection by Seed certification officer	Two inspections <table><tr><th>Inspection</th><th>Growth Stage</th><th>Parameters</th></tr><tr><td>I</td><td>Before flowering</td><td>Isolation, volunteer plants, off types, designated diseases, etc.,</td></tr><tr><td>II</td><td>Flowering</td><td>Isolation, offtypes etc.,</td></tr></table>			Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, off types, designated diseases, etc.,	II	Flowering	Isolation, offtypes etc.,
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	Foundation	Certified										
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Harvest & Seed Extraction	- Physiological maturity : 90 % of seeds turn golden yellow - Harvest the pods at harvestable maturity. 2- 3 pickings may be taken. - Admixture of male and female line should be avoided. - Seeds from the female parent constitute the hybrid seed.											
Seed yield	2000 to 2500 kg/ha.											
Size Grading	10/64 (BSS 5x5)											
Seed Standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Pure seed (minimum)</td><td>98 %</td><td>98 %</td></tr></table>		Foundation	Certified	Pure seed (minimum)	98 %	98 %					
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	Inert matter (maximum)	2.0 %	2.0 %	
	Other crop seed (maximum)	5/Kg	10/Kg	
	Total weed seeds (maximum)	5/Kg	10/Kg	
	Other distinguishable varieties (maximum)	10/Kg	20/Kg	
	Germination (minimum)	75 %	75 %	
	Moisture (minimum)	9.0 %	9.0 %	
	For vapour- proof containers (maximum)	7.0 %	7.0 %	

GREENGRAM (*Vigna radiata* L.)

Floral Biology	• Flowers bisexual, complete with papilionaceous corolla. • Keel petal spirally coiled. • Flowers open early in the morning ; they are completely open by 7-8 AM. • The anthers begin shedding pollen in the previous day evening before the flowers open and pollination is complete before mid-night. • Self pollination is the rule.						
Pollination	• Self pollination						
Land Requirement	• The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. • Unless the variety grown was the same and approved by a certification agency for varital purity. • The field should be properly leveled. • Well drained, light soil with neutral pH.						
Isolation	<table><tr><td>Foundation seed</td><td>10 m</td></tr><tr><td>Certified seed</td><td>5 m</td></tr></table>			Foundation seed	10 m	Certified seed	5 m
Foundation seed	10 m						
Certified seed	5 m						
Time of sowing	• June - July						

Seed rate	Crop spacing differs with varieties . <table><tr><td>Varieties</td><td>Kg ha ⁻¹</td></tr><tr><td>Kharif& Rabi</td><td>15</td></tr><tr><td>Spring & Summer</td><td>25</td></tr></table>	Varieties	Kg ha ⁻¹	Kharif& Rabi	15	Spring & Summer	25											
Varieties	Kg ha ⁻¹																	
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Seed treatment	- Thiram + Bavistin (1:1) @ 2.5 g kg ⁻¹or - Bavistin @ 2 g kg ⁻¹or <i>Trichodermaviridi</i>(organic fungicide) @ 4 g kg ⁻¹ - Fungicide treated seeds should be again treated with bacterial culture of Rhizobium . <i>Rhizobium</i> @ 600 g for seeds of one ha <i>Caution :Trichodermaviridi& Rhizobium</i> treatment should be done 24 h after the fungicidal treatment.																	
Methods of sowing	Dibble at 3-4 cm depth.																	
Spacing	Crop spacing differs with varieties <table><tr><td>Varieties</td><td>Spacing (cm)</td></tr><tr><td>Kharif& Rabi</td><td>30 x10</td></tr><tr><td>Spring & Summer</td><td>20 x 10</td></tr></table>	Varieties	Spacing (cm)	Kharif& Rabi	30 x10	Spring & Summer	20 x 10											
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Nutrient management	20: 50 : 0 kg NPK ha ⁻¹. - FYM -12.5 tonns ha ⁻¹. <table><tr><td rowspan="2">Method of application</td><td colspan="2">(Kg ha ⁻¹)</td></tr><tr><td>Urea</td><td>Diammonium Phosphate</td></tr><tr><td>Basal</td><td>25</td><td>100</td></tr><tr><td colspan="3">Foliar application</td></tr><tr><td>25 & 40DAS</td><td colspan="2">DAP @ 2 per cent</td></tr><tr><td>Pod formation stage and 15 days later</td><td colspan="2">Potassium sulphate @ 1 per cent</td></tr></table> 20 kg N + 50 kg P₂O₅ in the form of 100 kg DAP/ha. <i>Caution :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .</i>	Method of application	(Kg ha ⁻¹)		Urea	Diammonium Phosphate	Basal	25	100	Foliar application			25 & 40DAS	DAP @ 2 per cent		Pod formation stage and 15 days later	Potassium sulphate @ 1 per cent	
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25 & 40DAS	DAP @ 2 per cent																	
Pod formation stage and 15 days later	Potassium sulphate @ 1 per cent																	

Water management	- Once in 7-10 days. - Critical stage for irrigation- flowering stage. - Frequent irrigation for the spring and summer crop. For the kharif crop, one to two irrigations may be required.																
Weed management	I. Apply pre- emergence herbicide Atrazine @ 500 g ha⁻¹- 3rd day after sowing. II. Hand weeding on 30-35 DAS One hand weeding and hoeing may be done. Spray one kg of treflan in 1000 litres of water per ha on the soil, at the time of the final preparations, has also been effective for controlling weeds.																
Plant protection	Pest <table border="1"> <tr> <td>Aphid and white fly</td><td>Spray Metasystox @ 625 ppm ha⁻¹ 625 ml of in 625 litres of water /ha.</td></tr> <tr> <td>Caterpillars and pod borers</td><td> - Dust Sevin 10% @ 25 kg ha⁻¹. - Spray Endosulfan @ 0.03% or - Spray Malathian @ 0.05% </td></tr> <tr> <td colspan="2">Pre-harvest sanitation spray</td></tr> <tr> <td>Bruchid</td><td> - Done to prevent carry-over of bruchids - Spray Endosulphon @ 2 ml l⁻¹ 3-5 days prior to harvest. </td></tr> <tr> <td colspan="2">Diseases</td></tr> <tr> <td>Anthracnose and Cercospora leaf spot</td><td> - Spray Copper oxychloride (1.25 kg to 2.50kg in 625 litres of water per ha) or - Spray with Captan or Zineb @ 2g/litre or - Bavistin @ 0.5 kg in 1,000 litres of water per ha </td></tr> <tr> <td>Dry root rot</td><td> - Drench the soil with Copper oxychloride @ 5 kg ha⁻¹ in 250 l of water. </td></tr> <tr> <td>Powdery mildew</td><td> - Dust fine Sulphur @ 25 kg ha⁻¹ or - Spray wettablesulphur @ 2.5 kg ha </td></tr> </table>	Aphid and white fly	Spray Metasystox @ 625 ppm ha ⁻¹ 625 ml of in 625 litres of water /ha.	Caterpillars and pod borers	- Dust Sevin 10% @ 25 kg ha⁻¹. - Spray Endosulfan @ 0.03% or - Spray Malathian @ 0.05%	Pre-harvest sanitation spray		Bruchid	- Done to prevent carry-over of bruchids - Spray Endosulphon @ 2 ml l⁻¹ 3-5 days prior to harvest.	Diseases		Anthracnose and Cercospora leaf spot	- Spray Copper oxychloride (1.25 kg to 2.50kg in 625 litres of water per ha) or - Spray with Captan or Zineb @ 2g/litre or - Bavistin @ 0.5 kg in 1,000 litres of water per ha	Dry root rot	Drench the soil with Copper oxychloride @ 5 kg ha⁻¹ in 250 l of water.	Powdery mildew	- Dust fine Sulphur @ 25 kg ha⁻¹ or - Spray wettablesulphur @ 2.5 kg ha
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		⁻¹ or Spray Benlate @ 150 g ha ⁻¹										
	Rust	- Spray wettablesulphur @ 2.5 kg ha ⁻¹ or - Dust fine sulphur @ 2.5 kg ha ⁻¹ at 15 days internal.										
Foliar Fertilization	2% DAP spray during 25-35 days after sowing followed by another spray at 40-50 days. Potassium sulphate 1% spray during pod formation stage followed by another spray at 15 days after 1 st spray.											
Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Three rougings <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr></table> - Rogue the off–type plants and diseased plants affected by leaf spot and stem canker, yellow mosaic virus and sterility virus from time to time, as required. - Offtypes are identified based on leaf colour, plant stature, leaf shape, pod colour, flower colour, and seed colour.			First rouging	Vegetative stage	Second rouging	Flowering stage					
First rouging	Vegetative stage											
Second rouging	Flowering stage											
Field Inspection by Seed certification officer	Two inspections <table><tr><td>Inspection</td><td>Growth Stage</td><td>Parameters</td></tr><tr><td>I</td><td>Before flowering</td><td>Isolation, volunteer plants, offtypes, designated diseases, etc.,</td></tr><tr><td>II</td><td>Flowering and maturity</td><td>Isolation, offtypes etc.,</td></tr></table>			Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, offtypes, designated diseases, etc.,	II	Flowering and maturity	Isolation, offtypes etc.,
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Field Standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Off-types</td><td>0.10</td><td>0.20</td></tr><tr><td>Plants affected by seed borne diseases**</td><td>0.10</td><td>0.20</td></tr></table> ** Halo blight (<i>Pseudomonas phaseolicola</i>)				Foundation	Certified	Off-types	0.10	0.20	Plants affected by seed borne diseases**	0.10	0.20
	Foundation	Certified										
Off-types	0.10	0.20										
Plants affected by seed borne diseases**	0.10	0.20										

Pre-harvest sanitation spray	The field carry – over of Bruchids can be controlled by giving a pre – harvest sanitation spray with endosulphan 0.07% (2ml of insecticide per litre) 3-5 days prior to harvest.			
Harvest & Seed Extraction	• Physiological maturity :Pods turn to black /grey/ brown in colour Harvest the pods heads at Physiological maturity. • Summer and spring – Harvest when pods initiate to turn black • Kharif - Harvest when most pods have turned black • Ready for harvest at 85-90 DAS • The crop is left in the field to dry for about one week. • Threshing is done by beating the plants with sticks when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. • Threshed seeds are winnowed to remove chaff, dust and light grains. • Seeds are dried to optimum moisture content ie., 9 per cent.			
Seed yield	• 800 to 1000 kg/ha			
Seed Standards		Foundation	Certified	
	Pure seed (minimum)	98 %	98 %	
	Inert matter (maximum)	2.0 %	2.0 %	
	Other crop seed (maximum)	5/Kg	10/Kg	
	Weed seeds (maximum)	5/Kg	10/Kg	
	Other distinguishable varieties (maximum)	10/Kg	20/Kg	
	Germination (minimum)	75 %	75 %	
	Moisture (minimum)	9.0 %	9.0 %	
	For vapour- proof containers (maximum)	7.0 %	7.0 %	

BLACKGRAM (*Vigna mungo*)

Floral Biology	• Flowers bisexual, complete with papilionaceous corolla. • Keel petal spirally coiled. • Flowers open early in the morning ; they are completely open by 7-8 AM. • The anthers begin shedding pollen in the previous day evening before the flowers open and pollination is complete before mid-night.
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	Self pollination is the rule.								
Pollination	Self pollination								
Land Requirement	- The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. - Unless the variety grown was the same and approved by a certification agency for varital purity. - The field should be properly leveled. - Well drained, light soil with neutral pH.								
Isolation	<table><tr><td>Foundation seed</td><td>10 m</td></tr><tr><td>Certified seed</td><td>5 m</td></tr></table>	Foundation seed	10 m	Certified seed	5 m				
Foundation seed	10 m								
Certified seed	5 m								
Seed rate	Crop spacing differs with varieties . <table><tr><td>Varieties</td><td>Kg ha ⁻¹</td></tr><tr><td>Kharif& Rabi</td><td>15</td></tr><tr><td>Spring & Summer</td><td>25</td></tr></table>	Varieties	Kg ha ⁻¹	Kharif& Rabi	15	Spring & Summer	25		
Varieties	Kg ha ⁻¹								
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Spring & Summer	25								
Seed treatment	- Thiram + Bavistin (1:1) @ 2.5 g kg ⁻¹or - Bavistin @ 2 g kg ⁻¹or <i>Trichodermaviridi</i>(organic fungicide) @ 4 g kg ⁻¹ Fungicide treated seeds should be again treated with bacterial culture of Rhizobium. <i>Rhizobium</i> @ 600 g for seeds of one ha <i>Caution :Trichodermaviridi& Rhizobium treatment should be done 24 h after the fungicidal treatment.</i>								
Time of sowing	June – July								
Methods of sowing	Dibbling at 3-4 cm depth.								
Spacing	25 x 15 cm								
Nutrient management	20: 50 : 0 kg NPK ha ⁻¹. - FYM -12.5 tonns ha ⁻¹. <table><tr><td rowspan="2">Method of application</td><td colspan="2">(Kg ha ⁻¹)</td></tr><tr><td>Urea</td><td>Diammonium Phosphate</td></tr><tr><td>Basal</td><td>25</td><td>50</td></tr></table>	Method of application	(Kg ha ⁻¹)		Urea	Diammonium Phosphate	Basal	25	50
Method of application	(Kg ha ⁻¹)								
	Urea	Diammonium Phosphate							
Basal	25	50							

	Foliar application															
	25 & 40 DAS	DAP @ 2 per cent														
	Pod formation stage and 15 days later	Potassium sulphate @ 1 per cent														
20 kg N + 50 kg P₂O₅ in the form of 100 kg DAP/ha. <i>Caution</i> :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .																
Water management	• Once in 7-10 days. • Critical stage for irrigation- flowering stage. • Frequent irrigation for the spring and summer crop. For the kharif crop, one to two irrigations may be required.															
Weed management	I . Apply pre- emergence herbicide Atrazine @ 500 g ha⁻¹- 3rd day after sowing. II. Hand weeding on 30-35 DAS One hand weeding and hoeing may be done. Spray one kg of treflan in 1000 litres of water per ha on the soil, at the time of the final preparations, has also been effective for controlling weeds.															
Plant protection	<table><tr><td colspan="2">Pest</td></tr><tr><td>Aphid and white fly</td><td>Spray Metasystox @ 625 ppm ha⁻¹ 625 ml of in 625 litres of water /ha.</td></tr><tr><td>Caterpillars and pod borers</td><td> • Dust Sevin 10% @ 25 kg ha⁻¹. • Spray Endosulfan @ 0.03% or • Spray Malathian @ 0.05% </td></tr><tr><td colspan="2">Pre-harvest sanitation spray</td></tr><tr><td>Bruchid</td><td> • Done to prevent carry-over of bruchids • Spray Endosulphon @ 2 ml l⁻¹ • 3-5 days prior to harvest. </td></tr><tr><td colspan="2">Diseases</td></tr><tr><td>Anthracnose and Cercospora leaf spot</td><td> • Spray Copper oxychloride (1.25 kg to 2.50kg in 625 litres of water per ha) or • Spray with Captan or Zineb @ 2g/litre or • Bavistin @ 0.5 kg in 1,000 litres of water per ha </td></tr></table>		Pest		Aphid and white fly	Spray Metasystox @ 625 ppm ha ⁻¹ 625 ml of in 625 litres of water /ha.	Caterpillars and pod borers	• Dust Sevin 10% @ 25 kg ha⁻¹. • Spray Endosulfan @ 0.03% or • Spray Malathian @ 0.05%	Pre-harvest sanitation spray		Bruchid	• Done to prevent carry-over of bruchids • Spray Endosulphon @ 2 ml l⁻¹ • 3-5 days prior to harvest.	Diseases		Anthracnose and Cercospora leaf spot	• Spray Copper oxychloride (1.25 kg to 2.50kg in 625 litres of water per ha) or • Spray with Captan or Zineb @ 2g/litre or • Bavistin @ 0.5 kg in 1,000 litres of water per ha
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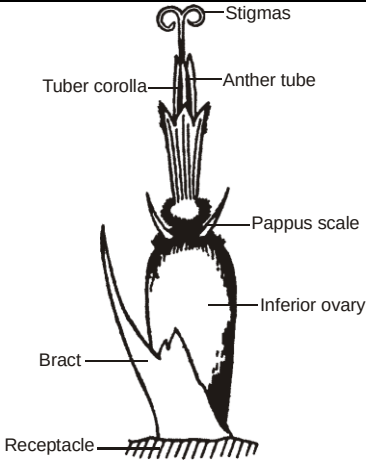
	Dry root rot	• Drench the soil with Copper oxychloride @ 5 kg ha⁻¹ in 250 l of water.										
	Powdery mildew	• Dust fine Sulphur @ 25 kg ha⁻¹ or • Spray wettable sulphur @ 2.5 kg ha⁻¹ or • Spray Benlate @ 150 g ha⁻¹										
	Rust	• Spray wettable sulphur @ 2.5 kg ha⁻¹ or • Dust fine sulphur @ 2.5 kg ha⁻¹ at 15 days interval.										
Roguing by Seed producer	• Remove off-types, diseased and pest infested plants. • Three rougings <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr></table> • Rogue the off-type plants and diseased plants affected by leaf spot and stem canker, yellow mosaic virus and sterility virus from time to time, as required. • Offtypes are identified based on leaf colour, plant stature, leaf shape, pod colour, flower colour, and seed colour.			First rouging	Vegetative stage	Second rouging	Flowering stage					
First rouging	Vegetative stage											
Second rouging	Flowering stage											
Field Inspection by Seed certification officer	• Two inspections <table><tr><th>Inspection</th><th>Growth Stage</th><th>Parameters</th></tr><tr><td>I</td><td>Before flowering</td><td>Isolation, volunteer plants, offtypes, designated diseases, etc.,</td></tr><tr><td>II</td><td>Flowering and maturity</td><td>Isolation, offtypes etc.,</td></tr></table>			Inspection	Growth Stage	Parameters	I	Before flowering	Isolation, volunteer plants, offtypes, designated diseases, etc.,	II	Flowering and maturity	Isolation, offtypes etc.,
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Field Standards			
		Foundation	Certified
	Off-types Plants affected by seed borne diseases**	0.10 0.10	0.20 0.20
	** Ashy stem blight (<i>Macrophomina phaseolii</i>) ; Anthracnose (<i>Colletotrichum lindemithianum</i>)		
Harvest & Seed Extraction	- Physiological maturity : Pods turn to black / grey / brown in colour - Harvest the pods heads at Physiological maturity. - Summer and spring – Harvest when pods initiate to turn black - Kharif - Harvest when most pods have turned black - Ready for harvest at 85-90 DAS - The crop is left in the field to dry for about one week. - Threshing is done by beating the plants with sticks when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. - Threshed seeds are winnowed to remove chaff, dust and light grains. - Seeds are dried to optimum moisture content ie., 9 per cent.		
Seed yield	1000 to 1500 kg/ha.		
Size Grading			
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	5/Kg	10/Kg
	Weed seeds (maximum)	5/Kg	10/Kg
	Other distinguishable varieties (maximum)	10/Kg	20/Kg
	Germination (minimum)	75 %	75 %
	Moisture (minimum)	9.0 %	9.0 %
	For vapour- proof containers (maximum)	7.0 %	7.0 %
Seed storage period	One year		

LECTURE 12-13

SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF SUNFLOWER AND GROUNDNUT VARIETIES, CASTOR & SESAME

SUNFLOWER (*Helianthus annuus*)

Origin	Southern United States
Floral Biology	- Inflorescence is called capitulum or head. - The flowers of the capitulum are of two types: i) the ray florets are sterile and form the outer row, ii) the disc florets (Figure 1) are fertile and originate from the centre of the inflorescence. - The outer whorls of disc florets mature first and proceed inwards. Protandry - The flowers are thus protandrous and are normally cross-fertilized; - The anthers mature and shed their pollen into the tube before the stigmatic surfaces are receptive. - Pollination is effected by insects walking over the surface of the inflorescence. - Pollen viability is for about 12 h, hence fertile pollen is available throughout the day. - Flowering in most hybrid sunflowers is remarkably uniform, with 80-90 % of the heads opening within 3-4 days. - Self-incompatibility is the rule therefore, if honey bee population is inadequate, seed set is poor. - Short days favour floral initiation.
	 The disc floret of sunflower
Pollination	Cross pollination

Mode of propagation	Seed				
Method of seed production	Varieties :Open pollination among plants raised in isolation. Hybrids: Two true breeding strains are developed and used as male and female parent.				
Land requirement	The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. The field should be properly leveled. Well drained, deep soil with neutral pH.The crop should not be rotated with sorghum to avoid charcoal rot.				
Isolation	<table border="1"> <tr> <td>Foundation seed</td><td>400 m</td></tr> <tr> <td>Certified seed</td><td>200 m</td></tr> </table>	Foundation seed	400 m	Certified seed	200 m
Foundation seed	400 m				
Certified seed	200 m				
Time of sowing	<table border="1"> <tr> <td>Rainfed</td><td>June- July; October- November</td></tr> <tr> <td>Irrigated</td><td>April – May; December - January</td></tr> </table>	Rainfed	June- July; October- November	Irrigated	April – May; December - January
Rainfed	June- July; October- November				
Irrigated	April – May; December - January				
Seed rate	15 kg ha⁻¹				
Seed treatment	- Thiram + Bavistin (1:1) @ 2.5 g kg⁻¹ or Bavistin @ 2 g kg⁻¹ or <i>Trichoderma viridi</i>(organic fungicide) @ 4 g kg⁻¹ - Fungicide treated seeds should be again treated with bacterial culture of <i>Rhizobium</i>. <i>Rhizobium</i>&Phospho bacterium @ 600 g ha⁻¹ <i>Caution :Trichoderma viridi , Rhizobium &Phospho bacterium treatment should be done 24 h after the fungicidal treatment.</i> Seed Dormancy breaking - Soak the seeds in water for 12-16 h [or] - Leach the inhibitors in seeds by placing in running water - Seed treatment with Ethrel (2 chloro ethyl phosphoric acid @ 300 ppm for 8 h , by placing in air tight container - Soak the seeds in KNO₃ @ 0.5 % , for 16 h				
Method of sowing	- Seeds are sown with seed drill or manually in rows - Depth of seeding - 2 to 4 cm.				
Spacing	80 x 20 cm				

Nutrient management	12.5 tonns ha⁻¹- FYM 65: 45 : 45 kg NPK ha⁻¹.			
	Method of application	(Kg ha ⁻¹)		
		Urea	Diammonium Phosphate	Potash
	Basal	45	45	45
	Top dress 30-55 DAS	15	-	-
	Foliar application			
	30 & 60 DAS	2 % DAP 20 ppm NAA (280 g NAA in 625 l of water)		
<i>Caution</i> :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .				
Special Operation I	Supplementary Pollination - Since, sunflower is self incompatible, supplementary Pollination is required to increase the seed set. - In varieties, two adjacent capitulumare gently rubbed [OR] by gently passing the palm over the capitulum. - Supplementary pollination should be done during mid flowering stage @ 58-60 DAS in long duration varieties and @ 45- 48 DAS in short duration varieties . - Repeat it on alternate days between 7 AM and 11 AM for 2 weeks period. - Also spray Neera (containing sucrose and minerals) to attract insects. <i>In hybrids</i> , it is done by gently passing the palm covered with muslin cloth over the male capitulum first and then on female capitulum so as to transfer the pollen effectively to favour cross pollination.			
Water management	- One irrigation at the time of sowing - Life irrigation on third day. - Critical stage for irrigation- flowering and seed development stage			
Weed management	- Pre- emergence application of Fluchlorin @ 800 ml ha⁻¹ in 200 l of water on 3rd DAS and irrigate the crop [OR] spray pre-emergence herbicide - First weeding after one to two weeks of sowing .			
Plant protection	Pest			
	Cutworms	Soil application of 5% Heptachlor dust @ 15 kg		

		ha ⁻¹ .							
	Head borer and Jassid	• Spray Endosulfan @ 1000ml or • SprayMetasystox (25 EC) @ 0.025 per cent.							
	Bird Damage	• Manual scaring • Tie glittering white and red paper criss crossing across the field. • Use automatic gun to produce noise at constant intervals.							
	Diseases								
	Stem or Head rot	Seed treatment • Mancozeb and Carbendazim each @ 2g kg⁻¹ seed OR • Warm water (50-60°C) treatment for 10-15 min							
	Alternaria Blight	• Seed treatment with Tiram, Mancozeb and Carbendazim each @ 2g kg⁻¹ • Spray <u>Mancozeb @ 0.3 %</u>, four times at 7-10 days interval.							
	Leaf spot and Rust	• Spray Mancozeb @ 1000 g ha⁻¹ , 2 to 3 times at 10 days interval after 20 DAS.							
	Root rot	• Spot drenching of Copper oxychloride @ 0.2% .							
Roguing by Seed producer	• The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. • Rouging is done based on plant height, head size, branching habit, number of heads and colour the seed. <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr><tr><td>Third rouging</td><td>Preharvest stage</td></tr></table>			First rouging	Vegetative stage	Second rouging	Flowering stage	Third rouging	Preharvest stage
First rouging	Vegetative stage								
Second rouging	Flowering stage								
Third rouging	Preharvest stage								

Field Inspection by Seed certification officer	Three inspections			
	Inspection	Growth Stage	Parameters	
	I	6-7 pair leaf stage	Isolation, volunteer plants, offtypes, designated diseases, etc.,	
	II	Flowering	Isolation, offtypes etc.,	
	III	Maturity	Designated diseases, true nature of plant and head, characteristics fo seed etc.,	
Field Standards			Foundation	Certified
	Off-types (<i>Caution: Sterile Plants shall not be considered as offtype</i>)		0.10	0.20
	Objectionable weed plants (Wild <i>Helianthus</i> spp.)		None	None
	Plants infected by downy mildew (<i>Plasmoparahalstedii</i>)		0.05	0.50
	Plants infected by downy mildew (<i>Orobanche Cumana</i>)		None	None
Harvest & Seed Extraction	- Physiological maturity: Back of the thalamus turns lemon yellow in colour. - Ready for harvest at 85-90 DAS - Heads may be removed with shears (or) knife.			
	Seed Extraction			
	- Heads are left in the field to dry in the threshing floor in a thin layer for about one week. - Manual / Mechanical Threshing is done with pliable sticks when seed moisture is between 15 and 18 per cent to minimize the mechanical injury. - Dry the seeds upto 8 per cent moisture content.			
Seed yield	2 t ha⁻¹.			
Size Grading	9/64 “ round perforated sieve as middle seive			
Seed Standards			Foundation	Certified
	Pure seed (minimum)		98 %	98 %
	Inert matter (maximum)		2.0 %	2.0 %

	Huskless (Maximum)	a. % (By number)	2.0 % (By number)
	Other crop seed (maximum)	None	None
	Total weed seeds (maximum)	5/Kg	10/Kg
	Objectionable weed seeds (maximum)	None	None
	Seeds infested with <i>Orobanche Cumana</i> (maximum)	None	None
	Germination (minimum)	70 %	70 %
	Moisture (minimum)	9.0 %	9.0 %
	For vapour- proof containers (maximum)	7.0 %	7.0 %

SUNFLOWER HYBRIDS

Tool	Cytoplasmic Genic Male Sterility (CGMS)								
Pollination	• Cross pollination								
Problems in Hybrid seed production	1. Insufficient pollination 2. Development of axillary flowers in seed parent 3. Self incompatibility nature								
Method of seed production	• Three steps <table><tr><td>A line (CMS) multiplication</td><td>A x B line</td></tr><tr><td>B and R line multiplication</td><td>Open pollination</td></tr><tr><td>Hybrid seed production</td><td>A x R line</td></tr></table>			A line (CMS) multiplication	A x B line	B and R line multiplication	Open pollination	Hybrid seed production	A x R line
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B and R line multiplication	Open pollination								
Hybrid seed production	A x R line								
Steps in hybrid seed production	<table><tr><td>Breeder seed</td><td>A, B & R line multiplication</td></tr><tr><td>Foundation seed</td><td>A, B & R line multiplication</td></tr><tr><td>Certified (Hybrid) seed</td><td>A x R line</td></tr></table>			Breeder seed	A, B & R line multiplication	Foundation seed	A, B & R line multiplication	Certified (Hybrid) seed	A x R line
Breeder seed	A, B & R line multiplication								
Foundation seed	A, B & R line multiplication								
Certified (Hybrid) seed	A x R line								
Special Operation I	Nipping in Male line (R) multiplication • Male parents produce side branches which supply more pollen and facilitate higher seed set.								

Isolation	<table border="1" data-bbox="542 247 1083 333"> <tr> <td>Foundation seed</td><td>600 m</td></tr> <tr> <td>Certified seed</td><td>400 m</td></tr> </table>	Foundation seed	600 m	Certified seed	400 m
Foundation seed	600 m				
Certified seed	400 m				
Seed rate	<table border="1" data-bbox="542 451 1107 541"> <tr> <td>Female parent</td><td>12 kg ha⁻¹</td></tr> <tr> <td>Male parent</td><td>4 kg ha⁻¹</td></tr> </table>	Female parent	12 kg ha ⁻¹	Male parent	4 kg ha ⁻¹
Female parent	12 kg ha ⁻¹				
Male parent	4 kg ha ⁻¹				
Special Operation II	Staggered sowing - Flowering in male and female parents should not vary by more than 3 to 4 days. - If the variation is higher , staggered sowing is required to ensure synchronization of flowering - In the case of KBSH 1, R line has to be sown 8 days earlier and in APSH 11, 4 days prior sowing is required.				
Planting ratio	8:1 or 4:1(Female : Male), Border rows:4				
Method of sowing	- Seeds are sown with seed drill or manually in rows - Depth of seeding - 2 to 4 cm.				
Spacing	<table border="1" data-bbox="557 1150 1107 1291"> <tr> <td>Female parent</td><td>60 x 30 cm</td></tr> <tr> <td>Male parent</td><td>45 x 20 cm</td></tr> </table>	Female parent	60 x 30 cm	Male parent	45 x 20 cm
Female parent	60 x 30 cm				
Male parent	45 x 20 cm				
Special Operation II	Supplementary Pollination 1. Hand pollination <i>In hybrids</i>, it is done by gently rubbing the palm covered with muslin cloth over the male capitulum first and then on female capitulum so as to transfer the pollen effectively to favour cross pollination. <i>In varieties</i>, supplementary pollination is done by gently rubbing the two adjacent capitulum or by gently rubbing the palm over the capitulum. It can be done on 58 - 60 days and 45 - 48 days for long and short duration varieties, respectively. 2. Keeping Bee hives:In sunflower, for enhancing pollination and seed set, bee				

	hives @ 5 ha⁻¹ have to be kept. Under such cases spraying of insecticide must be used sparingly. 3. Sib mating:The pollination can also be promoted both in hybrid (male and female) and varieties by manually rubbing the heads nearby and is known as sib mating. 4.Also spray Neera (containing sucrose and minerals) to attract insects.																							
Roguing by Seed producer	- The off-type plants, weeds and diseased plants should be rogued out at appropriate stage, preferably before flowering. - Rouging is done based on plant height, head size, branching habit, number of heads and colour the seed. <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr><tr><td>Third rouging</td><td>Maturity stage</td></tr></table>			First rouging	Vegetative stage	Second rouging	Flowering stage	Third rouging	Maturity stage															
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	Foundation	Certified																						
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Plants infested with <i>Orobanchecumana</i>	None	None																						

Harvest & Seed Extraction	- Male and female parents should be harvested separately. - The male plants are harvested first. - Admixture of male and female line should be avoided - Seeds from female parent constitute the hybrid seed.		
Seed yield	2 t ha⁻¹.		
Size Grading	Seeds can be processed by using 9/64" round perforated sieve		
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Huskless (Maximum)	2.0 % (By number)	2.0 % (By number)
	Other crop seed (maximum)	None	None
	Total weed seeds (maximum)	5/Kg	10/Kg
	Objectionable weed seeds (maximum)	None	None
	Seeds infested with <i>Orobanche Cumana</i> (maximum)	None	None
	Germination (minimum)	70 %	70 %
	Moisture (minimum)	9.0 %	9.0 %
	For vapour- proof containers (maximum)	7.0 %	7.0 %

CASTOR (*Ricinus communis*)

Interesting Facts	Reversion to monoecious - The proportion of pistillate and staminate flowers varies with genotypes, environment, age of the plant and nutrition. Bloom - The white waxy coating on the plant is called bloom. - It offers protection against extreme weather conditions and jassids. - An important character to identify offtypes - There are 4 types of bloom <table border="1" data-bbox="435 636 1198 852"> <tr> <td></td><td>Parts of plant with waxy coating</td></tr> <tr> <td>No bloom</td><td>No parts above ground</td></tr> <tr> <td>Single bloom</td><td>Stem alone</td></tr> <tr> <td>Double bloom</td><td>Stem, petioles, lower side of leaves</td></tr> <tr> <td>Triple bloom</td><td>On all parts above ground</td></tr> </table>		Parts of plant with waxy coating	No bloom	No parts above ground	Single bloom	Stem alone	Double bloom	Stem, petioles, lower side of leaves	Triple bloom	On all parts above ground
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Climate & Soil	- Season has profound influence on sex expression. <table border="1" data-bbox="529 898 1385 1146"> <tr> <td>Kharif /Summer</td><td> - Promotes male flowers. - Ideal for seed production of varieties, male and female parents of hybrids </td></tr> <tr> <td>Rabi/ Winter</td><td> - Hybrid seed production </td></tr> </table> - Performs in medium to deep sandy loam and heavy loam soils - Does not tolerate alkalinity and salinity.	Kharif /Summer	- Promotes male flowers. - Ideal for seed production of varieties, male and female parents of hybrids	Rabi/ Winter	Hybrid seed production						
Kharif /Summer	- Promotes male flowers. - Ideal for seed production of varieties, male and female parents of hybrids										
Rabi/ Winter	Hybrid seed production										
Floral Biology	✓ Castor is highly polymorphic species. ✓ The plant is monoecious with pistillate flowers on the upper part of the raceme (inflorescence) and staminate flowers on the lower part. ✓ Castor is highly protogynous ✓ Protogyny (Female organ mature before male organ) Sex Expression - The proportion of pistillate and staminate flowers varies with genotypes, environment, age of the plant and nutrition. - Female tendency is highest in winter, while male tendency predominates in summer and rainy seasons. - Femaleness is higher in young plants with high level of nutrition than										

	that in old plants with low levels of nitrogen.						
Pollination	• Cross Pollination (Wind pollinated)						
Crop duration	• 90-120 days						
Land requirement	The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. Unless the variety grown was the same and approved by a certification agency for varital purity. The field should be properly leveled. Well drained, sandy loam soil rich in organic content.						
Isolation	<table border="1"> <tr> <td>Foundation seed</td><td>300 m</td></tr> <tr> <td>Certified seed</td><td>150 m</td></tr> </table>	Foundation seed	300 m	Certified seed	150 m		
Foundation seed	300 m						
Certified seed	150 m						
Time of sowing	• June to July.						
Seed rate	• 10 kg ha⁻¹						
Method of sowing	• Sowing with seed drill or behind country plough. • Depth of seeding - 8 cm.						
Spacing	<table border="1"> <tr> <td>Variety</td><td>Spacing (cm)</td></tr> <tr> <td>Long duration</td><td>90 x 90 cm [OR] 90 x 60</td></tr> <tr> <td>Short duration</td><td>60 x 45 [OR] 60 x 30</td></tr> </table>	Variety	Spacing (cm)	Long duration	90 x 90 cm [OR] 90 x 60	Short duration	60 x 45 [OR] 60 x 30
Variety	Spacing (cm)						
Long duration	90 x 90 cm [OR] 90 x 60						
Short duration	60 x 45 [OR] 60 x 30						
Nutrient management	• 12.5 tonns ha⁻¹- FYM • 30: 15 : 15 kg NPK ha⁻¹.						
Water Management	• First irrigation at the time of sowing. • Life irrigation on third day. • Critical stage for irrigation are primodial initiation and flowering in differential segmental order branches <i>Caution : Moisture stress in critical growth stages may lead to more number of male flowers.</i>						
Weed Management	• Pre- emergence application of Fluchlorin @ 2 lit ha⁻¹ in 900 l of water • First weeding after two to three weeks of sowing at flowering stage • Second weeding at pegging stage.						

Plant protection	Pest		
	Semi looper:	• Spray 0.35 percent Thiodan or 0.03 percent Dimecron.	
	Castor pod borer	• Spray Thiodon @ 200ml / ha	
	Cater pillar:	• Spray @ 0.04 percent malathion.	
	Disease		
	Phytophthora blight and Cercosporaleafspot	• Spray Bordeaux mixture (4:4:50) at 15 day intervals, or spray 2 to 3 time 0.25 percent Dithane M-45.	
Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Four stages		
	Stage	Characters	
	10 days prior to flowering	Stem colour, internode length	
	Flowering	No. of nodes upto primary raceme and sex expression	
	A week before picking	Spike and capsule character, reversion to monoecious in second order	
	After 1 st picking	Reversion to monoecious or flower initiation in third order raceme	
Field Inspection by Seed certification officer	Two inspections		
	Inspection	Growth Stage	Parameters
	I	Flowering	Isolation, volunteer plants, offtypes, etc.,
	II	Maturity	Isolation, volunteer plants, offtypes, etc.,
Field Standards			
		Foundation	Certified
	Off-types	0.10	0.20
Harvest & Seed	- Ready for harvest at after 90-120 DAS - Castor produces 4- 5 sequential order spikes, which can be harvested in 3-4 pickings at an interval of 30 days.		

Extraction	• Harvestable maturity : Fruits start turning yellow • Shattering is not a problem in any variety • Gather the fruits and place in piles • Allow sun drying until fruits blacken • Threshing is done with castor Sheller [or] Manual threshing is done with pliable sticks. • Winnow the seeds and clean with screens to remove husks, dry skins, clods and stones. • Dry the seeds to 8 per cent moisture content		
Seed yield	1.5 to 2 tonns / ha		
Size Grading	8/64 “ round perforated sieve		
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	None	None
	Weed seeds (maximum)	None	None
	Other distinguishable varieties (maximum)	5/Kg	10/Kg
	Germination (minimum)	70 %	70 %
	Moisture (minimum)	8.0 %	8.0 %
	For vapour- proof containers (maximum)	5.0 %	5.0 %
Seed storage period	Upto 2 years		

HYBRID CASTOR

Tool	Two line breeding using Pistillate mechanism - In addition to monoecism a sub form of dioecism exists in castor, which has led to the identification of pistillate mechanisms. N Type - It is governed by a recessive sex switching gene. This can be maintained by sibmating. The progeny from seed produced on pistillate plants segregates in 1: 1 ratio of pistillate and monoecious plants. S Type - Racemes start out as female and then revert to normal monoecism any time after the first raceme. NES Type - This line is normally pistillate under moderate temperature but produces interspersed staminate flowers under high temperature. In crossing fields (hybrid seed production plot) usually one or two roguing of the female line are sufficient to ensure that all flowering plants are pistillate to remove off types that appear.
Floral Biology	- Castor is a cross pollinated crop, protogynous and wind pollinated. - Inflorescences (racemes) are borne terminally on the main and lateral branches. - The racemes of castor are monoecious with the pistillate flowers on the upper 30-50% and staminate flowers on the lower part of the inflorescence. - The proportion of pistillate and staminate flowers among the racemes varies a great deal both within and among genotypes. It is influenced by the environment of the plant, genotypes and nutrition. - Female tendency is the highest in winter, while male tendency predominates in summer and rainy seasons. - Also, the femaleness in young plants with high levels of nutrition is stronger than in old plants with low levels of nutrition.
Pollination	Cross pollination

Problems in Hybrid seed production	Sex expression • Delay or advancement in planting beyond optimum season will lead to alteration in sex expression of female parent. • During roughing seed producer has to examine every female plant regularly for any possible reversion to monoecism at primary, secondary, tertiary and quarternary orders, • It is necessary to remove whole plant if reversion to monoecious nature occurs upto tertiary order of branches. • If reversion occurs after tertiary only, remove the specific spike well before anthesis and destroy or throw them beyond the permissible isolation distance.																											
Method of seed production	• Two line breeding <table><tr><td>Female parent (Pistillate)</td><td>Open pollination</td></tr><tr><td>Male parent</td><td>Open pollination</td></tr><tr><td>Hybrid seed production</td><td>Female x male</td></tr></table> • Pistillate lines were maintained conventionally by retaining 20 per cent monoecious or sex revertants as pollen source Popular hybrids <table><tr><th>Hybrids</th><th>Female parent</th><th>Male parent</th></tr><tr><td>GCH 3</td><td>TSP 10 R</td><td>JI 15</td></tr><tr><td>GAUCH 1</td><td>VP1</td><td>V19</td></tr><tr><td>GCH 2</td><td>VP1</td><td>JI 35</td></tr><tr><td>GCH 4</td><td>VP1</td><td>48 -1</td></tr><tr><td>TMVCH 1</td><td>LRES 17</td><td>TMV 5</td></tr><tr><td>YRCH 1</td><td>DPC 9</td><td>TMV 5</td></tr></table>	Female parent (Pistillate)	Open pollination	Male parent	Open pollination	Hybrid seed production	Female x male	Hybrids	Female parent	Male parent	GCH 3	TSP 10 R	JI 15	GAUCH 1	VP1	V19	GCH 2	VP1	JI 35	GCH 4	VP1	48 -1	TMVCH 1	LRES 17	TMV 5	YRCH 1	DPC 9	TMV 5
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Steps in hybrid seed production	<table><tr><td>Foundation seed</td><td colspan="2">Multiplication of female and male parents</td></tr><tr><td>Certified (Hybrid) seed</td><td colspan="2">Crossing of male and female parents</td></tr></table>			Foundation seed	Multiplication of female and male parents		Certified (Hybrid) seed	Crossing of male and female parents	
	Foundation seed	Multiplication of female and male parents							
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	Female seed production - Female parents are raised in male promoting environment - Plants produce environmentally sensitive staminate flowers - Crossing between the pistillate and staminate flowers leads to self-production of the female parents								
Male seed production Open pollination of monoecious plants leads to seed production of male parents									
Isolation	<table><tr><td>Foundation seed</td><td>300 m</td></tr><tr><td>Certified seed</td><td>150 m</td></tr></table>			Foundation seed	300 m	Certified seed	150 m		
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Time of sowing	<table><tr><td>Parental seed multiplication</td><td colspan="2">January - February</td></tr><tr><td>Hybrid seed production</td><td colspan="2">1st fortnight of September</td></tr></table>			Parental seed multiplication	January - February		Hybrid seed production	1st fortnight of September	
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Seed rate	<table><tr><td>Female parent</td><td>5 kg ha⁻¹</td></tr><tr><td>Male parent</td><td>2 kg ha⁻¹</td></tr></table>			Female parent	5 kg ha ⁻¹	Male parent	2 kg ha ⁻¹		
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Special operation I	Planting ratio & Border rows <table><tr><td></td><td>Planting ratio</td><td>Border rows</td></tr><tr><td>Single cross</td><td>3:1 or 4-6:1</td><td>4</td></tr></table>				Planting ratio	Border rows	Single cross	3:1 or 4-6:1	4
	Planting ratio	Border rows							
Single cross	3:1 or 4-6:1	4							
Special Operation II	Staggered sowing - Flowering in male and female parents should not vary much , flowering of male and female parents will not be synchoronized - If the variation is higher , staggered sowing is required . - In the case of GCH 4 , male parent has to be sown 7-8 days earlier .								

Spacing	90 X 40 cm																	
Nutrient management	12.5 tonns ha⁻¹- FYM 40: 50 : 40 kg NPK ha⁻¹. - Apply 10 Kg N ha⁻¹ at 30 DAS of female parent and after first picking . - This will encourage development of latter order branches and spikes																	
Roguing by Seed producer	- Remove off-types, diseased and pest infested plants. - Four stages <table><tr><th>Stage</th><th>Male/Female</th><th>Characters</th></tr><tr><td>10 days prior to flowering</td><td>Both</td><td>Stem colour, leaf shape, internode length, bloom</td></tr><tr><td>Flowering (Primary spike initiation)</td><td>Both</td><td>No. of nodes upto primary raceme and sex expression ie., proportion of male to female flowers (Continue for 10-15 days until completion of flower) initiation</td></tr><tr><td>A week before picking</td><td>Female</td><td>Spike and capsule character, reversion to monoecious in second order</td></tr><tr><td>After 1st picking</td><td>Female</td><td>Reversion to monoecious or flower initiation in third order raceme</td></tr></table>			Stage	Male/Female	Characters	10 days prior to flowering	Both	Stem colour, leaf shape, internode length, bloom	Flowering (Primary spike initiation)	Both	No. of nodes upto primary raceme and sex expression ie., proportion of male to female flowers (Continue for 10-15 days until completion of flower) initiation	A week before picking	Female	Spike and capsule character, reversion to monoecious in second order	After 1 st picking	Female	Reversion to monoecious or flower initiation in third order raceme
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Field Standards			
		Foundation	Certified
	Off-types (including plants found to flower over the main stem)	0.50	1.00
	Male variants (three secondary rachis from base and above possessing only male flowers are considered as mostly male)	1.00	2.00
	Female variants	1.00	2.00
	Monoecious plants and the racemes reverted to monoecism on female plants before anthesis	In female parent , at least 25 % of plants required for pollen source	2.00
Harvest & Seed Extraction	• Castor produces 4 or 5 sequential order spikes, which can be harvested in 3-4 pickings starting from 90-120 days at 25-30 days interval. • The male parent is harvested first. • Admixture of male and female line should be avoided • Seeds from female parent constitute the hybrid seed		
Seed yield	15-20 qtl/ha seed yield		
Seed Standards		Foundation	Certified
	Pure seed (minimum)	98 %	98 %
	Inert matter (maximum)	2.0 %	2.0 %
	Other crop seed (maximum)	None	None
	Weed seeds (maximum)	None	None
	Other distinguishable varieties (maximum)	5/Kg	10/Kg
	Germination (minimum)	70 %	70 %
	Moisture (minimum)	8.0 %	8.0 %
	For vapour- proof containers (maximum)	5.0 %	5.0 %

GROUNDNUT (*Arachis hypogaea*)

Interesting Facts	• There are three types of groundnut: Spreading, Semi-spreading and Bunch type. • Seed coat colour varies from light pink, pink, red or red mottled with white.				
Floral Biology	• Flowers possess long, tubular calyx. • Corolla consists of five free petals which includes standard, wing and keel petals. • Stigma remains enclosed in keel petal even in fully opened flowers. • Hence self pollination is the rule.				
Pollination	• Self pollination (Cross pollination 0-5 %)				
Mode of propagation	• Seed				
Seed Multiplication Ratio	• 1: 4 (lowest seed multiplication ratio among the field crops)				
Seed Multiplication Stages	• Breeder Seed – Foundation seeds (5 stages) - Certified seeds • This is followed because of very low seed multiplication ratio and since self pollination is the rule.				
Land requirement	• The same variety of same seed class should not have been cultivated in the previous season to avoid volunteer plants. • Unless the variety grown was the same and approved by a certification agency for varietal purity. • The field should be properly leveled. • Well drained, sandy loam soil rich in humus content, calcium and boron with neutral pH. • Do not rotate with solanaceous crops such as tomato, potato and brinjal if bacterial wilt diseases is a serious problem.				
Isolation	<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>Foundation seed</td><td>3 m</td></tr> <tr> <td>Certified seed</td><td>3 m</td></tr> </table>	Foundation seed	3 m	Certified seed	3 m
Foundation seed	3 m				
Certified seed	3 m				
Time of sowing	• June – July and December – January				

Seed rate	<table border="1" data-bbox="508 285 1153 459"> <tr> <th>Varieties</th><th>kg/ha</th></tr> <tr> <td>Bunch type</td><td>100</td></tr> <tr> <td>Spreading type</td><td>80</td></tr> <tr> <td>Semi-spreading type</td><td>80</td></tr> </table>	Varieties	kg/ha	Bunch type	100	Spreading type	80	Semi-spreading type	80
Varieties	kg/ha								
Bunch type	100								
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Semi-spreading type	80								
Seed treatment	• Thiram + Bavistin (1:1) @ 2.5 g kg⁻¹ or • Bavistin @ 2 g kg⁻¹ or • <i>Trichoderma viridi</i> (organic fungicide) @ 4 g kg⁻¹ Fungicide treated seeds should be again treated with bacterial culture of Rhizobium. • <i>Rhizobium</i> & Phospho bacterium @ 600 g each for seeds of one ha <i>Caution :Trichodermaviridi, Rhizobium & Phospho bacterium treatment should be done 24 h after the fungicidal treatment.</i>								
Method of sowing	• Sowing – with seed drill or behind country plough. • Depth of seeding - 5 to 8 cm.								
Spacing	• Crop spacing differs with varieties. <table border="1" data-bbox="508 951 1153 1125"> <tr> <th>Varieties</th><th>Spacing</th></tr> <tr> <td>Bunch type</td><td>25 x 15 cm</td></tr> <tr> <td>Spreading type</td><td>60 x 15 cm</td></tr> <tr> <td>Semi-spreading type</td><td>45 x 15</td></tr> </table>	Varieties	Spacing	Bunch type	25 x 15 cm	Spreading type	60 x 15 cm	Semi-spreading type	45 x 15
Varieties	Spacing								
Bunch type	25 x 15 cm								
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Semi-spreading type	45 x 15								
Nutrient management	• 12.5 tonns ha⁻¹- FYM • 10 kg - Borax • 12.5 kg - Micro nutrient mixture (at the surface of the soil) • 40: 40 : 60 kg NPK ha⁻¹. application of Calcium Sulphate (Gypsum) • Apply gypsum @ 400 kg/ha by the side of the plants on 40th to 70th day depending upon soil moisture. • Apply gypsum, hoe and incorporate it in the soil and then earth up. • Avoid gypsum in calciferous soils. • Gypsum is effective in soils deficient in calcium and sulphur.								

	<table><tr><td rowspan="2">Method of application</td><td colspan="3">(Kg ha⁻¹)</td></tr><tr><td>Urea</td><td>Diammonium Phosphate</td><td>Potash</td></tr><tr><td>Basal</td><td>40</td><td>40</td><td>60</td></tr><tr><td colspan="4">Top dress</td></tr><tr><td>40-45 DAS Peg formation stage</td><td colspan="3">Gypsum 400 kg</td></tr><tr><td>Foliar Spray</td><td colspan="3">DAP 0.5 % at flowering stage</td></tr><tr><td colspan="4">Caution :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .</td></tr></table>	Method of application	(Kg ha ⁻¹)			Urea	Diammonium Phosphate	Potash	Basal	40	40	60	Top dress				40-45 DAS Peg formation stage	Gypsum 400 kg			Foliar Spray	DAP 0.5 % at flowering stage			Caution :Apply the fertilizer during sowing at 10-15 cm deep or on sides so as to keep it away from seed to avoid scorching effect on the seeds .			
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Water management	• One irrigation at the time of sowing • Life irrigation on third day. • Critical stage for irrigation- flowering, pegging stage, pod formation and development stage																											
Weed management	• Pre- emergence application of Fluchlorin @ 2 lit ha⁻¹ in 900 l of water • First weeding after two to three weeks of sowing at flowering stage • Second weeding at pegging stage.																											
Plant protection	<table><tr><td colspan="2">Pest</td></tr><tr><td>Red hairy cater pillar</td><td> • Spray Dichlorovos or Fenitrothion @ 750 ml ha⁻¹. </td></tr><tr><td>Aphids</td><td> • Spray Metasystox 25 EC @ 1 lit ha⁻¹ in 1000 l of water . </td></tr><tr><td>Prodinia&Hel iathis</td><td> • Apply Endosulfan 10% dust @ 25 kg ha⁻¹or • Spray Endosulfan @ 1250 ml ha⁻¹. </td></tr><tr><td>Pod borer</td><td>Soil application (prior to sowing) • Malathion 5% dust @ 25 kg ha⁻¹or • Endosulfan 4% dust 25 kg ha⁻¹ </td></tr><tr><td>White grubs</td><td>Soil application (prior to sowing) • Carbofuran 10 % granules @ 12 kg ha⁻¹or • Lindane 1% dust @ 40 kg ha⁻¹ </td></tr><tr><td colspan="2">Disease</td></tr><tr><td>Tikka leaf spot</td><td> • Spray Dithane M –45 @ 0.25 % or • Duter solution @ 0.075 % </td></tr><tr><td>Rust</td><td> • SparyMancozeb @ 1 kg ha⁻¹or • Chlorothalonil @ 1 kg ha⁻¹or </td></tr></table>			Pest		Red hairy cater pillar	• Spray Dichlorovos or Fenitrothion @ 750 ml ha⁻¹.	Aphids	• Spray Metasystox 25 EC @ 1 lit ha⁻¹ in 1000 l of water .	Prodinia&Hel iathis	• Apply Endosulfan 10% dust @ 25 kg ha⁻¹or • Spray Endosulfan @ 1250 ml ha⁻¹.	Pod borer	Soil application (prior to sowing) • Malathion 5% dust @ 25 kg ha⁻¹or • Endosulfan 4% dust 25 kg ha⁻¹	White grubs	Soil application (prior to sowing) • Carbofuran 10 % granules @ 12 kg ha⁻¹or • Lindane 1% dust @ 40 kg ha⁻¹	Disease		Tikka leaf spot	• Spray Dithane M –45 @ 0.25 % or • Duter solution @ 0.075 %	Rust	• SparyMancozeb @ 1 kg ha⁻¹or • Chlorothalonil @ 1 kg ha⁻¹or							
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		• Tridemorph @ 500 ml ha⁻¹.							
	Root rot	Seed treatment • Carbendazim @ 2g kg⁻¹ or • Carbendazim @ 1g l⁻¹							
Rouging by Seed Producer	• Remove off-types, diseased and pest infested plants. • Three rougings <table><tr><td>First rouging</td><td>Vegetative stage</td></tr><tr><td>Second rouging</td><td>Flowering stage</td></tr><tr><td>Third rouging</td><td>Maturity stage</td></tr></table> • Offtypes are identified based on foliage colour, spreading habit, flowering.			First rouging	Vegetative stage	Second rouging	Flowering stage	Third rouging	Maturity stage
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Field Inspection by Seed certification officer	• Two inspections <table><tr><td>Inspection</td><td>Growth Stage</td><td>Parameters</td></tr><tr><td>I</td><td>Flowering to maturity</td><td>Isolation, volunteer plants, offtypes, , etc.,</td></tr></table>			Inspection	Growth Stage	Parameters	I	Flowering to maturity	Isolation, volunteer plants, offtypes, , etc.,
Inspection	Growth Stage	Parameters							
I	Flowering to maturity	Isolation, volunteer plants, offtypes, , etc.,							
Pod verification / Inspection (Post harvest inspection)	On receipt of groundnut pods at the processing unit, the SCO (or) AO (SC) should examine the physical purity of the lot by testing the presence of ill filled pods, single seeded pods, pest affected pods, discoloured seeds and damaged pods in one kg sample. The presence of ill filled pods should not exceed 4 percent . Pod inspection is done to verify the varietal characters. It should be done, with in one month of final inspection.								
Field Standards	• Two inspections <table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Off-types</td><td>0.10</td><td>0.20</td></tr></table>				Foundation	Certified	Off-types	0.10	0.20
	Foundation	Certified							
Off-types	0.10	0.20							

Harvest & Seed Extraction	- Physiological maturity : Inside shell turn to black/grey colour. Production of mottling / rattling sound. - Ready for harvest at 85-90 DAS - Irrigate the field prior to harvest to facilitate easy harvest. - Pull out the plants and heap it before stripping of pods. - Strip off the pod from the plants manually or with groundnut stripper. - The crop is left in the field to dry for about one week. - Dry the pods upto 8-9 per cent moisture content- rattling sound indicates proper drying. - Shell the pods manually or with groundnut decorticator. - Dry the seeds to 7-8 % moisture content.																										
Seed yield	2 t ha⁻¹																										
Seed Standards	<table><tr><td></td><td>Foundation</td><td>Certified</td></tr><tr><td>Pure seed (minimum)</td><td>98 %</td><td>98 %</td></tr><tr><td>Inert matter (maximum)</td><td>2.0 %</td><td>2.0 %</td></tr><tr><td>Other crop seed (maximum)</td><td>None</td><td>None</td></tr><tr><td>Weed seeds (maximum)</td><td>None</td><td>None</td></tr><tr><td>Germination (minimum)</td><td>70 %</td><td>70 %</td></tr><tr><td>Moisture (minimum) (Kernels hand shelled)</td><td>9.0 %</td><td>9.0 %</td></tr><tr><td>For vapour- proof containers (maximum) (Kernels hand shelled)</td><td>5.0 %</td><td>5.0 %</td></tr></table>				Foundation	Certified	Pure seed (minimum)	98 %	98 %	Inert matter (maximum)	2.0 %	2.0 %	Other crop seed (maximum)	None	None	Weed seeds (maximum)	None	None	Germination (minimum)	70 %	70 %	Moisture (minimum) (Kernels hand shelled)	9.0 %	9.0 %	For vapour- proof containers (maximum) (Kernels hand shelled)	5.0 %	5.0 %
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Moisture (minimum) (Kernels hand shelled)	9.0 %	9.0 %																									
For vapour- proof containers (maximum) (Kernels hand shelled)	5.0 %	5.0 %																									
Seed storage period	Kernels for 6 months and Pods for 18 months																										

Micronutrient Deficiency Symptoms-groundnut

Calcium	- Early abortion of embryo leading to normal pods with nil or minute shriveled seed. - Normal plumule developed is of cream colour but, calcium deficiency results in 'Dark plumule'.
Boron	- High incidence of single seeded pods - Presence of 'Hollow heart', which leads to pathogen invasion. - Apply borax 10 Kg + gypsum 200 Kg ha⁻¹ at 45 DAS
Zinc	- Initially, light yellow stripes are visible along the veins. In acute conditions vein chlorosis and cessation of growth of terminal bud is seen. - Soil application of Zn So₄ @ 25 kg ha⁻¹ [OR] Foliar Spray or 0.5 % of Zn So₄

Iron	• Interveinalchlorosis is observed. Plants are stunted. • Foliar Spray of 1 % of Fe So₄ on 30, 40 and 50 DAS
Sulphur	• Stunted growth, uniformly chlorotic plants, thin stemmed and spindle appearance

LECTURE 14

SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF COTTON

Cotton botanically termed as *Gossypium* sp. is a fibre yielding crop. It is known as the queen of fibre crops. It serves as a cash crop to the farmer as the lint serves as the raw material for the textile industry. The seed is used both for multiplication and as animal feed. The success of commercial crop depends on the quality of the basic seed.

Floral Biology	Cotton is an often cross-pollinated crop where the extent of cross-pollination is > 60%. In cotton 4 different species are in popular usage, viz. <i>G. arboreum</i> (eg. K 10) <i>G. herbaceum</i> (e.g. Uppam) <i>G. hirsutum</i> (e.g. MCU varieties) and <i>G. barbadense</i> (e.g. Suvin and Suguna). There is much variation in case of flower opening. Asiatic cotton opens between 8 and 10. a.m. American cotton opens much earlier. Temperature affects the flower opening. After flower opening the cream yellow colour of corolla turns pink within a day and later changes to red. The receptivity of the stigma is 8 to 10 a.m.
Pollination	Often cross-pollinated
Method of seed production	Varieties: Under isolation, by open pollination, the varieties are multiplied. For nucleus seed production, selfing of flowers is done with cotton (lint) dipped in clay or red earth. Hybrids: In cotton both inter and intraspecific hybrids are available. Interspecific Hybrid: Varalakshmi : Lakshmi x SB298 E (<i>G. hirsutum</i> x <i>G. barbadense</i>) DCH 32 / Jayalakshmi: DS 28 x SB 425 (<i>G. hirsutum</i> x <i>G. barbadense</i>) TCHB 213: TCH 1218 x TCB 209 Intraspecific hybrid : Suguna, Savitha (T7 x M12)

Land requirement	The field should be fertile and formed into ridges and furrows. Black cotton soils are highly preferable than other soils. Land should be free from volunteer plants and designated diseases especially the wilt disease.		
Isolation	Foundation seed		Certified seed
	Varieties	50	30
	Hybrids	50	30
Time of sowing	Season Winter crop : Aug - Sep Summer crop : Feb - March Seeds and Sowing: Seeds should be obtained from an authenticated source with tag and bill.		
Seed rate	Varieties : 15 kg/ha (fuzzy seed) 7.5 kg/ha (delinted seed) Hybrids : Male : 2 kg /ha and Female: 4 kg /ha.		
Seed treatment	Treat the seeds with Azospirillum at 3 packets (600 g/ha) and 2 kg of Azospirillum / ha mixed with 25 kg of FYM and 25 kg of soil and applied on the seed line. This saves 25 % nitrogen besides increasing yield.		
Hybrid seed production tool	The hybrid seed production in cotton is achieved through emasculation and dusting technique. Emasculation: Removal of stamens or staminal column in a bisexual flower of female parent. Dusting: Pollen grains from the male parent in a hybrid seed production plot are collected and dusted over the stigma of the emasculated flowers of female parent. This method is also referred as Doak Method.		
Spacing	Spacing – Varieties		Hybrids
	1. Long duration	:90 x 30 cm	♀ : 120 x 60 cm
	2. Short duration	:60 x 30 cm	♂ : 90 x 60 cm
Planting ratio	8:2. But here it is block system is followed where flowers of 2 parts of male are sufficient to dust 8 parts of female parent. The male and female parents are raised at an isolation of 5 m to avoid genetic and physical contamination.		

Nutrient management	Manures and fertilizers Compost : 12.5 tons/ha Total : 100:50:25 NPK kg/ha Basal : 50:50:25 NPK kg/ha Top dressing: 25:0:0 NPK kg/ha (40-45 days after sowing) 25:0:0 NPK kg/ha (70-75 days after sowing)
Special Operation I	Foliar spray: Spray DAP 2% (for female parents, spray on 60, 70, 80 and 90th days after sowing. (Soak 5 kg of DAP in 25 liters of water over night and supernatant liquid should be taken and mixed with 475 liters of water for spraying 1 hectare). Micronutrient application: Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture Tamil Nadu with enough sand to make a total quantity of 50 kg for one hectare. NAA application: Spray 40 ppm of NAA (40 mg of NAA dissolved in 1 liter of water) at 40 / 45th day using high volume spray liquid in 1125 liter /ha. Repeat the same dose after 15 days of first spray. Advantage of NAA spray: to overcome the boll shedding will occur either due to extreme dry climate or lesser frequency of irrigation or physiological disorder. Topping: Arrest terminal growth by nipping the terminal 10-12th node for controlling excessive vegetative growth.
Water management	Once in 10 days. Critical periods are boll formation to boll maturation stages.
Roguing by Seed producer	The crop should be rogued for off types, selfed plants, from vegetative phase to harvest phase depending on plant stature, leaf size, leaf colour, hairiness, stem colour, flower colour, petal spot, pollen colour, number of sympodia, boll size, boll shape, pittedness etc. to maintain genetic purity.
Field Inspection by Seed certification officer	A minimum of two inspections shall be made from the time crop approaches flowering until it is ready for harvesting.

Field Standards	Field standards				
	Maximum permitted (%)				
	Foundation seed		Certified seed		
	Varieties	Hybrids	Varieties	Hybrids	
	Off types	0.1	0.1	0.2	0.5
Harvest & Seed Extraction	✓ The seed attains physiological maturation 45 days after anthesis. ✓ The initiations of hair line cracks on the dried bolls are the physical symptoms of physiological maturation. ✓ At that time, the moisture content will be 30-35%. ✓ The bolls are harvested as pickings in cotton. ✓ Due to continuous flowering habit once over harvest is not practiced. ✓ As and when the bolls burst with hairline cracks the bolls are collected and dried. ✓ Normally five to seven pickings can be practiced in a crop. ✓ But early 4-5 pickings are recommended for seed purpose. ✓ Harvest in the morning hours upto 10 to 11 a.m. only when there is moisture so that dry leaves and bracts do not stick to the kapas and lower the market value. ✓ Pick kapas from well burst bolls only. ✓ Remove only the kapas from the bolls and leave the bracts on the plants. ✓ As kapas is picked, sort out good puffy ones and keep separately. ✓ Keep stained, discoloured and insect attacked kapas separately.				
Special operation II	Kapas sorting Kapas is sorted manually to pick good quality seeds. Hard locks are to be removed (Kapas without proper bursting and lint is light yellow in colour), since these kapas mostly result in poor quality seeds, due to boll worm or other insect attack. Skewed bolls or ill filled or nonviable seeds are formed if stigmatic lobes are not pollinated. Acid delinting ✓ Fuzzy seeds will clog with one another. So for easy handling the seeds are delinted using H_2SO_4 @ 100 ml/kg of seed for 2-3 minutes. ✓ After acid treatment, the seed should be washed thoroughly for 3 to 4 times with fresh water. ✓ From the floaters, mature seeds without any visible damage can be picked and added to the sinkers.				

Seed yield	Physiological maturity:Cracking of bolls and kapas bursting. 1800-2500 kg/ha		
Size Grading	The free flowing delinted seeds can be graded using 10/64” round perforated metal sieve.		
Seed Standards	Characters	Foundation seed	Certified seed
	Physical purity % (min)	98	98
	Inert Matter % (max)	2.0	2.0
	Other crop seeds (max)	5 kg ⁻¹	10 kg ⁻¹
	Weed seeds (max)	5 kg ⁻¹	10 kg ⁻¹
	Genetic purity (%)	100	100
	Germination (min) %	65	65
	Moisture content (max) %		
	a. Moisture pervious	10	10
	b. Moisture vapour proof	6	6
Seed storage	The seeds can be stored upto 8-9 months in moisture pervious container and upto 12-15 months in moisture vapour proof containers. The seed treatment with thiram @ 2.5 kg⁻¹ or chlorine based halogen mixture @ 3g kg⁻¹ will protect the seed from storage fungi <i>Aspergillus</i> spp and preserve the storability.		
Mid storage correction	✓ The fuzzy and delinted seeds can be soaked in double the volume of 10⁻⁴ molar solution of Na₂HPO₄ for 2 and 1 hr respectively (3.59 g / 100 l of water.) ✓ Then the seeds are shade and sun dried to bring back to the moisture content of 10-12%. The mid storage correction improves the planting value of old seeds. ✓ Dead seeds may be removed by soaking acid delinted cotton seeds in monolayer for 3 h and drying back to original moisture content. ✓ The seeds when put into potable water will separate into sinkers and floaters. Dead seeds become buoyant and float.		

LECTURE 15

SEED PRODUCTION TECHNIQUES IN FORAGE CROPS (CENCHRUS SPECIES AND LUCERNE)

SEED PRODUCTION TECHNIQUES OF *Cenchrus spp*

Cenchrus is a widespread genus of plants in the grass family. Its species are native to many countries in Asia, Africa, Australia, the Americas, and various oceanic islands. Common names include **buffel grasses**, **birdwood grass**, **sandburs**, and **sand spur**. Such names allude to the sharp, spine-covered burrs characterizing the inflorescences of the members of the genus.

***Cenchrus ciliaris* (Buffel-grass or African foxtail grass; syn. *Pennisetum ciliare* (L.) Link)** is a species of grass native to most of Africa, southern Asia (east to India), southern Iran, and the extreme south of Europe (Sicily). Other names by which this grass is known include dhaman grass, anjan grass and white kolukkatai grass.

***Cenchrus setigerus* (Birdwood grass).** Other names by which this grass is known include black anjan grass and black kolukkatai grass.

***Cenchrus glaucus* (Blue buffel grass).** Other names by which this grass is known include Neelakolukattai.

Season	:	Oct – Dec		
Variety	:	CO 1 (Neela kolukkatai)		
Soil and field preparation	:	Land to be used for seed production of <i>Cenchrus</i> shall be free of volunteer plants. Plough the field twice or thrice with an iron plough to ensure good tilth..Well drained soil with high calcium content is suitable.		
Seed rate	:	6 - 8 kg/ha or 40,000 rooted slips/ha.		
Dormancy breaking treatments		Fresh seeds have dormancy for 6 - 8 months. To break dormancy, soak the seeds in 1 % potassium nitrate solution for 48 hours prior to sowing.		
Spacing	:	50 x 30 cm		
Isolation distance		Contaminants	FS	CS
		Fields of other varieties of the same spp. Fields of the same variety not conforming to varietal purity requirements for certification	20m	10m
		Fields of another <i>Cenchrus</i> sp. known to cross or suspected of being able to cross	200m	100m
Weed	:	Hand weeding can be done as and when necessary.		

management		
Water management	:	Immediately after sowing and life irrigation on 3 rd day. Then once in 10 days depending on soil type and weather condition. Sprinklers can also be installed.
Manures and fertilizers	:	Apply NPK fertilizers as per soil test recommendations. If the soil test is not done, follow the blanket recommendation of 25: 40: 20 kg N, P and K per hectare.
Field inspection	:	A minimum of three inspections shall be made, the first before flowering, the second during flowering and the third at maturity and prior to harvesting.
Seed harvest	:	The physiological maturity of seeds was attained at 18 DAA accompanied by maximum dry weight, germination and vigour.
Seed yield	:	55 – 60 kg/ha

Field standards

	Maximum permitted (%)	
Factor	Foundation	Certified
Off-types	0.10	1.0

Seed Standards

Factor	Standards for each class	
	Foundation	Certified
Pure seed (minimum)	80.0%	80.0%
Inert matter (maximum)	20.0%	20.0%
Other crop seeds (maximum)	20/kg	40/kg
Weed seeds (maximum)	20/kg	40/kg
Germination (minimum)	30%	30%
Moisture (maximum)	10.0%	10.0%
For vapour-proof containers (maximum)	8.0%	8.0%

LUCERNE (*Medicago sativa* L.)

Lucerne (*Medicago sativa* L.) has been probably the first crop under cultivation for forage production since pre-historic times. Lucerne is popularly known as alfalfa, rightly called as **queen of forage crops** in India and considered as one of the important forage legumes. The existing *M. sativa* is largely a tetraploid species ($2n=4x=32$) and it belongs to **leguminosae**

family. It can be grown under wide range of soil conditions throughout the world. In Tamil, it is known as **Kudirai masal**. It is very palatable and nutritious forage legume containing 15 - 20% crude protein on dry matter basis. Lucerne adds nitrogen to soil and improves soil fertility.

Season	:	Seed production once in a year during summer months. The crop should be harvested during first week of March and allowed for seed production in such a way that the peak period of flowering should coincide with summer days.
Variety	:	Varieties include Anand 2, Sirsa - 9, IGFR S - 244, CO 1 and CO2
Soil and field preparation	:	Land to be used for seed production of lucerne shall be free of volunteer plants. Well drained black cotton soil is good. Can be raised in alkaline soils also. Plough 2 to 3 times to obtain a good tilth and form beds and channels of convenient size
Seed rate	:	15 kg/ha
Spacing	:	25 cm apart in lines
Isolation		Foundation seed:400m Certified seed:100m
Weed management	:	Hand weeding whenever necessary
Water management	:	Immediately after sowing and life irrigation on 3 rd day. Then once in 10 days depending on soil type and weather condition. Sprinklers can also be installed
Manures and fertilizers	:	NPK - 15: 120 : 40 kg/ha.
Plant protection	:	Spraying 0.05% Methyl Dematon or Dimethoate is suggested for the control of Aphids/Ear head bug.
Micronutrient application	:	Spraying of 3% Borax + ZnSO ₄ solution at flower initiation and 50 % flowering stage.
Field inspection	:	A minimum of two inspections shall be made from the time the crop approaches flowering until it is ready for harvesting.
Seed harvest	:	Hand picking of pods at physiological maturity stage (Dark green colour of pod turned to yellowish green)
Seed yield	:	200-240 kg/ha

Field standards

Factor	Maximum permitted (%)	
	Foundation	Certified

Off-types	0.20	1.0
Objectionable weed plants	None	0.050

Seed Standards

Factor	Standards for each class	
	Foundation	Certified
Pure seed (minimum)	98.0%	98.0%
Inert matter (maximum)	2.0%	2.0%
Other crop seeds (maximum)	10/kg	20/kg
Total Weed seeds (maximum)	10/kg	20/kg
Objectionable weed seeds (maximum)	5/kg	10/kg
Germination including hard seeds (minimum)	80%	80%
Moisture (maximum)	10.0%	10.0%
For vapour-proof containers (maximum)	7.0%	7.0%

LECTURE 16-17

SEED PRODUCTION TECHNIQUES IN VARIETIES AND HYBRIDS OF TOMATO, BRINJAL, CHILLIES, BHENDI AND ONION

TOMATO (*Lycopersicum esculentus*)

Botany: Tomato is a typical day neutral plant. It requires temperature of 15-20° C for fruit setting. Tomato is self pollinated crop. Self fertilization is favoured by the position of receptive stigma within the cone anthers and the normal pendant position of the flower.

Method of seed production :Seed to Seed.

Stages of seed production : Breeder seed 4 Foundation Seed I 4 Foundation Seed II 4 Certified Seed

Varieties :

Indeterminate varieties:Pusa Ruby, SolanGola, Yaswant (A-2), Sioux, Marglobe, Naveen, Ptom-9301, Shalimar- 1, Shalimar-2. Angurlata, SolanBajr, SolanSagun, ArkaVikas. AritaSaurbh.

Determinate varieties: Roma (EC-13513), Rupali, MTH-15, Ptom-18, VL-1, VL-2, HS 101, HS 102, HS 110, Pusa Early Dwarf, PusaSheetal, Floradade, ArkaMeghli, Co.1, Co.2, Co.3 (Marutham), PKM.1, Py1,

Hybrids: COTH-1, Pant Hybrid-2, Pant Hybrid-10, Kt-4. Pusa Hybrid-I-4, Arka Shreshta, Arka Vardan, Arka Abhijit, Navell 1 &2 (Sandoz), Rupali, Sonali, MTH 6

Season : May - June and November - December

Land requirement : Selection of suitable land for tomato seed production is important where the previous crop should not be the same variety to avoid the contamination due to the volunteer plants.

Isolation requirement : For Seed production of tomato, varieties require minimum of 50 M for foundation seed and 25 M for certified seed. For hybrid seed production, it requires minimum of 200 M for foundation (parental line increase) and 100 M for certified hybrid seeds.

Seed rate : For i) Varieties - 300- 400 g/ha ii) For F₁ hybrid - Male parent 25 g/ha; Female parent 100 g/ha.

Nursery : Sow the seeds in raised nursery bed of 20 cm height, in rows of 5 cm gap and covered with sand. Eight and ten nursery beds will be sufficient to transplant one acre. Apply 2 kg of DAP 10days before pulling out of seedling.

Transplanting: Transplanting should be done with the seedlings are 20-25 days old, preferably at evening time. Spacing is 60 x 45 cm (90 x 60 cm for female parent and 60 x 45 cm for male parent of hybrids).

Manuring : After thorough preparation of a field to fine tilth, apply 25 tons of FYM per ha. Apply 100 : 100: 100 Kg of NPK/ha of which, 50% of the N is applied as basal dressing and remaining 50% of N as top dressing in two split doses at just before flowering and fruit formation stages.

Roguing : The roguing should be done based on the plant characters (determinate / indeterminate), leaf, branching and spreading characters and also based on fruit size, shape and color. The plants affected by early blight, leaf spot and mosaic (TMV) diseases should be removed from the seed production field.

Planting ratio: For hybrid seed production, the female and male parents are normally planted in the ratio of 12:1 or 12:2.

Pest and disease management: The major pests attacking tomato crop are leaf eating caterpillars and fruit borers, which can be controlled by spraying. The major diseases in tomato are early blight and mosaic virus. The early blight rot can be controlled by spraying Benlate or Dithane M-45.

Crossing (for hybrids): In tomato the hybrid seed production is normally done by 'Emasculation and Hand Pollination'. However use of chemical hybridizing agents (MH-1000 ppm) or CMS lines are also practiced.

Harvesting seed extraction and processing : The fruits are harvested after full maturity of the fruit when turn in to red color fruits from _first and last one or two harvests should not be used for seed extraction. The fruits from in between 6-7 harvest should be used for seed extraction. The seed viability is depends on the method on which the seeds were extracted and hence, it is more important to choose proper methods of seed extraction. Before seed extraction, the fruits are to be graded for true to type and selection of medium to large size fruits for getting higher recovery of quality seeds.

The acid method of seed extraction is the best method for tomato seed extraction. In this method, the fruits are to be crushed into pulp and taken in a plastic containers (or) cement tank. And then add 30 ml of commercial Hydrochloric acid per kg of pulp, stir well and allow it for ½ hour. In between this duration the pulp may be stirred well for one or two times. This facilitates the separation of seed and pulp. After ½ hour, the seeds will settle down at the bottom and then the floating fraction is to be removed. The collected seeds should be washed with water for three or four times.

Comparison of different seed extraction methods

	Fermentation	Acid	Alkali
Method	Crushed pulp along with seed are mixed with excess of water and kept in a container for 20-24 hours	Hydrochloric acid (HCl) is added to the pulp @ 20 ml / kg of pulp. This is kept as such for 20-25 minutes with 3-4 frequent stirring	In one litre of water one g of sodium bicarbonate is to be added. In one litre of water. The cool alkali mixture is added to one kg of pulp
Salient features	• Low cost. • Unskilled labour. • More Time taken • Low Seed recovery (0.5 to 0.6 %) • Dull seed colour. • Seed..borne.. pathogens	• Cost is more. • Skilled labour • Lesser Time • High seed recovery (0.8 to 1 %) • Bright colour market value higher. • Seed borne pathogen – removed • Improper washing leads to injury to seeds	• Recovery 0.7 to 0.8 per cent • Luster of the seeds will be lost. • Improper washing leads to injury to seeds

While following acid method we must use only plastic or stainless steel containers or cement tank. Care must be taken to avoid the usage of iron or zinc containers, which will affect the viability potential of the seeds and as well damage to the containers due to chemical reaction with acid.

For large scale seed extraction we can use the tomato seed extractor developed by Tamil Nadu Agricultural University. The seeds extracted by this machine may again be treated

with commercial Hydrochloric acid @ 2-3 ml/kg seed with equal volume of water for 3-5 minutes with constant stirring. And then seed should be washed with water for to four times.

It is easy to dry the seeds extracted by acid method and also remove the fungus growth over the seed coat, thus seeds possess golden yellow colour and high vigour. The seed extracted by fermentation method posses poor vigour and off colour due to fungal activity.

Hybrid seed production: Emasculation is done before the anthers are mature and the stigma has become receptive to minimize accidental self pollination. Thus emasculation is generally done in the evening, between 4 PM and 6 PM one day before the anthers are expected to dehisce or mature and the stigma is likely to become fully receptive.

Emasculate the bud by hand with the help of needle and forceps. Remove the calyx, corolla and staminal column or anthers, leaving gynoecium i.e., stigma and style intact in the flower. Emasculated flowers should be covered immediately with red coloured paper cover to protect against contamination from foreign pollen and also for easy identification of emasculated bud during dusting. Remove the red paper cover of the emasculated bud and dust the pollen gently over the stigmatic surface using cotton or camel brush, etc., After dusting, the emasculated flowers are again covered with white or other coloured paper cover for two to three days. Pollen collected from one male flower can be used for dusting 5 to 7 emasculated flowers.

Seed Yield : 100-120 Kg/ha

Specific requirements :

Factors	Foundation	Certified
Off types – variety	0.1 %	0.2%
Hybrid	0.01%	0.05%
Plants affected by seed borne diseases	0.1 %	0.5%

Seed standard (variety and hybrid)

Factors	Foundation	Certified
Pure seed (mini)	98%	98%
Inert matter (maxi)	2%	2%
Other crop seeds (maxi) 5/kg	5/kg	10/kg
Weed seeds (maxi)	None	None

Germination (mini)	70%	70%
Moisture (maxi)	8%	8%
For VP container	6%	6%

CHILLI (*Capsicum frutescense*)

Botany: Cross pollinated vegetable. The flower is protogynous. Anther dehisces only half to 5 ½ hr after stigma becomes receptive. Anthesis in chilli occurs between 6.00 and 9.00 hr. Flower remains open for 2 to 3 days, receptivity of stigma was the highest at the day of flower anthesis.

Method of seed production : Seed to seed

Stages of seed production : Breeder seed → Foundation seed → Certified seed.

Varieties: K.1, K.2, K.3, Co.1, Co.2, PKM.1, MDU.1, PKM.1, Bagyalakshmi.

Hybrids: KT.1, (Pusa Deepti), Solar Hybrid 1, Solar Hybrid 2. Early Bounty, Indira, Lario, Hira, Bharat.

Season : June-July, February-March, September- October.

Land requirement : There are no land requirement as of previous crops, but the land should be free from volunteer plants. Generally areas affected by wilt or root rot may be avoided. Crop rotation must be followed to avoid endemic Solanaceous pests.

Isolation requirement : Minimum isolation distance of 400 M for foundation and hybrid seed and 200 M for certified seed production are necessary.

Seed rate : Seed required for one hectare is 500 g to 1 kg for variety; for hybrids - Female = 200 g and male = 50 g

Nursery : Sow the seeds in raised nursery bed of 20 cm height, in rows of 5 cm gap and covered with sand. Eight and ten nursery beds will be sufficient to transplant one acre. Apply 2 kg of DAP 10 days before pulling out of seedling.

Transplanting: The seedlings of 30-35 days old are ready for transplanting. Transplanting may be done on the ridges in the evening. Spacing: Varieties : 60 x 45 cm and Hybrids : 75 x 60 cm

Foliar spray: To arrest the flower drop, NAA (Planofix) can be sprayed @ 4ml/L. Very light irrigation is also done arrest the flower drop.

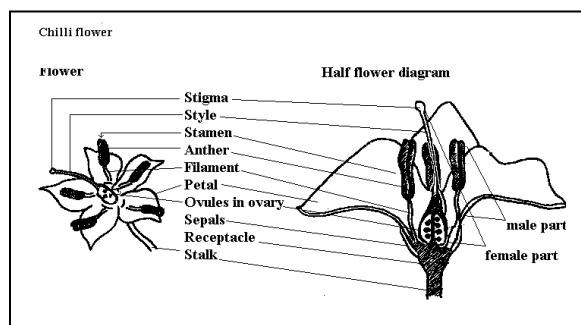
Manuring : Apply 50 tones of FYM/ha for irrigated crop. Basal 0:70:70 kg of NPK and 50 kg of N at 15 days after transplanting and 50 kg N at 45th days after transplanting.

Roguing: Field inspection and roughing should be done both for varieties and hybrid at different stages based on the plant height and its stature, flower colour and pod characters. The plants affected with leaf blight, anthracnose and viral diseases should be removed from the seed field.

Pest and disease management: The important pest attacking chilli and capsicum are thrips, aphids, pod or fruit borer and mites. The thrips and aphids can be controlled by spraying Dimecron (systemic pesticide), pod borer can be controlled by spraying Nuvacron and the mites can be controlled by spraying Kelthane. The major diseases affecting the plants are die back or fruit rot, powdery mildew and bacterial leaf spot. Spray Dithane M-45 for control of die back, Karathane for powdery mildew and Agromycin for leaf spot disease control.

Hybrid seed production: The crossing operation can be performed as per the methods outlined for tomato and brinjal hybrid seed production. However, hand emasculation and pollination is some what difficult since the flowers are minute. Hence use of male sterile lines can also be employed for hybrid seed production.

Structure of chilli flower



Harvesting and processing : Harvesting should be done in different pickings. First and last one or two pickings can be harvested for vegetable purpose. The well ripened fruits with deep, red colour alone should be collected in each picking. After harvest, fruit rot infected fruits are to be discarded. The harvested pods are to be dried under shade for one (or) two days and then under sun for another 2 or 3 days. Before drying pods are to be selected for true to type and graded for seed extraction. The seed are extracted from graded dried pods. The pods are taken in gunny bag and beaten with pliable bamboo sticks. The seeds are cleaned by winnowing and

dried to 10% moisture content over tarpaulin. Then seeds are processed with BSS 8 wiremesh screens. For large scale seed extraction, the TNAU model chilli seed extractor may be used.

Seed Yield : 50-80 Kg/ha

Specific standards :

Factors	Foundation	Certified
Off types	0.1%	0.2%
Designated diseased plant	0.1%	0.5%

Plants affected by diseases

The designated diseases are caused by *Collertotictumcapsici* and leaf blight caused by *Alternariasolari*.

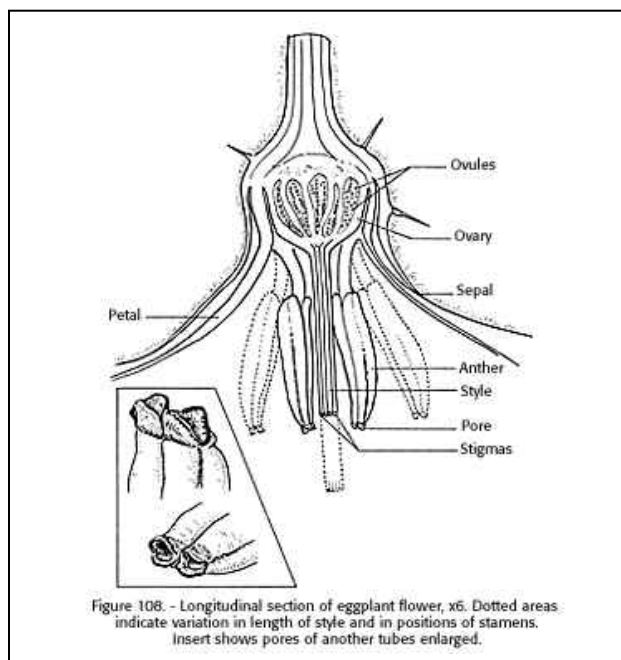
Seed standards (Variety & Hybrid)

Factors	Foundation	Certified
Pure seed	98%	98%
Inert matter	2%	2%
Other crop seeds	5/kg	10/kg
Weed seeds	5/kg	10/kg
Germination	60%	60%
Moisture content	8%	8%
For VP Container	6%	6%

BRINJAL (*Solanum melongena*)

Botany: Brinjal is often cross pollinated crop. Brinjal flower opens mainly in morning. A few flower open at 16 hr. Anther dehiscence occurs 15-20 minutes after flowers have opened. The period of receptivity ranges from a day prior to flower opening to 4 days after opening. Brinjal produces 4 types of flowers with different style length. (Long style, short style, medium style and pseudo short style). For seed production and better yield, the long and medium style is desirable. To increase the production of long and medium style application of more nitrogen or spraying of growth regulators during pre-flowering and flowering stages may be followed.

Structure of Brinjal flower



Method of seed production : Seed to Seed.

Stages of seed production : Breeder seed → Foundation Seed → Foundation Seed II → Certified Seed.

Varieties : Co.1, Co.2. MDU.1, PKM.1, KKM.1, PLR. 1. AU1, Pusa purple long, Arkanidhi, Pant smart, Arkaneelkanth, Arkashrish.

Hybrids: CoBH1, Arka Navneet (IIHR 22-1 x supreme), Pusa H-5, Pusa H-6, MHB 10, MHB 39 (Mahyco), Azad Hybrid.

Season : The brinjal seed production can be taken up in the following 2 seasons. May-June and December- January.

Land requirement : The land should be free of volunteer plants.

Isolation : For varieties, 200 M or 100 M of isolation distance is required for foundation and certified seed, respectively. For hybrid seed production minimum of 200 M isolation distance should be maintained.

Seed rate : Varieties - 400 - 500 g/ha
Hybrids - 200 g/ha (Female)
- 50 g/ha (Male)

Nursery : Sow the seeds in raised nursery bed of 20 cm height, in rows of 5 cm gap and covered with sand. Eight and ten nursery beds will be sufficient to transplant one acre. Apply 2 kg of OAP 10 days before pulling out of seedling.

Transplanting: Seedlings are transplanted when they are 30-35 days old (12-15 cm height) preferably in the evening time. Spacing of 75 x 60 cm (non spreading) and 90 x 60 cm (spreading) varieties, 90 x 60 cm for female parent and 60 x 45 cm for male parent of hybrids.

Manuring : The field should be thoroughly ploughed for fine filth and apply 25 tons of FYM/ha. The other fertilizer requirement for brinjal variety and hybrid are same as followed for tomato seed production.

Roguing: The roguing should be done based on the plant characters, leaf, branching and spreading characters and also based on fruit size, shape and color. The plants affected by Phomopsis blight, leaf spot and little leaf virus disease should be removed from the seed production field.

Pest and disease management: The pests like fruit borer, shoot borer, beetles, aphids, mealy bug and jassids can be controlled by spraying Nuvacron or Methyl parathion. The red spider mite can be controlled by spraying in Kelthane. The important diseases are damping off and little leaf which can be controlled by spraying fungicide and systemic insecticides, respectively. Powdery mildew, leaf spot and anthracnose diseases can be controlled by spraying Benlate.

Hybrid seed production: The planting ratio of female and male parents adopted for hybrid seed production is normally 5:1 or 6:1. For production of hybrid seeds, crossing programme is done using emasculation and dusting methods as followed in tomato.

Harvesting and processing : Harvesting is done when fruits are fully ripe (when the fruits turn into yellow colour) i.e., 405 days after flowering. The harvested fruits are to be graded for true to type and off type and fruit borer infested fruits are discarded. The graded fruits are cut in 2-3 pieces or whole fruits will be put in a cement tank with water and crushed manually and then allow it for fermentation for 1-2 days. Then the floating pulp portions are to be removed, the seeds settled at the bottom should be collected and washed with water and then the seeds with

equal volume of water is treated with commercial Hydrochloric acid @ 3-5 ml/kg of seed. The mixture is kept for 10-15 minutes with frequent stirring. Then the treated seeds are to be washed with water for 3-4 times. Afterwards seeds are dried under shade for 2-3 days over a tarpaulin and followed by sun drying for 1-2 days to reduce the seed moisture content to 8 per cent. Then the seeds are cleaned and graded with BSS 12 sieve. The processed seeds are treated with fungicides or Halogen mixture @ 5g/kg of seed.

Seed Yield : 100-200 Kg/ha

Specific standards :

Factors	Foundation	Certified
Off types – Variety	0.1%	0.2%
Hybrid	0.01%	0.05%
Designated diseased plant	0.1%	0.5%

The designated diseases in brinjal are Phomopsis blight caused by *Phomopsis vexans* and little leaf caused by Datura virus -2.

Seed standards (Variety & Hybrid)

Factors	Foundation & Certified
Pure seed	98%
Inert matter	2%
Other crop seed	None
Weed seed	None
Germination	70%
Moisture content	8%
For VP Container	6%
Genetic purity required for tomato & brinjal hybrids is	90%

BHENDI (*Abelmoschus esculentus*)

Botany: Anthesis is between 9 and 10 hr and is preceded by maximum anther dehiscence between 8 and 9 hr. The stigma remains receptive on the day of anthesis. Bhendi is often cross pollinated crop. Cross pollination to an extent of 12 per cent is due to protogynous.

Method of seed production : Seed to seed

Stages of seed production : Breeder seed → Foundation seed → Certified seed.

Varieties : Co.1, Co.2, MDU.1, ParbhaniKranti, ArkaAnamika, Pusa A-4, PusaSavani

Hybrids: o. 3, Mahyco hybrid, Shoba

Season : June-July, September- October and February- March

Land requirement : Select field on which bhendi crop was not grown in the previous season, unless the crop was of the same variety and certified. Field should be free from wild bhendi (*Abelmoschus sp.*)

Isolation requirement : Seed field must be isolated from other varieties at least by 400 M for foundation and hybrid seed production and 200 M for certified seed production.

Seed rate : Varieties : 8-10 kg/ha
Hybrids : 4 kg/ha (Female)
: 1 kg/ha (Male)

Manuring : Apply 12.5 tons of FYM/ha before ploughing. Apply 150:75:75 kg NPK/ha, of which 50% of the N should be applied as top dressing in two split doses at flowering and 10 days later.

Planting ratio: For hybrid seed production, female and male parents are normally planted in the ratio of 4:1.

Spacing: Ridges & furrow type of layout is use. Sometime bhendi is also sown at the border of main crop & layout is same as for main crop. For summer crop spacing is 45x30 cm. & Rainy season crop spacing is 60 x 45 cm.

Rouging: Minimum of three inspections for varieties and 4 inspections for hybrids, one at Vegetative, two at Flowering, and one at Fruit maturity stages. The rouging should be based on the plant characters, hairiness, fruit character like fruit colour, number of ridges, fruit length etc., and the off type and mosaic attacked plants should be removed from the seed field. Wild bhendi if present should be removed before flowering.

Pest and disease management: The major pest attacking bhendi are jassids, aphids and white fly, which can be controlled by spraying Rogar or Dimecron or Endosulphon. The pod borer and red spider mites can be controlled by spraying Endosulphon and Kelthane, respectively. The diseases such as yellow vein mosaic and powdery mildew can be controlled by spraying systemic insecticides and Karathane, respectively.

Hybrid seed production: In bhendi, since the flowers are large in size, hand emasculation and pollination is the best suitable method for seed production. The emasculation and dusting can be done as per the methods outlined in tomato. The male and female parents are raised in blocks at the ratio of 9:1 (Female : Male).

Harvesting and threshing : Fruits should be harvested when they have dried (30-35 days after crossing). The pods which expose hairline crack and turn to brown colour on drying alone are cut using sickle manually. The pods are dried and threshed using pliable sticks. Separated seeds are winnowed to remove plant debris and dried over a tarpaulin to 10% moisture content. Dried seeds are subject to water floatation in which, good seeds sink while poor seeds float. The floaters are removed, while sinkers are dried under shade followed by sun drying. Then the seed are cleaned, dried and treated with Captan/ Thiram.

Processing: Seeds are to be processed with BSS 7 wiremesh sieve.

Seed Yield : 1000-1200 Kg/ha

Specific standards :

Factors	Foundation	Certified
Off types	0.1%	0.2%
Objectionable weed	None	None
Disease affected plants	0.1%	0.5%

Objectionable weed : wild *Abelmoschus* sp., *A. manihot* and *A. moschatus*

Designated diseases: Yellow Vein Clearing Mosaic (Hybiscus virus-1)

Seed standards

Factors	Foundation	Certified
Pure seed	99%	99%
Inert matter	1%	1%
Other crop seed	None	5/kg
Total weed seed	None	None
Objectionable weed	None	None
Other Distinguishable Varieties (ODV)	10/kg	20/kg
Germination	65%	65%
Moisture	10%	10%
For VP Container	8%	8%

ONION (*Allium cepa*)

Botany: Onion (*Allium cepa*) is known only in cultivation, but related wild species occur in Central Asia. The most closely-related species include *Allium vavilovii* Popov & Vved. and *Allium asarense*. Onion is the biennial crop and takes two full seasons to produce seeds. In the first year bulbs are formed and in the second year stalks develop and seed is produced. Inflorescence of onion is known as Umbel. The crop strongly protandrous. It is a long-day plant. The day length influences bulb onion, but has little effect on induction of seeding. It appears to be day-neutral for seed production. It requires cool conditions during early development of the bulb crop and again prior to and during early growth of seed stalk. Varieties bolt readily between 10 to 15 degree C. In the early stages of growth, a good supply of moisture is required and temperatures should be fairly cool. During bulbing, harvesting and curing of seed, fairly high temperatures and low humidity is desirable. Seed production is widely adapted to temperate and sub-tropical regions. Onion is largely cross-pollinated crop with up to 93 per cent natural crossing but some self-pollination does occur. It is chiefly pollinated by honey-bees

Stages of seed production : Breeder seed → Foundation seed → Certified seed

I. Land requirement: Onion can be grown in different types of soils. The PH of the plot should vary from 6-8. Clay soils are not suitable for onion cultivation. Field should not have been under the same crop in the previous seasons.

II. Varieties and Hybrids: CO 1, CO 2, CO 3, CO 4 and MDU 1, CO(On) 5 free flowering and seed setting type.

III. Isolation:

Contaminants	Minimum Distance (meters)			
	Mother bulb Production		Seed Production Stage	
	Foundation	Certified	Foundation	Certified
Fields of other variety	5	5	1000	500
Fields of the same variety conforming to varietal purity requirements for certification	5	5	1000	500

IV. Seed, Season and Sowing:

Propagation material: Bulb and seed

A.Method of seed production: For onion seed production two methods are followed

1.Seed to seed method: In this method bulb crop is let to over winter in the field so as to produce seed in the following season. This is also called “In situ” Method.

Seed rate: 6 - 8 Kg of graded seed/ha.

Season: Middle of June to Middle of July in the plains

Transplanting: Seedlings of 35 - 40 days may be transplanted in the main field on both sides of the ridges.

Spacing: 20 x 10 cm.

Preparation of field: Plough the land to a fine tilth. Form ridges and furrows at 45 cm spacing. Sow the bulbs on both the sides of the ridges at 10 cm apart.

2.Bulb to Seed Method: The bulb produced in the previous season are lifted, Selected and stored. In the Second year replanted the bulbs for seed production.

Bulb treatment: Remove the top 1/4th of the bulb and treat with fungicides either Carbendazim or Mancozeb at 20 g in 10 lit of water for 5 – 10 minutes.

Bulb sowing: Sow the treated bulbs on the ridges during October - November.

Spacing: 50 x 20 cm.

Hybrid Techniques: Hand emasculatation and pollination

V. Fertilizer: FYM : 25 t/ha.

Application	N	P	K
Basal kg/ha	30	60	30
Top dress (30th day) kg/ha	30	-	-

VI.Irrigation: After sprouting, irrigate at weekly intervals. Withhold irrigation 10 days before harvest.

VII.Plant protection

Thrips and onion fly: Methyl demeton 25 EC 1 ml/lit or dimethoate 30 EC 1 ml/lit or monocrotophos 36 WSC 1 ml/lit with Teepol 0.5 ml/lit.

Cutworm: Drench the soil with chlorpyrifos @ 2 ml/lit.

Leaf spot: Spray Mancozeb 2 gm/lit or Copper oxychloride 2.5 gm/lit. Add Teepol 0.5 ml/lit of spray fluid.

VIII.Rouging: Depending upon the foliage colour, plant type or late maturing off types as thick necks, bottle necks which do not conform to the varietal character.

IX.Harvest: Harvest the well-matured bulbs at 110 - 115 days.

Bulb grading: Select medium sized bulbs of weighing 40 - 50 g.

Harvest: Harvest the earheads during March when 50 % of the seeds in umbels turn into black colour, to prevent the shattering loss.

Seed extraction: Dry the earheads under sun and extract the seeds.

Seed grading: Size grade the seeds by using BSS : 10 x 10 wire mesh sieve (2.5 mm) or 6/64" round perforated metal sieve.

Seed yield: 500 - 600 Kg/ha.

Seed storage: Dry the seeds to reduce the moisture content to 6 - 8 % and treat with Carbendazim 50 % WP @ 2 g /Kg seed and Halogen formulation (Bleaching powder + CaCo₃ + Arappu leaf powder @ 5:4:1) @ 3 g/Kg seed and store the seeds in polythene bags.

X. Specific standards

Factor	Maximum permitted	
	Foundation	Certified
*Bulbs not conforming to varietal Characteristics	0.01 % (by number)	0.05 % (by number)
Off-types (umbels) in seed parent at and after flowering	0.10%	0.05 %
Offtypes (umbels) in pollinator at and after flowering	0.01%	0.05 %
Pollen shedding (umbels) in seed parent at flowering	0.05 %	0.10 %

*Maximum permitted at second inspection at mother bulb production stage.

Seed standards

Factor	Standards for each class	
	Foundation	Certified
Pure seed (minimum)	98.0%	98.0%
Inert matter (maximum)	2.0%	2.0%
Other crop seeds (maximum)	5 per kg	10 per kg
Weed seeds (maximum)	5 per kg	10 per kg
Germination (minimum)	70.0 %	70.0 %

Moisture (maximum)	8.0 %	8.0 %
Moisture vapour proof containers (maximum)	6.0%	6.0%

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LECTURE 19

Seed Production Techniques in Varieties and Hybrids of Gourds - Bittergourd, Ashgourd, Snakegourd, Ribbedgourd, bottlegourd and Pumpkin

S. No.	Common Name	Botanical Name
1.	Bittergourd	<i>Momordica charantia</i>
2.	Ashgourd	<i>Benincasa cerifera</i>
3.	Snakegourd	<i>Trichosanthus anguina</i>
4.	Ribbedgourd	<i>Luffa acutangula</i>
5.	Bottlegourd	<i>Lagenaria siceraria</i>
6.	Pumpkin	<i>Cucurbita moschata</i>

Land Requirements: There are no land requirements as to previous crop, but the land should be free of volunteer plants. Generally the soil should be well drained and aerated.

Isolation Requirements: Most of the cucurbits are monoecious in character and a few are dioecious. A number of hermaphrodite and andromonoecious cultivars are also available in

some crops. Pollination is largely done by insects. For pure seed production and isolation distance all around seed field is necessary to separate it from fields of other varieties, fields of the same variety not conforming to varietal purity requirements for certification, from wild cucurbit species, and to separate musk melon from long melon and vice versa, and pumpkin from summer and winter squashes and vice versa as follows

Class	Variety (Minimum distance)	Hybrid (Minimum distance)
Foundation seed	1000 m	1500 m
Certified seed	500 m	1000 m

Seed production details in Cucurbitaceous vegetables

Particulars	Bittergourd	Snakegourd	Ribbedgourd	Bottlegourd	Ashgourd	Pumpkin
Season	June - July and Feb – March					
Varieties	CO1, MDU1, Coimbatore long green and Coimbatore long white	CO1, CO2, PKM1, MDU1, PLR 1 and PLR 2	CO1, CO2, PKM 1	Pusa Summer Prolific Long, Pusa Summer Prolific Round, Pusa Manjari, Pusa Megdoot and Arka Bahar	CO1, CO2	CO 1, CO 2, Arka Suryamuki and Arka Chandan
Seed rate (kg/ha)	2.5	2.5	2.5	1.5	2.5	1.0
Treatment for increasing female flower	Spraying of Ethrel 200 - 250 ppm at two true leaf stage and after a week of 1st spray					
Spacing (cm)	Take pits of size 45×45×45 cm at 2.5×2.0 m distance					
Fertilizers / (NPK g/pit)	6:12:6	12:24:12	9:15:9	6:12:12	6:12:6	6:12:6
Physiological maturity	Change of fruit colour in any part or 1/3 of fruit tip to yellow to red		Complete drying of fruits and rattling sound when shaken		Appearance of ashy coating and metallic Sound	Change of fruit colour to orange brown
Processing	Hand picking	Hand picking	BSS 4 wire mesh		16/64 round perforated sieve	
Fruit to seed recovery (%)	30	15-16	13-14		1.0-1.3	1.0-1.3
Seed yield (kg/ha)	120-150	220-250	200-250		120-150	120-150

Seed standards

Factor	Standards for each class	
	Foundation	Certified
Pure seed (minimum)	98.0%	98.0%
Inert matter (maximum)	2.0%	2.0%
Other crop seeds (maximum)	None	None
Total Weed seeds (maximum)	None	None
Germination including hard seeds (minimum)	60%	60%
Moisture (maximum)	7.0%	7.0%
For vapour-proof containers (maximum)	6.0%	6.0%

LECTURE 20

Principles of GM Crop and Organic Seed Production

GM Crops

The term genetically modified (GM), refers to the transfer of genes between organisms using a series of laboratory techniques for cloning genes, splicing DNA segments together, and inserting genes into cells. Collectively, these techniques are known as recombinant DNA technology. Most current GM crops grown in the India are engineered for insect resistance or herbicide tolerance. Potential future applications of the technology include nutritional enhancements, stress tolerance, disease resistance, biofuel efficiency, and remediation of polluted sites.

Regulatory measures specific to transgenic seeds

Recognizing the potential of Genetic Engineering and its relevance to India, Ministry of Science and Technology gave sufficient impetus for research and monitoring of transgenic seed development. The measures of transgenic regulation fall under the Environment and Protection Act, 1986 and EPA rules, 1989. Establishment of Department of Biotechnology in 1986 exclusively to apply biotechnological approaches in agriculture and human health

- Establishment of Review Committee on Genetic Manipulation (RCGM) in 1989 for effective monitoring and evaluation which lay guide lines for assessment of GM crops
- Establishment of Institute Bio-safety Committee at the organization level, to monitor r-DNA technology work
- Establishment of Special Monitoring cum Evaluation Committee under RCGM to monitor the impact of transgenic plants on the environment with following members:

- Establishment of Genetic Engineering Approval Committee (GEAC) which will recommend the Government for approval of a transgenic variety for commercial cultivation.
- An All India Crop Biotechnology Association (AICBA) was also formed to represent seed industry in addition to existing ones such as Seed Association of India at New Delhi and Association of Seed Industries at Mumbai
- Under Seeds Rules, 1968, Central Government has declared CICR laboratory of ICAR as Referral Laboratory for Bt. cotton seeds in 2003. The detection kits developed at CICR has been recognized by the Government as a standard test for Bt. detection in GM seeds. As per the rule all seed testing laboratories are to perform Bt. purity testing along with genetic purity testing and the following standards have been specified for the same:

Submitted sample size: 25g

Working sample size: 10 seeds

Minimum level of Bt. toxin to be present: 450 nano g / sq.cm. or /gm seed

The purity in terms of quantum of gene expression of Bt. Protein should be **90% for labeling of Bt.** cotton seed.

Bt DETECTION METHODS:

A working sample of 25 g should be taken in a random manner from the seed packet. Three kinds of tests are used commonly in India to detect the purity of GM seeds.

1. Antibody based tests:

- a. ELISA (Enzyme Linked ImmunoSorbent Assay) and b. Dip-stick-strip immuno-sorbent test

2. PCR (polymerase Chain Reaction) based tests:

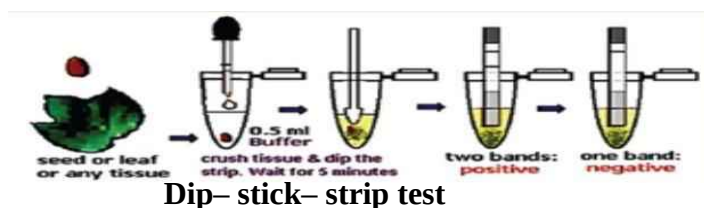
- a. Regular PCR
- b. Real-Time qPCR (quantitative PCR)

3. Biochemical tests

- a. GUS (Glucuronidase Assay) to detect the GM reporter enzyme

1. ELISA test / dip-stick-strip test: 1. ELISA test / dip-stick-strip test:

90 seeds to be tested from working sample size of 25 g drawn from a single packet of 450 g. A minimum number of 81 seeds tested positive for the test protein, Cry1Ac, Cry2Ab, Cry1C and fusion-gene protein Cry1AbCry1Ac, may be taken as the acceptable value for 90% gene purity.



2. PCR test:

30 seeds to be tested from working sample size of 25 g drawn from a single packet of 450 g. A minimum number of 27 seeds tested positive for primers specific to the gene (cry1Ac, cry2Ab, cry1C and fusion cry1Ac gene (cry1Ab+cry1Ac)) may be taken as the acceptable value for 90% gene purity. If only 25-26 seeds are positive, 30 freshly drawn seeds from the same working sample may be re-tested again on PCR. The total number of positive seeds from the two tests should be equivalent to or more than 54 out of the total 60 seeds tested from the working sample.

3. Event specific PCR (only for referral purposes):

10 seeds to be tested from working sample size of 25 g drawn from a single packet of 450 g. A minimum number of 9 seeds tested positive for the event may be taken as the acceptable value for 90% event purity.

4. Gus-Reporter test:

90 seeds to be tested from working sample size of 25 g drawn from a single packet of 450 g. A minimum number of 81 seeds tested positive for the test protein GUS (β -Glucuronidase) which is the reporter gene for Bollgard-II may be taken as the acceptable value for 90% gene purity of Cry2Ab.

PRINCIPLES OF ORGANIC SEED PRODUCTION

Organic farming system rely upon crop rotation, crop residues, animal manures, off- farm organic wastes, mineral bearing rocks and aspects of biological pest control to maintain soil productivity and tilth, to supply plant nutrients and to control pest and diseases. The primary objective of organic agriculture is to optimize the health and productivity of interdependent communities of soil life, plants, animals and people.

Organic Certification is a labeling term with written assurance that denotes products that have been produced in accordance with organic production standards and certified by a duly constituted certification body or authority.

Organic Farming Techniques

Green manuring: Green manuring is growing in the field plants usually belonging to leguminous family and incorporating into the soil after sufficient growth. The plants that are grown for green manure known as green manure crops. The most important green manure crops are sunnhemp, dhaincha, pillipesara, clusterbeans and Sesbania rostrata. It supply nitrogen supply, improves soil organic matter, phosphate and micro-nutrient which are mobilized, concentrated in the top soil and made available for plant growth.

Organic manure: FYM, sheep manure, crop residues, poultry manure, oil cakes, composts-coir pith compost and other farm wastage.

Vermicompost: It is the compost prepared by earthworms. Biologically degradable and decomposable organic wastage are used as earthworm feed.

Biofertilizer: Microorganisms to fix atmospheric nitrogen, to release and mobilize phosphorous and other nutrients. Natural fertilizers containing carrier based micro-organisms viz., Rhizobium, Azotobacter, Azospirillum, Blue Green Algae, Mycorrhizae and Phosphobacteria Plant products (botanicals): Seed hardening done with garlic extract, leaf extracts of Prosopis, Pungam, Acacia, Calotropis etc. The decayed plant extracts can be used as liquid manure for promoting plant growth.

Mulching: It is an important technique for improving soil microclimate, enhancing soil life, structure and fertility, conserving soil moisture and energy, reducing weed growth, preventing damage by impact from solar radiation and rainfall (erosion control) and reducing the need for tillage. Widely used traditional mulches include layers of dry grass, crop residues (straw, leaves *etc.*), fresh organic material from trees, bushes, grasses and household refuse green manures. Panchmukhi: Five factors of agricultural treatments adopted by the natural farmers as seed treatment, soil treatment, water treatment, environmental treatment and crop treatment are cumulatively known as panchmukhi farming process for boosting agricultural yields.

Panchagavya: It is a foliar spray prepared mainly with cow ingredients as follows

Biogas slurry/cow dung 5 kg, cow urine 3 litres, cow milk 2 litres, curd 2 litres, clarified butter/ghee 1 litres, sugarcane juice 3 litres, palm sugar 1 kg, tender coconut water 3 litres, banana are the ingredients. The ingredients are mixed in a mud pot after stirring them well. Then, it is kept in a shady place for one week for fermentation. Then 3 litres of Panchagavya are diluted in 100 litres of water. This mixture is sufficient for spraying four acres. The diluted mixture has to be thoroughly stirred for 20 minutes before spraying. It can be stored for one month. It reduces vegetative growth and enhances quick flowering and also gives resistance against pests and diseases.

Organic Seed Production Practices

Land selection: Should be organically managed. Avoid the low lying area to restrict the runoff water contamination from conventional farming system. To avoid contamination from wind, the organic farm shall be separated from conventional farm by live fence or manmade organically managed crop can be maintained as buffer zone. A buffer zone of at least 3 meters shall be maintained between conventional and organic management land. The equipment or implements used for organic management shall be cleaned before use.

Crops should be rotated to reduce pest problems and any potential for seed contamination by open pollination with similar species types. The seed production field should not have known weed problems that are too difficult to control through organic means

Land preparation: Soil should be tilled to ensure a fine seed bed, which is critical for germination, particularly with small-seeded crops. The soil should have good water-holding capacity to allow for uniform germination and continued vegetative growth. The beds should be raised and shaped depending on rainfall. Uniformity of seedbeds is especially important because the seeds are often precision planted and uniform emergence and seedling developments are required for optimum management

Soil Fertilization: Soil fertility should be maintained by any combination of above mentioned organic farming practices. Synthetic fertilizers should not be used.

Choice of crop and varieties: Any crop of variety/hybrid except genetically modified organisms/crop which suits to the location shall be used or grown. Pest and disease resistant varieties are mostly preferred.

Seeds and planting material: shall be used from organically certified source. In case of unavailability of organic seed, untreated seeds from conventional farm shall be used for first year and for subsequent years organic seeds shall be used. In case of growing other varieties which are not grown in the first year, chemically untreated conventional material shall be used. Genetically engineered seeds, pollen, transgenic plants or plant materials shall not be allowed.

Planting techniques: Seeds are generally planted directly by drilling in the field or transplanting from a greenhouse-grown seedling. These seed transplants should be organically produced. The seeds must be planted in such a way that proper vegetative development occurs that will support fruit and seed development with proper spacing and depth in the bed. Row spacing and plant density must allow for maximum plant development of the flower and unrestricted access to inflorescences for pollinators to ensure proper seed set. Proper spacing will also allow for improved air movement, reducing pathogens and providing space for harvest operations at the end of the season.

Rouging: Rouging at periodical intervals to remove the off types from both in male and female lines during hybrid seed production.

Weed, pest and disease management: Weed can be managed through mulching with plant residues and other fully biodegradable materials, livestock grazing and hand weeding coupled with mechanical cultivation. The seed crop is in the field for a long period of time, there are many opportunities for multiple pathogens to interact with a single crop. To control these pathogens organically is complex and requires proper growing conditions. Biological pest control: By using and following biological methods pest can be effectively controlled.

- Biocontrol agent like *Pseudomonas*, *Trichoderma*.
- Introduction of predators or parasites of the pest.

- Natural enemies like spiders, insects, mites, nematodes and birds
- Non-synthetics control such as lures, traps and repellent.
- Mulches and nets.
- Sanitation to remove disease vectors, weed seeds and habitat for pest organism.
- Development of habit for natural enemy of the pest.
- Use of botanical pesticides, Insect trap pheromones
- Crop rotation, trap crops and alternate host crop..

Physical method: In this method there is use of human effort to control insect pest and disease. The different physical method is listed below:

- Regulation of temperature: applicable for stored place
- Regulation of light: applicable for field crop
- Regulation of moisture: Use for stored insect pest control
- Use of sound waves

Mechanical method: it includes Hand picking, Sieving and winnowing, Shaking and beating, Netting, Wrapping, Painting, banding.

Harvesting and post harvest handling: Harvested, threshed and cleaned seeds should be stored without treatment of any synthetic pesticides. But it can be treated with organic sources like botanicals, biofertilizers, cows' product and biocontrol agent.

Organic Seed Certification

Tamil Nadu Organic Certification Department (TNOCD) created in the month of May 2007 function under the control of the Director of Seed Certification and its headquarters is based at Coimbatore. The main purpose is to do Inspection and Certification in the state in accordance with National Programme for Organic production (NPOP).

In simplified terms, the National Organic Program Standards require for crop farms:

- 3 years (36 months prior to harvest) with no application of prohibited materials (no synthetic fertilizers, pesticides, or GMOs) prior to certification.
- Distinct, defined boundaries for the operation.
- Proactive steps to prevent contamination from adjoining land uses.

- Implementation of an organic system plan, with proactive fertility management systems; conservation measures; and environmentally sound manure, weed, disease, and pest management practices.
- Monitoring of the operation's management practices to assure compliance
- Use of natural inputs and/or approved synthetic substances on the National List, provided that proactive management practices are implemented prior to use of approved inputs
- No use of prohibited substances.

LECTURE 21

Post Harvest Handling of Seeds - Threshing Methods - Drying - Methods of Seed Drying

THRESHING METHODS

“Threshing” is the operation of separating the grains from the plants. These operations may be carried out in the field or on the threshing floor, by hand or with the help of animals or machines. Whatever the system used, it is very important that threshing be done with care. Otherwise, these operations can cause breakage of the seeds or protective husks, thus reducing the quality of SEED and subsequent losses from the action of insects and moulds. Care should also be taken while transporting the harvested crops from the field to threshing floor to avoid any loss.

Hand threshing

It is generally done in case of pigeonpea. One of the simplest systems for threshing of pigeonpea crop is to strike the sheaves of crop spread over threshing floor with a flail or a stick. While this method can be adopted for other pulse crops like chickpea, mungbean, urdbean, fieldpea, rajmash and lentil but a more convenient method is threshing with animals or vehicles. The threshing-floors on which the sheaves are spread must have a hard, clean surface.

Threshing with hand driven machines

Normally a hand-operated machine like Olped thresher, which is basically used for threshing of paddy can also be used for threshing of cut stocks of pigeonpea. By means of the handle or pedal, a big drum fitted with metal rings or teeth is made to rotate.

Threshing with hand driven machines

The use of motorized threshers or threshers operated by tractor power has become very common for threshing. Equipped with a rotating threshing drum (with beaters or teeth) and a stationary counter-thresher, these machines often have devices to shake out the straw and to clean and bag the grain. Whether self-propelled or tractor-drawn, these threshers are often mounted on rubber wheels for easy movement to the field.

SEED DRYING

The crop is harvested once the seed reaches physiological maturity. At that time the seeds possess about 20 % moisture content, which is not ideal for safe storage. Every year large volumes of high quality seed are lost because of excess moisture. Different levels of seed moisture content have its own impact on biological properties of the seed. At high moisture content of 40-60 per cent - germination occurs, at 18-20 per cent - heating occurs, 12-14 per cent favours fungal invasion while 8-9 per cent is conducive for the insects activity and reproduction.

The process of elimination of moisture from the seed is called drying.

Stage of moisture elimination

The moisture from the seed is eliminated in 2 stages

Surface moisture of the seed that initially removed by the drying air.

The removal of the moisture in the surface cause an imbalance in the moisture potential in the surface of the seed and the inner portion of the seed which leads to the migration of moisture from the inner organ to the surface.

The migration of moisture to the surface is slower than the evaporation and a moisture gradient is developed in the seed.

Elimination of moisture from the seed depends upon the relative humidity and temperature of the environment surrounding the seed. When RH of the atmosphere is less than the seed, moisture is eliminated from the seed. While drying, care should be taken to minimize /prevent oxidation and decomposition and volatilization. In this process there will be loss of dry weight of seed which is widened when the processes take place at high temperature. Hence, high moisture seeds should be dried at low temperature.

Equilibrium moisture content

Seed is in equilibrium with the environment when the rate of moisture loss from the seed to the surrounding atmosphere is equal to the rate of moisture gained by the seed from the atmosphere.

Drying temperature

Greater the seed moisture content lesser should be the drying temperature and vice versa.

Importance of seed drying

According to Harrington (1972) in the case of orthodox or 'desiccation insensitive' seeds, if the moisture contents above 14 percent begin to exhibit increased respiration, heating, and fungal invasion which destroy seed viability more rapidly than indicated by the moisture content rule. Below 5 percent seed moisture, a breakdown of membrane structure hastens seed deterioration probably due to membrane damage caused by lipid peroxidation and resulted in loss of the water molecules necessary to retain their structural configuration. Between 6 and 14 percent moisture content, lipid peroxidation is likely to be minimum because sufficient water is available to serve as a buffer against autoxidation, but insufficient to activate lipoxygenase-mediated free radical production.

Invasion of fungi and bacteria in and on the seed and insect population, which may rapidly build up on seeds in high moisture conditions, leads to high respiration and heating of seeds. High temperature and high moisture will kill seeds rapidly, and so does the invasion of microorganisms and insects. Further, seeds may also suffer mechanical damage during seed processing if, their moisture content is too high. Seeds with high moisture content will be adversely affected by seed fumigation, done to control insects. Therefore, it is very important to dry the seeds carefully to optimum moisture content which will facilitate better seed storage life.

Seed drying process

Seed drying is a process in which moisture present in the seed is removed in vapour by the surrounding air due to moisture equilibration. The hygroscopic nature of the seed allows it reach equilibrium with the ambient relative humidity of the air by absorbing or desorbing moisture.

Air is the principle source of drying the seed, because of its ability to evaporate the moisture on the surface of the seeds and create a vapour pressure gradient away from the seed by carrying the evaporated moisture away from the seed mass. Therefore, to facilitate drying of seeds with air, the air surrounding the seed should be dryer than the seed.

Main factors affecting seed drying:

The temperature of the seed, physical structure and chemical composition of the seed and seed coat permeability, influence the rate of moisture migration from the centre of the seed, to the surface. Even if seeds are not dried too fast, they may be injured by over drying or longer *duration of drying*. In addition to above mentioned seed related parameters, the drying conditionlike exposure of seeds to very high temperature, very rapid drying and over drying can cause internal damage to the seed system and this in turn increase the susceptibility of seeds to mechanical damage.

1. Initial seed moisture content

At high seed moisture levels water is removed more rapidly from seed. The lower the moisture level, water is more tightly held by the seed, therefore more energy is required to remove it.

2. Relative humidity of the air

It was mentioned earlier on that dry air is required for seed drying. The lower the relative humidity of air, the faster drying will occur. In deep bed drying, relative humidities of 40 - 55% are commonly used.

3. Temperature

When relative humidity is low, the simplest way of drying seeds is to spread them in thin layers to provide complete exposure to the air. Under conditions of high relative humidity, heat is used to reduce the relative humidity and increase the capacity of air to hold water. Thus the *temperature* required for drying depends on the moisture content of seeds to be dried and the relative humidity of air surrounding the seed.

This will lead to two detrimental effects on seed i) mechanical damage ii) physiological damage. As a general rule, though, drying air temperature should not exceed 40⁰C (or) 110⁰F.

4. Air flow rate

Airflow rates requirements will vary according to seed kind, initial moisture level and seed depth. Drying in the humid regions requires the use of higher air flows compared to dry regions. Using high airflow rates will also reduce over-drying in the bottom section of a batch, since drying time is reduced and exposure of seed to low relative humidity air is also decreased.

5. Permeability of seed to moisture migration

The rate at which moisture moves from the inside to the periphery of the seed depends upon the permeability of the seed to moisture migration. In rice seed the permeability to moisture movement is low. If drying is accelerated by using high temperature air, a stress develops due to

a difference of moisture within the seed and mechanical checking or cracking can occur. Rice drying requires ‘tempering’ to allow the equalization of moisture within the seed.

Methods of drying

1. Sun drying

It is commonly used in the dry season in tropical, subtropical or warm temperate climates. As a first step of drying, the seed with higher moisture content, seeds are spread as a thin layer under shade condition or room condition where in which the relative humidity of the air is low. After a day, the seeds are heaped and covered. During this time, seed will try to establish moisture equilibrium condition between the inner core and seed surface. As a second step, the seed will be exposed to morning sunlight in between 8.00 am to 12.00 noon. After this seeds will be heaped and covered until 3.00 pm. Seeds should not be dried between 12:00 to 3:00pm, to avoid exposure of seeds to UV rays. This process of natural drying will enable the seeds undergo stage by stage drying, with sufficient intermittent tempering period, which is crucial to maintain the seed viability.

Main requirements are the following.

- i) Frequent stirring and turning to promote uniform drying.
- ii) Arrangements for immediate covering of seeds in the event of rain.
- iii) Overheating should be avoided while the seeds still have a high moisture content.
- iv) Protection against ants, birds, rodents and insects.

2. Artificial drying

Drying with artificial heat source should imitate the natural drying process so that the seeds are subjected to progressive drying, which causes a continuous release of moisture. The principle of artificial driers has a provision of a regulated flow of warm dry air, so that all the seeds dry uniformly and as quickly as possible without the risk of overheating.

Types of Dryers

In many dryers, the first two or three hours of drying are at a considerably lower temperature than the effective drying temperature. Seed moisture content, depth of seeds in bin, air temperature and air volume are factors that must be controlled to obtain good drying performance.

Commercial equipment is available for drying by any of the three basic methods i) Bin dryers, ii) Batch dryers, and iii) Continuous flow dryers.

I. Bin dryers

i) Circular metal bin dryers

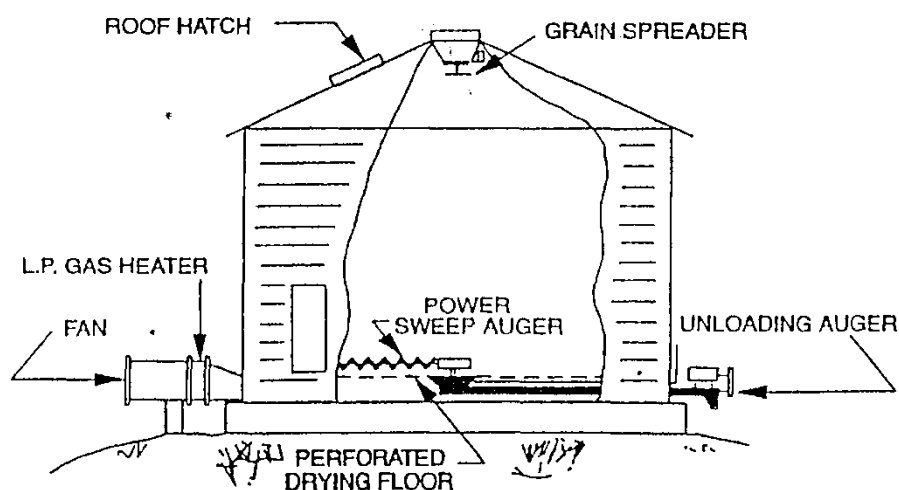
Such dryers usually consist of perforated floor, fan, heater and seed spreader. Here the seeds are spread in a thin layer over a perforated metal sheet while the heated air is blown to pass through the seeds. This heated air removes the moisture from the seed.

II) Batch dryer

i) Batch in bin dryers

In bin batch-dryers, the seed is placed in a (usually round) bin and ambient or slightly-heated air is blown through it by a fan. The maximum thickness of the seed layer in the bin depends on the initial moisture content, the type of seed, the air temperature and relative humidity, and fan horsepower. To obtain a uniform airflow through the seeds, a fully perforated floor is required.

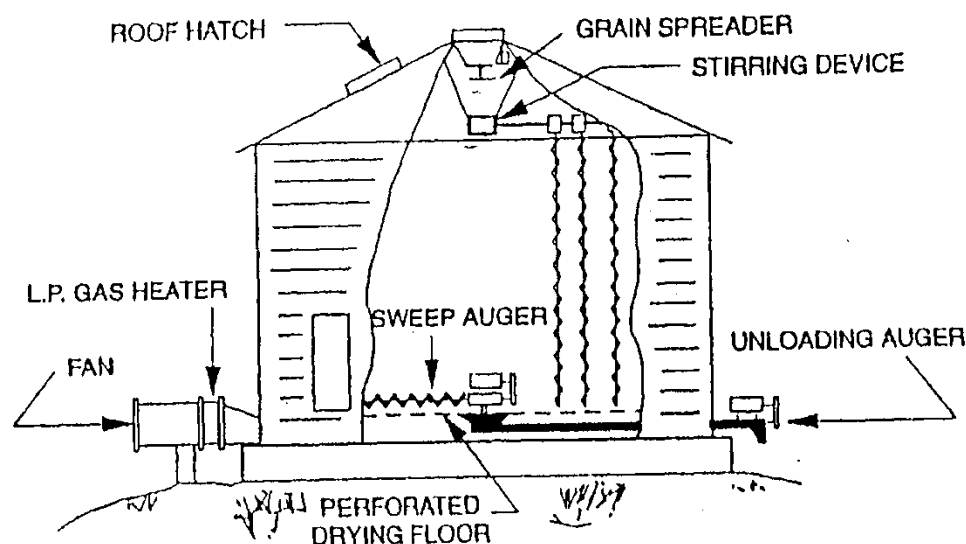
After the seed in a bin has reached the acceptable average moisture content, a moisture gradient will remain from the top to the bottom of the seed. The surface layer will have a moisture content above the average and the bottom layer of the bin will be lower than average. Thus, proper mixing of the seeds is essential before further storage or packaging. This can be addressed by installing one or more grain stirrers to mix the entire content of a bin for 3 to 12 hours.



Batch in Bin dryer

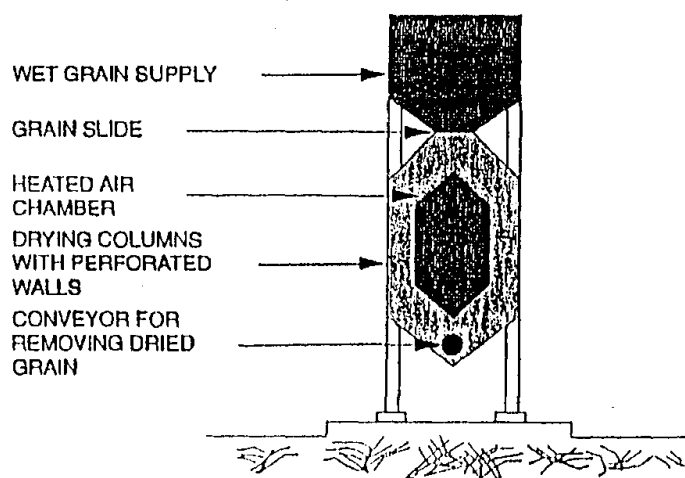
iii) Bin Layer-Dryer

In the process of bin batch drying, a batch of seed is placed in a bin and the entire batch is dried. In bin layer drying, the seed is dried in layers; successive layers of moist seed are added periodically to the drying bin after the previous layers have been partially dried. The final seed depth in the layer-drying bin is much greater than in a bin hatch-dryer.



iii. Column Batch-Dryer

In a column batch dryer, drying air is forced perpendicularly from an air chamber through the wet seed held in a relatively thin column (0.25-0.45 m). The airflow ranges from 15-30 m of slightly-heated air per minute per m of screen area. The temperature of the drying air should not exceed 35-40°C. After the seeds in a column batch-dryer have been dried to near the desired average moisture content, the seeds are cooled by ambient air after turning off the heating unit. As in other batch-type dryers, the seeds in a column batch-dryer are not uniformly dried. Thus, the seeds have to be well mixed prior to further storage or packaging.



Column batch dryer

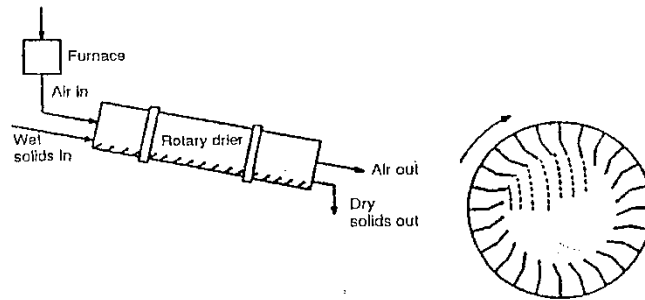
III. Continuous flow dryers

i) *Louisiana State University dryer (L.S.U. dryers)*

This is a continuous column heater air drier largely used for paddy. The paddy seeds are fed from the top with the help of gravity force in zigzag manner and heated air is blown from the bottom usually at right angles to the direction of seed motion. The falling seeds get dried up by the heated air and this process is repeated till the reduction of moisture content to the expected level.

ii) *Rotary dryer*

Rotary seed dryers are frequently batch-type units because of small lot sizes. Dryers often consist of a perforated rotating drum in which the seeds are tumbled while being heated with drying air which is injected perpendicularly to the drum axis. A batch-type rotary dryer is particularly suitable for drying small lots of high moisture vegetable and/or flower seeds. Unlike in sun/bin/wagon/portable-batch dryers, seeds are dried uniformly in rotary dryers. This is the great advantage of this dryer type. However, it has relatively high fixed and maintenance costs



Continuous flow dryers are superior to bin dryers since they register short drying period as well as more uniform drying.

Advantages of artificial drying

1. Quick method, timely and uniform drying is possible
2. Makes early harvest possible
3. It reduces the chances of losses due to over ripening and shattering of seed
4. Losses due to rodents and birds are prevented.
5. Less damage during processing operation.
6. Permits long time storage by preventing sun checks and other damages.

Disadvantages

1. Initial cost of drying equipment is high
2. Fuel is expensive
3. It produces possible fire hazards
4. Considerable supervision is necessary.

3. Chemical desiccants

Various chemical desiccants can be used such as silica gel, calcium oxide, lithium chloride and anhydrous calcium chloride to reduce the relative humidity of the surrounding atmosphere and facilitate drying of seeds.

SEED PROCESSING

Seed lots received from the field are often at high moisture content and contain trash and other inert material, weed seeds, deteriorated and damaged seeds, off-size seeds, etc. Seed processing is necessary in order to dry the seeds to safe moisture level; remove or reduce to the extent possible the various undesirable material, weed seeds, other crop seeds, deteriorated or damaged seeds.

Other than this the seed lot heterogeneity in its physical characters like size, colour, shape etc. The seed lot is heterogeneous due to the following reasons

1. The soil is heterogeneous and there is a lot of variability in the fertility status of the soil due to the availability of nutrients, physical, chemical and biological properties.
2. Variability is introduced due to the position of seed set on the plant/fruit , time of pollination and fertilization over a period of time
3. Variability is created by biotic factors like pest and variability infestation.
4. Variability is also due to the management practices like water, land preparation, leveling, staggered sowing, and uneven distribution of fertilizer and irrigation water, uneven plant protection sprays and uneven maturity at harvest.

The inherent qualities such as germinability and vigour are exemplified by certain physical characteristics of the seed i.e., large size, a denser seed, optimum length etc., So, if grading is done to obtain a particular range of size, shape, length and density of the seeds, the quality of the lot is upgraded.

In its common usage in India, seed processing refers to all the steps necessary for preparation of harvested seed for marketing, namely, handling, drying, shelling, preconditioning cleaning, size grading, treating and packaging, etc.

Seed Processing Plant Layout Planning

Layout plan for construction of a seed processing plant should be carefully planned to ensure that the thorough seed cleaning, upgrading, seed treatment and other seed processing operations are carried out efficiently, without mixing and damaging seed lots, with a minimum of equipment, personnel, time and at minimum cost. The following factors should be considered in planning and designing a seed processing plant:

1. Kinds of crop seeds to be handled and kinds of contaminating crop and weed seeds usually present in the seed lots
2. Size of operation
3. Whether drying facilities should be required
4. Selection of suitable equipment
5. Location of the plant
6. Source of power for running machinery
7. System of seed delivery to processing plant and
8. Availability of labour

The key to efficient plant layout is a thorough knowledge of what needs to be done, and sound planning. First, the general sequence of processes involved between the time seeds enter the processing plant and the time they are cleaned, packaged and ready for shipment, must be charted. The sequence of operations depends upon the kind of crop and the initial quality of seed lot, type of contaminants, moisture content of the seed lot, etc. The layout planner must have an intimate knowledge of the seed to be processed, its physical characteristics, the contaminants in it, and also of the selection of machines used to bring the seed to acceptable marketing standards.

Seed Processing Plant Building Layout

Seed processing plant building will comprise of following components:

1. Receiving-cum-drying platform
2. Processing area
3. Auxiliary building

Receiving-cum-drying platform

This area will be utilized to receive the raw seed and to sun dry small lots of crop seeds. This area can also be utilized for storage of seeds on wooden palettes. The platform will be connected to processing shed through a rolling shutter.

Processing area

The processing area should be situated between the shed and ventilated storage building. The hall should be connected to ventilated flat stores through a covered gallery for easy movement of processed and packaged seed to seed stores. The hall should have a big rolling shutter in the processing plant to permit entry of seed processing equipment into the hall for installation.

Height will be kept to facilitate installation of the seed processing equipment and machinery. A sequence of processing machines to be installed is shown in Fig. 1. Floor of the processing hall should be above the ground level.

The shed should have sufficient provision for natural as well as forced ventilation in order to maintain congenial atmosphere inside the shed. The shed should accommodate seed scalping, seed processing and packaging equipment and will have sufficient space for weighing and packaging.

Analysis of Operation

a) Processing sequence: After the machines needed have been identified, the next step is to determine the proper processing sequence. The seed separators, elevators, conveyors and storage bins should be so arranged that seeds flow continuously from beginning to end, and yet be flexible enough to bypass a machine or return to a part for re-cleaning.

b) Matching capacity: Equipment size of capacity must be carefully planned to prevent bottlenecks. When the overall operating capacity needs have been determined, all machines must be able to handle that capacity with some reserve capacity for problem lots. Surge bins can handle variations in individual machine capacities. But when differences are great, either larger models, or more than one machine installed in parallel flow, must be used to maintain uninterrupted flow.

c) Conveying: The type of conveying system is also a very important factor. The conveying system must be able to handle the capacity needed in a particular spot. And it must be carefully adapted to the seed handled.

Type of Layouts

There are three main types of processing plant layouts: multistorey, single level and combination.

Multistorey: In this system, seed is carried by elevators to the top floor and emptied into large bins. Cleaning machines are then arranged in a vertical series on the lower floors. Seed flows from one machine down into the next by gravity.

Single level: In the single storey plant, seed is moved from one machine to the next by elevators placed between the machines. A great advantage of the single level system is that one man can supervise the processing line without running up and downstairs. He can thus maintain closer supervision of all operations.

Combined designs: A compromise between the single and multistorey system could also be adapted.

Planning

After the proper machines, elevator capacities, cleaning sequences, and lay out design have been selected, detailed layout planning can begin. Careful layout planning can identify and remedy bottlenecks and trouble spots before the plant is built, and thus prevent trouble later.

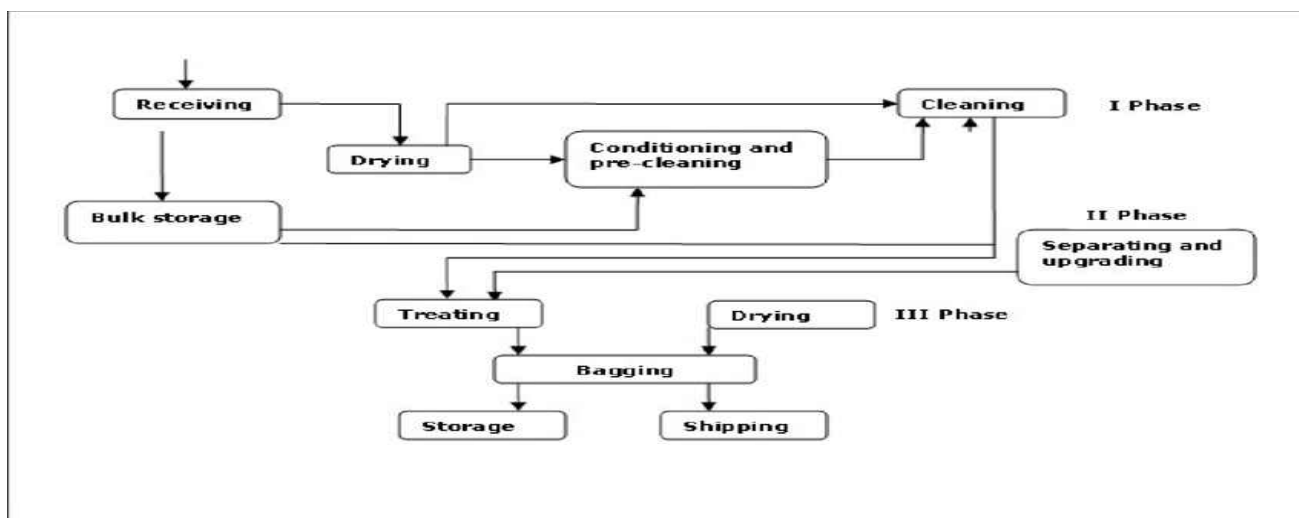
As the lay out or design develops, it should be drawn on paper. A good method is to draw lines of flow first and then convert these flow lines into machine lines. After appropriate revisions, detailed drawings can be made to show exact locations of equipment and distances. Scale drawings are the most widely used method of layout planning. Scale models and scale templates are also very effective, but are more expensive.

Requirements in seed processing

1. There should be complete separation
2. Minimum seed loss
3. Upgrading should be possible for any particular quality
4. Efficiency
5. It should have only minimum requirement

Movement of seed in a processing plant

Handling of seed at the processing plant adheres to a definite path irrespective of crop for easy management of seed which is sensitive at each and every step of handling and ready to lose or gain its quality all through the steps.



Physical characteristics used to separate seeds are

1.	Size grader	: Based on size it can be separated with air screen cleaner cum grader
2.	Length	: Disc or indented cylinder separator
3.	Weight	: Specific gravity separator
4.	Shape	: Spiral separator or draper separator for round and flat Seeds
5.	Surface texture	: Rough from smooth surface seed- dodder mill
6.	Colour	: Electronic colour separator
7.	Electrical conductivity	

Seed differing in their ability to conduct electrical charge can be separated with electronic separator.

8. Affinity to liquid

The seed coat of seed will absorb water, oils etc., which provides a means of separating seed on the magnetic separator.

Seed processing equipments

I. Air screen cleaner

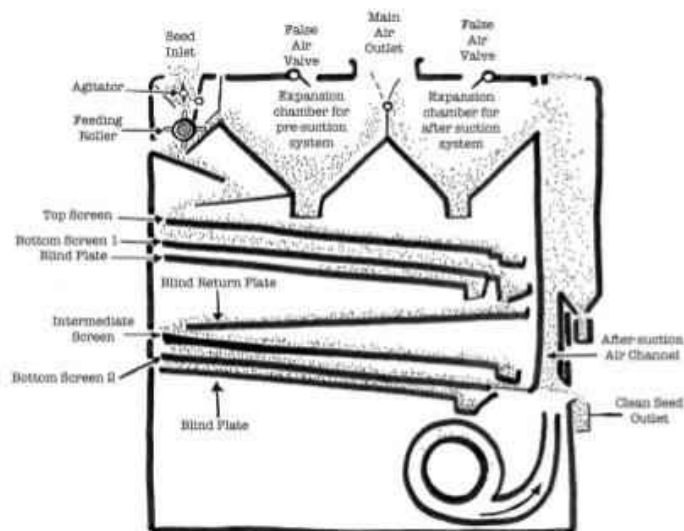
This is the most important machine of every cleaning plant. It uses screens and aspiration (air blow) for two separations. A coarse upper screen removes larger material, a lower fine screen stops the seeds and lets through fine matter and then the seed fraction passes through a transverse or nearly vertical air stream which can separate light impurities such as empty or partly filled seeds, husks and glumes from the seed. In most cases a number of sieves with different sized perforations are used and the cleaning is a process of gradually shifting out smaller particles. Factors which determine the quality and quantity of seed cleaned include (i) size of the perforations, (ii) the precision of the perforation, (iii) the angle at which the sieves operate, (iv) the amplitude and speed of movement of the sieves and (v) correct cleaning and maintenance of the equipment.

– Air Screen cleaner

- Flat screen
- Rotational :

- Intake
- Pressure of feeding roller
- Air pressure
- Sizes of screen's holes
- Inclination of screens
- Speed of oscillation
- Attention to exit of air!!

Attention to discard system when buying equipment

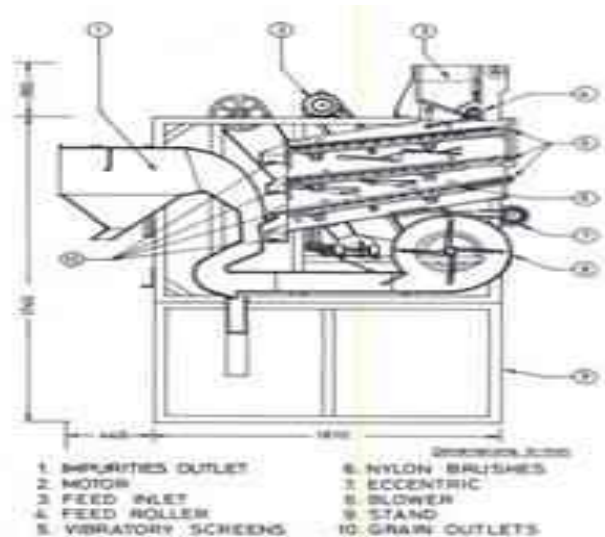


II. Cleaner cum grader

The dried seeds should be cleaned and graded with help of a cleaner cum grader. For large scale cleaning and grading the commonly available machine is the “Crippen Model Seed Cleaner cum Grader”.

It consists of the following parts

- A hopper in the top for seed filling. A fluted roller below the hopper to regulate the seed flow to the screen.
- Screen (or) sieves: Perforated metal sheet with specific size of perforation in which there are two types.



- Screen shaking unit : for oscillating the sieves to move the seeds on the screens
- Screen brushes to remove the blocked seeds
- Air blower with adjustments for air outlet
- Collecting outlet
- Air duct for directing the blown up light particles to outside
- Collecting bins.

Working principle

The seeds are fed into the hopper and they are guided to fall on the first sieve. The first sieve is a scalping screen which scalps all the foreign materials larger and heavier than seed and the entire quantity of seed passes through the first sieve. The second sieve is a cleaning sieve which removes all the unwanted particles larger in size than the seed. The third sieve is actually the grading sieve which size grade the seed lot and bring into a uniform size and which also screen the undersized, shriveled and immature seed, dust and dirt. The seeds are then rolled and passed through air column, where they are relieved of the light chaffy and other materials by the blowing air.

Adjustments

Fluted roller

The speed of this roller can be adjusted so as to increase (or) decrease the flow of seeds to the hopper of the sieves.

Slope (or) inclination of the screen

The angle of inclination of the screens can be adjusted according to the nature of seeds.

Rate of vibration of sieve

This can be adjusted either to increase or to decrease the speed of the rolling seeds on the screen.

Volume of air flow: By increasing (or) decreasing the air inlet.

Choice of screens: According to variety we have to change the screen

Screen dams

Small check dams, which can be provided here and there on the screens so that the seeds can be stopped a while and takes the charge either to pass or to roll.

Types of seed cleaner cum grader

1. Crippen model cleaner cum grader
2. Clipper model cleaner cum grader
3. Petkas cleaner cum grader

III. Disc separator

It consists of a series of discs, which revolve together on a horizontal shaft inside the cylindrical body. Each disc contains many under cut pockets. The seed enter the intake end of the separator and move through the open centers of the discs towards the discharge end of machine. As the discs revolve through the seed mass the pockets lift out short seed but rejects longer seed. Longer seeds are conveyed by flights on the disc spokes towards the discharge end of the machine where they go out through the tailings gate. The rate of seed travel through the open disc centers is controlled by conveyor or blades attached to the spokes of the discs. The disc separator makes a very precise separation. No factor other than seed length and shape affects its separation. Flexibility is obtained by varying size of the pockets.

IV. Indented cylinder separator

The indented cylinder separator is a rotating, almost horizontal cylinder with movable, horizontal separating adjustments which are mounted inside it. Indent lines are there inside the surface of the cylinder. The indented cylinder revolves, turning the seed mass to give each seed a chance to fit into indent. Short seeds are lifted out of the seed mass and are dropped into the lifting and long seeds remain in the cylinder and are discharged out via., a separate spout at the end of the cylinder.



Lab Model

As the cylinder revolves, it creates centrifugal force which helps to hold the seed in the indent. Short seeds are held in the indent until the cylinder turns to the point where the indent is inverted enough for gravity to cause the seed to fall out of the indent. The length, surface texture and size of seeds determine how they fit into the indent, so that it can be lifted out of the seed mass. The speed of the cylinder creates centrifugal force which holds the seeds in the indent as it are lifted upward. Thus the shape and size of the seed to cause some seeds to fall out after being lifted only a short distance, while other seeds are lifted closer to the top of the cylinder before they fall out.

As the seeds enter the cylinder, the small, short, easy to separate seeds are quickly removed. The center cylinder section removes the intermediate sizes of seeds still in the cylinder. All indents in a cylinder are the same size, only the progressively declining amount of material to be lifted causes this difference in separating action.

V. Specific gravity separator

Seeds of the same size and general shape can often be separated because they differ in specific gravity or relative weight. This difference is very useful in removing light, immature seeds or heavy sand and rocks to improve the purity and germination of crop seeds.

If seeds which differ in specific gravity (relative weight / unit of volume) are placed on substrate of intermediate density, seeds of higher specific gravity will fall down through the substrata, while seeds of lower specific gravity will be buoyed up the substrata. Here air is used as separation substrata.

Working principle of the specific gravity separator

As seeds flow on the deck of the gravity separator, they enter a column of air coming up through the porous surface of the deck. The pressure of terminal velocity of the air rising through the deck can be controlled very closely to separate two kinds of seeds differing in specific gravity, the air is adjusted so that only the lighter seeds are lifted up off the deck surface. These lighter seeds are held up by air pressure and tend to float on the deck surface. The heavier seed possess a velocity greater than that of the air columns so

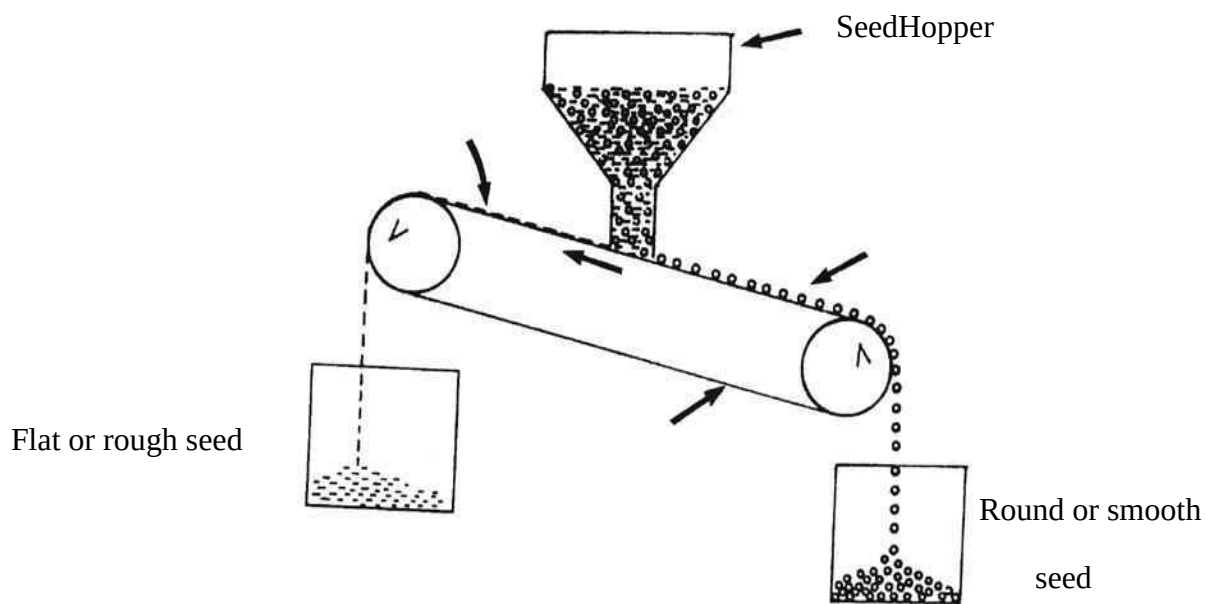
they are not lifted and so will lie on the deck surface. The air column thus stratifies the seed mixture into vertical zones of relative weight with the heavier seed lying on the deck and the lighter seeds lifted up to the top of the seed mass.

Adjustments

1. Feed rate 2) Air flow 3) End slope 4) Side slope 5) Deck oscillation speed 6) Deck speed.

VI. Roll mill or dodder mill or velvet roll mill

It is used to separate the seeds based on surface texture and shape. This separator should be used only after the seed has been carefully cleaned and separated from the chaff. These are effective in separating seeds with a rough seed coat or shape angles from smooth seeds.



Working principle of the roll mill

The roll mill consists basically of two roller, covered with flannel or velvet, placed side by side, so that they touch each other down their entire length. The rollers are mounted on an incline and they turn in opposite directions. A curved adjustable shield is mounted above the rollers.

Separating action

The mixture of smooth and rough seeds is fed into the place, where the rollers touch each other, at the high end of the machine. As the rollers turn up and out, seeds that are rough or have sharp or broken edges are caught by the nap of the fabric covering the rollers. These seeds are thrown up against the curved shield. They strike the shield at an angle, bounce back down to the roller and are again thrown up against the shield. Smooth seeds bounce down the inclined position forward between the rollers, and discharge at the lower end of the machine. They are not affected by the fabric roller covering, and are not pitches over the side of the rollers.

Adjustments

1) Rate of feed 2) Speed 3) Clearance between shield and rolls 4) The angle of inclination of rolls.

VII. Magnetic separator

The separation is mainly based on the affinity for liquids which is used for separation. Since seeds contains no free iron and are not attracted by a magnet they must be selectively pretreated with a magnetic material such as finely ground iron powder. Rough seed coats, cracked or broken seed coats, dirt lumps, chaff or seed with a sticky residue on the surface will hold the liquid and become sticky, so that iron powder will adhere to them. Smooth coated seeds will not absorb liquid. So no iron powder will adhere to them.

The seed are then discharged from mixing chamber and brought into contact with a powerful magnet, which removes the iron coated seeds. Most magnetic cleavers pass the seeds over a revolving drum which has a high intensity magnetic field. Seeds with an affinity for liquids which are now coated with iron powder are attracted by the magnet and adhere to the drum until they are removed by a brush or scraper. Seeds which are relatively free of iron powder are not attracted by the magnet and will fall into a separate discharge spout.

The first requisite of magnetic seed separation is that the seed to be separated must possess different seed coat characters. Crop seeds should have a smooth surface, while the seeds to be removed should have a rough surface which will retain liquid and can accept the iron powder. Success in separating the components depends upon the magnitude of seed coat differences and thoroughness with which the moistened seeds and the iron powder are blended.

VIII. Colour separator

Many large crop seeds such as peas and beans differ in colour between varieties. Colour variation may also occur due to immaturity or disease. Electronic colour sorting machines can separate such seeds by difference in colour and also remove mud balls and discoloured seeds in the same operation.



The electronic colour sorter views each seed individually with photo electric cells. The seed is compared with a selected background or colour range and is discharged from the machine according to its colour. If it is the great desired colour, the seed is discharged through the good seed spout. If its colour or shade falls within the reject range, a blast of compressed air deflects the seed and sends it in to the reject discharge spout. These are highly sensitive. Since the machine views each seed individually, capacity is low, but the initial cost is high and operating cost is less. The usefulness of machine is greater with large seeded crops.

IX. Spiral separator

The separator, which classifies seed according to its shape and rolling ability, consists of sheet metal strips fitted around a central axis in the form of a spiral. The unit resembles an open screw conveyor standing in a vertical position. The seed is introduced

at the top of the inner spiral. Round seeds roll faster down the incline than flat or irregularly shaped seeds, which tend to slide or tumble.

The orbit of round seed increases with speed on its flight around the axis, until it rolls over the edge of the inner flight into the outer flight where it is collected separately. The slower moving seed does not build up enough speed to escape from the inner flight. Most spirals have multiple inner flights arranged one above the other to increase the capacity.



Processing equipments used for improving the quality of the seed

From harvest upto final stage of seed storage, the seeds are to pass through various seed processing equipments depending upon the speciality and specificity. But some equipment like driers and seed cleaner cum graders are common for all types of seed. The processing machineries and equipments used in the seed handling are as hereunder.

S.No.	Processing Equipments	Usage with reference to specific seed Management
A.	Threshing with extraction equipments	
1.	Thresher	To remove the seeds from the inflorescence especially in cereals
2.	Ginning machine	To separate the lint and seed from kapas in Cotton

3.	Maize sheller	To shell the seed from the cobs
4.	Pulse thresher	To remove seed from the pods
5.	Tomato seed Extractor	To extract tomato seed from fruit without wasting the pulp
6.	Chilli seed extractor	For easy removal of seed from chilli fruits
7.	Groundnut Decorticator	To shell the kernel (seed from the pods
8.	Sunflower thresher	For removal of seeds from the head
9.	Debearder	To remove the awns form (Barley) the seed
10.	Mechanical scarifier	To scarify the hard seed mechanically to improve the germination of seeds
11.	Pebble mill	To remove webby hairs from grasses
12.	Timothy bumper mill	To remove weed seed from timothy seed
13.	Hammer mill	To remove the hook or appendages from the seeds (i.e. Stylosanthus)
B.	Driers	To reduce the moisture content to lower or needed level for safe handling both for processing and for storage at the final stage

C.	Grading equipments	
14.	Cleaner cum grader	This homogenize the precleaned seed based on size and is known as basic grading in seeds. The sieve sizes requirement vary with Crop
15.	Precleaner and Aspirator	This remove the inert material and dust particles from seed and improve the grading Efficiency
D.	Upgrading machines	
16.	Specific gravity Separator	Improve the quality of graded seed further using its weight or specific gravity. Heavier seeds are good storers and expresses maximum field establishment
17.	Indent cylinder	In lengthier seeds it maintains the size of seed (breadth and length). The broken / damaged are removed and good seeds are Selected
18.	Disc separator	It is for removal of weed seeds and to improve the general appearance of seed
19.	Roll mill	To separate smooth seed from rough seed based on the surface texture especially the weed seed
20.	Magnetic separator	Removal of weed seed from clovers, alfalfa, trefoils and vetch
21.	Inclined draper	Separation of smooth or round seeds from rough, flat or elongated seeds
22.	Electronic Colour	Separation of off-coloured seed

	Sorter	
23.	Electrostatic	Based on electrical properties removes

	Separator	Johnson grass from sesamum seeds (Specific utility)
24.	Spiral separator	Separation of seeds based on shape (eg.) separation of rape, vetch and soybean seed from wheat, oat or rye grass
25.	Polishers	To improve the luster of seed
26.	Picker belts	To remove undesirable ears / pods from shelled seeds (eg.) Groundnut, Corn
27.	Vibratory separator	Removal of weed seed
28.	Seed treater	To treat the seed with fungicide and pesticide
29.	Seed packing Machine	To easier the work and to avoid human error of mixing
30.	Conveyors / Elevators (Belt , Bucket)	Easier the transfer of seed from machine to machine and avoids the contamination of seed at various level.

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Though all the machines are highly useful in improving the seed quality, specific machines are utilized for specific crop. The sequential usage of machineries varies with crop seeds (Gregg, 1967).

Precautions in handling processing equipment

All machine adjustments like the speed, oscillation and duration should be perfect. Otherwise, it will result in mechanical damage of seed which reduces the quality of seed in terms of vigour, viability, storability and field stand. Drying of seed should be designed properly as the moisture content needed for threshing; grading, treating and bagging vary with operations. Dosage, exposure period and choice of chemical are important in mechanical seed treatment.

SEED QUALITY ENHANCEMENT

INTRODUCTION

Basically, seed treatments are intended to improve the quality of seeds. Because, nothing will work upon a poor quality seed, no matter how lavishly other inputs are spent on the crop to be established from such seeds. Even for a resource poor farmer, the high quality of seed would be a great advantage, because it ensures 15 to 20% increased yield.

The genetic improvements are continuously incorporated into crops by the plant breeders and molecular biologists. Any achievements in the crop improvement can be propagated and established in field only through seeds. Hence, the production of high quality seed is necessary and important to the agricultural industry.

WHY SEED QUALITY ENHANCEMENT TREATMENTS?

For any crop, the time from sowing to seedling establishment is crucial, in which the seeds are exposed to wide range of environmental stresses viz., harmful micro-organisms, destruction by seed, soil and air borne pathogens, damage by insects, cold, frost, temperature and moisture effects, monsoon failure, severe drought, problem soils, competition from weeds and toxic effects of chemicals present in soil, which pose problems of dormancy, germination and survival.

Value addition to seed is necessary to tolerate prolonged exposure to adverse environment during imbibition and seedling establishment so as to improve the crop stand. A judicious and comprehensive package of crop specific seed management technologies starting from dormancy breaking treatments, seed germination and vigour augmenting treatments, seed protection treatments, and finally seed handling and conditioning treatments need to be developed for sustainable good crop growth and productivity.

WHAT IS SEED QUALITY ENHANCEMENT?

Seed quality enhancement means the application of physical, physiological, biological and chemical agents to the seed in order to enhance the physical, physiological, genetical, biochemical and health qualities of seed.

HISTORY

Seed treatments of different nature and types are practiced to promote rapid uniform and higher emergence as well as to produce vigorous seedling for better stand establishment in the field. According to Theophrastus (C-373-287 BC), pre-soaked cucumber seeds in milk or water germinated quicker.

Keeping the seeds in moist cloth or soaking in water prior to sowing is a common practice among farmers of India. Place of seeds in luke-warm water overnight increased speed of emergence. Will (1883) made the first report that the repeated soaking and drying of the seeds result in drought and frost resistance. Henckel pursued intensive research on this technology from 1930 onwards.

Henckel (1964) reported that soaking seeds in water for about 48 h (30 % moisture by weight) at 10-15°C and drying back to original weight induced drought tolerance in plants. The process is termed as “Pre-sowing drought hardening”. Short term soaking of seeds resulted in increased yield in wheat and barley (Basu, 1976). Therefore, pre-sowing seed treatments have many kinds and types on the situations for which the treatments are designed. Table 1 shows the various terms which are different from each other and/ or synonymous.

Different terminologies of pre-sowing seed treatments.

Malnassy (1971)	Coined the term “Seed Priming”
Heydecker (1975)	Acknowledged the term seed priming which is synonymous with seed invigouration; introduced the terms halopriming and osmopriming
Taylor <i>et al.</i> (1988)	Introduced the term “Solid Matrix Priming” (SMP)
Khan <i>et al.</i> (1990)	Proposed “Matricconditioning” as alternative for SMP
Callan <i>et al.</i> (1990)	Coined the word “Biopriming”

History of seed protective treatments.

Middle Ages	Soaking of seeds in chlorine salts and manure
1600’s	Soaking in salt water

1700's	Introduction of copper salts
1740	Introduction of arsenic chemicals
1765	Soaking in hot water
1808	Ban on arsenic chemicals in Germany
1915	Introduction of organo – mercurials
1960	Introduction of first systemic fungicide
1970	First systemic fungicide against air borne pathogens
1982	Ban on organo – mercurial compounds in western Europe
1990	Introduction of modern fungicides and insecticides

List of seed quality enhancement treatments.

1. Dormancy breaking treatments

Soaking in water, hot water soaking, leaching of inhibitors using tap water, Wetting and thawing, alternate wetting and drying, mechanical scarification, acid scarification, bioscarification, scorching / Burning, warm stratification, cold stratification, electrical seed treatment, seed treatment with growth stimulants and seed treatment with growth retardants

2. Germination augmenting treatments

Seed fortification, seed infusion, pre- germinative sprouting, **Osmopriming**, osmoconditioning, seed invigouration, solid matrix priming, seed conditioning, fluid drilling, dry permeation, seed hardening, seed priming, pre-hydration, irradiation, magnetic seed treatment, electrical seed treatment, biofertilizer treatment, seed treatment with bioinoculants, mid – term storage correction treatment and hydration – dehydration treatment

3. Seed coating treatments

Seed pelleting, seed coating, seed colouring and polykote film technology

4. Protective seed treatments

Seed dressing with insecticide (Both dry and slurry treatments), seed dressing with fungicide (Both dry and slurry treatment), botanical seed treatment and **biocontrol agents**

SEED FORTIFICATION

INTRODUCTION

One of the earliest reports describing the usage of seed fortification is by Theophrastus (372 – 287 BC) who recommended soaking of cucumber seeds in milk or water to make them germinate quicker. Seed fortification was attempted as early as 1937 for raddish by Amlog and Naundorf. In the early years, seed fortification was eagerly attempted using growth regulators.

WHAT IS SEED FORTIFICATION

Seed fortification is a physiological method of seed invigouration that aids in improving the initial stamina of the seed that helps in improving the initial field stand and that of the final yield. Seed fortification is the impregnation of the needy substance into the seed through the imbibition phase and enriches the endogenous level of the needy bioactive substances. Therefore, the main objective of seed fortification is to invigourate the seed for better establishment and improved productivity.

METHODOLOGY

In seed fortification, seeds are soaked in solutions of equal volume /in different seed to solution ratio for duration of 6 – 24h depending upon the crop to raise the moisture content of the seed to 20 – 25% just enough for endogenous impregnation of chemicals through exogenous application. Then the seeds are sown after surface drying for easy handling or as such (not specified for drying back to original moisture content).

PRINCIPLE INVOLVED

During fortification, the first phase of germination ends with completion of imbibition process and hence the time taken from sowing to emergence is much reduced (Hegarty, 1970). Fortification of seed increased the field germination by promoting embryo growth. The improvement in field emergence due to seed fortification could also be ascribed to activation of cells, which results in the enhancement of mitochondrial activity leading to the formation of more high energy compounds and vital biomolecules, which were made available during the early phase of germination.

FACTORS INFLUENCING SEED FORTIFICATION

The exploration on seed fortification experiments reveals that bioactive chemicals/ substances including growth regulators, organic acids, nutrients, seed protectants etc., can be successfully used in enriching the seeds which are capable of improving the seed germination and seedling vigour. The success of seed fortification treatment depends on

- The choice of chemicals (growth regulators, macro and micro nutrients)
- The concentration of chemicals (molar to percentage solution)
- Duration of soaking (2-24h)
- Method of treatment (equal volume / lesser volume /in between moist medium)

Seed fortification for different crops

Crop	Chemical	Concentration	Duration
Wheat	Potassium dihydrogen phosphate	0.5%	12h
Groundnut	CaCl ₂	1.0%	6h
Sunflower	Distilled water or KNO ₃ or FeSO ₄ or ZnSO ₄	0.1%	12h
Sunflower	Ammonium acetate	400 mg l ⁻¹	10h
Field bean	Ammonium molybdate	500 or 1000 ppm	24h
Sorghum and Cotton	KCl	2%	8h
Paddy	Potassium dihydrogen chloride and KCl	0.5%and 1.0%	12h
Cowpea	Zinc sulphate + Manganese sulphate + Na ₂ MO ₄	each 100 mg/kg of seed	3h
Mustard	Magnesium nitrate or MgSO ₄	7.5 mμ	12h or 96h
Cluster bean	FeSO ₄	1.0%	3h
Blackgram	ZnSO ₄	100 ppm	3h

Seed fortification with vitamins

Crop	Chemical	Concentration	Duration
Greengram	Ascorbic acid	0.004%	4h
Mung bean	Ascorbic acid	20 ppm	-

Seed fortification with organic acids

Crop	Chemical	Concentration	Duration
Soybean	Succinic acid	23 mg lit ⁻¹	4h
Greengram	Succinic acid	0.0025%	

Seed fortification with growth regulator

Crop	Chemical	Concentration
Paddy	GA ₃	
Maize	Triacntanol	10 ppm
Ragi	Brassinolide	0.1 ppm

Seed fortification with organics and botanicals

Crop	Chemical	Concentration	Duration
Paddy	Neem kernel extract	2.5%	12h
Paddy	Cowdung solution	10%	
Sorghum	Prosopis and pungam leaf extract 1:1 ratio	1%	
Ragi	Prosopis and pungam leaf extract	1%	12h
Cowpea and bhendi	Prosopis leaf extract	1%	
Cotton	Pungam leaf extract Prosopis leaf extract	1% , 0.5%	
Blackgram	Prosopis leaf extract	10%	6h
Sorghum	Prosopis leaf extract	10%	
Sesamum	Tamarind leaf extract	1-2%	

BENEFITS OF SEED FORTIFICATION

- ✓ The low and medium quality seeds can be upgraded.
- ✓ The method promises not only accelerated germination and vigorous seedling growth, but also early flowering and higher yield.
- ✓ Seed fortification can serve as a low cost technology compared to soil and foliar application which involves more chemical, labour, wastage as well as cost.

- ✓ The beneficial effects of chemicals may depend on species, concentration and duration of sowing.

Seed priming

Seed priming is a pre-sowing treatment that involves the controlled hydration of seeds sufficient to allow pre-germinative metabolic events to take place but insufficient to allow radicle protrusion through the seed coat (Heydecker *et al.*, 1973).

2. SEED PRIMING

2.1. What is it?

It is the process of controlled hydration of seeds to a level that permits pre-germinative metabolic activity to proceed, but that prevents actual emergence of the radicle.

2.2. Why it is done?

In order to increase the speed and uniformity of germination particularly under adverse conditions of temperature, moisture and salinity.

2.3. How it is done?

Seeds are soaked in variety of solutes, including solutions of various inorganic salts, sugars and polyethylene glycol (PEG) a chemically inert, high molecular weight compound that unlike the lower molecular weight compound probably does not penetrate the cell walls. The temperature suggested during priming is between 10°C – 15°C. The duration of priming varies with the crop from 1 day (sorghum) to 2-3 day (onion). The seeds are then rinsed and re-dried to original water content.

3. METHODS OF SEED PRIMING

Heydecker *et al.* (1973) used different terms depends upon the methods adopted for priming (Fig.1).

3.1. Osmopriming

Soaking the seeds in osmotic solutions. Polyethylene glycol (PEG) is also commonly used as a priming osmoticum. This process is known as osmopriming. PEG is advantageous because it is chemically inert and does not have adverse effects on the embryo. The large size of the PEG molecules prevent its penetration into seed tissues, thus reducing toxic side effects on seeds. The disadvantages of using PEG are its relatively high cost, problems with removing it from the seed coat after priming, obtaining uniform aeration during priming, and environmental hazards with PEG disposal.

3.2. Halopriming

Soaking the seeds in salt solutions. Halopriming involves soaking of seed in salt solutions. Compared to PEG, salts are easier to aerate and to remove from the seed after treatment, they are less costly and may also provide a nutritional effect. The seed size, structure, biochemical constitution, position of seed protecting layers, the type of salt and soaking time are the principal factors influencing ion penetration into the embryo. High ion concentration in the embryo may affect germination adversely when inorganic salts are used as osmoticum in the seed priming process.

3.3. Hydropriming

Soaking in water. In hydropriming, the water availability is not limited to the seeds, so they eventually germinate. Therefore, the process must be arrested at a specific time to prevent the onset of phase III of germination.

3.4. Solid matrix priming

This consists of mixing seeds with an organic or inorganic carrier and water for a period of time. The moisture content of the matrix is brought to a level just below what is required for radicle protrusion. Seed water potential is regulated by the matrix potential of the seed and during priming the water is largely held by the carrier. Seeds can imbibe water from the carrier till the equilibrium is reached. Taylor *et al.* (1988) introduced the term solid matrix priming (SMP), in which a solid matrix instead of an osmotic solution is employed. According to Khan (1992) solids used for matrix priming should have i) a proportionately high matrix to osmotic component ii) negligible water solubility iii) low chemical reactivity iv) high water holding

capacity v) variable particle size, structure and porosity vi) high surface area viii) high bulk value and low bulk density and viii) ability to adhere to seed surfaces. These characters of the solid matrix system actually minimize aeration problems.

3.5. Biopriming

Coating the seeds with biological agents like bacteria and soaking in warm water until the seed moisture content increased to 35-40% (e.g. maize seeds) and redrying.

3.6. Drum priming

This is misting of seed with water and redrying before they complete germination. Seeds are rotated in a drum with specific amount of water introduced as a fine mist. As the seeds absorb water, a sensitive scale monitors increase in seed weight until the desired wet weight is achieved.

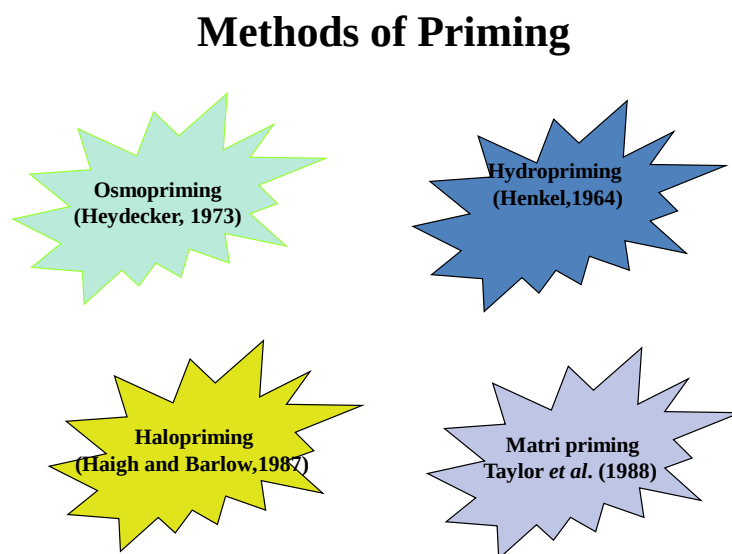


Fig.1. Methods of priming

4. FACTORS AFFECTING SEED PRIMING

Aeration

Several factors must be controlled during the priming treatment in order to improve seed performance. Proper aeration of the osmotic solution was essential to living seeds and helped in seed germination. It was showed that the response to aeration during priming varied according to species. Onion seeds primed in an aerated PEG solution, using enriched oxygen, had greater per cent germination after treatment compared to seeds primed in nonaerated solutions.

Light

The beneficial effects of seed priming could be modified by light quality during treatment. When seeds of endive were primed in KNO_3 solution under continuous red light, they had a greater germination percentage and more uniform germination than seeds primed in the dark. Low light levels from fluorescent and incandescent bulbs during priming did not affect germination .

Duration of the Priming Process

The ideal duration of the priming process varies according to the type of osmotica, osmotic potential of the solution, temperature during the treatment, and species. If radicle protrusion occurs during seed priming, irreversible embryo damage could be expected during dehydration after priming.

Temperature

Duration of seed priming depends on the species and seed quality. However, temperature during treatment and osmotic potential of the solution affect the length of the soaking period. If the soak temperature is maintained below the optimum range for rapid germination, radicle growth during priming may be restricted. At 20°C, a high percentage (63%) of carrot seeds germinated in the priming solution (-0.5 MPa PEG), whereas no germination was observed with solution at 15°C.

Osmotic Potential

Osmotic potential of the solution is another factor affecting the length and effectiveness of the priming treatment. seed priming in high osmotic potential solutions may allow rapid seed germination and radicle growth because seed water contents become too great. Decreasing the osmotic potential from -0.5 to -1.0 MPa prevented the germination of carrot seeds in the priming solution.

Addition of promotive substance

The addition of promotive substance to seeds during priming can further enhance seed performance. GA_3 is the most successfully incorporated in seed priming treatments. Abscicic acid, Kinitin, ethephon have also produced good results.

Seed Quality

A differential response to seed priming among seed lots of the same species has been reported. Seed vigour is another factor influencing the response to seed priming. However, those seeds, which were still viable after ageing, had improved germination rate after priming. Priming accelerated the germination rate of both high and low vigour seed lots, but not to the same extent. The high vigour lots generally performed more consistently. The differential response was attributed to seed genotype rather than seed vigour. Seed companies generally conduct some type of seed vigour analysis before priming a seed lot because of the effects of seed quality on priming.

Dehydration after Priming

To facilitate handling and storage, the seeds should be dried to an acceptable moisture level for long-term storage after the priming treatment. Water uptake during priming is influenced by the osmotic potential of the solution, type of osmoticum, duration of the treatment, and the physical/chemical characteristics of the seeds.

Seed Storage

After seed priming and dehydration, seeds usually are stored before planting. The conditions and duration of storage may alter the advantages achieved from priming. Primed spinach seed maintained high germination at stressful high temperatures after 1 month of storage.

MODE OF ACTION

Mechanisms by which seed priming treatment improves germination performance have been discussed by several workers.

- i. Seeds complete phase I (hydration) and II (lag phase) of germination, and only require a favourable gradient for water uptake in order to begin radical growth. This situation is therefore more comparable to that of pre-germinated seeds (Bradford, 1986) in that the hydrated, primed seeds enter immediately into phase III of imbibitions (growth).
- ii. Rearrangement of cell membrane structure lost during seed drying and increase in membrane integrity (Knyple *et al.*, 1980; Simon and Harun, 1972).
- iii. Repair of damage to nucleic acids and proteins acquired during storage (Burgass and Powell, 1984; Bino *et al.*, 1992)
- iv. Higher respiration and ATP production (Mazor *et al.*, 1984)
- v. Increase in hydrolytic enzyme activity especially α -amylase (Wilson, 1971) and strong increase of super oxide dismutase and catalase activities (Bailly *et al.*, 2000).
- vi. Expression of certain proteins related to water stress and heat shock (Czarnecka *et al.*, 1984).
- vii. Increased B- tublin accumulation in the radicle tip during priming. Tublin is the main functional protein of microtubules, which is necessary for mitotic spindle formation (Bino *et al.*, 1992).
- viii. Accumulation of more sugar which could readily support metabolic activity (Sung and Chang, 1993), leading to faster seedling emergence.
- ix. Increase in UDP glucose, which plays central role in production of nucleotide sugars.

- x. Increased DNA content (Coolbear *et al.*, 1980) as a result of activation or synthesis of enzymes of nucleic acid or both.
- xi. Reduction in time of imbibitions required for the onset of RNA, protein synthesis and polyribosome formation and increase in total amount of RNA and protein synthesized (Khan *et al.*, 1978).
- xii. The shorter lag-time of germination of reduced primed seeds may therefore be related to higher scavenging of AOS (active oxygen species) during the first stages of inhibition.

6. PHYSIOLOGICAL BASIS OF SEED PRIMING

Events during Germination

Seed imbibition

Fundamental to the success of seed priming is the uptake of water that occurs in three phases in germinating seed.

Phase I

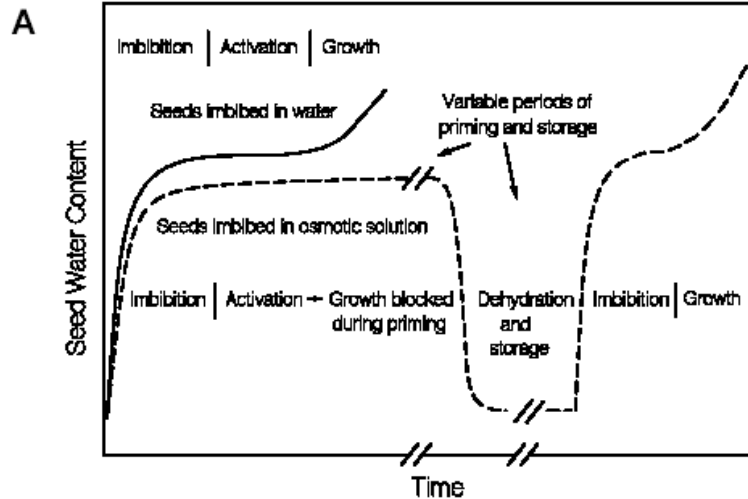
Rapid initial water uptake into seed due to the seed's low water potential (water moves from high to low water potential). During this phase, DNA and mitochondria are repaired and proteins synthesized using existing messenger ribonucleic acids (mRNAs).

Phase II

It is represented only in a gradual increase in seed water content, but new physiological activities associated with germination are initiated including the synthesis of mitochondria and proteins relying on the translation of mRNAs. Phase I and II represent the process of germination and are foundation for successful seed priming, where the seed is brought to a seed moisture content that is just short of radicle production

Phase III

This signifies the completion of germination and the start of seedling growth by resumption of radicle growth and is identified by another rapid intake of water. This provides the turgor necessary for radicle cell elongation.



Seed priming can enhance seed germination. Seed priming involves allowing seeds to absorb sufficient water to initiate metabolic processes, but insufficient water to allow completion of germination. After the priming period, the seeds can be dried, stored, and transported to the farmer. When seeds are imbibed again, the lag period before radicle emergence occurs is considerably reduced, improving the rate and uniformity of germination.

The Cell cycle

The cell cycle is the series of molecular events that allows cells to duplicate and segregate their chromosomes to form new cells. The important phases are G₁, G₁/S checkpoint, S, G₂, G₂/M checkpoint and finally the M phase (Fig. 3).

G₁-Cell doubles in size, new organelles and materials formed

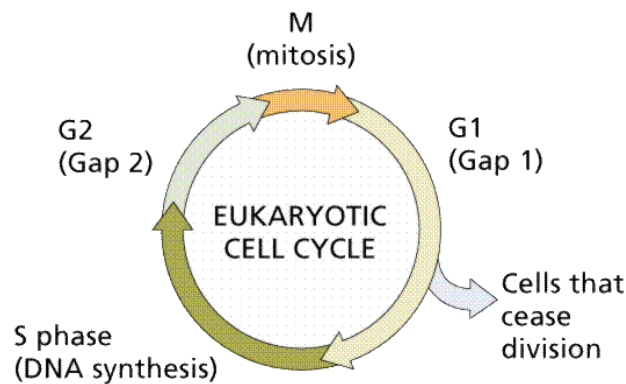
G₁ / S-Check point; Cell cycle can stop here or proceed to S phase

S-DNA and associated nuclear protein replicated; cells contain 2 full copies of its genetic information.

G₂-Chromosomes begin to condense, band of microtubules forms.

G₂ / M-Checkpoint

M-If conditions are suitable (Mitosis) cell divides.



The Cell Cycle

8. PHYSIOLOGICAL EVENTS DURING PRIMING

Protein Synthesis

Protein synthesis occurs during priming and is increased upon subsequent germination compared with the levels seen in untreated controls, while the activities of several germination associated enzymes have been reported to be increased upon germination after priming.

Increase in β -tubulin in the radicle tip has been observed during priming by De Castro *et al* (1995) and Bino *et al* (1992). Tubulin is the main functional protein of microtubules which is necessary for mitotic spindle formation.

Nucleic acids

Nucleic acid accumulation has been reported in many seeds after priming. Coolbear *et al* (1980) reported that there is very large accumulation of RNA in ungerminated seeds and there is close correlation between this net synthesis and subsequent germination performance. Bray *et al* (1989) detected DNA synthesis in leek embryos during priming, which may reflect DNA repair, thus contributing to rapid germination and embryo growth upon priming.

Mitochondria

Germination is an energy demanding process that requires the operation of mitochondria, which must survive desiccation in the quiescent seed and become rapidly functional after imbibition to meet the ATP demand.

POLYMER SEED COATING

INTRODUCTION

A high quality seed with better germination, vigour and seed health is the pre-requisite for a good crop. However in the field, various diseases and pests may still attack seeds. To overcome this, Integrated Pest Management strategy has to be adopted. The farmer is very much interested in the best seed management practices, which are safe, environmentally sound and scientifically proven to overcome the adverse situations. A strong reduction in the amounts of pesticides being used can be achieved by applying crop protection agents directly to the seed instead of spraying them in the field and can be incorporated in a pellet or a film coat layer, where it will be most effective. The Agriculture industry is now responding to this trend with the development of more and more molecules that must be applied to the seeds and which require high quality application technologies. This technology may also be combined with other seed enhancement and protection technologies.

WHAT IS SEED POLYMER COATING

A seed coating is the substance applied to the seed that does not obscure its shape. Seed coating examples are fungicides, microbiological treatments and micronutrients. The major benefit of seed coating is that the seed enhancement material is placed directly on to the seed. Smaller amounts of chemicals are needed as compared to broadcasting or surface dressing onto the growing medium. Seed coating is one of the most economical approach for improving seed performance.

Film coating is normally used to apply a thin, uniform layer of polymer over seeds without significantly increasing seed size or weight. The plasticizer polymer forms a flexible film that adheres and protects the fungicide, preventing dusting off and loss of fungicide during handling. The film is readily water-soluble (hydrophilic) so as not to impede germination.

The more precise and uniform application of fungicides, insecticides, colours and other additives can be accomplished by a typical seed slurry treatment. Film coating helps to smooth and round the seed surface, which improves flow ability and machine planting.



Polymer coating of seeds

Polymer and colourant products can be added to the standard treatment slurry. They are compatible with the fungicides and insecticides commonly used in seed treatment. The polymer products can be used at near full strength or can be diluted with 1 - 4 parts water plus seed treatment chemicals. They contain excellent surfactants and spreaders and have a very quick drying time.

Different colours – Pink, red, green , yellow, blue , black and colours are specific to crops.



Different colours of polymer

WHY USE COLOURANTS ON SEED?

Polymer coated seed can be easily identified which prevents accidental consumption. It helps to differentiate seed of different competitors and is the marketing strategy of the seed company. There are unlimited options to this uniquely branding technology. This technology are important to know about Herbicide treated seeds or variety differentiation (A line / B line). It is

used to enhance the seed qualities such as improving plantability, stand establishment, seed flow in seed planters, etc. Uniform distribution of the pesticides on the seeds and thereby prevents the dusting off chemicals by strong binding capacity. Reduces pollution of pesticides at the processing plant and also at farm level. Freeze sensitive seed coating is useful for fall planting in cold climates. It is a user friendly and environmentally safer technology.

BENEFITS

The major benefits of seed film coating are mainly related to safety and easy handling. The other advantages of this method over the conventional one are

- Improves plantability and emergence of seeds to benefit seed companies and farmers.
- Enables accurate and even dosage of chemicals and reduces chemical wastage.
- It makes room for including all the required ingredients like inoculants, protectants, nutrients, plant growth promoters, hydrophobic/hydrophilic substances, herbicides, oxygen suppliers etc.
- By encasing the seed within a thin film of biodegradable polymer, the adherence of the seed treatment to the seed is improved.
- It ensures dust free handling, making treated seed both user and environment friendly.
- Addition of colourant helps visual monitoring of placement accuracy.
- Polymer coating results in more even seeding rate due to the smooth flow of the seeds
- Polymer acts as a temperature switch and protective coating by regulating the water uptake and subsequent germination of seed.

Thereby the two goals of seed enhancement like seed designing and seed functioning can be improved through seed film coating.

STEPS IN POLY COAT FILM COATING

The steps involved in the polymer coating to seeds are given in Fig. 1 and Plate 3.

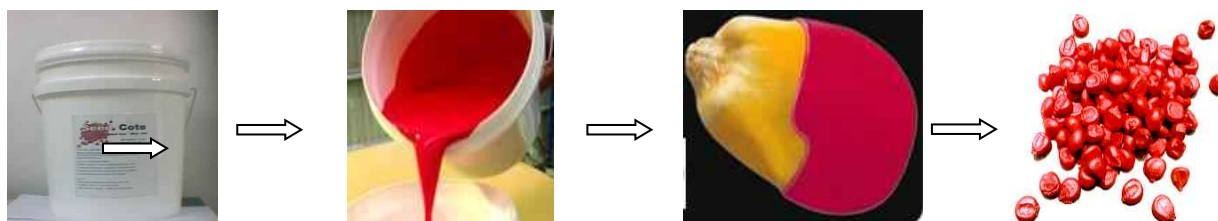
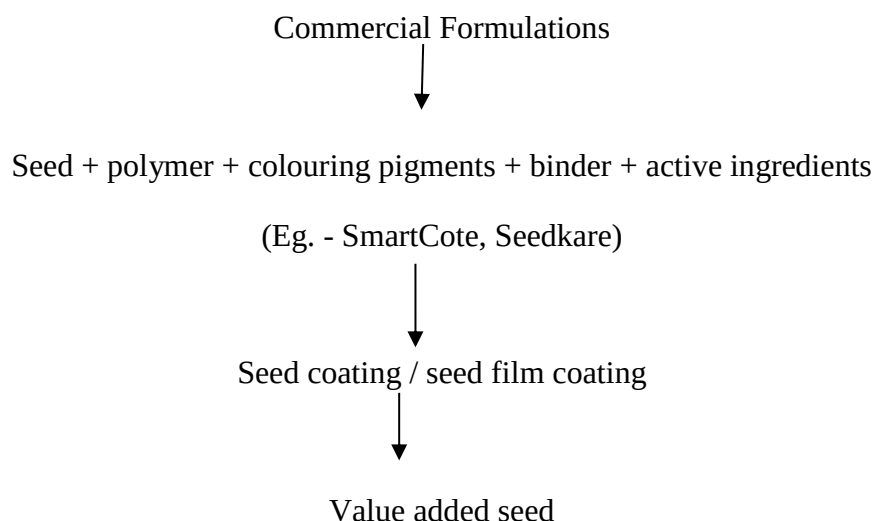


Plate 4. Steps in seed polymer coating



Steps involved in polymer coating

DIFFERENTIATE BETWEEN SEED POLYMER COATING, COLOURING AND PELLETTING TECHNOLOGIES

Particulars	Seed polymer coating	Seed pelleting
Adhesive	Seed coating polymer itself has a plasticizer and binding ability.	Adhesives , nutrients,filler materials , biofertilizers are required
Fungicide and insecticide	Can be combined	Can be combined
Shape of the seed	Not obscured	Shape is modified as globular unit
Size of the seed	Not modified	Modified due to addition of filler materials
Filler materials	Not required	Required.Which may be gypsum (or) botanical leaf powder.
Nutrients and biofertilizers	Usually combined	Can be combined during the process of pelleting
Process	Have coating process	Have stamping, coating and rolling process
Safety and handing	More safe	No safety

Resistance to mechanical injury	More resistance	More resistance
Stability	More stable	Less stable
Prevention of dusting of fungicides	Prevents dusting off pesticide and fungicide	Does not prevent
Wastage of pesticide and fungicide	Completely prevented	Wastage is more during handling
Improvement in the physical appearance of seed	Very much improved	Improved but having no preference
Handling	Easy handling	Results in disintegration and breakage
Identification	Useful for brand identify	<i>Not useful</i>
Effect on seed germination and seedling vigour	Improved	Improved, but not preferred by consumers
Storability	Good storer	Pelleted seed cannot be stored for more than 3 to 4 months with better shelf life
Applicability	Suitable for dry, garden and wet land situations	Suitable for only garden / wet land situation
Inter linking of technologies	Compatible	Compatible

SEED COATING TECHNOLOGIES



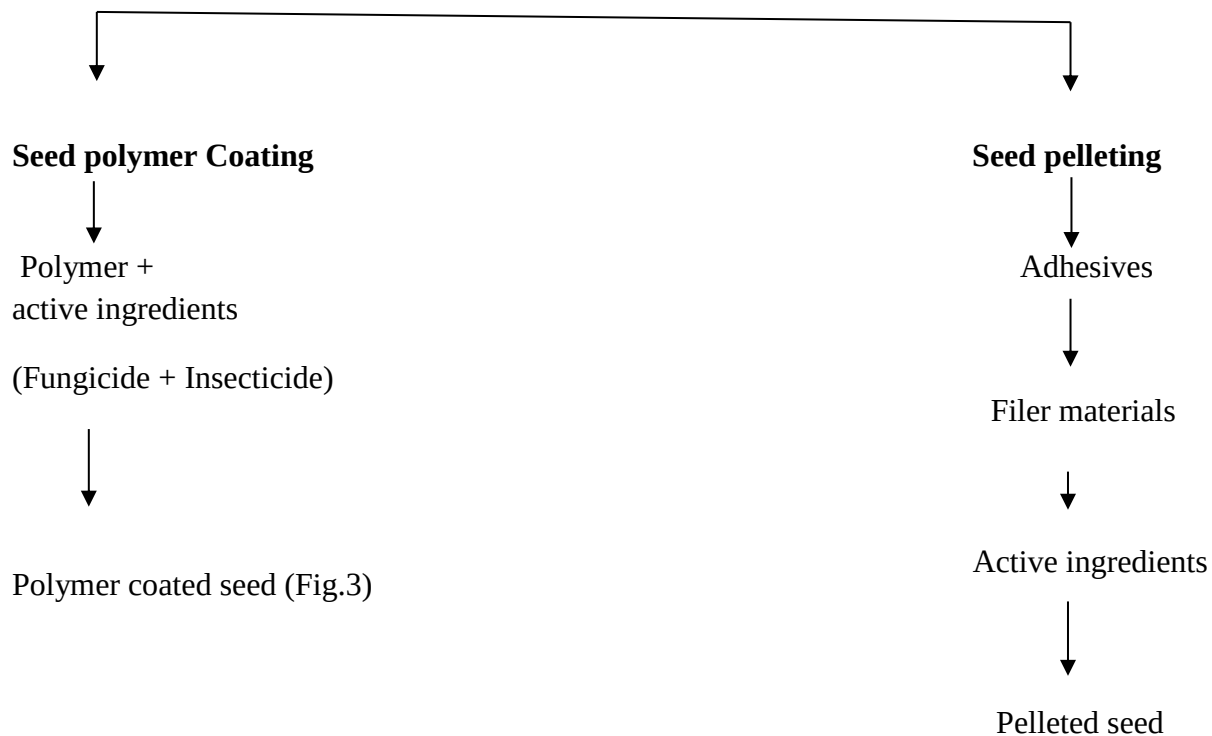
Seed polymer coating

Sophisticated process of applying precise amount of active ingredients along with a liquid polymer directly on to the seed surface without obscuring its shape.

Seed pelleting

Process of enclosing the seed with small quantity of ingredients along with filler material to produce a globular unit of standard size to facilitate precision planting.

Steps of various seed coating technologies



COMMERCIAL FORMULATIONS IN MARKET

N-Hance: containing specially selected micronutrients and macronutrients for difficult soil conditions.

N- Rich: rich concentration of specifically isolated strains of rhizobia for maximising nitrogen fixation.

Grasscote - Pasture grass & Turf

Features

- ✓ Fluffy florets, awns and appendages.
- ✓ Flexible coating material bends with the seed.
- ✓ Thiram fungicide as standard.

Benefits

- ✓ Smooths fluffy grass florets to prevent clumping (Plate 6).
- ✓ Increases seed weight for accurate delivery.
- ✓ Unique coating material flexes with seed to reduce breaking.
- ✓ A range of bright colours to assist identification in the field.

- ✓ Enhanced protection from detrimental fungal attack.

Flowcote Carrot

Features

- ✓ 200% build up Encrusting in a wide range of colours and finishings.
- ✓ Smooth, low-dust coating.
- ✓ Powerful fungicide partnership as standard - Thiram, Metalaxyl-M.

Benefits

- ✓ Smooths the carrot seed "beard" and appendages for significantly better flow from bag (Plate 7).
- ✓ Increases seed weight for accurate delivery.
- ✓ A range of bright colours to assist identification in the field.
- ✓ Enhanced protection from detrimental fungal attack.

Polycote – Horticultural

Features

- ✓ Increases seed flow ability, reduces static and clumping.
- ✓ Your choice of fungicides customized for your situation.
- ✓ Thiram fungicide included as standard.

Benefits

- ✓ A range of bright colours to assist identification in the field and reduces the risk of batch mix-up (Plate. 8).
- ✓ Precise, even application of chemicals and colour.

SEED PELLETING

WHAT PELLETING IS?

Seed pelleting is the process of enclosing a seed with a small quantity of inert material just large enough to produce a globular unit of standard size to facilitate precision planting (Plate.1). The inert material creates natural water holding media and provides small amount of nutrients to young seedlings. Seed pelleting also serves as mechanism of applying needed

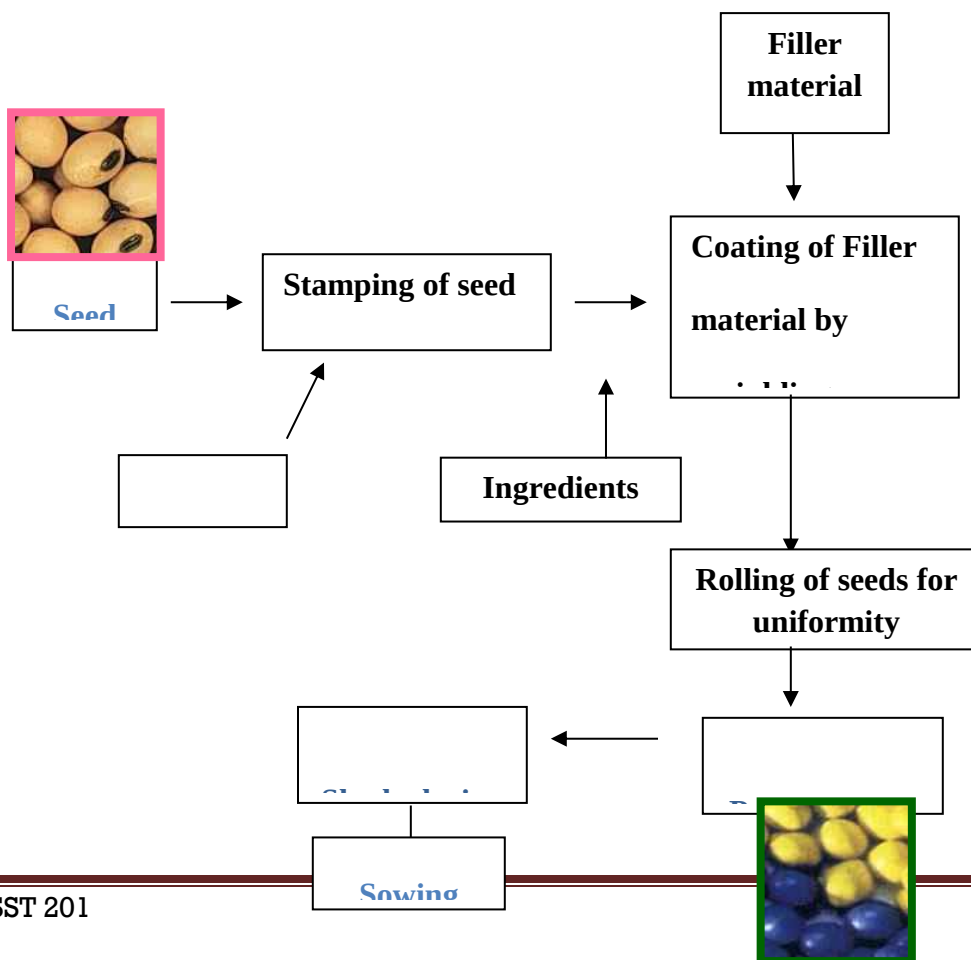
materials in such a way that they affect the seed or soil at the seed soil interface. The main objective of seed pelleting is precision sowing with added advantages of better establishment and increased productivity.



Pelleted seeds

STEPS IN PELLETING

The three basic steps involved in pelleting are stated as stamping, coating and rolling. The materials needed for pelleting are seed, adhesive and filler materials. Initially the seeds are uniformly coated with adhesive in correct quantity . Then the filler materials are sprinkled on the coated seeds and are rolled on the filler material for effective and uniform coating (Fig.1).



Steps in seed pelleting

5. MATERIALS NEEDED FOR SEED PELLETING

Adhesive

Gum arabic, methyl cellulose, gelatin, casein, plastic resins, polyvinyl acetate, methyl ethyl cellulose, polyurethane polyvinyl alcohol, polyvinyl acetate, poly electrolyte or dextran and poly ethylene oxide.

Some of the cheap and low cost binding materials are the rice gruel, maida gruel, sago gruel and starch (revive) gruel.

Characteristics of adhesive

1. The coating must be porous to allow movement of air to the seed.
2. The coating must weaken or breakdown when it comes in contact with soil moisture to prevent any physical impedance to seed germination.
3. Material used must not have any toxicity to the seeds.
4. It must be possible to apply the coating on commercial basis.
5. Must have affinity (combinability) for both seed coat and selected filler material
6. Should have the required degree of water solubility or insolubility for easy emergence
7. Should have the required strength and plasticity to prevent dusting and breakage
8. Should have the most appropriate viscosity for each application

Filler

The materials used as filler materials for pelleting must be beneficial and harmless to both seed and the rhizosphere. The most common materials used for protectants for *Rhizobia* include lime, gypsum, dolomite and rock phosphate. Other materials include clay minerals such as montmorillonite and vermiculite. Besides, blood, peat, poultry manure, moss and mucilage are also used.

The size of particle selected for filler vary with the type of filler material. The particle size is important for resistant coating on the seed material. The activated clay, lime and

dolomite should pass through 300 mesh sieve. The dried blood, wood charcoal, milk powder and yeast extract should pass through 150 mesh sieve size. In case of dried blood, milk powder and yeast extract alone are used; they are used in combination with dolomite in different proportions

Characters of filler

1. It should be non-toxic
2. Friendly to both seed, adhesive and environment
3. Easily soluble in water
4. Easily available for commercial production
5. Low cost
6. Must be porous to allow movement of air to the seed
7. The coating must weaken or break down easily when it comes in contact with soil moisture to prevent any physical impedance
8. Size of the particles should pass through 300 mesh sieve.

TYPES OF PELLETING

The pelleting is grouped into different types based on the ingredients used for pelleting.

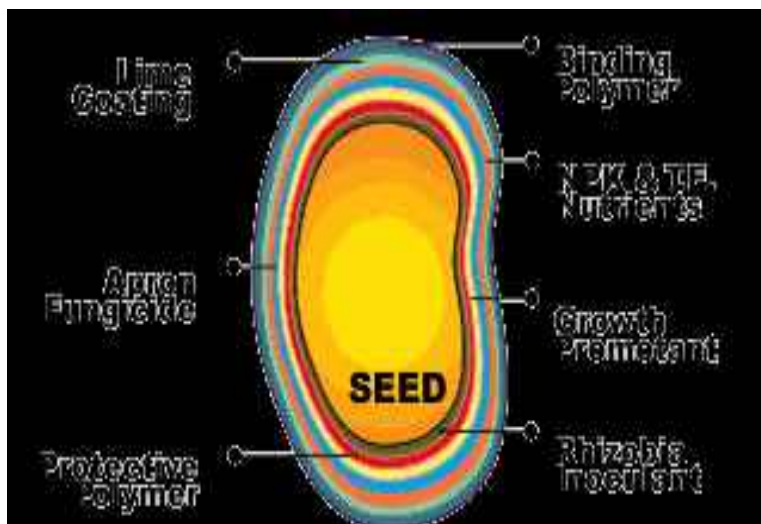
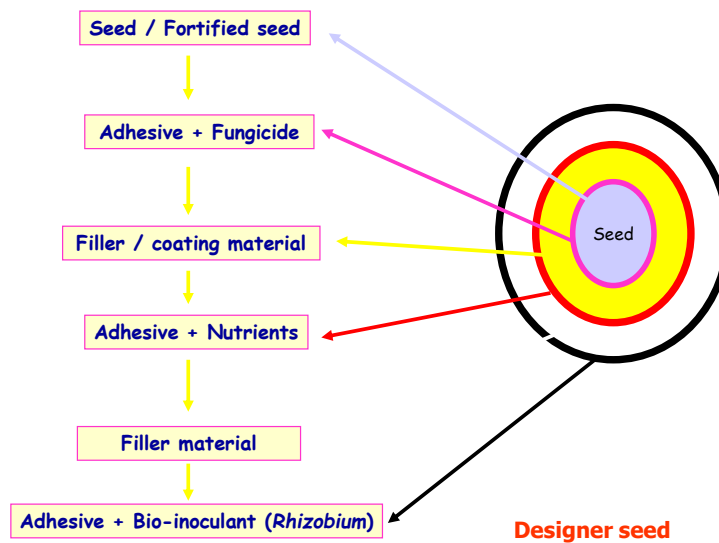
Types of pelleting

Types of pelleting	Ingredients for pelleting
Inoculant pelleting	Biofertilizer,viz, <i>Rhizobium</i> , <i>Phosphobacteria</i> , <i>Azospirillum</i> , <i>Azotobactor</i> , Vesicular Arbuscular Mychorhizae, <i>Azophos</i>
Protective pelleting	Biocontrol agents like <i>Pseudomonas</i> , <i>Trichodermma</i> , <i>Rhizobacteria</i> , <i>Bacillus</i> , <i>Streptomyces</i> ,and insecticides and fungicides
Herbicide pelleting	Filler antitode or absorbent coating. Herbicide antitode like 1,8 naphthalic anhydride (NA)
Nutrient pelleting	Coating with micro and macronutrients namely ZnSO ₄ , FeSO ₄ , CuSO ₄ , KCl, borax, diammonium phosphate (DAP) and other salts suitable for each crop.
Hydrophilic coating (Water stress condition)	Starch graft polymers, magnesium carbonate

Oxygen supplier coating (Flooded condition)	Peroxides of zinc and calcium
Organic pelleting	Leaf powders (arappu, pungam, acacias) and rhizome powders (turmeric, <i>Acorus</i>)

Procedure

There are three basic steps in pelleting namely stamping, coating and rolling. Seeds are introduced into a coating drum or pan that resembles a cement mixture. An amalgam of pelleting materials (clays, limestone, calcium carbonate, talc, vermiculate) and cementing adhesives (gum arabic, gelatin, methyl cellulose, polyvinyl alcohol) are used to form the pellet and other compounds such as inoculants, fungicides, insecticide, nutrients and botanicals etc. may be added to enhance seed performance. As drum rotates, the seeds are first sprayed with water followed by the addition of the pelleting materials and the pellet gradually increases in size with each turn of coating drum. Longer rotation times with greater amounts of pelleting materials lead to greater pellet size and roundness. At the end of the pelleting process, a binder is added to harden the outer layer of the pellet. This also reduces the amount of dust produced during handling, shipping and planting. After pelleting is completed, the pelleted seeds are dried and handled in the same way as unpelleted seeds (Fig. 2 and 3) (Copeland and Mc Donald, 1999)



Desinger seed

ADVANTAGES

1. The mechanical planting of seeds is facilitated by seed characteristics such as uniform size and shape, sufficient size and weight which aid in easy mechanical separation and easy flow and prevent clumping together. Seed pelleting has been employed to achieve all these features.



Mechanical planting of pelleted seeds

2. Small and irregular shaped seeds can be handled easily by pelleting which changes the shape of seed.
3. Aids in precision sowing by increasing the size and weight of seeds is particularly useful for very small seeds, thus permitting precision plating which results in uniform seedling emergence and elimination of empty containers in nursery.
4. Pelleting also has the potential for accurately dosing seeds with chemicals.
5. Pelleting helps in obtaining correct plant distribution and better protection for each seedling against fungi or insect pests at lower application of protectants per hectare, thus reducing environmental pollution.
6. Improved the ballistic properties because of increase in weight of seed, therefore increase the capacity of aerially sown seed to penetrate into standing vegetation.
7. Pelleting can be used to pack an inoculants, growth regulators, protectants, nutrient, herbicide and hydrophilic substance as well as to increase supply of oxygen, thereby reducing the cost of production.
8. The pellet can have a colour so that birds and small animals cannot recognize the seeds.
9. Soil acidity which inhibits multiplication of rhizobia in the rhizosphere of the developing seedlings may be overcome by lime pelleting of seed. Lime pelleted seed improves nodulation and dry matter yield in acid soils by overall improvement in legume

performance. Remedial measures can also be infused through pelleting to tide over problem soil.

10. Handling of small seeds is made easy which in turn reduces the seed rate as well as cut the work of thinning.
11. Singling of seeds and prevention of clogging.
12. Attraction of moisture.
13. Supply of plant growth regulators and micronutrients.
14. Influence of micro-environment.
15. Saving of chemicals / fertilizers applied to the soil.
16. Ensure uniform field establishment.
17. Increased yield.

9. DISADVANTAGES

There are some drawbacks in pelleting the seeds as listed below

1. When the seed is pelleted, its cost and weights are more than bare seeds.
2. If proper pelleting techniques / machine is not used, there would be missing of seed during pelleting or some times more than one seed may contain in a pellet.
3. More moisture is required to germinate since pellet must be dissolved first.
4. Delayed field emergence at low moisture level.

10. PRECAUTIONS

Certain technical problems must be monitored during the potential process (Copeland and Mc Donald, 1999).

1. The pelleting material must be compatible with the seed, so that seed quality is maintained and germination is not hindered.
2. If the pelleting material is too hard after drying, it may be difficult for radical emergence through the pelleted material.

3. Some pellets may form around multiple seeds. In order cases, particularly when the seed lot is not thoroughly cleaned, pellets may form around inert matter such as sand or undersized plant material. The size of the pellet can also be an issue to seed purchasers.
4. Seed analysts also have a greater responsibility when conducting a purity test with pelleted seeds.

Despite the technical problems, seed pelleting is recognized as an important addition to the precision planting in many agricultural crops. In forestry, seed pelleting is an attractive tool to implement aerial seedling for wasteland afforestation programmes.

LECTURE 27

Seed legislations and The Seeds Act, 1966

Seed legislations

Legislations are an important aspect of a democratic society. They help in preventing unnecessary hindrance when one is attempting to do a job legally which is useful to the society.

The Seed Act is also called as **Sanctioning Legislation** and the Seed Rules as **Governing Legislation**. The Seeds Act sanctioned the definition for seeds and the formation of Central Seeds Laboratory, Certification Agency, Seed Analyst, Seed Inspector, Central Seed Committee and powers to the above.

The Seeds Rule provided the legal frame work for day-to-day working of the above and hence they were known as governing legislations. The Seeds Act is unique as it deals with an agricultural input that is alive. Hence, the modalities involved in production and maintenance of quality are to be as stringent as possible with a practical approach. The Seeds Act and Seed Rules formed the basis for future acts.

The Seeds Act, 1966

- An Act to provide for regulating the quality of certain seeds for sale, and for matters connected therewith.

- Seed Act passed in the Parliament on 29th Dec., 1966 and was published in issue No.66 of part II Sec.1- Gazette of India dated 30.12.1966 and supplemented with Seeds Rules 1968 on 29.08.1968.
- The total draft of Seeds Act 1966 and Seeds Rules 1968 were brought into effect on 02.10.1969.

Objective

- In the interest of increased agricultural production in the country, it is considered to regulate the quality of seeds, such as seeds of food crops, cotton, jute etc., to be sold for purpose of agriculture (including horticulture)

How / method to achieve this objectives

- Constitution of Central Seed Committee consisting of representative of both Central and State Governments to advise the governments on all matters arising out of the proposed legislation.
- Fixing minimum standards for Germination, Physical purity, varietal purity and maximum standard for moisture and ODV and other quality factors.
- Testing seeds for quality factors at the seed testing laboratories to be established by the Central government and State governments.
- Creation of seed inspection and certification service in each state and grant of license's and certificates to dealers in seeds
- Compulsory labeling of seed container to indicate the quality of seeds offered for sales
- Restricting the export, import and inter state movement of non-descriptive seed

The Seeds Act 1966 – Salient features

- Establishment of Central Seed Committee
- Certification voluntary but labeling is compulsory
- Label should contain necessary informations
- Establishment of Central and State Seed Testing Laboratory
- Establishment of Central Seed Certification Board

- Establishment of State Seed Certification Agencies
- Provision of Seed Inspectors and Seed Analysts
- Investigation and prosecution of the offences under the Act and Rules.
- To impose certain penalties for the offences committed under the Act.

Seed Act consists of 25 sections

- **General** (Sec.1,2)
- **Sanctioning legislation** (Sec. 3,4,8,18)
- **Regulatory legislation** (Sec.5,6,7,9,10,11,12,13,14,15,16,17,19,20,21,22,23,24, &25)

Sec.1. Short Title, Extent and Commencement

It extends to the whole of India. It shall come into force on such date as the Central Government may, by notification in the Official Gazette

Sec.2. Definitions

"Seed" means any of the following classes of seeds used for sowing or planting-

Seeds of food crops including edible oil seeds and seeds of fruits and vegetables; cotton seeds; seeds of cattle fodder; and includes seedlings, and tubers, bulbs, rhizomes, roots, cuttings, all types of grafts and other vegetatively propagated material, of food crops or cattle fodder.

"Variety" means a sub-division of a kind identifiable by growth, yield, plant, fruit, seed, or other characteristic.

Sec.3. Central Seed Committee

1. Central Government constitutes - to advise the Government and the State Governments on matters arising out of the administration of this Act and to carry out the other functions assigned to it by or under this Act.

2. The Committee shall consist of the following members,

- (i) A Chairman to be nominated by the Central Government;
- (ii) Eight persons to be nominated by the Central Government to represent such interests as that Government thinks fit, of whom not less than two persons shall be representatives of growers of seed

- (iii) One person to be nominated by the Government of each of the states.
3. The members of the Committee be entitled to hold office for two years and shall be eligible for re-nomination.
 4. The Committee may subject to the previous approval of Central Govt. make by-laws fixing the quorum and regulating its own procedure and conduct of all business
 5. The committee may appoint one or more sub-committees for the purpose of discharging such of its functions
 6. The functions of the committee or sub-committee thereof may be exercised notwithstanding any vacancy therein
 7. The Central Govt. may appoint a person to be the secretary of the Committee.

Sec.4.Central Seed Laboratory and State Seed Laboratory

(1) The Central Government may, by notification in the Official Gazette, establish a Central Seed Laboratory or declare any seed laboratory as the Central Seed Laboratory to carry out the functions entrusted to the Central Seed Laboratory by or under this Act.

(2) The State Government may, by notification in the Official Gazette, establish one or more State Seed Laboratories or declare any seed laboratory as a State Seed Laboratory where analysis of seeds of any notified kind or variety shall be carried out by Seed Analysts under this Act in the prescribed manner.

Sec.5.Power to notify kinds or varieties of seeds

If the Central Government, after consultation with the Committee, is of opinion that it is necessary or expedient to regulate the quality of seed of any kind or variety to be sold for purposes of agriculture, it may, by notification in the Official Gazette, declare such kind or variety to be a notified kind or variety for the purposes of this Act and different kinds or varieties may be notified for different States or for different areas thereof.

Sec.6. Power to specify minimum limits of germination and purity, etc.

The Central Government may, after consultation with the Committee and by notification in the Official Gazette, specify-

- the minimum limits of germination and purity with respect to any seed of any notified kind or variety;

- the mark or label to indicate that such seed conforms to the minimum limits of germination and purity specified under clause (a) and the particulars which such mark or label may contain.

Sec.7. Regulation of sale of seeds of notified kinds or varieties

No person shall, himself or by any other person on his behalf, carry on the business of selling, keeping for sale, offering to sell, bartering or otherwise supplying any seed of any notified kind or variety, unless-

- Such seed is identifiable as to its kind or variety;
- Such seed conforms to the minimum limits of germination and purity specified under clause (a) of section 6;
- The container of such seed bears in the prescribed manner, the mark or label containing the correct particulars thereof, specified under clause (b) of section 6; and
- He complies with such other requirements as may be prescribed.

Sec.8. Certification agency

The State Government or the Central Government in consultation with the State Government may, by notification in the Official Gazette, establish a certification agency for the State to carry out the functions entrusted to the certification agency by or under this Act.

Sec.9. Grant of certificate by certification agency

1. Any person selling, keeping for sale, offering to sell, bartering or otherwise supplying any seed of any notified kind or variety may, if he desires to have such seed certified by the certification agency, apply to the certification agency for the grant of a certificate for the purpose.
2. Every application under sub-section (1) shall be made in such form, shall contain such particulars and shall be accompanied by such fees as may be prescribed.
3. On receipt of any such application for the grant of a certificate, the certification agency may, after such enquiry as it thinks fit and after satisfying itself that the seed to which the application relates conforms to the minimum limits of germination and purity specified for that seed under

clause (a) of section 6, grant a certificate in such form and on such conditions as may be prescribed.

Sec.10. Revocation of certificate

If the certification agency is satisfied, either on a reference made to it in this behalf or otherwise, that-

- The certificate granted by it under section 9 has been obtained by misrepresentation as to an essential fact; or
- The holder of the certificate has, without reasonable cause, failed to comply with the conditions subject to which the certificate has been granted or has contravened any of the provisions of this Act or the rules made thereunder; then, without prejudice to any other penalty to which the holder of the certificate may be liable under this Act, the certification agency may, after giving the holder of the certificate an opportunity of showing cause, revoke the certificate.

Sec. 11. Appeal

1. Any person aggrieved by a decision of a certification agency under section 9 or section 10, may, within thirty days from the date on which the decision is communicated to him and on payment of such fees as may be prescribed, prefer an appeal to such authority as may be specified by the State Government in this behalf:

Provided that the appellate authority may entertain an appeal after the expiry of the said period of thirty days if it is satisfied that the appellant was prevented by sufficient cause from filing the appeal in time.

2. On receipt of an appeal under sub-section (1), the appellate authority shall, after giving the appellant an opportunity of being heard, dispose of the appeal as expeditiously as possible.
3. Every order of the appellate authority under this section shall be final.

Sec. 12. Seed Analysts

The State Government may, by notification in the Official Gazette, appoint such persons as it thinks fit, having the prescribed qualifications, to be Seed Analysts and define the areas within which they shall exercise jurisdiction.

Sec. 13. Seed Inspectors

(1) The State Government may, by notification in the Official Gazette, appoint such persons as it thinks fit, having the prescribed qualifications, to be Seed Inspectors and define the areas within which they shall exercise jurisdiction.

(2) Every Seed Inspector shall be deemed to be a public servant within the meaning of section 21 of the Indian Penal Code (45 of 1860) and shall be officially subordinate to such authority as the State Government may specify in this behalf.

Sec. 14. Powers of Seed Inspector

(1) The Seed Inspector may-

- Take samples of **any seed of any notified kind or variety** from-
- any person selling such seed; or
- any person who is in the course of conveying, delivering or preparing to deliver such seed to a purchaser or a consignee; or
- a purchaser or a consignee after delivery of such seed to him;

Send such sample for analysis to the Seed Analyst for the area within which such sample has been taken;

- **Enter and search at all reasonable times**, with such assistance, if any, as he considers necessary, any place in which he has reason to believe that an offence under this Act has been or is being committed and order in writing the person in possession of any seed in respect of which the offence has been or is being committed, not to dispose of any stock of such seed for a specific period not exceeding thirty days or, unless the alleged offence is such that the defect may be removed by the possessor of the seed, seize the stock of such seed;
- **Examine any record, register, document** or any other material object found in any place mentioned in clause (c) and **seize the same** if he has reason to believe that it may furnish evidence of the commission of an offence punishable under this Act; and
- **Exercise such other powers** as may be necessary for carrying out the purposes of this Act or any rule made thereunder.

(2) Where any sample of any seed of any notified kind or variety is taken under clause (a) of sub-section (1), **its cost, calculated at** the rate at which such seed is usually sold to the public, shall be paid on demand to the person from whom it is taken.

(3)) Seed inspectors are empowered to **break-open any container** in which any seed of any notified kind or variety may be contained or to break-open the door of any premises where any such seed may be kept for sale

(4) Where the Seed Inspector takes any action under clause (a) of sub-section (1), he shall, as far as possible, **call not less than two persons to be present at the time** when such action is taken and take their signatures on a memorandum to be prepared in the prescribed form and manner.

(5) The provisions of the **Code of Criminal Procedure, 1898 (5 of 1898)**, shall, so far as may be, apply to any search or seizure under this.

Sec.15. Procedure to be followed by Seed Inspectors

- Whenever a Seed Inspector intends to take sample of any seed of any notified kind or variety for analysis, he shall-

(a) **Give notice** in writing, then and there, of such intention to the person from whom he intends to take sample

(b) Except in special cases take **three representative samples** in the prescribed manner and mark and seal or fasten up each sample in such manner as its nature permits.

(2) When samples of any seed of any notified kind or variety are taken the Seed Inspector shall-

(a) deliver one sample to the person from whom it has been taken;

(b) send another sample for analysis to the Seed Analyst

(c) retain the remaining sample for production in case any legal proceedings are taken or for analysis by the CSTL

(3) If the person from whom the samples have been taken **refuses** to accept one of the samples, the Seed Inspector shall send intimation to the Seed Analyst of such refusal and thereupon **Seed Analyst** receiving the sample for analysis shall divide it into two parts and shall seal or fasten up

one of those parts and shall cause it, when he delivers his report, to be delivered to the Seed Inspector who shall retain it for production in case legal proceedings are taken.

(4) Where a Seed Inspector takes any action

(a) If he **seizes the stock** of the seed, he shall, as soon as may be, inform a **magistrate** and take his orders as to the custody thereof;

(b) He shall revoke the order if it is ascertained that the seed does not so contravene

5) Where a Seed Inspector **seizes any record**, register, document or any other material object he shall, as soon as may be, inform a **magistrate** and take his orders as to the custody thereof.

Sec.16.Report of Seed Analyst

1) The Seed Analyst shall, as soon as may be after the receipt of the sample under sub-section (2) of section 15, **analyse the sample at the State Seed Laboratory** and deliver, **in such form** as may be prescribed, one copy of the report of the result of the analysis to **the Seed Inspector** and another copy thereof to **the person** from whom the sample has been taken.

2) After the institution of a prosecution under this Act, the accused vendor or the complainant may, on payment of the prescribed fee, make an application to the court for sending any of the samples mentioned in clause (a) or clause (c) of sub-section (2) of section 15 to the Central Seed Laboratory for its report and on receipt of the application, the court shall first ascertain that the mark and the seal or fastening as provided in clause (b) of sub-section (1) of section 15 are intact and may then despatch the sample under its own seal to the Central Seed Laboratory which shall thereupon send its report to the court in the prescribed form within one month from the date of receipt of the sample, specifying the result of the analysis.

3) The report sent by the Central Seed Laboratory under sub-section (2) shall supersede the report given by the Seed Analyst under sub-section (1).

4) Where the report sent by the Central Seed Laboratory under sub-section (2) is produced in any proceedings under Section 19, it shall not be necessary in such proceedings to produce any sample or part thereof taken for analysis.

Sec.17. Restriction on export and import of seeds of notified kinds or varieties

No person shall, for the purpose of sowing or planting by any person (including himself), export or import or cause to be exported or imported any seed of any notified kind or variety, unless-

- it conforms to the minimum limits of germination and purity specified for that seed under clause (a) of section 6; and
- its container bears, in the prescribed manner, the mark or label with the correct particulars thereof specified for that seed under clause (b) of section 6.

Sec.18. Recognition of seed certification agencies of foreign countries

The Central Govt. may, on the recommendation of the Committee and by notification in the Official Gazette, recognise any seed certification agency established in any foreign country, for the purposes of this Act.

Sec.19. Penalty

If any person-

- contravenes any provision of this Act or any rule made thereunder; or
- prevents a Seed Inspector from taking sample under this Act;**or**
- prevents a Seed Inspector from exercising any other power conferred on him by or under this Act;

He shall, on conviction, be punishable-

- for the first offence with fine which may extend to five hundred rupees, and
- in the event of such person having been previously convicted of an offence under this section, with imprisonment for a term which may extend to six months, or with fine which may extend to one thousand rupees, or with both.

Sec.20. Forfeiture of property

When any person has been convicted under this Act for the contravention of any of the provisions of this Act or the rules made thereunder, the seed in respect of which the contravention has been committed may be forfeited to the Government.

Sec.21. Offences by companies

(1) Where an offence under this Act has been committed by a company, every person who at the time the offence was committed was in charge of, and was responsible to the company for the conduct of the business of the company, as well as the company, shall be deemed to be guilty of the offence and shall be liable to be proceeded against and punished accordingly:

Provided that nothing contained in this sub-section shall render any such person liable to any punishment under this Act if he proves that the offence was committed without his knowledge and that he exercised all due diligence to prevent the commission of such offence.

(2) Notwithstanding anything contained in sub-section (1), where an offence under this Act has been committed by a company and it is proved that the offence has been committed with the consent or connivance of, or is attributable to any neglect on the part of, any director, manager, secretary or other officer of the company, such director, manager, secretary or other officer shall also be deemed to be guilty of that offence and shall be liable to be proceeded against and punished accordingly.

Explanation. – For the purpose of this section,-

- "company" means any body corporate and includes a firm or other association of individuals; and
- "director", in relation to a firm, means a partner in the firm.

Sec.22. Protection of action taken in good faith

No suit, prosecution or other legal proceeding shall lie against the Government or any officer of the Government for anything which is in good faith done or intended to be done under this Act.

Sec. 23. Power to give directions

The Central Government may give such directions to any State Government as may appear to the Central Government to be necessary for carrying into execution in the State any of the provisions of this Act or of any rule made thereunder.

Sec.24. Exemption

Nothing in this Act shall apply to any seed of any notified kind or variety grown by a person and sold or delivered by him on his own premises direct to another person for being used by that person for the purpose of sowing or planting.

Sec. 25. Power to make rules

(1) The Central Government may, by notification in the Official Gazette, make rules to carry out the purpose of this Act.

(2) In particular and without prejudice to the generality of the fore-going power, such rules may provide, for-

- the functions of the Committee and the travelling and daily allowances payable to members of the Committee and members of any sub-committee appointed under sub-section (5) of section 3;
- the functions of the Central Seed Laboratory;
- the functions of a certification agency;
- the manner of marking or labelling the container of seed of any notified kind or the requirements which may be complied with by a person carrying on the business
- the form of application for the grant of a certificate under section 9, the particulars it may contain, the fees which should accompany it,
- the form and manner in which and the fee on payment of which an appeal may be preferred and the procedure to be followed by the appellate authority in disposing of the appeal;
- the qualifications and duties of Seed Analysts and Seed Inspectors;
- the manner in which samples may be taken by the Seed Inspector, the procedure for sending such samples to the Seed Analyst or the Central Seed Laboratory and the manner of analysing such samples;
- the form of report of the result of the analysis and the fees payable in respect of such report.
- the records to be maintained by a person carrying on the business and
- any other matter which is to be or may be prescribed.

(3) Every rule made under this Act shall be laid as soon as may be after it is made, before each House of Parliament while it is in session for a total period of thirty days which may be comprised in one session or in two successive sessions, and if, before the expiry of the session.

THE SEEDS RULES, 1968

- Provided wider scope for understanding various provisions of the Seeds Act 1966.
- The function of the Central Seed Testing Laboratory, Seed Certification Agency are elaborately dealt with.
- Labeling of any notified kind of variety of seed is made compulsory.
- It provides certain requirements to be complied with the person carrying on the business of selling seed

Seed Rules 1968 – Salient features

- Categorized the certified seed in to three distinct classes.
- Certification procedures are dealt in detail.
- Qualification of seed Analysts and his duties are specified.
- Qualification of seed inspectors and his duties are widely defined.
- Procedures for dealing with a written complaint lodged to the seed inspector is laid down.
- Seed sampling procedures are dealt in detail.
- Maintenance of various records and issuing of memorandum in Form-VII by Seed Inspector to the dealer.

Consists of 11 parts and 39 rules

PART I – PRELIMINARY

1. Short title: These rules may be called the Seeds Rules, 1968

2. Definitions

a. "**Act**" - the Seeds Act, 1966

b. "**Advertisement**" - a. representations other than those on the label, disseminated in any manner or by any means relating to seed for the purposes of the act;

- c. **"Certification sample"** - a sample of seed drawn by a certification agency or by a duly authorized representative of a certification
- d. **"Certification tag"** - a tag or label of certain design to be specified by the certification agency and shall constitute the certificate granted by the certification agency;
- e. **"Certified seed"** - seed that fulfills all requirements for certification provided by the Act and these rules and to the container of which the certification tag is attached.
- f. **"Certified seed producer"** - a person who grows or distributes certified seed in accordance with the procedure and standards of the certification agency;
- g. **"Complete record"** - the information which relates to the origin, variety, kind germination and purity of seed of any notified kind or variety offered for sale, sold or otherwise supplied:
- h. **"Form"** - a form appended to these rules:
- i. **"Origin"** - the State, Union, Territory or foreign country where the seed is grown and in case seeds of different origin are blended the label shall show the percentage of seed of each origin;
- j. **"Processing"** - cleaning, drying, treating, grading and other operations which would change the purity and germination of the seed and thus requiring re-testing to determine the quality of the seed, but does not include operations such as packaging and labeling"
- k. **"Section"** - a section of the Act'
- l. **"Service sample"** - a sample submitted to the Central Seed Laboratory or to a State Seed Laboratory for testing, the results to be used as information for seeding selling or labeling purposes;
- m. **"Treated"** - that the seed has been subjected to an applications of a substance or process in such a manner as to reduce, control or repel certain disease organisms, insects, or any other pests attacking such seeds or seedlings growing therefore and for other purposes.

PART II - CENTRAL SEED COMMITTEE

3. Functions of the Central Seed Committee:

In addition to the functions entrusted to the committee by the Act, the committee shall,

- a. Recommend the rate of fees to be levied for analysis of samples by the Central and States Seed Testing Laboratories and for certification by the certification agencies;
- b. Advice the Central or State Governments on the suitability of seed testing laboratories;

- c. Send its recommendations and other concerning records to the Central Government;
- d. Recommend the procedure and standards for certification, tests and analysis of seeds ; and
- e. Carry out such other functions as are supplemental, incidental or consequential to any of the functions conferred by the Act or these rules.

4. Traveling and daily allowances payable to Members of the Committee and its Sub-committees:

The members of the committee and its sub-committee shall be entitled to draw traveling and daily allowances as specified below when they are called upon to attend a meeting of the committee or a sub-committee thereof:

- a. An official member of the committee or its sub-committee shall be entitle to draw traveling and daily allowances in accordance with the rules of the Government under which he is for the time being
- b. A non official member shall be allowed traveling and daily allowances in accordance with the general orders issued in this behalf by the Central Government from time to time.

PART III - Central Seed Laboratory

5. Functions:

In addition to the functions entrusted to the /central Seed Laboratory by the Act, the Laboratory shall carry out the following functions, namely;

- a. Initiate testing programmes in collaboration with the State Seed Laboratories designed to promote uniformity in test results between all seed laboratories in India;
- b. Collect data continually on the quality of seeds found in the market and make this data available to the Committee; and
- c. Carry out such other functions as may be assigned to it by the Central Government from time to time.

PART IV- SEED CERTIFICATION AGENCY

6. Functions of the certification agency:

In addition to the functions entrusted to the certification agency by the Act, the Agency shall certified seeds of any notified kinds or varieties;

- a. Certify seeds of any notified kinds or variety;
- b. Outline the procedure of submission of applications and for growing, harvesting, processing, storage and labeling of seeds intended for certification till the end to ensure that seed lots finally approved for certification are true to variety and meet prescribed standards for certification under the Act or these rules;
- c. Maintain a list of recognized breeders of seeds;
- d. Verify, upon receipt of an application, for certification, that the variety is eligible for certification that the seed source need for planting was authenticated and the record of purchase is in accordance with these rules and the fees has been paid;
- e. Take sample and inspect seed lots produced under the procedures laid down by the certification agency and have such samples tested to ensure that the seed conforms to the prescribed standards of certification;
- f. Inspect seed processing plants to see that the admixtures of other kinds and varieties are not introduced.
- g. Ensure that action at all stages e.g. field inspection, seed processing, plant inspection, analysis of samples taken and issue of certificates (including tags, marks, labels and seals) is taken expeditiously;
- h. Carry out educational programmes designed to promote the use of certified seed including a publication listing certified seed growers and sources of certified seed;
- i. Grant certificate (including tags, marks, labels and seals etc.) in accordance with the provisions of the Act and these Rules;
- j. Maintain such records as may be necessary to verify that seed plants for the production of certified seed were eligible for such planting under these rules;
- k. Inspect fields to ensure that the minimum standards for isolation, rouging (where applicable) use of male sterility (where applicable) and similar factors are maintained at all times, as well as ensure that seed borne diseases are not present in the field to a greater than those provided in the standards for certification.

PART V - MARKING OR LABELING

7. Responsibility for marking or labeling:

When seed of a notified kind or variety is offered for sale under section 7 each container shall be marked or labeled in the manner hereinafter specified.

The person whose name appears on the mark or label shall be responsible for the accuracy of the information required to appear on the mark or label so long as seed is contained in the unopened original container.

Provided, however, that such person shall not be responsible for the accuracy of the statement appearing on the mark or he shall not be responsible for the accuracy of the germination statement beyond the date of validity indicated on the mark or label.

8.Contents of the mark or label :

There shall be specified on every mark or label

- i. Particulars, as specified by the Central Government under Clause (b) of section 6 of the act.
- ii. A correct statement of the net content in terms of weight and expressed in metric system.
- iii. Date of testing.
- iv. If the seed in container has been treated.
 - a. Statement indicating that the seed has been treated.
 - b. The commonly accepted chemical or abbreviated chemical (generic) name of the applied substance; and
 - c. If the substance of the chemical used for treatment, and present with the seed is harmful to human beings or other vertebrate animals, a caution statement such as "Do not use for food, feed or oil purposes". The caution for mercurial and similarly toxic substance shall be the word "Poison" which shall be in type size, prominently displayed on the label in red.
- v. The name and address of the person who offers for sale, sells or otherwise supplies the seed and who is responsible for its quality;
- vi. The name of the seed as notified under section 5 of the act.

9. Manner of marking or labeling the container

The mark or label containing the particulars of the seed as specified shall appear on each container of seed or on a tag or mark or label attached to the container in a conspicuous place on the inner most container in which the seed is packed and on every other covering in which that container is packed and shall be legible.

10. Mark or label not to contain false or misleading statement:

The mark or label shall not contain any statement, claim, design, device, fancy name or abbreviation which is false or misleading in any particular concerning the seed contained in the container.

11. Mark or label not to contain reference to the act or rules contradictory to required particulars:

The mark or label shall not contain any reference to the Act, or any of these rules or any comment on, or reference to, or explanation of any particulars or declaration required by the Act or any of these rules which directly or by implication contradicts, qualifies or modifies such particulars or declaration.

12. Denial of responsibility for mark or label content prohibited:-

Nothing shall appear on the mark or label or in any advertisement pertaining to any seed of any notified kind or variety which shall deny responsibility for the statement required by or under the Act to appear on such mark, label or advertisement.

PART VI- REQUIREMENTS

13. Requirements to be complied with by a person carrying on the business

1. No person shall sell, keep for sale, and offer to sell or otherwise supply any seed of any notified kind or variety, after the date recorded on the container
2. No person shall alter any mark or label attached to the container of any seed.
3. Every person selling, keeping for sale, offering to sell, any seed of notified kind or variety shall keep **over a period of three years** a complete record of each lot of seed sold except that any seed sample may be discarded one year after the entire lot represented by such sample has been disposed off.

14. Classes and sources of certified seed :-

To produce the following classes of seed the source seed would be

Class of seed	Source seed
Breeder seed	Nucleus Seed
Foundation seed Stage One	Breeder seed
Foundation seed Stage Two	Foundation seed Stage One
Certified seed Stage One	Foundation seed Stage Two

Certified seed Stage Two	Certified seed Stage One
Commercial multiplication	Certified seed Stage Two

PART VII - Certification of Seeds

15. Application for the grant of a certificate:

Every application for the grant of a certificate shall be made in Form in accordance with the procedure outlined by the certification agency for submission of applications and contains the following particulars, namely:

- a. the name, profession and place of residence of the applicant;
- b. the name of the seed to be certified, its notified kind or variety;
- c. class of the seed;
- d. source of the seed;
- e. limits of germination and purity of the seed;
- f. label of the seed

16. Fees:

Every application under sub-section (1) of the section 9 shall be accompanied by a fee of Rs,25 in cash.

17. Certificate:

Every shall be granted by the certifications agency after making enquiries and satisfying itself to the specified by the certification agency namely

- 1.The person to whom the certificate is granted shall attach a certification tag to every container of the certified seed and shall follow the provisions in respect of marking or labeling provided by or under the act.
2. The certification tag shall contain the following particulars namely.
 - a. Name and address of the certification agency.
 - b. Kind and variety of the seed.
 - c. Lot no or other mark of the seed.
 - d. Name and address of the certified seed producer.
 - e. Date of issue of the certificate and of its validity.
 - f. An appropriate sign to designate certified seed.

- g. An appropriate word denoting the class designation of the seed.
3. The colour of the certification tag shall be white for foundation seed and blue for certified seed.
 4. The container of the certified seed shall carry a seal of such material and in form as the certification agency may determine and no container carrying a certification tag shall be sold by the person if the tag or seal has either been tampered with or removed.
 5. The certification tag on the container shall specify.
 - a. The period during which the seed shall be used for sowing or planting.
 - b. That the use of seed after the expiry of the validity period by any person is entirely at his risk and the holder of the certificate shall not be responsible for any damage to the buyer of the seed.
 - c. That no one should purchase the seed if the seal or the certification tag has been tampered with.
 6. The holder of the certificate shall keep record of the details of each lot of the seed which is issued for sale in such form as to be available for inspection and to be easily identified by reference to the number of the lot as shown in the certification tag of each container and such records shall be retained in the case of a seed for which expiry date is fixed for a period of two years from the expiry of such date.
 7. The holder of the certificate shall allow any Seed Inspector, authorized in writing by the certification agency in that behalf, to enter with or without prior notice, the premises where the seeds are grown, processed and sold and to inspect premises, plant and the process of processing at all reasonable hours.
 8. The holder of the certificate shall allow the Seed Inspector, authorized in writing by the certification agency, to inspect all registers and records maintained under these rules and to take samples of the seeds and shall supply to the Seed Inspector such information as he may require for the purposes of ascertaining whether the conditions subject to which the certificate has been granted, have been complied with.
 9. The holder of the certificate shall on request furnish to the certification agency from every lot of the seed or from such lot or lots as the said agency may from time to time specify, a sample of such quantity as the agency may consider adequate for any examination required to be made.

10. If the certification agency so directs, the holder of the certificate shall not sell or offer for sale any lot in respect of which a sample is furnished under the proceeding clause until the agency authorizes the sale of such lot.
11. The holder of the certificate shall, on being directed by the certification agency that any part of a lot has been found by the said agency not to conform to prescribed standards of quality or purity specified by or under the Act, withdraw the remainder of that lot from sale and so far as may, in the particular circumstances of the case, be practicable, recall all issues already made from that lot.
12. The holder of the certificate shall comply with the provisions of the Act and these Rules and with the directions given after not less than one month's notice by the certification agency to such holder.

The Certification agency shall, before granting the certificate, ensure that the Seed conforms to the standards laid down in the Manual known as "Indian Minimum Seed Certification Standards" published by the Central Seed Committee, as amended from time to time". (Amendment No.18-48/81-SD, dated 10th June, 1981).

PART VIII- APPEALS

18. The form and manner in which and the fee on payment of which the Appeal may be preferred:-

1. Every memorandum of appeal under sub-section (1) of section 11 shall be in writing and shall be accompanied by a copy of the decision of the certification agency against which it has been preferred and shall set forth concisely and under distinct heads the grounds of objections to such decision without any argument, or narrative.
2. Every such memorandum of Appeal shall be accompanied by a treasure receipt for sum of 100/- rupees.
3. Every such memorandum of appeal may be presented either in person or through an agent duly authorized in writing in this behalf by the appellant or may be sent by the registered post.

19. Procedure to be followed by the appellate authority:-

In deciding appeals under the Act the Appellate authority shall exercise all the powers which a Court has and shall follow the same procedure with a Court follows in deciding appeals from the decree or order of an original Court under the Code of Civil Procedure, 1908.

PART IX - SEED ANALYSTS AND SEED INSPECTORS

20. Qualifications of Seed Analysts: A person shall not be qualified for appointment as Seed Analyst unless he.

- i. Possesses a Master's or equivalent degree in Agriculture or Agronomy or Botany or Horticulture of a University recognized for this purpose by the Government and has had not less than one year's experience in seed technology; or
- ii. Possessed a Bachelor's degree in Agriculture or Botany of a University recognized for this purpose by the Government and has had not less than three year's experience in seed technology.

22. Duties of a seed analyst :

1. On receipt of a sample for a analysis the Seed Analyst shall first ascertain that the mark and the seal as provided are intact and shall not the condition of the seals thereon.
2. The Seed analyst shall analyze the samples in accordance with the procedures laid down in the Seed Testing Manual published by the Indian Council of Agriculture Research as amended from time to time. (Amendment No.7(17)/69-Seeds-Dev., dated 30-6-1973)
3. The Seed Analyst shall deliver in Form VII, a copy of report of the result of the analysis to the persons as soon as ,may be but not later than 30 days from the date of receipt of samples sent by the Seed Inspector
4. The Seed Analyst shall from time to time forward to the State Government the reports giving the result of analytical work done by him.

22. Qualification of Seed Inspectors:

A person shall not be qualified for appointment as Seed Inspector unless he is a graduate in Agriculture of a University recognized for the purpose by the Government and has had not less than one year's experience in seed production, or seed development in seed analysis or testing in seed testing laboratory.

23. Duties of a Seed Inspector:

In addition to the duties specified by the Act, the Seed Inspector shall

- a. **Inspect** as frequently as may be required by certification agency all places used **for growing storage or sale of any seed of any notified kind or variety;**

- b. **Satisfy himself** that the conditions of the certificates are being observed;
- c. **Procedure and send for analysis**, if necessary, samples of any seeds, which he has reason to suspect are being produced, stocked or sold or exhibited for sale on contravention of the provisions of the Act or these rules;
- d. **Investigate any complaint**, which may be made to him in writing in respect of any contravention of the provisions of the Act or these rules.
- e. **Maintain a record of all inspections** made and action taken by him in the performance of his duties including the taking of samples and the seizure of stocks and submit copies of such record to the Director of Agriculture or the certification agency as may be directed in this behalf.
- f. When so authorized by the State Government detain imported containers which he has reason to suspect contain seeds, **import of which is prohibited** except and in accordance with the provisions of the Act or these rules.
- h. Perform such other duties as may be entrusted to him by the State Government

PART X - Sealing, fastening, despatch and analysis of samples

24. Manner of taking samples :

Samples of any seed of any notified kind of variety for the purpose of analysis shall be taken in a clean dry container which shall be closed sufficiently tight to prevent leakage and entrance of moisture and shall be carefully sealed.

25. Containers to be labeled and addressed: -

All containers containing samples for analysis shall be properly labeled and the parcels shall be properly addressed. The label on any sample of seed sent for analysis shall bear.

- a. Serial number;
- b. Name of the sender with official designation, if any;
- c. Name of the person from whom the sample has been taken.
- d. Date and place of taking the sample;
- e. Kind or variety of the seed for analysis;
- f. Nature and quantity of preservative, if any, added to the sample.

26. Manner of packing, fastening and sealing the samples:-

All samples of seed sent for analysis shall be packed, fastened and sealed in the following manner.

- a. The stopper shall first be securely fastened to as to prevent leakage of the containers in transit;
- b. The container shall then be completely wrapped in fairly strong thick paper. The ends of the paper shall be neatly folded in and affixed by means of gum or other adhesive.
- c. The paper cover shall be further secured by means of strong twine or thread both above and across the container, and the twine or thread shall then be fastened on the paper cover by means of sealing wax on which there shall be test four distinct and clear impression of the seal of the sender of which one shall be at the top of the packet, one at the bottom and the other two on the body of the packet. The knots of the twine or thread shall be covered by means of sealing wax bearing the impression of the seal of the sender.

27. Form of order:

The order to be given in writing by the Seed Inspector in Form III

28. Form of receipt of records:

When a Seed Inspector seizes any record, register, document or any other material he shall issue a receipt in Form IV to the person concerned.

29. Samples how to send to the Seed Analyst:

The container of sample for analysis shall be sent to the Seed Analyst by registered post or by hand in a sealed packet enclosed together with a memorandum in Form V in an outer cover addressed to the Seed Analyst.

30. Memorandum and impression of seal to be sent separately:

A copy of the memorandum and a specimen impression of the seal used to seal the packet shall be sent to the Seed Analyst separately by registered post or delivered to him or to any person authorized by him.

31. Addition of preservatives to samples:

Any person taking a sample of seed for the purpose of analysis under the Act may add a preservative.

32. Nature and quantity of the preservative to be noted on the label:

Whenever any preservative is added to a sample the nature and quantity of the preservative added shall be clearly noted on the label to be affixed to the container.

33. Analysis of the sample:

On receipt of the packet, it shall be opened either by the Seed Analyst or by an officer authorized in writing in that behalf by the Seed Analyst, who shall record the condition of the seal on the packet. Analysis of the sample shall be carried out at the State Seed Laboratory in accordance with the procedure laid down by the Central Government.

34. Form of notice:

The notice to be given to the person from whom the Seed Inspector intends to take sample shall be in Form VI.

35. Form of report:

The report of the result shall be delivered or sent in Form VII.

36. Fees:

The fees payable in respect of the report from the Central Seed Laboratory shall be Rs.30/- per sample of the seed analyzed.

37. Retaining of the sample:

The sample of any seed shall, be retained under a cool, dry environment to eliminate the loss of viability and in insect proof or rat proof containers. The containers shall be dusted with suitable insecticides and the storage room fumigated to avoid infestation of samples by insects. The samples shall be packed in good quality containers of uniform shape and size before storage.

PART XI - Miscellaneous**38. Records:**

A person carrying on the business referred to shall maintain the following records, namely:

- a. Stock record of seed;
- b. Record of the sale of seeds;

39. Form of Memorandum: - The memorandum to be prepared shall be in Form VIII.

The Seeds (Control) Order, 1983 & Essential Commodities Act, 1955**Seeds (Control) Order, 1983**

- The inclusion of seeds as an essential commodity item under the Essential Commodity Act, 1955 brought the Seeds (Control) Order.

- A person carrying on the business of selling, exporting and importing of seeds needs to obtain a license
- The Essential Commodity Act, 1955 gives powers to State governments to regulate various aspects of trading in essential commodities under the supervision of Central Government.
- The seed dealer has to essentially display the stock position (opening and closing) on daily basis along with a list indicating prices or rates of different seeds.

Seed Control order was enacted on 30th December, 1983

Deals with four things.

- (i) Preliminary (2 clauses)
- (ii) Dealer in seeds to be licensed (8 clauses)
- (iii) Enforcement authority (6 clauses)
- (iv) Miscellaneous (2 clauses)

PRELIMINARY

1. Short title and extent

- i. This Order may be called the Seeds (Control) Order, 1983.
- ii. It extends to the whole of India.
- iii. It shall come into force on the 30th December, 1983

2. Definitions

“Act” means Essential Commodities Act, 1955 (10 of 1955).

“Inspector” - inspector of seeds appointed under clause 12

“Registering authority” means a licensing authority appointed under clause 11

DEALER IN SEEDS TO BE LICENSED

3. Dealer to obtain license

No person shall carry on the business of selling exporting or importing seeds at any place except under and in accordance with the terms and conditions of license granted to him under this order.

4. Application for license

Every person desiring to obtain a license for selling, exporting or importing seeds shall make an application in duplicate in Form A together with a fee of rupees fifty for license to the licensing Authority

5. Grant or refusal of license

1. The licensing authority may, after making such inquiry as it thinks fit, grant a license in form B to any person who applies for it under Clause 4. Provided that a license shall not be issued to a person:-

A. Whose earlier license granted under this order is under suspension , during the period of such suspension.

B. Whose earlier license granted under this order has been cancelled, within a period of one year from the date of such cancellation.

C. Who has been convicted under the Essential Commodities Act, 1955 or any order issued there under within three years proceeding the date of application.

2. When the licensing authority refuses to grant a license to a person who applies for it under clause 4, he shall record his reasons for doing so.

6. Period of validity of license

Every license granted under this Order, shall unless previously suspended or cancelled, remain valid for three years from the date of its issue.

7. Renewal of license

1. Every holder of license desiring to renew the license, shall, before the date of expiry of the license, make an application for renewal in duplicate, to the licensing authority in Form C together with a fee of rupees twenty for renewal. On receipt of such application, together with such fee, the licensing authority may renew the license.

2. If any application for renewal is not made before the expiry of the license, but is made within one month from the date of expiry of the license, the license may be renewed on payment of additional fee of rupees twenty-five, in addition to the fee for renewal of license.

8. Dealers to display stock and price list

Every dealer of seeds shall display in his place of business:-

A. The opening and closing stocks, on daily basis, of different seeds held by him.

B. A list indicating prices of rate of different seeds.

9. Dealers to give memorandum to purchases.

Every dealer shall give a cash or credit memorandum to a purchaser of seeds.

10. Power to distribute seeds

Where it is considered necessary to do so in public interest, the Controller may, by an order in writing direct any producer or dealer to sell or distribute any seed in such a manner as may be specified therein.

ENFORCEMENT AUTHORITY

11. Appointment of licensing authority

The State Government may after notification in the Official Gazette appoint such number of persons as it thinks necessary to the licensing authority and may also define in that notification the area within which each such licensing authority shall exercise his jurisdiction.

12. Appointment of Inspector

The State Government may by notification in the Official Gazette appoint such number of persons as it thinks necessary as the Inspectors and may in such notification define the local area within which each such Inspector shall exercise his jurisdiction.

13. Inspection and Punishment

1. An inspector may with a view to securing compliance with this Order:-

A. Require any **dealer to give any information** in his possession with respect to purchase, storage and sale of seeds by him;

B. Enter upon and **search any premises** where any seed is stored or exhibited for sale to ensure compliance with the provisions of this Order.

C. **Draw samples** of seeds meant for sale, export and seeds imported, and send the same in accordance with the procedure laid down in Schedule I, to a laboratory notified under the to ensure that the sample conforms to standard of quality claimed.

D. **Seize or detain any seed** in respect of which he has reason to believe that a contravention of this Order has been committed or is being committed.

E. **Seize books of accounts** or document relating to any seed in respect of which he has reason to believe that a contravention of this Order has been committed or is being committed. Provided that the Inspector shall give a receipt, in respect of the books of accounts or documents seized, to the person from whom they have been seized.

1. Provided further that the seized books of accounts or documents shall be returned to the persons from whom the same had been seized after copies thereof or extracts there from as certified by such person have been taken.

2. Subject to the provision of paragraph (d) of sub-clause (1) the provision, of section 100 of the **Code of Criminal procedure 1973**(2 of 1974) relating to search and seizure shall so far as may be apply to searches and seizures under this clause.

3. Where any seed is seized by an Inspector under this clause, he shall forthwith report the fact of such seizure to a **Magistrate** where upon the provisions of Sections 457 and 458 of the Code of Criminal Procedure 1973(2 of 1974) shall, so far as may be applied to the custody and disposal of such seed.

4. Every person, if so required by an Inspector, shall be bound to offer all necessary facilities to him for the purpose of enabling him to exercise power under this clause.

14. Time limit for analysis

The laboratory, to which a sample has been sent by an Inspector for analysis under this Order, shall analyse the said samples and send the analysis report to the concerned Inspector **within 60 days** from the date of receipt of the sample in the laboratory.

15. Dispension /Cancellation of license

The licensing authority may, after giving the holder of license an opportunity of being heard suspend or cancel the license on the following grounds:

- a. That the license had been obtained by **misrepresentation** as to the material particulars, or ,
- b. That any of the provisions of this Order or any conditions of license has been **contravened**.

16. Appeal

Any person aggrieved by an order:-

- a. Refusing to grant, amend or renew the licence for sale, export or import of seeds;
- b. Suspending or canceling any licence may within sixty days from the date of order, appeal to the authority as the State Government may specify in this behalf and the decision of such authority shall be final; provided that an application for appeal shall accompany an appeal fee of rupees fifty.

MISCELLANEOUS

17. Amendment of license

The licensing authority may on receipt of a request in writing together with a fee of rupees ten from a dealer amend the license of such dealer.

18. Maintenance of records and submission of returns etc.,

1. Every dealer shall maintain such book accounts and records relating to his business as may be directed by the State Government.

2. Every dealer shall submit monthly returns relating to his business for the preceding month in Form D to the licensing authority by the 5th day of every month.

SEED LAW ENFORCEMENT

The seed laws with which we are concerned therefore impose certain restrictions on at least some of the various stages through which seed passes up to the point of sale to the farmer (production, processing, testing, packing, and trading). These restrictions are enforced by penal sanctions.

Seed Law Enforcement

- **Seed Inspectors appointed by State Governments** (under section 13 of Seeds Act-1966).
- **Seed Sampling by Seed Inspectors and verification of records or any other material and seize the same if not as per Seeds Act to ensure the quality of seeds.**
- **Seed testing – Seed Testing Laboratories.**
- **Seed certification – State Seed Certification Agency.**
- Suitable instructions must be issued to draw targeted seeds samples from the market before the commencement of sowing season as per the provisions of Seeds Act/ Seed (Control) Order.

LECTURE 28

Salient features of PPV&FRA, 2001 - National Seed Policy, 2002 - Seed Bill, 2004

Introduction

In order to provide for the establishment of an effective system for the protection of plant varieties, the rights of farmers and plant breeders and to encourage the development of new varieties of plants it has been considered necessary to recognize and to protect the rights of the farmers in respect their contributions made at any time in conserving, improving and making available plant genetic resources for the development of new plant varieties.

The Government of India enacted “The Protection of Plant Varieties and Farmers’ Rights (PPV &FR) Act, 2001” adopting *sui generis* system. Indian legislation is not only in conformity with International Union for the Protection of New Varieties of Plants (UPOV), 1978, but also

have sufficient provisions to protect the interests of public sector breeding institutions and the farmers.

The legislation recognizes the contributions of both commercial plant breeders and farmers in plant breeding activity and also provides to implement TRIPs in a way that supports the specific socio-economic interests of all the stakeholders including private, public sectors and research institutions, as well as resource-constrained farmers.

To implement the provisions of the Act the Department of Agriculture and Cooperation, Ministry of Agriculture established the Protection of Plant Varieties and Farmers' Rights Authority on 11th November, 2005.

The Chairperson is the Chief Executive of the Authority. Besides the Chairperson, the Authority has 15 members, as notified by the Government of India (GOI). Eight of them are ex-officio members representing various Departments/Ministries, three from SAUs and the State Governments, one representative each for farmers, tribal organization, seed industry and women organization associated with agricultural activities are nominated by the Central Government. The Registrar General is the ex-officio Member Secretary of the Authority.

Objectives of the PPV & FR Act, 2001

1. To establish an effective system for the protection of plant varieties, the rights of farmers and plant breeders and to encourage the development of new varieties of plants.
2. To recognize and protect the rights of farmers in respect of their contributions made at any time in conserving, improving and making available plant genetic resources for the development of new plant varieties.
3. To accelerate agricultural development in the country, protect plant breeders' rights; stimulate investment for research and development both in public & private sector for the development new of plant varieties.
4. Facilitate the growth of seed industry in the country which will ensure the availability of high quality seeds and planting material to the farmers.

General functions of the Authority

- Registration of new plant varieties, essentially derived varieties (EDV), extant varieties;
- Developing DUS (Distinctiveness, Uniformity and Stability) test guidelines for new plant species;

- Developing characterization and documentation of varieties registered
- Compulsory cataloging facilities for all variety of plants
- Documentation, indexing and cataloguing of farmers' varieties;
- Recognizing and rewarding farmers, community of farmers, particularly tribal and rural community engaged in conservation, improvement, preservation of plant genetic resources of economic plants and their wild relatives
- Maintenance of the National Register of plant Varieties and
- Maintenance of National Gene Bank.

Farmers variety

Farmer" means any person who-

- i. Cultivates crops by cultivating the land himself; or
 - ii. Cultivates crops by directly supervising the cultivation of land through any other person; or
 - iii. Conserves and preserves, severally or jointly, with any person any wild species or traditional varieties or adds value to such wild species or traditional varieties through selection and identification of their useful properties;
1. "Farmers' variety" means a variety which-
- i. Has been traditionally cultivated and evolved by the farmers in their fields; or
 - ii. Is a wild relative or land race of a variety about which the farmers possess the common knowledge;

Extant Variety” means a variety, which is–

- Notified under section 5 of the Seeds Act, 1966 (54 of 1966).
- A farmers' variety.
- A variety about which there is common knowledge.
- Any other variety which is in the public domain.

Essentially derived varieties

(i) It is predominantly derived from the initial variety, or from a variety that is itself predominantly derived from the initial variety, while retaining the expression of the essential characteristics that result from the genotype or combination of genotypes of the initial variety,(italics added by author).

(ii) it is clearly distinguishable from the initial variety and

Rights under the Act

Breeders' Rights

Breeders will have exclusive rights to produce, sell, market, distribute, import or export the protected variety. Breeder can appoint agent/ licensee and may exercise for civil remedy in case of infringement of rights.

Researchers' Rights

Researcher can use any of the registered variety under the Act for conducting experiment or research. This includes the use of a variety as an initial source of variety for the purpose of developing another variety but repeated use needs prior permission of the registered breeder.

Farmers' Rights

- A Farmer who has evolved or developed a new variety is entitled for registration and protection in like manner as a breeder of a variety;
- Farmers variety can also be registered as an extant variety;
- A farmer can save, use, sow, re-sow, exchange, share or sell his farm produce including seed of a variety protected under the PPV & FR Act, 2001 in the same manner as he was entitled before the coming into force of this Act provided farmer shall not be entitled to sell branded seed of a variety protected under the PPV&FR Act, 2001;
- Farmers are eligible for recognition and rewards for the conservation of Plant Genetic Resources of land races and wild relatives of economic plants;
- There is also a provision for compensation to the farmers for non-performance of variety under Section 39 (2) of the Act, 2001 and

- Farmer shall not be liable to pay any fee in any proceeding before the Authority or Registrar or the Tribunal or the High Court under the Act.

Registration

A variety is eligible for registration under the Act if it essentially fulfills the criteria of **Distinctiveness, Uniformity and Stability (DUS)**.

Distinct, if it is clearly distinguishable by at least one essential characteristic from any other variety whose existence is a matter of common knowledge in any country at the time of filing of the application.

Uniform, if subject to the variation that may be expected from the particular features of its propagation it is sufficiently uniform in its essential characteristics;

Stable, if its essential characteristics remain unchanged after repeated propagation or, in the case of a particular cycle of propagation, at the end of each such cycle.

The Central Government issues notification in official Gazettes specifying the genera and species for the purpose of registration of varieties. So far, the Central Government has notified 57 crop species for the purpose of registration. The PPV & FR Authority has developed “Guidelines for the Conduct of Species Specific Distinctiveness, Uniformity and Stability,” tests or “Specific Guidelines’, for individual crop species. These include bread wheat, rice, maize, sorghum, pearl millet, chickpea, pigeon pea, green gram, black gram, field pea/garden pea, kidney bean/French bean, lentil, diploid cotton (two species), tetraploid cotton (two species), jute (two species), sugarcane, ginger, turmeric, Indian mustard, karan rai, rapeseed, gobhi sarson, sunflower, safflower, castor, sesame, linseed, groundnut, soybean, black pepper, small cardamom, rose, chrysanthemum, mango, potato, eggplant, tomato, lady’s finger, cauliflower, cabbage, onion, garlic, durum, dicoccum and triticum species of wheat, jsabgol, menthol mint, damask rose, periwinkle brahmi, orchids (Venda, cymbidium & Dendrobium) and 20 different crops are to be notified very soon.

Varieties can be registered with the Plant Variety Authority for 18 years in case of trees / vines and 15 years for extant varieties from the date of notification of that variety by government under Sec 5 of Seeds Act,1966 or 15 years for any variety. List of such varieties will be

published by the Authority for public knowledge. Upon registration, a specified quantity of seeds should be deposited by breeder with National Gene Bank.

Publications of Authority

- Plant variety journal of India
- General and crop specific DUS test guidelines
- Technical Bulletin
- Gene Bank Manual
- Agro-biodiversity (Hotspots Book (Two Volumes)
- A video CD entitled ‘Seed of Sustenance’ highlighting various provisions of the PPV & FR Act, 2001
- Annual Reports
- A video CD on Krishak Adhikar
- A compendium of varieties registered under PPV & FR Act 2001 (from 2009 to 2012)
- A book entitled ‘Cultivated plants & their wild relatives in India’ an Inventory
- **Fees for registration**
- Application for registration of plant varieties should be accompanied with the fee of registration prescribed by the authority. Fee for registration for different types of variety is as under:

Type of variety	Fees of Registration	
Extant variety notified under section 5 of the seeds Act, 1996	Rs.1000/-	
New variety /Essentially Derived Variety (EDV)	Individual	Rs.5000/-
	Educational	Rs.7000/-
	Commercial	Rs.10000/-
Extant variety about which there is common knowledge (VCK)	Individual	Rs.2000/-
	Educational	Rs.3000/-
	Commercial	Rs.5000/-

- The Registration of a variety is renewable subject to payment of annual fee as notified in the Plant Variety Journal of India of the Authority.

DUS Test Centers

Authority has 122 DUS test Centers for different crops with a mandate for maintaining and multiplication of reference collection, example varieties and generation of database for DUS descriptors as per DUS guidelines of respective crops. The list of DUS test Centers is available on the official website of the Authority.

Plant Variety Journal of India

- Authority publishes its official journal “Plant Varieties Journal of India” (PVJI) as a monthly bilingual (Hindi & English) publication and made available to public on the first working day of each month on its official website. This journal has the equivalent status of a gazette under the Regulations, 2006. The contents of Journal includes official and public notices, passport data of plant varieties, DUS test guidelines of crop species, details of certificate of registration and other related matters.

Certificate of Registration

- Applications which have fulfilled all requirements and have been finally accepted by the Registrar for registration were issued Certificates of Registration. 747 Certificates have been issued, out of which 91 have been issued for new varieties, 633 for extant varieties notified under the seeds Act, 1966, 22 for farmers’ varieties and 1 for Essentially Derived Variety (EDV). The certificate of registration issued will be valid for nine years in case of trees and vines and six years in case of other crops. It may be reviewed and renewed for the remaining period on payment of renewal fees subject to the condition that total period on payment of renewal fees subject to the condition that total period of validity shall not exceed eighteen years in case of trees and vines from the date registration of the variety, fifteen years from the date of notification of variety under the Seeds Act, 1996 and in other cases fifteen years from the date registration of the variety. So far the Authority has received 4934 applications as on October, 2013 for registration for different categories of varieties including farmer’s varieties.

National Register of Plant Varieties

- National Register of Plant Varieties has been kept at the head office of the Registry, containing the names of all the registered plant varieties with the names and addresses of their respective breeders, the rights of such breeders in respect of the registered varieties, the particulars of the denomination of each registered variety, its seed or other propagating material along with specification of salient features thereof and such other matters as may be prescribed.

National Gene Bank

- Authority has established National Gene Bank to store the seed material including parental lines submitted by the breeders of the registered varieties. The seed lot is stored under low temperature conditions at 5° C for the entire registration period, and if necessary after few years of storage in the National Gene Bank, the seed lot will be rejuvenated and replenished at the cost of the applicant. The seed stored in the National Gene Bank will be used for dispute settlement or when an exigency arises for invoking compulsory licensing provision. Such a seed deposition in the National Gene Bank would dissuade market malpractices or violations as the sample in custody can be drawn to verify the facts. When the period of registration granted lapses, the material automatically moves to public domain.

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National Gene Fund

A National Gene Fund has been established by the Authority to receive the contributions from:

- The benefit sharing received in the prescribed manner from the breeder of a variety or an essentially derived variety registered under the Act, or the propagating material of such variety or essentially derived variety, as the case may be;
- The annual fee payable to the Authority by way of royalty;
- The compensation deposited by breeders and
- The contribution from any National and International organizations and other sources.

The Gene Fund shall be utilized for

- Any amount to be paid by way of benefit sharing,

- The compensation payable to the farmers/community of farmer',
- The expenditure for supporting the conservation and sustainable use of genetic resources including in-situ and ex-situ collections and for strengthening the capability of the panchayat in carrying out such conservation and sustainable use,
- The expenditure of the schemes relating to benefit sharing.

Benefit sharing

- The benefit sharing is one of the most important ingredients of the farmers' rights. Section 26 provides benefits sharing and the claims can be submitted by the citizen of India or firms or non-governmental organization (NGOs) formed or established in India. Depending upon the extent and nature of the use of genetic material of the claimant in the development of the variety along with commercial utility and demand in the market of the variety breeder will deposit the amount in the Gene Fund. The amount deposited will be paid to the claimant from National Gene Fund. The Authority also publishes the contents of the certificate in the PVJI for the purpose of inviting claims for benefits sharing.

Rights of Community

- It is compensation to village or local communities for their significant contribution in the evolution of variety which has been registered under the Act.
- Any person/group of persons/governmental or non-governmental organization, on behalf of any village/local community in India, can file in any notified centre, claim for contribution in the evolution of any variety.

Convention countries

- Convention country means a country which has acceded to an international convention for the protection of plant varieties to which India has also acceded or a country which has law of protection of plant varieties on the basis of which India has entered into an agreements for granting plant breeders' rights to the citizen of both the countries. Any person if applies for the registration of variety in India within twelve months after the date on which the application was made in the convention country and that date shall be deemed for the purpose of this act to be the date of registration.

Plant Varieties Protection Appellate Tribunal

The Act provides for establishment of Plant Varieties Protection Appellate Tribunal (PVPAT). All orders or decisions of the Registrar of Authority relating to registration of variety and orders or decisions of the Registrar relating to registration as agent or licensee can be appealed in the Tribunal. Further, all orders or decisions of Authority relating to benefit sharing, revocation of compulsory license and payment of compensation can also be appealed in the Tribunal. There is transitory provision by which it is provided that till the PVPAT is established the Intellectual Property Appellate Board (IPAB) will exercise the jurisdiction of PVPAT. The decisions of the PVPAT can be challenged in high Court. The Tribunal shall dispose of the appeal within one year.

Plant genome savior awards

The community of farmers who have engaged in conservation, improvement and preservation of genetic resources of economic plants and their wild relatives particularly in areas identified as agro-biodiversity hotspots are awarded annually from Gene Fund. The name of the award is Plant Genome Saviour Community Award and the amount is INR. 10,00,000 for each community. A maximum of five awards are conferred in a year.

THE SEEDS BILL, 2004

A BILL to provide for regulating the quality of seeds for sale, import and export and to facilitate production and supply of seeds of quality and for matters connected therewith or incidental thereto. **BE** it enacted by Parliament in the Fifty-Fifth Year of the Republic of India as follows:-

1. Short title, extent, application and commencement :

- (1) This Act may be called the Seeds Act, 2004.
- (2) It extends to the whole of India.

(3) Save as otherwise provided in this Act, it shall apply to-

(a) every dealer; and

(b) every producer of seed except when the seed is produced by him for his own use and not for sale.

(4) It shall come into force on such date as the Central Government may, by notification, appoint.

2. Definitions.

Extant variety - "extant variety" means a variety available in India which is-

(a) notified under section 5 of the Seeds Act, 1966; or

(b) farmers' variety as defined in PVP Act; or

(c) a variety about which there is common knowledge; or

(d) any other variety which is in public domain.

Chapter II

The Central Seed Committee, registration and other sub-committees

3. Constitution of Central Seed Committee.

The Central Government shall, by notification, constitute, for the purpose of this Act, a Committee to be called the Central Seed Committee.

4. Composition of the Committee.

(1) The Committee shall consist of a Chairperson, members, *ex-officio* and other members, to be nominated by the Central Government.

(2) The Secretary to the Government of India in the Department of Agriculture and Co-operation, Ministry of Agriculture, shall be Chairperson, *ex officio*.

(3) The Committee

5. Powers and functions of the Committee.

The Committee shall be responsible for and shall have all the powers for the effective implementation of this Act and shall advise the Central Government and the State Governments on matters relating to-

- (a) seed programming and planning;
- (b) seed development and production;
- (c) export and import of seeds;
- (d) standards for registration, certification and seed testing;
- (e) seed registration and its enforcement;
- (f) such other matters as may be specified by the Central Government.

6. Powers of Committee to specify minimum limits of germination, purity, seed health, etc.

The Committee may, by notification, specify—

- (a) the minimum limits of germination, genetic and physical purity, and seed health, with respect to any seed of any kind of variety;
- (b) the mark or label to indicate that such seed conforms to the minimum limits of germination, genetic and physical purity, and seed health specified under clause (a), and other particulars, such as expected performance of the seed in accordance with the information provided by the producer under section 14 which such mark or label may contain.

7. Registration and other Sub-Committees of the Committee and their functions.

(1) The Committee shall constitute a Sub-Committee to be called the Registration Sub-Committee consisting of a Chairman and such number of other members, to assist him in the discharge of the functions of the Committee, as may be prescribed.

(2) It shall be the duty of the Registration Sub-Committee-

(a) to register seeds of varieties after scrutinizing their claims as made in the application in such manner as may be prescribed;

(b) to perform such other functions as are assigned to it by the Committee.

(3) The Committee may appoint as many other Sub-Committees including a Sub-Committee on Seed Certification as it deems fit consisting wholly of the members of the Committee or wholly of other persons or partly of members of the Committee and partly of other persons as it thinks fit to exercise such powers and perform such duties as may be delegated to them.

8. Procedure of the Committee and its Sub-Committees.

The Committee may, subject to the previous approval of the Central Government, make regulations for the purpose of regulating its own procedure and the procedure of any Sub-Committee thereof.

9. Secretary and other officers of the Committee.

The Central Government shall –

(a) appoint a person to be the Secretary of the Committee; and

(b) provide the Committee with such technical and other officers and employees as may be necessary for the efficient performance of the functions of the Committee under this Act.

10. Meetings of the Committee.

(1) The Committee shall meet as and when necessary at such time and place and shall observe such procedure in regard to transaction of business at its meetings (including the quorum at meetings) as may be provided by regulations.

(2) The Chairperson or, in his absence, the Agricultural Commissioner or, in the absence of both the Chairperson and the Agriculture Commissioner, any member chosen by the members present from amongst themselves, shall preside at a meeting of the Committee.

(3) All questions at a meeting of the Committee shall be decided by a majority of votes of the members present and voting and in the case of an equality of votes, the Chairperson or, in his

absence, the Agriculture Commissioner or, in the absence of both the Chairperson and the Agriculture Commissioner the person presiding shall have and exercise a second or casting vote

11. State Seed Committee.

Every State Government shall establish a State Seed Committee to –

- (a) advise the Committee on registration of regional or local seeds of any kind or variety;
- (b) advise the State Government on registration of seed producing units, seed processing units, seed dealers and horticulture nurseries;
- (c) maintain, in each district, a list of seed dealers, seed producers, seed processing units and horticulture nurseries;
- (d) seek information from persons engaged in the production, supply, distribution, trade or commerce in seeds of any kind or variety regarding stocks, prices, sales and other information in the manner as may be prescribed;
- (e) advise the State Government and the Committee on all matters arising out of the administration and implementation of this Act; and
- (f) carry out other functions assigned to, by, or under this Act.

Chapter III - Registration of kinds and varieties of seeds, etc.

12. Maintenance of National Register of seeds of kinds and varieties.

(1) For the purposes of this Act, a register of all kinds and varieties of seed to be called the National Register of Seeds shall be kept by the Registration Sub-Committee wherein all specifications, as may be prescribed, shall be maintained.

(2) Subject to the directions of the Committee, the Register shall be kept under the control and management of the Registration Sub-Committee.

(3) The Registration Sub-Committee shall, within such intervals and in such manner as it thinks appropriate, publish the list of kinds and varieties of seed which have been registered during that interval.

13. Registration of seeds of any kind or variety.

(1) No seed of any kind or variety shall, for the purpose of sowing or planting by any person, be sold unless such seed is registered under sub-section (2) by the Registration Sub-Committee in such manner as may be prescribed.

(2) Subject to the provisions of sections 14 and 15, the Registration Sub-Committee may register, or refuse any kind or variety of seed on the basis of information furnished by the producer on the results of multi-locational trials for such period as may be prescribed to establish the performance of that seed.

(3) The Registration Sub-Committee may grant provisional registration as prescribed to the varieties of seeds which are available in the market on the date of commencement of this Act.

(4) Registration made under this Act shall be valid for a period of fifteen years in the case of annual and biennial crops, and eighteen years for long duration perennials.

(5) At the expiry of the period granted under sub-section (4), the kind or variety of seed may be re-registered for a like period by the Registration Sub-Committee on the basis of information furnished by the producer on the results of such trials as may be prescribed under sub-section (2) to re-establish performance of the kind or variety of seed.

(6) The Registration Sub-Committee shall have the power to issue such directions to protect the interests of a producer against any abusive act committed by any third party during the period between the date of filing of application for registration and the date of decision by the Committee on such application.

14. Procedure for registration.

(1) Every application for registration under sub-section (1) section 13 shall be made in such form and contain such particulars and be accompanied by such fees as may be prescribed.

(2) On receipt of any such application for the registration of a kind or variety of seed, the Registration Sub-Committee may, after such enquiry as it deems fit and after satisfying itself that the kind or variety of seed to which the application relates conforms to the claims made by the importer or by the seller, as the case may be, as regards the efficacy of the kind or variety of seed and its safety to human beings and animals, register the kind or variety, as the case may, of the seed on such conditions as may be specified by it and allot a registration number thereto and issue a certificate of registration.

(3) The Registration Sub-Committee may, having regard to the efficacy of the seeds and its safety to human beings and animals, vary the conditions subject to which a certificate of registration has been granted and may, for that purpose, require the certificate holder by notice in writing to deliver the certificate to it within such time as may be specified in the notice.

15. Special provision for registration of transgenic varieties.

(1) Notwithstanding anything contained in section 14, no seed of any transgenic variety shall be registered unless the applicant has obtained clearance in respect of the same as required by or under the provisions of the Environment (Protection) Act, 1986:

Provided that the Registration Sub-Committee may, subject to clearance under the said Act, grant provisional registration, for a period not exceeding two years on the basis of information furnished by the producer on the results of multi-locational trials in the prescribed manner.

(2) Save as otherwise provided in sub-section (1), the form and manner in which and procedure for registration of transgenic variety of seed and the fee payable thereto shall be the same as applicable in case of registration under section 14.

16. Cancellation of registration of seeds of kinds and varieties.

(1) The Registration Sub-Committee may cancel any registration granted under section 13 or section 15 or any one or more of the following grounds, namely:-

(a) that the holder of the certificate has violated any of the terms and conditions of the registration; or

(b) that the registration has been obtained by misrepresentation or concealment of essential data; or

(c) that the variety is not performing in accordance with the information provided by the producer under sub-section (3) of section 14 or has become obsolete or has outlived its utility; or

(d) that prevention of commercial exploitation of such variety of seeds is necessary.

(i) in the public interest;

(ii) to protect public order or public morality; or

(iii) to protect human beings, animal and plant life and health to avoid serious prejudice to the environment.

(2) No order of cancellation of registration under this section shall be made unless the holder thereof or the affected person concerned has been given a reasonable opportunity of showing cause in respect of the grounds for such cancellation.

17. Notification of cancellation of registration of seeds of kinds and varieties.

The Registration Sub-Committee shall notify the cancellation of registration of a seed of any kind or variety made under section 13 or any registration made under section 15 in the Official Gazette

18. Exclusion of certain kinds or varieties of seed from registration.

(1) Notwithstanding anything contained in this Act, no registration of any kind or variety of seed shall be made under this Act, if prevention of commercial exploitation of such kind or variety is necessary to protect public order or public morality or human, animal or plant life and health, or to avoid serious prejudice to the environment.

(2) A kind or variety of seed containing any technology, which is harmful, or potentially harmful, shall not be registered.

Explanation.- For the purposes of this sub-section, the expression “technology” includes genetic use restriction technology and terminator technology.

19. Evaluation of performance.

The Committee may, for conducting trials to assess performance, accredit centers of the Indian Council of Agricultural Research, State Agricultural Universities and such other organizations fulfilling the eligibility requirements as may be prescribed, to conduct trials to evaluate the performance of any kind or variety of seed.

20. Compensation to farmer.

Where the seed of any registered kind or variety is sold to a farmer, the producer, distributor or vendor, as the case may be, shall disclose the expected performance of such kind or variety to the farmer under given conditions, and if, such registered seed fails to provide the expected performance under such given conditions, the farmer may claim compensation from the producer, distributor or vendor under the Consumer Protection Act, 1986.

21. Seed producers and seed processing units to be registered.

(1) No producer shall grow or organize the production of seed unless he is registered as such by the State Government under this Act.

(2) No person shall maintain a seed processing unit unless such unit is registered by the State Government under this Act.

(3) The State Government shall register a producer or seed processing unit if he or it meets the specifications prescribed by the Central Government in terms of infrastructure, equipment and qualified manpower.

(4) Every application for registration under sub-section (3) shall be made in such form and manner and shall be accompanied by such fee as may be prescribed.

(5) The State Government may, after making such enquiry and subject to such conditions as it thinks fit, grant a certificate for maintaining a seed producing or a seed processing unit in such form as may be prescribed.

(6) Every seed producing and processing units shall furnish periodic returns to the Seed Certification Agency in such form and at such time as may be prescribed on the quantity of seeds of different kinds or varieties processed by them.

(7) The State Government may, after giving the holder of certificate of registration under sub-section (1), or sub-section (2), as the case may be, suspend or cancel the registration if —

(a) such registration has been obtained by misrepresentation as to a material particular relating to the specification in terms of infrastructure, equipment or availability of qualified manpower; or

(b) any of the provisions of this Act or the rules made thereunder has been contravened.

22. Seed dealers to be registered.

(1) Every person who desires to carry on the business of selling, keeping for sale, offering to sell, bartering, import or export or otherwise supply any seed by himself, or by any other person on his behalf shall obtain a registration certificate as a dealer in seeds from the State Government .

(2) Every applicant for dealership under sub-section (1) shall be required to furnish information about seed stocks, sales and other related information as may be prescribed.

(3) Even application for registration under sub-section (1) shall be made in such form and manner and shall be accompanied by such fee as may be prescribed.

(4) The State Government may, after making such enquiry and subject to such conditions as it thinks fit, grant a certificate of registration as a dealer in seeds in such form as may be prescribed.

(5) Every dealer registered under this section shall furnish such information and returns regarding seed stocks, seed lots, expiry date of seed lots and other related information as may be prescribed to the State Government.

(6) The State Government may, after giving the dealer an opportunity of being heard, suspend or cancel a certificate granted under this Act if-

(a) such registration had been obtained by misrepresentation of any material fact;

(b) contravenes any of the provisions of this Act or the rules made thereunder.

23. Horticulture nursery to be registered.

(1) No person shall conduct or carry on the business of horticulture nursery unless such nursery is registered with the State Government.

(2) Every application for registration under sub-section (1) shall be made in such form and contain such particulars and shall be accompanied by such fee as may be prescribed.

24. Duties of registration holders of horticulture nursery.

Every person who is a holder of a registration of a horticulture nursery under section 23 shall-

(a) keep a complete record of the origin or source of every planting material and performance record of mother trees in the nursery;

(b) keep a layout plan showing the position of the root-stocks and scions used in raising the horticulture plants;

(c) keep a performance record of the mother trees in the nursery;

(d) keep the nursery plants as well as the parent trees used for the production or propagation of horticulture plants free from infectious or contagious insects, pests or diseases affecting plants.

(e) furnish such information to the State Government on the production, stocks, sales and prices of planting material in the nursery as may be prescribed.

Chapter IV - Regulation of sale of seed and Seed Certification Agencies

25. Regulation of sale of seeds of registered kinds and varieties.

No person shall himself, or by any other person on his behalf, carry on the business of selling, keeping for sale, offering to sell, bartering, import or export or otherwise supply any kind of seed of any registered kind or variety unless-

(a) such seed is identifiable as to its kind or variety;

(b) such seed conforms to the minimum limit of germination and genetic, physical purity, seed health specified under clause (a) of section 6;

- (c) the container of such seed bears in the prescribed manner, the mark or label bearing the correct particulars thereof, specified under clause (b) of section 6;
- (d) the container of such seed, in the case of transgenic varieties, bears a declaration to this effect as specified in sub-clause (2) of section 15;
- (e) he complies with such other requirements as may be prescribed.

26. State Seed Certification Agency.

The Committee may, in consultation with the State Government, by notification, establish a State Seed Certification Agency for the State to carry out the functions entrusted to the State Seed Certification Agency by or under this Act:

27. Accreditation of Seed Certification Agencies.

(1) The Committee may in consultation with the State Government and the State Seed Committee, accredit –

- (a) organizations to carry out certification, on the fulfillment of such criteria, as may be prescribed, or
- (b) individuals or seed producing organisations to carry out self- certification, in such manner as may be prescribed.

(2) The accredited individuals and seed producing organisations shall be subject to such inspection and control of the Committee, the concerned State Government and State Seed Certification Agency, as may be prescribed.

(3) The accreditation may be withdrawn by the Committee, for reasons to be recorded in writing and after giving to the concerned organization or individual, as the case may be, a reasonable opportunity of being heard.

28. Grant of certificate by the State Seed Certification Agency.

(1) Any person selling, keeping for sale, offering to sell, bartering or otherwise supplying any seed of any registered kind or variety may, if he desires to have such seed certified by the State Seed Certification Agency, apply to that Agency for the grant of a certificate for the purpose.

(2) Every application under sub-section (1) shall be made in such form, shall contain such particulars and shall be accompanied by such fee as may be prescribed.

(3) On receipt of an application under sub-section (1), the State Seed Certification Agency may, after such enquiry as it thinks fit and after satisfying itself that the seed to which the application relates conforms to the prescribed standards, grant a certificate in such form and on such conditions as may be prescribed:

Provided that such standards shall not be lower than the minimum limit of germination, genetic and physical purity specified for that seed under clause (a) of section 6.

29. Revocation of certificate.

If the State Seed Certification Agency is satisfied, either on a reference made to it in this behalf or otherwise, that-

(a) the certificate granted by it under section 28 has been obtained by misrepresentation as to an essential fact; or

(b) the holder of the certificate has, without reasonable cause, failed to comply with the conditions subject to which the certificate has been granted or has contravened any of the provisions of this Act or the rules made thereunder, then, without prejudice to any other penalty to which the holder of the certificate may be liable under this Act, the State Seed Certification Agency may, after giving the holder of the certificate an opportunity of showing cause, revoke the certificate.

30. Recognition of seed certification agencies in foreign countries.

The Central Government may, on the recommendation of the Committee and by notification, recognise any seed certification agency established in any foreign country, for the purposes of this Act.

Chapter V

31. Appeals.

(1) Any person aggrieved by a decision of the Registration Sub-Committee under section 14, section 16 or section 27 or of the State Seed Certification Agency under section 28 or section 29 may, within thirty days from the date on which the decision is communicated to him prefer an appeal to such authority (hereinafter referred to as the appellate authority) as the Central Government may think fit to constitute:

Provided that the appellate authority may entertain an appeal after the expiry of the said period of thirty days if it is satisfied that the appellant was prevented by sufficient cause from filing the appeal in time.

(2) An appellate authority shall consist of a single person or three persons as the Central Government may think fit, to be appointed by that Government.

(3) The form and manner in which an appeal may be preferred under sub-section (1), the fee payable for such appeal and the procedure to be followed by the appellate authority shall be such as may be prescribed.

(4) On receipt of an appeal preferred under sub-section (1), the appellate authority shall, after giving the appellant and the other party an opportunity of being heard, dispose of the appeal as expeditiously as possible.

Chapter VI - Seed analysis and Seed testing

32. Central and State Seed Testing Laboratories.

(1) The Central Government may, by notification, establish a Central Seed Testing Laboratory or declare any seed-testing laboratory as the Central Seed Testing Laboratory to carry

out the functions entrusted to the Central Seed Testing Laboratory by or under this Act in the prescribed manner

(2) The State Government may, in consultation with the Committee, and by notification, establish one or more State Seed Testing Laboratories or declare any seed testing laboratory in the Government or non-Government sector as a State Seed Testing Laboratory where analysis of seed of any kind or variety shall be carried out under this Act in the prescribed manner.

(3) Every Seed Testing Laboratory referred to in sub-section (1) shall have as many Seed Analysts as the Central Government may consider necessary.

(4) Every Seed Testing Laboratory referred to in sub-section (2) shall have as many Seed Analysts as the State Government may consider necessary.

33. Seed Analysts.

(1) In case of the Central Seed Laboratory, the Central Government and in other cases the State Government may, by notification, appoint such persons as the Government thinks fit and having the prescribed qualifications to be Seed Analysts and define the local limits of their jurisdiction.

(2) Every Central Seed Testing Laboratory established or declared under sub-section (1) of section 32 and every State Seed Testing Laboratory established or declared under sub-section (2) of that section shall have as many Seed Analysts as the Central Government or the State Government, as the case may be, specify.

34. Seed Inspectors.

(1) The State Government may, by notification, appoint such persons as it thinks fit, having the prescribed qualifications, to be Seed Inspectors and define the areas within which they shall exercise jurisdiction.

(2) Every Seed Inspector shall be subordinate to such authority as the State Government may specify in this behalf.

35. Powers of Seed Inspectors.

(1) The Seed Inspector may-

(a) take samples of any seed of any kind or variety from-

(i) any person selling such seed; or

(ii) any person who is in the course of conveying, delivering or preparing to deliver such seed to a purchaser or a consignee; or

(iii) a purchaser or a consignee after delivery of such seed to him;

(b) send such sample for analysis to the Seed Analyst of the area within which such sample has been taken;

(c) enter and search, at all reasonable times, with such assistance, if any, as he considers necessary, any place in which he has reason to believe that an offence under this Act has been or is being committed and order in writing the person in possession of any seed in respect of which the offence has been or is being committed, not to dispose of any stock of such seed for a specific period not exceeding thirty days or, unless the alleged offence is such that the defect may be removed by the possessor of the seed, seize the stock of such seed;

(d) examine any record, register, document or any other material object found in any place mentioned in clause (c) and seize the same if he has reason to believe that it may furnish evidence of the commission of an offence punishable under this Act; and

(e) exercise such other powers as may be necessary for carrying out the purposes of this Act or any rule or regulation made thereunder.

(2) The power conferred by this section includes the power to break-open any container in which any seed of any kind or variety may be contained or to break-open the door of any premises where any such seed may be kept for sale:

Provided that the power to break-open the door shall be exercised only after the owner or any other person in occupation of the premises, if he is present therein, refuses to open the door on being called upon to do so.

(3) Where the Seed Inspector takes any action under clause (a) of sub-section (1), he shall, as far as possible, call not less than two persons to be present at the time when such action is taken and take their signatures on a memorandum to be prepared in such form and manner as may be prescribed.

(4) The provisions of the Code of Criminal Procedure, 1973, or in relation to the State of Jammu and Kashmir, the provisions of any corresponding law in force in that State, shall, so far as may be, apply to any search or seizure under this section as they apply to any search or seizure made under the authority of a warrant issued under section 94 of the said Code, or, as the case may be, under the corresponding provisions of the said law.

Chapter VII - Export and Import of seeds

36. Import of seeds.

(1) All import of seeds –

(a) shall be subject to the provisions of the Plants, Fruits and Seeds (Regulation of Import into India) Order, 1989, or any corresponding order made under section 3 of the Destructive Insects and Pests Act, 1914;

(b) shall conform to minimum limits of germination, genetic and physical purity, and seed health as prescribed under section 6; and

(c) shall be subject to registration as may be granted on the basis of information furnished by the importer on the results of multi-locational trials for such period as may be prescribed to establish performance.

(2) The Central Government may, by notification, permit to import an unregistered variety in such quantity and subject to fulfilling such conditions as may be specified in that notification for research purposes.

37. Export of seeds.

The Central Government may, on the advice of the Committee, restrict, by notification, the export of seeds of any kind or variety if it is deemed that such export may adversely affect the food security of the country, or if it is felt that the reasonable requirements of the public will not be met, or on such other grounds as may be prescribed.

Chapter VIII - Offences and punishment

38. Offences and punishment.

(1) If any person –

(a) contravenes any provision of this Act or any rule made thereunder; or

(b) imports, sells, stocks or exhibits for sale or barter; and or otherwise supplies any seed of any kind or variety deemed to be misbranded ; or

(c) imports, sells, stocks or exhibits for sale or barter, or otherwise supplies any seed of any kind or variety without a certificate of registration; or

(d) obstructs the Committee, Registration Sub-Committee or Seed Certification Agency or Seed Inspector or Seed Analyst or any other authority appointed or duly empowered under this Act in the exercise of its powers or discharge of their duties under this Act or the rules made thereunder, he shall, on conviction, be punishable – with fine which shall not be less than five thousand rupees but which may extend to twenty five thousand rupees.

(2) If any person sells any seed which does not conform to the standards of physical purity, germination or health or does not maintain any records required to be maintained under this Act or the rules made thereunder he shall, on conviction, be punishable with fine which shall not be less than five thousand rupees but which may extend to twenty- five thousand rupees.

(3) If any person furnishes any false information relating to the standards of genetic purity, misbrands any seed or supplies any spurious seed or spurious transgenic variety, sells any non-

registered seeds he shall, on conviction be punishable with imprisonment for a term which may extend to six months or with fine which may extend to fifty thousand rupees or with both.

39. Forfeiture of property.

When any person has been convicted under this Act for the contravention of any of the provisions of this Act or the rules made thereunder, the seed in respect of which the contravention has been committed shall be forfeited to the Central Government.

40. Offences by companies.

(1) Where an offence under this Act has been committed by a company, every person who at the time the offence was committed was in charge of, and was responsible to the company for the conduct of the business of the company, as well as the company, shall be deemed to be guilty of the offence and shall be liable to be proceeded against and punished accordingly:

Provided that nothing contained in this sub-section shall render any such person liable to any punishment under this Act if he proves that the offence was committed without his knowledge and that he exercised all due diligence to prevent the commission of such offence.

(2) Notwithstanding anything contained in sub-section (1), where an offence under this Act has been committed by a company and it is proved that the offence has been committed with the consent or connivance of, or is attributable to any neglect on the part of, any director, manager, secretary or other officer of the company, such director, manager, secretary or other officer shall also be deemed to be guilty of that offence and shall be liable to be proceeded against and punished accordingly.

Explanation. – For the purpose of this section,-

(a) “company” means any body corporate and includes a firm or other association of individuals; and

(b) “director”, in relation to a firm, means a partner in the firm.

Chapter IX - Power of Central Government

41. Power of Central Government to give directions to the State Governments.

The Central Government may give such directions to any State Governments as may appear to the Central Government to be necessary for carrying into execution in the State any of the provisions of this Act or of any rule made there under.

42. Power of Central Government to issue directions to the Committee.

(1) Without prejudice to the foregoing provisions of this Act, the Committee shall, in the discharge of its functions and duties under this Act, be bound by such directions on questions of policy as the Central Government may give in writing to it from time to time.

(2) The decision of the Central Government whether a question is one of policy or not shall be final.

43. Exemption from registration.

(1) Nothing in this Act shall restrict the right of the farmer to save, use, exchange, share or sell his farm seeds and planting material, except that he shall not sell such seed or planting material under a brand name or which does not conform to the minimum limit of germination, physical purity, genetic purity prescribed under clause (a) or clause (b) of section 6.

(2) The Central Government may, by notification, and subject to conditions, if any, as it may specify therein, exempt from all or any of the provisions of this Act or the rules made thereunder, any educational, scientific or research or extension organization.

Chapter X - Miscellaneous

44. Protection of action taken in good faith.

No suit, prosecution or other legal proceeding shall lie against the Government or any person for anything which is in good faith done or intended to be done under this Act.

45. Power to remove difficulties.

(1) If any difficulty arises in giving effect to the provisions of this Act, the Central Government may, by order published in the Official Gazette, make such provisions not inconsistent with the provisions of this Act as may appear to be necessary for removing the difficulty:

Provided that no order shall be made under this section after the expiry of two years from the date of commencement of this Act.

(2) Every order made under sub-section (1) shall be laid before each House of Parliament.

46. Power of Central Government to make rules.

(1) The Central Government may by notification, make rules to carry out the provisions of this Act.

47. Power of Committee to make regulations.

(1) The Committee may, with the previous approval of the Central Government, by notification, make regulations not inconsistent with the provisions of this Act and the rules made thereunder, to provide for all matters for which provision is necessary or expedient for the purpose of giving effect to the provisions of this Act.

(2) In particular and without prejudice to the generality of the foregoing power, such regulations may provide for all or any of the following matters, namely:-

(a) the procedure for conduct of business to be transacted by the Committee or any Sub-Committee thereof under section 8;

(b) the procedure in regard to transaction of business at meetings of the Committee (including the quorum at meetings) under sub-section (1) of section 10.

48. Rules and regulations to be laid before Parliament.

Every rule and every regulation made under this Act shall be laid as soon as may be after it is made, before each House of Parliament, while it is in session, for a total period of thirty days

which may be comprised in one session or in two or more successive sessions, and if, before the expiry of the session immediately following the session or the successive sessions aforesaid, both Houses agree in making any modification in the rule or regulation or both Houses agree that the rule or regulation should not be made, the rule or regulation shall, thereafter, have effect only in such modified form or be of no effect, as the case may be; so, however, that any such modification or annulment shall be without prejudice to the validity of anything previously done under that rule or regulation.

49. Repeal and savings.

(1) On the commencement of this Act, the Seeds Act, 1966 shall stand repealed;

Provided that such repeal shall not affect,-

(a) the previous operation of the law so repealed or anything duly done or suffered thereunder; or

(b) any right, privilege, obligation or liability acquired, accrued or incurred under the law so repealed; or

(c) any penalty, forfeiture or punishment incurred in respect of any offence committed against the Act so repealed; or

(d) any investigation, proceeding, legal proceeding or remedy in respect of any such right, privilege, obligation, liability, penalty, forfeiture or punishment as aforesaid; and any such investigation, proceedings, legal proceeding or remedy may be instituted, continued or enforced; any such penalty forfeiture or punishment may be imposed as if this Act had not been passed:

Provided further that, subject to the first proviso and any saving provisions made elsewhere in this Act anything done, any action taken, any rule made, any notifications or orders issued under the provisions of the Act so repealed shall, in so far as they are not inconsistent with the provisions of this Act, be deemed to have been done, taken, made or issued under the corresponding provisions of this Act, and shall continue to be in force accordingly, unless and

until expressly or implied repealed by any thing done, action taken, rules made or, notification or orders issued under this Act.

(2) Notwithstanding such repeals any kind or variety of seeds that has been notified under the law as so repealed shall be deemed to have been registered under this Act, and any seed certification agency established under section 18 of the Seeds Act, 1966 shall be deemed to have been established or recognized, as the case may be, under this Act.

The Schedule --[See section 4(4)(i),(ii) and (iii)]

GEOGRAPHICAL ZONES

ZONE-I	Andhra pradesh, karnakata, Kerala, Lakshadweep, Pondicherry and Tamil nadu.
ZONE-II	Andaman and Nicobar Islands, Bihar, Chatisgarh, Jharkhand, Madhya Pradesh, Orissa AND West Bengal.
ZONE-III	Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura.
ZONE-IV	Dadra and Nagar Haveli, Daman and Diu, Goa, Gujarat, Rajasthan and Maharashtra.
ZONE-V	Chandigarh, Haryana, Himachal Pradesh, Jammu And Kashmir, National Capital Territory of Delhi, Punjab, Uttranchal And Uttar Pradesh.

NATIONAL SEEDS POLICY, 2002

Important Objectives

- Provision of an appropriate climate for the seed industry to utilize available and prospective opportunities
- Safeguarding the interest of Indian farmers
- Conservation of agro-biodiversity
- Department of Agriculture & Co-operation, GOI will monitor the progress of implementation of the policy
- Now, GOI will set up National Seed Board (NSB) and PVP&FR Authority

- Encourages private sector participation in research and development of new plant varieties.
- The rights empowered to various bodies for regulating the quality of seeds produced, distributed and for providing variety protection as per the Seeds Act, 1966 and PPV & FR Act, 2001 have been retained in the policy
- Promotion of seed village scheme to increase the production and make available the seeds in time as well as upgrading the quality of farmers' saved seeds
- Establishment of seed banks for ensuring supply in times of calamity and storage facility at village level
- Establishment of a National Seed Board in place of Central Seed Committee and Central Seed Certification Board to undertake seed certification and advising Government on all matters related to seed planning and development. NSB will serve as the apex body in the seed sector
- Setting up of National Seed Research and Training Centre to impart training in seed technology
- Development of a National Seed Grid to provide information on availability of different varieties of seeds with production details
- Both public and private sector will be encouraged to join the grid for a clear assessment of demand and supply of seeds
- Few of Policy's other recommendations have been addressed in PPV & FR, Act, 2001 also.
- Maintenance of a National Register on seeds of varieties
- Establishing a national gene fund,
- Disclosure of the variety's expected performance and provision for farmer to claim compensation in case of crop failure.
- Further, aims of National Seed Policy such as development of infrastructure, ensuring supply of good quality seeds and facilitating the International seed trade are sought to be addressed through the proposed Seeds Bill, 2004.

NATIONAL SEEDS POLICY, 2002

AIMS AND OBJECTIVES

1. It has become evident that in order to achieve the food production targets of the future, a major effort will be required to enhance the seed replacement rates of various crops. This would require a major increase in the production of quality seeds, in which the private sector is expected to play a major role. At the same time, private and Public Sector Seed Organisations at both Central and State levels, will be expected to adopt economic pricing policies which would seek to realise the true cost of production.
2. The creation of a facilitative climate for growth of a competitive and localised seed industry, encouragement of import of useful germplasm, and boosting of exports are core elements of the agricultural strategy of the new millennium. Biotechnology will be a key factor in agricultural development in the coming decades. Genetic engineering/modification techniques hold enormous promise in developing crop varieties with a higher level of tolerance to biotic and abiotic stresses.
3. A conducive atmosphere for application of frontier sciences in varietal development and for enhanced investments in research and development is a pressing requirement. At the same time, concerns relating to possible harm to human and animal health and bio-safety, as well as interests of farmers, must be addressed. Globalization and economic liberalization have opened up new opportunities as well as challenges.

The main objectives of the National Seeds Policy, therefore, are the provision of an appropriate climate for the seed industry to utilize available and prospective opportunities, safeguarding of the interests of Indian farmers and the conservation of agro-biodiversity. While unnecessary regulation needs to be dismantled, it must be ensured that gullible farmers are not exploited by unscrupulous elements. A regulatory system of a new genre is, therefore, needed, which will encompass quality assurance mechanisms coupled with facilitation of a vibrant and responsible seed industry.

THRUST AREAS

1. VARIETAL DEVELOPMENT AND PLANT VARIETY PROTECTION

- 1.1 The development of new and improved varieties of plants and availability of such varieties to Indian farmers is of crucial importance for a sustained increase in agricultural productivity.
 - 1.1.1 Appropriate policy framework and programmatic interventions will be adopted to stimulate varietal development in tune with market trends, scientific-technological advances, suitability for biotic and abiotic stresses, locational adaptability and farmers' needs.
- 1.2 An effective sui generis system for intellectual property protection will be implemented to stimulate investment in research and development of new plant varieties and to facilitate the growth of the Seed Industry in the country.
 - 1.2.1 A Plant Varieties & Farmers' Rights Protection (PVP) Authority will be established which will undertake registration of extant and new plant varieties through the Plant Varieties Registry on the basis of varietal characteristics.
 - 1.2.2 The registration of new plant varieties by the PVP Authority will be based on the criteria of novelty, distinctiveness, uniformity and stability.
 - 1.2.3 The criteria of distinctiveness, uniformity and stability could be relaxed for registration of extant varieties, which will be done within a specified period to be decided by the PVP Authority.
 - 1.2.4 Registration of all plant genera or species as notified by the Authority will be done in a phased manner.
 - 1.2.5 The PVP Authority will develop characterization and documentation of plant varieties registered under the PVP Act and cataloguing facilities for all varieties of plants.
- 1.3 The rights of farmers to save, use, exchange, share or sell farm produce of all varieties will be protected, with the proviso that farmers shall not be entitled to sell branded seed of a protected variety under the brand name.
- 1.4 The rights of researchers to use the seed/planting material of protected varieties for bonafide research and breeding of new plant varieties will be ensured.
- 1.5 Equitable sharing of benefit arising out of the use of plant genetic resources that may accrue to a breeder from commercialization of seeds/planting materials of a new variety, will be provided.
- 1.6 Farmers/groups of farmers/village communities will be rewarded suitably for their significant contribution in evolution of a plant variety subject to registration. The contribution of

traditional knowledge in agriculture needs to be highlighted through suitable mechanisms and incentives.

- 1.7 A National Gene Fund will be established for implementation of the benefit sharing arrangement, and payment of compensation to village communities for their contribution to the development and conservation of plant genetic resources and also to promote conservation and sustainable use of genetic resources. Suitable systems will be worked out to identify the contributions from traditional knowledge and heritage.
- 1.8 Plant Genetic Resources for Food and Agriculture Crops will be permitted to be accessed by Research Organisations and Seed Companies from public collections as per the provisions of the 'Material Transfer Agreement' of the International Treaty on Plant Genetic Resources and the Biological Diversity Bill.
- 1.9 Regular interaction amongst the Private and Public Researchers, Seed Companies/Organisations and Development Agencies will be fostered to develop and promote growth of a healthy seed industry in the country.
- 1.10 To keep abreast of global developments in the field of Plant Variety Protection and for technical collaboration, India may consider joining Regional and International Organisations.
- 1.11 The PVP Authority may, if required, resort to compulsory licensing of a protected variety in public interest on the ground that requirements of the farming community for seeds and propagating material of a variety are not being met or that the production of the seeds or planting material of the protected variety is not being facilitated to the fullest possible extent.

2. SEED PRODUCTION

- 2.1 To meet the Nation's food security needs, it is important to make available to Indian farmers a wide range of seeds of superior quality, in adequate quantity on a timely basis. Public Sector Seed Institutions will be encouraged to enhance production of seed towards meeting the objective of food and nutritional security.
- 2.2 The Indian seed programme adheres to the limited three generation system of seed multiplication, namely, breeder, foundation and certified seed. Breeder seed is the progeny of nucleus seed.

- 2.2.1 Nucleus seed is the seed produced by the breeder to develop the particular variety and is directly used for multiplication as breeder seed.
- 2.2.2 Breeder seed is the seed material directly controlled by the originating or the sponsoring breeder or Institution for the initial and recurring production of foundation seed.
- 2.2.3 Foundation seed is the progeny of breeder seed. Foundation seed may also be produced from foundation seed. Production of foundation seed stage-I and stage-II may thus be permitted, if supervised and approved by the Certification Agency and if the production process is so handled as to maintain specific genetic purity and identity.
- 2.2.4 Certified seed is the progeny of foundation seed or the progeny of certified seed. If the certified seed is the progeny of certified seed, then this reproduction will not exceed three generations beyond foundation stage-I and it will be ascertained by the Certification Agency that genetic identity and genetic purity has not been significantly altered.
- 2.3** Public Sector Seed Production Agencies will continue to have free access to breeder seed under the National Agriculture Research System. The State Farms Corporation of India and National Seeds Corporation will be restructured to make productive use of these organisations in the planned growth of the Seed Sector.
- 2.4 Private Seed Production Agencies will also have access to breeder seed subject to terms and conditions to be decided by Government of India.
- 2.5 State Agriculture Universities/ICAR Institutes will have the primary responsibility for production of breeder seed as per the requirements of the respective States.
- 2.6 Special attention will be given to the need to upgrade the quality of farmers' saved seeds through interventions such as the Seed Village Scheme.
- 2.7 Seed replacement rates will be raised progressively with the objective of expanding the use of quality seeds.
- 2.8 DAC, in consultation with ICAR and States, will prepare a National Seed Map to identify potential, alternative and nontraditional areas for seed production of specific crops.
- 2.9 To put in place an effective seed production programme, each State will undertake advance planning and prepare a perspective plan for seed production and distribution over a rolling (five to six year) period. Seed Banks will be set up in non-traditional areas to meet demands for seeds during natural calamities.

- 2.10 The 'Seed Village Scheme' will be promoted to facilitate production and timely availability of seed of desired crops/varieties at the local level. Special emphasis will be given to seed multiplication for building adequate stocks of certified/quality seeds by providing foundation seed to farmers.
- 2.11 For popularising newly developed varieties and promoting seed production of these varieties, seed minikits of pioneering seed varieties will be supplied to farmers. Seed exchange among farmers and seed producers will be encouraged to popularise new/non-traditional varieties.
- 2.12 Seeds of newly developed varieties must be made available to farmers with minimum time gap. Seed producing agencies will be encouraged to tie up with Research Institutions for popularization and commercialization of these varieties.
- 2.13 As hybrids have the potential to improve plant vigour and increase yield, support for production of hybrid seed will be provided.
- 2.14 Seed production will be extended to agro-climatic zones which are outside the traditional seed growing areas, in order to avoid unremunerative seed farming in unsuitable areas.
- 2.15 Seed Banks will be established for stocking specified quantities of seed of required crops/varieties for ensuring timely and adequate supply of seeds to farmers during adverse situations such as natural calamities, shortfalls in production, etc. Seed Banks will be suitably strengthened with cold storage and pest control facilities.
- 2.15.1 The storage of seed at the village level will be encouraged to facilitate immediate availability of seeds in the event of natural calamities and unforeseen situations. For the storage of seeds at farm level, scientific storage structures will be popularised and techniques of scientific storage of seeds will be promoted among farmers as an extension practice.
- 2.16 Seed growers will be encouraged to avail of Seed Crop Insurance to cover risk factors involved in production of seeds. The Seed Crop Insurance Scheme will be reviewed so as to provide effective risk cover to seed producers and will be extended to all traditional and non-traditional areas covered under the seed production programme.

3. QUALITY ASSURANCE

- 3.1 The Seeds Act will be revised to regulate the sale, import and export of seeds and planting materials of agriculture crops including fodder, green manure and horticulture and supply of quality seeds and planting materials to farmers throughout the country.
- 3.2 The National Seeds Board (NSB) will be established in place of existing Central Seed Committee and Central Seed Certification Board. The NSB will have permanent existence with the responsibility of executing and implementing the provisions of the Seeds Act and advising the Government on all matters relating to seed planning and development. The NSB will function as the apex body in the seed sector.
- 3.2.1 All varieties, both domestic and imported varieties that are placed on the market for sale and distribution of seeds and planting materials will be registered under the Seeds Act. However, for vegetable and ornamental crops a simple system of varietal registration based on “breeder’s declaration” will be adopted.
- 3.2.2 The Board will undertake registration of kinds/varieties of seeds that are to be offered for sale in the market, on the basis of identified parameters for establishing value for cultivation and usage (VCU) through testing/trialing.
- 3.2.3 Registration of varieties will be granted for a fixed period on the basis of multilocal trials to determine VCU over a minimum period of three seasons, or as otherwise prescribed as in the case of long duration crops and horticultural crops. Samples of the material for registration will be sent to the NBPGR for retention in the National Gene Bank.
- 3.2.4 Varieties that are in the market at the time of coming into force of the revised Seeds Act, will have to be registered within a fixed time period, and subjected to such testing as will be notified.
- 3.2.5 The NSB will accredit ICAR, SAUs, public/private organisations to conduct VCU trials of all varieties for the purpose of registration as per prescribed standards.
- 3.2.6 The NSB will maintain the National Seeds Register containing details of varieties that are registered. This will help the Board to coordinate and assist activities of the States in their efforts to provide quality seeds to farmers.
- 3.2.7 The NSB will prescribe minimum standards (of germination, genetic characteristics, physical purity, seed health, etc.) as well as suitable guidelines for registration of seed and planting materials.

- 3.2.8 Provisional registration would be granted on the basis of information filed by the applicant relating to trials over one season to tide over the stipulation of testing over three seasons before the grant of registration.
- 3.3 Government will have the right to exclude certain kinds or varieties from registration to protect public order or human, animal and plant life and health, or to avoid serious prejudice to the environment.
- 3.4 The NSB will have the power to cancel the registration granted to a variety if the registration has been obtained by misrepresentation or concealment of essential data, the variety is obsolete and has outlived its utility and if the prevention of commercial exploitation of such variety is necessary in the public interest.
- 3.5 Registration of Seed Processing Units will be required if such Units meet the prescribed minimum standards for processing the seed.
- 3.6 Seed Certification will continue to be voluntary. The Certification tag/label will provide an assurance of quality to the farmer.
- 3.6.1 The Board will accredit individuals or organisations to carry out seed certification including self-certification on fulfillment of criteria as prescribed.
- 3.7 To meet quality assurance requirements for export of seeds, Seed Testing facilities will be established in conformity with ISTA and OECD seed certification programmes.
- 3.8 The State Government, in conformity with guidelines and standards specified by the Board, will establish one or more State Seed Testing Laboratories or declare any Seed Testing Laboratory in the Government or non-Government Sector as a State Seed Testing Laboratory where analysis of seeds will be carried out in the prescribed manner.
- 3.9 Farmers will be encouraged to use certified seeds to ensure improved performance and output.
- 3.10 Farmers will retain their right to save, use, exchange, share or sell their farm seeds and planting materials without any restriction. They will be free to sell their seed on their own premises or in the local market without any hindrance provided that the seed is not branded. Farmers' right to continue using the varieties of their choice will not be infringed by the system of compulsory registration.
- 3.11 Stringent measures would be taken to ensure the availability of high quality of seeds and check the sale of spurious or misbranded seeds.

4. SEED DISTRIBUTION AND MARKETING

- 4.1 The availability of high quality seeds to farmers through an improved distribution system and efficient marketing set-up will be ensured to facilitate greater security of seed supply.
- 4.2 For promoting efficient and timely distribution and marketing of seed throughout the country, a supportive environment will be provided to encourage expansion of the role of the private seed sector. Efforts will be made to achieve better coordination between State Governments to facilitate free Inter-State movement of seed and planting material through exemption of duties and taxes.
- 4.3 Private Seed Sector will be encouraged and motivated to restructure and reorient their activities to cater to nontraditional areas.
- 4.4 A mechanism will be established for collection and dissemination of market intelligence regarding preference of consumers and farmers.
- 4.5 A National Seed Grid will be established as a data-base for monitoring of information on requirement of seed, its production, distribution and preference of farmers on a district-wise basis.
- 4.6 Access to term finance from Commercial Banks will be facilitated for developing efficient seed distribution and marketing facilities for growth of the seed sector.
- 4.7 Distribution and marketing of seed of any variety, for the purpose of sowing and planting will be allowed only if the said variety has been registered by the National Seeds Board.
- 4.8 National Seeds Board can direct a dealer to sell or distribute seeds in a specified manner in a specified area if it is considered necessary to the public interest.

5. INFRASTRUCTURE FACILITIES

- 5.1 To meet the enhanced requirement of quality/certified seeds, creation of new infrastructure facilities along with strengthening of existing facilities, will be promoted.
- 5.2 National Seed Research and Training Center will be set up to impart training and build a knowledge base in various disciplines of the seed sector.
- 5.3 The Central Seed Testing Laboratory will be established at the National Seed Research and Training Center to perform referral and other functions as required under the Seeds Act.
- 5.4 Seed processing capacity will be augmented to meet the enhanced requirement of quality seed.

- 5.5 Modernisation of seed processing facilities will be encouraged in terms of modern equipment and latest techniques, such as seed treatment for enhancement of performance of seed, etc.
- 5.6 Conditioned storage for breeder and foundation seed and aerated storage for certified seed would be created in different regions.
- 5.7 A computerized National Seeds Grid will be established to provide information on availability of different varieties of seeds with production agencies, their location, quality etc. This network will facilitate optimum utilisation of available seeds in every region.
- 5.7.1 Initially, seed production agencies in the public sector would be connected with the National Seed Grid, but progressively the private sector will be encouraged to join the Grid for providing a clear assessment of demand and supply of seeds.
- 5.8 State Governments, or the National Seeds Board in consultation with the concerned State Government, may establish Seed Certification Agencies.
- 5.9 State Governments will establish appropriate systems for effective execution and implementation of the objectives and provisions of the Seeds Act.

6. TRANSGENIC PLANT VARIETIES

- 6.1 Biotechnology will play a vital role in the development of the agriculture sector. This technology can be used not only to develop new crops/varieties, which are tolerant to disease, pests and abiotic stresses, but also to improve productivity and nutritional quality of food.
- 6.2 All genetically engineered crops/varieties will be tested for environment and bio-safety before their commercial release, as per the regulations and guidelines of the Environment Protection Act (EPA), 1986.
- 6.3 The EPA, 1986, read with the Rules, 1989 would adequately address the safety aspects of transgenic seeds/planting materials. A list will be generated from Indian experience of transgenic cultivars that could be rated as environmentally safe.
- 6.4 Seeds of transgenic plant varieties for research purposes will be imported only through the National Bureau of Plant Genetic Resources (NBPGR) as per the EPA, 1986.
- 6.5 Transgenic crops/varieties will be tested to determine their agronomic value for at least two seasons under the All India Coordinated Project Trials of ICAR, in coordination with the tests for environment and bio-safety clearance as per the EPA before any variety is commercially released in the market.

- 6.6 After the transgenic plant variety is commercially released, its seed will be registered and marketed in the country as per the provisions of the Seeds Act.
- 6.7 After commercial release of a transgenic plant variety, its performance in the field, will be monitored for at least 3 to 5 years by the Ministry of Agriculture and State Departments of Agriculture.
- 6.8 Transgenic varieties can be protected under the PVP legislation in the same manner as non-transgenic varieties after their release for commercial cultivation.
- 6.9 All seeds imported into the country will be required to be accompanied by a certificate from the Competent Authority of the exporting country regarding their transgenic character or otherwise.
- 6.9.1 If the seed or planting material is a product of transgenic manipulation, it will be allowed to be imported only with the approval of the Genetic Engineering Approval Committee (GEAC), set up under the EPA, 1986.
- 6.10 Packages containing transgenic seeds/planting materials, if and when placed on sale, will carry a label indicating their transgenic nature. The specific characteristics including the agronomic/yield benefits, names of the transgenes and any relevant information shall also be indicated on the label.
- 6.11 Emphasis will be placed on the development of infrastructure for the testing, identification and evaluation of transgenic planting materials in the country.

7. IMPORT OF SEEDS AND PLANTING MATERIAL

- 7.1 The objective of the import policy is to provide the best planting material available anywhere in the world to Indian farmers, to increase productivity, farm income and export earnings, while ensuring that there is no deleterious effect on environment, health and bio-safety.
- 7.1.1 While importing seeds and planting material, care will be taken to ensure that there is absolutely no compromise on the requirements under prevailing plant quarantine procedures, so as to prevent entry into the country of exotic pests, diseases and weeds detrimental to Indian agriculture.
- 7.1.2 All imports of seeds will require a permit granted by the Plant Protection Advisor to the Government of India, which will be issued within the minimum possible time frame.

- 7.2 All import of seeds and planting materials, etc. will be allowed freely subject to EXIM Policy guidelines and the requirements of the Plants, Fruits and Seeds (Regulation of import into India) Order, 1989 as amended from time to time. Import of parental lines of newly developed varieties will also be encouraged.
- 7.3 Seeds and planting materials imported for sale into the country will have to meet minimum seed standards of seed health, germination, genetic and physical purity as prescribed.
- 7.4 All importers will make available a small sample of the imported seed to the Gene Bank maintained by NBPGR.
- 7.5 The existing policy, which permits free import of seeds of vegetables, flowers and ornamental plants, cuttings, saplings of flowers, tubers and bulbs of flowers by certain specified categories of importers will continue. Tubers and bulbs of flowers will be subjected to post-entry quarantine.
- 7.5.1 After the arrival of consignments at the port of entry, quarantine checks would be undertaken; which may include visual inspection, laboratory inspection, fumigation and growout tests. For the purpose of these checks, samples will be drawn and the tests will be conducted concurrently.

8. EXPORT OF SEEDS

- 8.1 Given the diversity of agro-climatic conditions, strong seed production infrastructure and market opportunities, India holds significant promise for export of seeds.
- 8.2 Government will evolve a long term policy for export of seeds with a view to raise India's share of global seed export from the present level of less than 1% to 10% by the year 2020.
- 8.3 The export policy will specifically encourage custom production of seeds for export and will be based on long term perspective, dispensing with case to case consideration of proposals.
- 8.3 Establishment and strengthening of Seeds Export Promotion Zones with special incentives from the Government will be facilitated.
- 8.4 A data bank will be created to provide information on the International Market and on export potential of Indian varieties in different parts of the world.
- 8.5 A data base on availability of seeds of different crops to assess impact of exports on domestic availability of seeds will be created.

- 8.6 Promotional programmes to improve the quality of Indian seeds to enhance its acceptability in the International Market will be taken up.
- 8.7 Testing and certification facilities will be established in conformity with international requirements.

9. PROMOTION OF DOMESTIC SEED INDUSTRY

- 9.1 Incentives will be provided to the domestic seed industry to enable it to produce seeds of high yielding varieties and hybrid seeds at a faster pace to meet the challenges of domestic requirements.
- 9.2 Seed Industry will be provided with a congenial and liberalized climate for increasing seed production and marketing, both domestic and international.
- 9.3 Membership to International Organisations and SeedAssociations like ISTA, OECD, UPOV, ASSINSEL, WIPO, at the National level or at the level of individual seed producing agencies, will be encouraged.
- 9.4 Emphasis will be given to improving the quality of seed produced and special efforts will be directed towards improving the quality of farmers' saved seeds.
- 9.5 Financial support for capital investment, working capital and infrastructure strengthening will be facilitated through NABARD/ Commercial Banks/Cooperative Banks.
- 9.6 Tax rebate/concessions will be considered on the expenditure incurred on in-house research and development of new varieties and other seed related research aspects. In order to develop a competitive seed market, the States will be encouraged to remove unnecessary local taxation on sales of seeds.
- 9.7 To encourage seed production in non-traditional areas including backward areas, special incentives such as transport subsidy will be provided to seed producing agencies operating in these marginalised areas.
- 9.8 Reduction of import duty will be considered on machines and equipment used for seed production and processing which are otherwise not manufactured in the country.

10. STRENGTHENING OF MONITORING SYSTEM

- 10.1 The Department of Agriculture & Cooperation (DAC) will supervise the overall implementation and monitoring of the National Seeds Policy.

10.2 The physical infrastructure in terms of office automation, communication facilities, etc., in DAC will be augmented in a time bound manner.

10.3 The technical capacity of DAC need to be augmented and strengthened to undertake the additional work relating to implementation of National Seeds Policy, implementation of PVP&FR Bill, Seeds Act, Import and Export of Seeds, etc.

10.4 Capacity building, including National and International training and participation in Seminars/Workshops will be organized for concerned officials.

11. CONCLUSION

The Government of India trusts that the National Seeds Policy will receive the fullest support of State Governments/Union Territory Administrations, State Agricultural Universities, plant breeders, seed producers, the seed industry and all other stakeholders, so that it may serve as a catalyst to meet the objectives of sustainable development of agriculture, food and nutritional security for the population, and improved standards of living for farming communities. The National Seeds Policy will be a vital instrument in attaining the objectives of doubling food production and making India hunger free. It is expected to provide the impetus for a new revolution in Indian agriculture, based on an efficient system for supply of seeds of the best quality to the cultivator. The National Seeds Policy will lay the foundation for comprehensive reforms in the seed sector. Significant changes in the existing legislative framework will be effected accompanied by programmatic interventions. The Policy will also provide the parameters for the development of the seed sector in the Tenth and subsequent Plans. The progress of implementation of the Policy will be monitored by a High Level Review Committee.

Seed testing for quality assessment - importance – methods

Seed testing

Seed testing is determining the standards of a seed lot viz., physical purity, moisture, germination and ODV and thereby enabling the farming community to get quality seeds.

The Seed Testing Laboratory is the hub of seed quality control. Seed testing services are required from time to time to gain information regarding planting value of seed lots. Seed testing is possible for all those who produce, sell and use seeds.

Objective

1. To identify the quality problem and their probable cause
2. To determine their quality, that is, their suitability for planting
3. To determine the need for drying and processing and specific procedures that should be used
4. To determine if seed meets established quality standards or labelling specifications.
5. To establish quality and provide a basis for price and consumer discrimination among lots in the market. The primary aim of the seed testing is to obtain accurate and reproducible results regarding the quality status of the seed samples submitted to the Seed Testing Laboratories.

Importance

- The importance of seed testing was realized more than 100 years ago for assured planting values. The adulteration of vegetable seeds by stone dust which was packed in some parts of the world particularly in Europe.
- Seed testing has been developed to aid agriculture to avoid some of the hazards of crop production by furnishing the needed information about different quality attributes viz., purity, moisture, germination, vigour and health.
- Quality control of seed depends on the different seed testing protocols which determine the genuineness of the cultivar.

- Testing of seed to evaluate the planting value and the authenticity of the certified lot.
- Seed testing is required to assess the seed quality attributes of the seed lots which have to be offered for sale.
- These quality attributes are seed moisture content, germination and vigour, physical and genetic purity, freedom from seed borne diseases and insect infestation. In India, seed testing is done mainly for moisture, germination and physical purity of seeds.
- Standard seed testing procedures for the evaluation of the seeds were developed by ISTA. It is obligatory on the part of the seed analyst to follow rules prescribed by ISTA (ISTA, 1985) if the seed is moving to the International trade.
- The seed testing procedures which are described below are based mostly on the international rules because most of our rules (Chalam *et al.*, 1967) are based on, ISTA, 1996. Economic yield of a crop depends on the quality of seeds which can be evaluated by seed testing (ISTA, 1996).
- The testing of seed quality is carried out on seed samples drawn from seed lot to be used for cultivation. The quantity of seed sample taken for testing in laboratory is minute compared to that of seed lot it represents.

Role of Seed Testing Laboratories

Seed testing laboratories are essential organization in seed certification and seed quality control programmes. The main objective is to serve the producer, the consumer and the seed industry by providing information on seed quality. Test results may cause rejection of poor seed multiplication or low grade seed in a court of law.

Analysis of seed in the laboratory: Seed testing is possible for all those who produce, sell and use seeds. Seed testing is highly specialized and technical job. With a view to maintain uniformity in quality control the seed analysis laboratory includes for distinct sections.

1. Section for purity testing: Purity analysis of seed lot is considered under two factors
 - a) Testing the cleanliness of seed lot and
 - b) Testing the genuineness of the cultivar
2. Section for moisture testing

3. Section for viability, germination and section for vigour testing.

Sampling in Seed Testing Laboratory

The seed samples received in the laboratory (submitted sample) are required to be reduced to obtain working samples for carrying out various tests. A number of methods are available for obtaining working samples.

Mixing and dividing of seeds

The main objective of mixing and dividing of seeds is to obtain the representative homogenous seed sample for analysis by reducing the submitted sample to the desired size of working sample.

Method of mixing and dividing

- Mechanical dividing
- Modified halving method
- Hand halving method
- Random cup method
- Spoon method

Mechanical method

The reduction of sample size is carried out by the mechanical dividers suitable for all seeds except for chaffy and fuzzy seeds.

Objective of mechanical dividing

- To mix the seed sample and make homogenous as far as possible. To reduce the seed sample to the required size without any bias.
- The submitted sample can be thoroughly mixed by passing it through the divider to get 2 parts and passing the whole sample second time and 3rd time if necessary to make the seeds mixed and blended so as to get homogenous seed sample when the same seeds are passed through it into approximately equal parts.

- The sample is reduced to desired size by passing the seeds through the dividers repeatedly with one half remain at each occasion.

Types of mechanical dividers

Boerner divider

It consists of a hopper, a cone and series of baffles directing the seeds into 2 spouts. The baffles are of equal size and equally spaced and every alternate one leading to one spout. They are arranged in circle and are directed inward. A valve at the base of the hopper retains the seeds in the hopper. When the valve is opened, the seeds fall by **gravity** over the cone where it is equally distributed and approximately equal quantity of seeds will be collected in each spout. A disadvantage of this divider is that it is difficult to check for cleanliness.

Soil divider

It is a sample divider built on the same principles as the Boerner divider. Here the channels are arranged in a straight row. It consists of a hopper with attached channels, a frame work to hold the hopper, two receiving pans and a pouring pan. It is suitable for large seeds and chaffy seeds.

Centrifugal or Gamet divider

The principle involved is the **centrifugal force** which is used for mixing and dividing the seeds. The seeds fall on a shallow rubber spinner which on rotation by an electric motor, throw out the seeds by centrifugal force. The circle or the area where the seeds fall is equally divided into two parts by a stationary baffle so that approximately equal quantities of seed will fall in each spout.



Boerner divider



Soil divider



Centrifugal or Gamet divider

Random cup method

This is the method suitable for seeds requiring working sample upto 10 grams provided that they are not extremely chaffy and do not bounce or roll (e.g.) *Brassica* spp. Six to eight small cups are placed at random on a tray. After a preliminary mixing the seed is poured uniformly over the tray. The seeds that fall into the cup is taken as the working sample.

Modified halving method

The apparatus consists of a tray into which is fitted a grid of equal sized cubical cups open at the top and every alternate one having no bottom. After preliminary mixing the seed is poured evenly over the grid. When the grid is lifted, approximately half the sample remains on the tray. The submitted sample is successively halved in this method until a working sample size is obtained.

Spoon method

This is suitable for samples of single small seeded species. A tray, spatula and a spoon with a straight edge are required. After preliminary mixing, the seed is poured evenly over the tray. The tray should not be shaken thereafter. With the spoon in one hand, the spatula in the other and using both small portions of seed from not less than 5 random places on the tray should

be removed. Sufficient portions of seed are taken to estimate a working sample approximately but not less than the required size.

Hand halving method

This method is restricted to the chaffy seeds. The seed is poured evenly on to a smooth clean surface and thoroughly mixed into a mound. The mound is then divided into 1/2 and each half is mound again and halved into 4 portions. Each of the 4 portions is halved again giving 8 portions. The halved portions are arranged in rows and alternate portions are combined and retained. The process is repeated until the sample of required weight is obtained.

Physical Purity

Purity analysis

The purity analysis of a seed sample in the seed testing laboratory refers to the determination of the different components of the purity viz., pure seeds, other crop seeds, weed seeds and inert matter.

The objective of the purity analysis is to determine whether the submitted sample conforms to the prescribed physical quality standards with regard to physical components.

The purity analysis is done on the working sample of prescribed weight drawn from submitted sample. The analysis may be made on one working sample of the prescribed weight or on two sub-samples of atleast half of this weight, each independently drawn.

Weighing the working sample

The number of decimal places to which the working sample and the componenets of the working sample should be weighed is given below.

Weight of the working sample (g)	The number of decimal places required	Example
<1	4	0.7534

1-9.999	3	7.534
10-99.99	2	75.34
100-999.9	1	753.4
1000 or more	0	7534

Purity separation

The working sample after weighing is separated into its components viz., pure seed, other seed crop, weed seed and inert matter.

Pure seed

The seeds of kind / species stated by the sender. It includes all botanical varieties of that kind / species. Immature, undersized, shrivelled, diseased or germinated seeds are also pure seeds. It also includes broken seeds, if the size is $>1/2$ of the original size except in leguminacea, and cruciferae where the seed coat entirely removed are regarded as inert matter.

Other crop seed

It refers to the seeds of crops other than the kind being examined.

Weed Seed

It includes seeds of those species normally recognized as weeds or specified under Seed Act as a noxious weed.

Inert matter

It includes seed like structures, stem pieces, leaves, sand particles, stone particles, empty glumes, lemmas, paleas, chaff, awns, stalks longer than florets and spikelets.

Method of purity separation

Place the sample on the **purity work board** after sieving / blowing operations and separate into other crop seeds and inert matter. After separation, identify each kind of weed

seeds, other crop seeds as to genus and species. The names and number of each are recorded. The type of inert matter present should also be noted.



Calculation

All the four components must be weighed to the required number of decimal places. The percentages of the components are determined as follows.

$$\% \text{ of components} = \frac{\text{Weight of individual component}}{\text{Total weight of all components}} \times 100$$

If there is a gain or loss between the weight of the original samples and the sum of all the components is in excess of one percent, another analysis should be made.

Duplicate tests

If the analysis result is near the border line in relation to the seed standards, one more test is done and the average is reported. However, if a duplicate analysis is made of two half sample or whole samples, the difference between the two must not exceed the permissible tolerance. If the difference is in excess of the tolerance, analyze further (but not more than 4 pairs in all) until a pair is obtained which has its member within tolerance.

Seed germination test

Germination is defined as the emergence and development from the seed embryo, of those essential structures, for the kind of seed in question, indicates its ability to produce a normal plant under favourable conditions.

Principles

Germination tests shall be conducted with a pure seed fraction. A minimum of 400 seeds are required in four replicates of 100 seeds each or 8 replicates of 50 seeds each or 16 replicates of 25 seeds each depending on the size of seed and size of containers of substrate.

The test is conducted under favourable conditions of moisture, temperature, suitable substratum and light if necessary. No pretreatment to the seed is given except for those recommended by ISTA.

Materials required

Substratum

The substratum serves as moisture reservoir and provides a surface or medium for which the seeds can germinate and the seedlings grow. The commonly used substrate are sand, germination paper and soil.

1. Sand

Size of sand particle

Sand particles should not be too large or too small. The sand particles should pass through 0.80 mm sieve and retained by 0.05mm sieve.

Toxicity

Sand should not have any toxic material or any pathogen. If there is presence of any pathogen found then the sand should be sterilized in an autoclave.

Germination tray

When we use the sand, germination trays are used to carry out the test. The normal size of the tray is 22.5 x 22.5 x 4 cm. The tray may either zinc or stainless steel.

Method of seed placement

Seed in sand(S)

Seeds are planted in a uniform layer of moist sand and then covered to a depth of 1 to 2 cm with sand.

Top of sand (TS)

Seeds are pressed in to the surface of the sand.

Spacing

We must give equal spacing on all sides to facilitate normal growth of seedling and to avoid entangling of seed and spread of disease. Spacing should be 1-5 times the width or diameter of the seed.

Water

The amount of water to be added to the sand will depend on size of the seed. For cereals, except maize, the sand can be moistened to 50% of its water holding capacity. For large seeded legumes and maize sand is moistened to 60% water holding capacity.

2. Paper

Most widely used paper substrates are filter paper, blotter or towel (kraft paper). It should have capillary movement of water, at vertical direction (30 mm rise / min.). It should be free from toxic substances and free from fungi or bacteria. It should hold sufficient moisture during the period of test. The texture should be such that the roots of germinating seedlings will grow on and not into the paper.

Methods

Top of paper (TP)

Seeds are placed on one or more layers of moist filter paper or blotter paper in petriplates. These petriplates are covered with lid and placed inside the germination cabinet. This is suitable for those seeds which require light.

Between paper (BP)

The seeds are germinated between two layers of paper. The seeds are placed between two layers of paper and rolled in towels. The rolled towels are placed in the germinator in an upright position.

Germination apparatus

Germination cabinet / Germination room

This is called chamber where in temperature and relative humidity are controlled. We can maintain the temperature, relative humidity and light required for different crops.

Room germinator

It works with same principle as that of germinator. This is a modified chamber of larger one and the worker can enter into it and evaluate the seedlings. Provisions are made to maintain the temperature and relative humidity. This is used widely in practice.

Seed counting board

This is used for accurate counting and spacing of seeds. This consists of 2 plates. The basal one is stationary and top one is movable. Both top and basal plates are having uniform number of holes viz., 50/100, when the plates are in different position.

After taking the sample, the top plate is pulled in such a way that the holes are in one line so that the fixed number of seeds falls on the substratum.

Vacuum seed counter

Consists of a head, pipe and wall. There are plates of 50 or 100 holes which can be fitted to the head. When vacuum is created the plate absorbs seeds and once the vacuum is released the seeds fall on the substrate.

Impression board

Made of plastic / wood with 50 or 100 holes / pins. Here the knobs are arranged in equal length and space. By giving impression on the sand it makes uniform depth and spacing for seed.

Evaluation of germination test

The germination test is evaluated as

- Normal seedlings
- Abnormal seedlings
- Hard seeds
- Fresh and ungerminated seeds
- Dead seeds

ISTA classified the seedlings into different categories based on the development of essential structures.

Normal seedlings

Seedlings which has the capacity for continued development into normal plant when grown in favourable conditions of soil, water, temperature and light

Characters of normal seedlings

- A well developed root system with primary root except in certain species of gramineae which normally produce seminal root or secondary root.
- A well developed shoot axis consisting of elongated hypocotyls in seedlings of epigeal germination.
- A well developed epicotyl in seedlings of hypogeal germination.
- One cotyledon in monocotyledon and two in dicotyledons.
- A well developed coleoptiles in gramineae containing a green leaf.
- A well developed plumule in dicotyledons.

Normal seedlings

- Seedlings with following slight defects are also taken as normal seedlings.
- Primary root with limited damage but well developed secondary roots in leguminaceae (Phaseolus, Pisum), gramineae (Maize), cucurbitaceae (Cucumis) and malvaceae (cotton)
- Seedlings with limited damage or decay to essential structures but no damage to conducting tissue.

- Seedlings which are decayed by a pathogen with a clear evidence that the parent seed is not the source of infection

Abnormal seedlings

Seedlings which do not show the capacity for continued development into normal plant when grown in favourable condition of soil, water, temperature and light.

Types of abnormal seedlings

Damaged seedlings

Seedlings with any one of the essential structures missing or badly damaged so that the balanced growth is not expected.

Seedlings with no cotyledons, with splits, cracks and lesions or essential structures and without primary root

Deformed seedlings

Weak or unbalanced development of essential structures such as spirally twisted or stunted plumule or hypocotyls or epicotyls, swollen shoot, stunted roots etc

Decayed seedlings

Seedlings with any one of the essential structures showing diseased or decayed symptoms as a result of primary infection from the seed which prevents the development of the seedlings.

Hard seeds

Seeds which do not absorb moisture till the end of the test period and remain hard (e.g.) seed of leguminaceae and malvaceae

Fresh and ungerminated seeds

Seeds which are neither hard nor have germinated but remain firm and apparently viable at the end of the test period.

Dead seeds

Seeds at the end of the test period are neither hard or nor fresh or have produced any part of a seedling. Often dead seeds collapse and milky paste comes out when pressed at the end of the test.

Retesting

If the results of a test are considered unsatisfactory it will not be reported and a second test will be made by the same method or by alternative method under the following circumstances.

1. Replicates performance is out of tolerance
2. Results being inaccurate due to wrong evaluating of seedlings or counting or errors in test conditions
3. Dormancy persistence or phytotoxicity or spread of fungi or bacteria. The average of the two test shall be reported.

Use of tolerances

The result of a germination test can be relied upon only if the difference between the highest and the lowest replicates is within accepted tolerances.

To decide if two test results of the same sample are compatible again the tolerance table is used.

Reporting results

The result of the germination test is calculated as the average of 4x100 seed replicates. It is expressed as percentage by number of normal seedlings. The percentage is calculated to the nearest whole number. The percentage of abnormal seedlings, hard, fresh and dead seeds is calculated in the same way. These should be entered on the analysis of certificate under appropriate space. If the result is 'nil' for any of these categories it shall be reported as '0'.

Determination of moisture content

Definition

The moisture content of a seed sample is the loss in weight when it is dried. It is expressed as a percentage of the weight of the original sample. It is one of the most important factors in the maintenance of seed quality.

Method of moisture determination

1. Air oven method

In this method, seed moisture is removed by drying the seed sample at a specified temperature for a specified duration.

2. Moisture meters

Moisture meters estimate seed moisture quickly but the estimation is not as precise as by the air oven method.

Weight of the submitted sample

100 g for species that have to be ground. 50 g for all other species. The sample should be submitted in polythene bag of 700 gauge.

Air oven method for seed moisture estimation

Materials required

Grinding mill

It should be constructed of non-absorbent material. It should grind evenly and should be operated at such a speed that during grinding, it should not cause heating of the ground material. Air currents that might cause loss of moisture must be reduced to a minimum. The fineness of grinding should be adjustable.

Container

Container of glass or non-corrosive metal (e.g.) stainless steel should be used

Oven

A good quality electric air oven with a thermostatic electronic temperature control for maintaining temperature within $\pm 1^{\circ}\text{C}$ is required

Grinding

For some seeds (e.g. Cereals and Cotton) fine grinding is essential before the moisture content is determined. In such cases, at least 50% of the ground material should pass through a wire sieve with meshes of 0.5 mm and not more than 10% remain on a wire sieve with a mesh of 1.0 mm. For leguminous seeds, coarse grinding is recommended; at least 50% of the ground material shall pass through a wire sieve with meshes of 4.0 mm.

Pre drying

If the species is one for which grinding is necessary and the moisture content is more than 17%. (or 10% in the case of soy bean and 13% in rice) pre drying before grinding is necessary. For this purpose, two 50 g portions are weighed and placed on open trays at 130°C for 5-10 min. If seed moisture content is about 25% or more it should be pre-dried at 70°C for 2-5 hours, depending on the initial water content. The pre dried seeds should be kept in a closed desiccator for cooling. Then each of the duplicate quantities is weighed separately and about 20 g is ground. The ground material is then subjected to moisture testing using a hot air-oven as described below.

Moisture estimation

It should be carried out in duplicate on two independently drawn 5-10 g working samples, weighed with an accuracy of 1 mg. Most species are dried for 1 hr at 130°C , cereals for 2 hours (130°C) and maize for 4 hours (130°C). Seeds containing high percentage of oil should be dried at 103°C for 17 hours.

Steps

1. Empty container along with its cover should be weighed
2. The submitted sample should be mixed thoroughly and two small portions or seed sample are to be drawn and it should be ground as per the requirements.
3. Then fill the container with 5 grams of ground sample and weigh it.

4. After weighing, remove the cover or lid of the container and the open container should be kept in the oven which has already been heated to the prescribed drying temperature.
5. At the end of the drying period, container should be closed with its cover or lid. The container should be transferred into a Desiccator. The Desiccator should be closed and the sample should be allowed to cool for 30 minutes.
6. The sample should be weighed again and the moisture content may be calculated to one decimal place by the following formula.

$$m = \frac{m_2 - m_3}{m_2 - m_1} \times 100$$

Where, m = Seed moisture content
 m_1 = Weight of the empty container with its cover
 m_2 = Weight of the container with its cover and seeds before drying
 m_3 = Weight of the container with its cover and seeds after drying

The duplicate result of the determination may not differ by more than 0.2% otherwise the analysis should be repeated. If pre dried, the moisture content is calculated from the results obtained in the first (pre-drying) and second stages of the procedure. If S_1 is the moisture lost in the first stage and S_2 is the moisture lost in the second stage, each calculated as above and expressed as a percentage, the original moisture content of the sample is calculated as below.

$$m = S_1 + S_2 - \frac{S_1 \times S_2}{100}$$

m = moisture content

S_1 = Moisture percentage lost in predrying stage

S_2 = Moisture percentage lost in drying stage

Moisture meters: Universal (OSAW) digital moisture meters

The principle involved in these moisture meters is that wet grains are good conductors while dry grains are less conductors of electricity. So, the moisture content is directly proportional to the electrical conductivity of the seed.

It consists of a compression unit to compress the sample to pre-determined thickness. The thickness setting is very easily read on a vertical and circular scale. The seed material on test is taken in a test cup and is compressed. Then press the push type switch till the reading comes in the display. Here no temperature reading and correlated dial are required. The computer version of digital moisture meter automatically compensate for temperature corrections.

LECTURE 30

VARIETAL IDENTIFICATION - GROW OUT TEST - MOLECULAR AND BIOCHEMICAL TEST - DETECTION OF GENETICALLY MODIFIED CROPS

GROW - OUT TEST

Genetic purity refers to the percentage of contamination by seeds or genetic materials of other varieties. Contamination due to pollen shedders and physical admixtures during processing causes impurity. **Grow-out test** is often practiced in detecting the off types which time consuming (takes one full growing season for completion), land intensive, tedious and highly vulnerable to human and environmental error.

Grow-out test is the official measure for controlling the genetic purity of the seed lot. It serves as a pre-control as well as a 'post-control' test for avoiding genetic contaminations. According to the official regulations in India, it is pre-requisite for seed certification of hybrids such as **cotton, tomato, watermelon, castor, musk melon and brinjal**.

The test is conducted by the Director of Seed Certification or under his supervision. An SCO (GOT) is authorized to conduct the GOT. Upon receipt of a sample, the Director assigns a code number with intimation to ADA (SC) concerned. The SCO(GOT) will conduct the grow out test as per guidelines and report to the Director within 2 weeks regarding the crop.

Size of submitted sample

1000 g	Maize, Cotton, Groundnut, Soybean and species of other genera with seeds of similar size
500 g	Sorghum, Wheat, Paddy and species of other genera with seeds of similar size
250 g	Beta and species of other genera with seeds of similar size
100 g	Bajra, Jute and species of other genera with seeds of similar size
250 tubers / planting stakes / roots / corms	Seed potato, Sweet potato and other vegetatively propagating crops

Size of working sample

The working sample for grow out test shall be obtained through subsequent mixing and dividing of the submitted sample in accordance with the prescribed procedure for seed sampling.

The minimum population required for taking the observations shall be **400 plants**; however, it will also depend on the maximum permissible off-type plants prescribed for the species under consideration in the Indian Minimum seed Certification standards. The minimum population to be maintained in **4 replicates** are

Foundation seeds : 800 plants

Certified seeds : 400 plants

The number of seeds required for raising the crop to obtain the required number of plants shall depend on the germination percentage of the seed sample and hence seed rate should be adjusted accordingly.

Location of the grow out test

The grow out test shall be conducted in glass houses or in field having suitable agro-climatic conditions in off-season. As pre quality regulation the seed distributor can be prevented from distributing spurious seeds and as post quality regulation distributed seeds can be prevented from multiplication. For **Tamil Nadu** the seed farm at **Kannampalayan, Coimbatore** is used for GOT.

Standard sample

The standard sample of a cultivar (control) is used as a check and sown along with the sample to be tested. The standard sample must not differ significantly in any character and be obtained from the originating plant breeder / breeding institute and be stored under controlled temperature and humidity conditions so as to use it each year to sow control plots for cultivars under test. A comparison must be made between the two lots of the standard sample before changing from one standard sample to other.

Method of raising the crop

Standard and recommended agronomic / cultural practices such as field preparation, size of the plot, row length, distance between the rows, distance between the plants, irrigation and fertilization, etc., in respect of the specific crop shall be followed both for the sample in question and its control (standard sample).

The germination percentage of the sample and the standard sample must be determined to adjust the seed rate. The sowing should be done by **dibbling or small plot drill. Subsequent thinnings is not recommended.**

The samples of the same cultivars must be sown in succession and the standard samples are sown at suitable intervals. (one standard sample for every ten sample to be tested).

The size of the plot, row length and spacing shall differ according to the crop. Recommended specification for the above variables are provided in Table mentioned below which can suitably be modified if considered essential.

Recommended row length, distances, spacing for some important crops

S. No.	Crop	Row length (m)	Plant to plant distance (cm)	Space between rows (cm)	Space between plots (cm)
1.	Wheat, barley oats	6	2	25	50

2.	Pea, Cowpea	6	10	45	90
3.	Chickpea, green gram, black gram	6	10	30	60
4.	Maize	10	25	60	90
5.	Hybrid cotton	5	10	45	45
6.	Paddy	6	15	20	45
	a) Very early to medium	6	25	30	60
	b) Late and very late				
7.	Pearl millet	6	10	60	90
8.	Sorghum	6	10	45	60

Planting instructions

1. If a seed drill is used, it must be carefully checked to ensure that it contains no seed of a previous sample before another sample is poured into it.
2. Each sample should be sown in atleast **two replicate plots**. As insurance against failure, a replicate should be situated in different fields, or different parts of the same field.
3. The plots may be of any convenient size that will provide enough plants for the determination to the accuracy required.
4. If the seeds is sown *in situ*, it should be sown in rows, mechanically if possible.

Methods for taking observations

Grow-out test plots must be examined **throughout the growing season** with emphasis on the period from the flowering to ripening. While taking the observation, all plants must be examined and the plants showing deviations in characters against the control should be tagged and examined carefully at a later stage to confirm whether they are off-types or not for the cultivars

both in the test crop as well as the control. The number of the **total plants and the off-type plants** found should be recorded.

The observation will be made by SCO(GOT) and sent to the Director (SC) within 2 weeks. The DSC will communicate the results to the concerned ADA (SC) who will in-turn communicate to the SCO and producer within 24 hrs of the receipt of the result.

Minimum Number of plants to be observed in grow-out test

Maximum permissible off types (%)	Minimum genetic Purity (%)	Number of plants required per sample
0.10	99.9	4,000
0.20	99.8	2,000
0.30	99.7	1,350
0.50	99.5	800
1.00 and above	99.0 and below	400

GOT register

GOT register consists of 1. Serial number 2. Date of receipt of the sample 3. Code number 4. Kind and variety 5. Class of seed 6. Date of sowing 7. number of plants on 15th day 8. Number of plants on 30th day 9. Number of plants on 45th day 10. Number of plants on 60th day 11. Number of plants on 70th day 12. Dates of inspection 13. Number of plants observed 14. True to type 15. Off type / selfed plants 16. Genetic purity % 17. Name and signature 18. Date of dispatch of result 19. Remarks.

Calculation and interpretation of the results

- ✓ The results of the grow-out test shall be reported as % of other species, cultivars or off-type plants and may be calculated upto first place of decimal.

- ✓ Herbage plants results may be expressed as the number of divergent plants produced by the weight of seed sown.
- ✓ If the sample is found to be a cultivar other than stated by the sender, the results shall be reported as such.
- ✓ If plants of other cultivars are more than 15%, the report shall state that the sample consists of mixture of different cultivars.
- ✓ If nothing worthy of special comments is found, the report shall state that the results of the grow-out test of the sample in question revealed nothing to indicate that the name of the cultivar or species stated by the sender is incorrect.
- ✓ While interpreting the results, tolerances should be applied by using the reject number for prescribed standards with reference to sample size as provided in Table.

Reject number for prescribed standards and sample size

Class of seed	Standard	Sample size	Replication	
			800	400
Foundation seeds	99.0 (1 in 100)	16	8	2
Certified seeds	98.0 (2 in 100)	24	12	2
Certified hybrid cotton	90.0 (10 in 100)	88	44	2

Standard	Reject numbers for sample size of	
	800	400
99.5 (1 in 200)	8	*
99.0 (1 in 100)	16	8
95.0 (5 in 100)	48	24
90.0 (10 in 100)	88	44

85.0 (15 in 100)	128	64
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* indicates that the sample size is too small for a valid test

Merits

1. Reduces genetic contamination.
2. Improves the genetic quality of subsequent stages.
3. Cheapest way to examine reasonable number of plants
4. Reduces rouges and off types.

Demerits

1. Takes long time for assessment (this happens when the crop has to be inspected for genetic purity for the growing period. It becomes more difficult in case of a long duration crop like castor, cotton etc.).
2. Area selected for GOT may lack similar season or growing environment as that of the crop leading to production of off types. This happens in case of temperature sensitive or photo sensitive crops.
3. For cole crops, GOT can be conducted only during a particular season.
4. The person incharge of GOT must possess sufficient knowledge on the crop characters and habit to provide a fair judgement. If not it can lead to problems of discarding a good seed lot.

MOLECULAR MARKERS

GOT is time consuming (takes one full growing season for completion), land intensive, tedious and highly vulnerable to human and environmental error. Molecular markers facilitate and resolve complications of conventional GOT and DUS testing. Molecular markers methods are being adopted by the researchers for **fingerprinting of cultivars, assessment of genetic purity and varietal identification** in seed technology and breeding programmes. In genetic purity assessment, molecular markers detect the degree of contamination due to selfing and

outcrossing in hybrid seed lot, segregation of genotypes in DUS testing and trait confirmation in transgenics.

A molecular marker is a molecule contained within a sample taken from an organism or other matter. It can be used to reveal certain characteristics about the respective source. DNA (negatively charged), for example it is a molecular marker containing information about genetic disorders, genealogy and the evolutionary history of life.

Molecular marker is a DNA sequence in the genome which can be located and identified. As a result of genetic alterations, the base composition at a particular location of the genome may be different in different plants. These differences are called as polymorphisms which can be mapped and identified. These are used to identify a particular sequence of DNA in a pool of unknown DNA.

Types of genetic markers

There are three categories of markers. They are,

1. First generation markers
2. Second generation markers
3. Third generation markers

First generation markers

Hybridization based markers. These are so called because the DNA profile is visualized through hybridization of DNA with radioactively labeled probes of known sources. Example : RFLP - Restriction Fragment Length Polymorphism

Second generation markers

PCR based markers, as their assay was carried out through amplification using either arbitrary or sequence of specific primers. Example : RAPD - Random Amplified Polymorphic DNA , AFLP - Amplified Fragment Length Polymorphism , SSR – Simple Sequence Repeats , VNTR - Variable Number Tandem Repeat, SCAR – Sequence Characterised Amplification Region

Third generation markers

Their detection requires sequence information. With the advancement in the field of DNA sequencing, these have become very popular in the last few years. Example : SNP - Single Nucleotide Polymorphism

These types of markers may also identify dominance and co-dominance within the genome. Identifying dominance and co-dominance with a marker may help identify heterozygotes from homozygotes within the organism. **Co-dominant markers** are more beneficial because they **identify more than one allele**.

List of markers

1. OP - Oligonucleotide Polymorphism
2. ASAP - Allele Specific Associated Primers
3. ISTR - Inverse Sequence-tagged Repeats
4. IRAP - Inter-retrotransposon Amplified Polymorphism

Molecular markers are of two types,

1. Based on nucleic acid (DNA) hybridization (non PCR based approaches)
2. Based on PCR amplification

1. Markers based on DNA hybridization

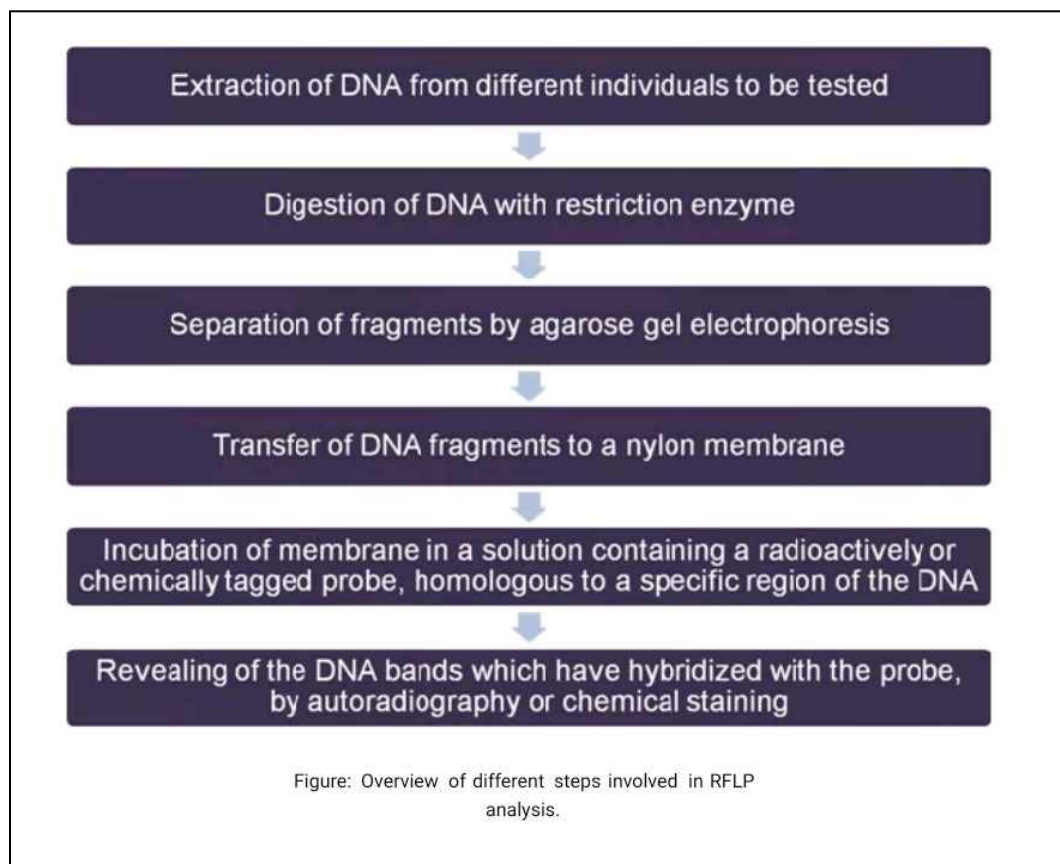
The DNA piece can be cloned, and allowed to hybridize with the genomic DNA which can be detected. Marker based DNA hybridization is widely used and major limitation is that it requires large quantities of DNA and the use of radioactivity (labeled probes).

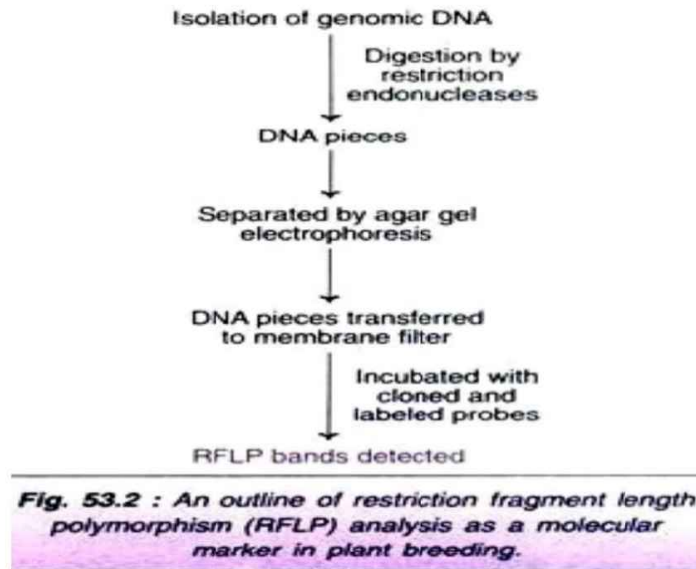
RFLP - Restriction Fragment Length Polymorphism

RFLP was the first marker used by Botstein *et al.*, in 1980 for linkage mapping in human. Later, it was widely accepted system for use in plants which is employed for the detection of polymorphism, based on the DNA sequence differences. It is mainly based on the **altered restriction enzyme sites**, as a result of mutations and recombinations of genomic DNA. The procedure basically involves the isolation of genomic DNA, its digestion by restriction enzymes,

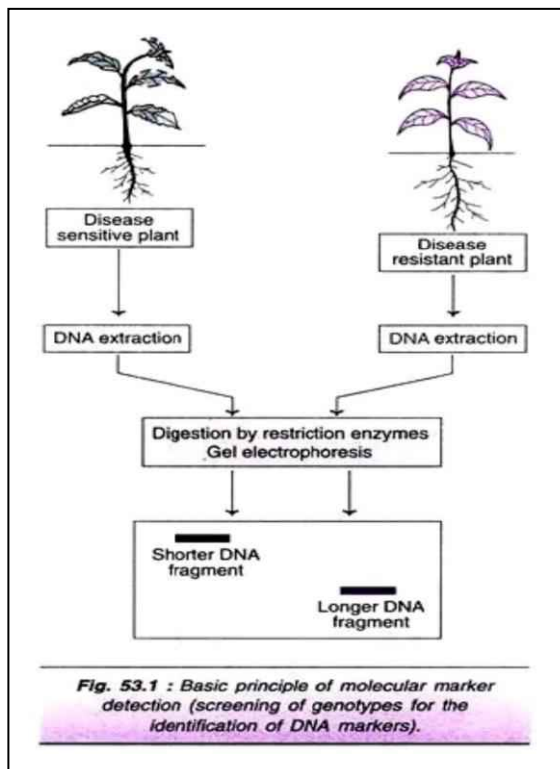
separation by electrophoresis, and finally hybridization by incubating with cloned and labeled probes. **RFLP** marker may require **10g of DNA**.

The restriction enzymes recognize and cut DNA specific sites called restriction sites. Polymorphism can be detected if there is any variation in any two individuals at the restriction sites of any particular restriction enzyme. Such polymorphism can be visualized as variation in length after cutting the DNA with the enzyme and carrying out southern blotting.





Two species, one with disease sensitivity and other with disease resistance, DNA marker than can identify these two alleles, then the genome can be extracted and separated by gel electrophoresis. For instant, the disease resistant plant may have longer DNA fragment while the disease sensitive plant may have shorter DNA fragment.



Merits

1. It is a codominant marker
2. The results are highly reproducible

Demerits

1. Long and tedious process.
2. Large amount of DNA required.
3. Use of radioactively labeled probes.

2. Based on PCR amplification

Polymerase chain reaction is a technique for the amplification of selected regions of DNA. Even a minute quantity of DNA can be amplified. PCR markers are of two types,

1. Locus non-specific markers – RAPD, AFLP
2. Locus specific markers – SSR, SNP

RAPD - Random Amplified Polymorphic DNA

First shown by Welsh and McClelland in 1990. The DNA isolated from the genome is denatured and annealed (heated and allowed to cool slowly or recombined in double stranded form) with primers, and amplified by PCR. Single short oligonucleotide primers can be selected and used for the amplification of DNA segments of the genome. The amplified products are separated on electrophoresis and identified. RAPD's may only need **0.02g of DNA**.

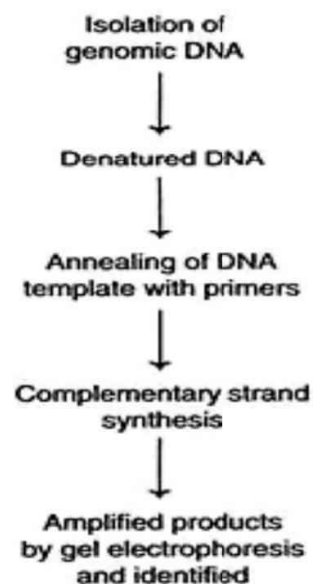
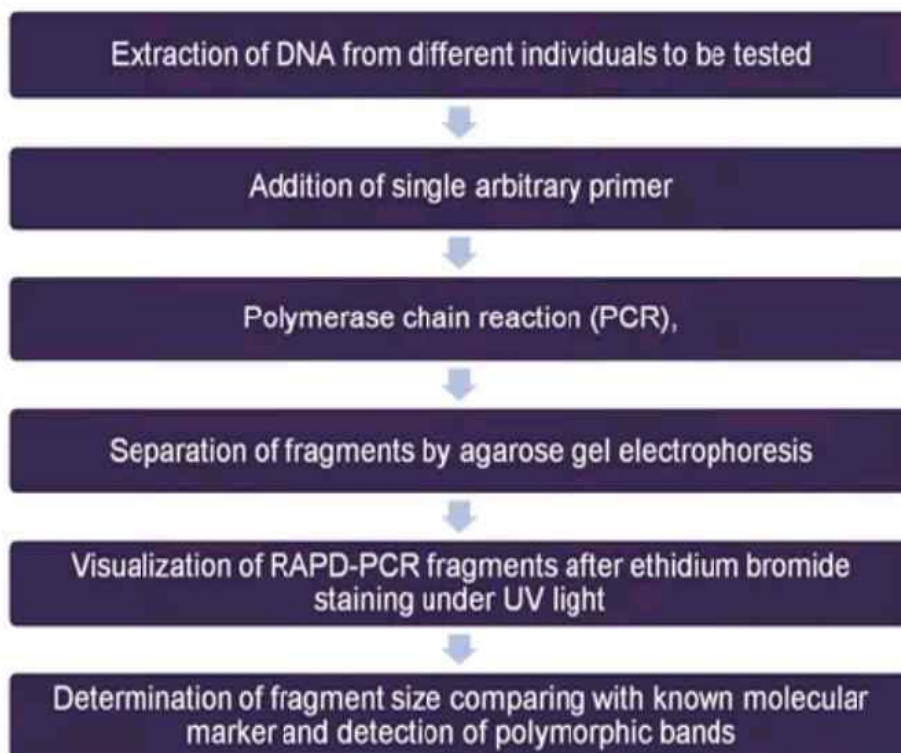


Fig. 53.4 : An outline of random amplified poly-



PCR is carried out using (10 nucleotide long) arbitrary primers which serves as both forward and reverse primer and amplify many genomic fragments simultaneously. Amplified fragments usually range from 500 base pair to 5 kb.

Genomic DNA from two different plants often results in different amplification patterns. This is based on the fact that particular fragment of DNA may be generated from one individual which represents polymorphism and can be used as a molecular marker of a particular species.

Merits

1. Quick and easy method.
2. Amount of DNA required is less.
3. No prior sequence information is needed.
4. Many polymorphic markers can be generated with a single primer.

Demerits

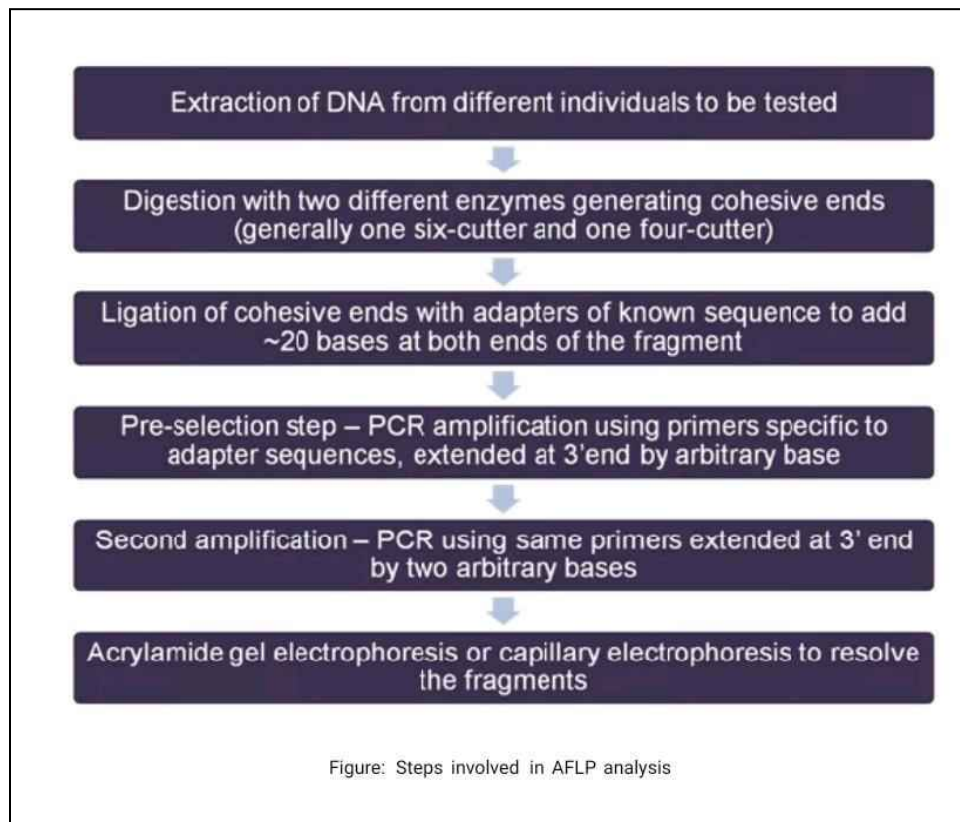
1. Dominant markers

2. Reproducibility is low as the results vary greatly.

AFLP - Amplified Fragment Length Polymorphism

It is a technique involving a combination of RFLP and RAPD first shown by Vos *et al.*, in 1995. Polymorphism originating from two sources is captured, first by difference in restriction sites of enzymes and second by difference in hybridization sites of arbitrary bases.

AFLP is based on the principle of generation of DNA fragments using restriction enzymes and oligo-nucleotide adaptor and their amplification by PCR. Thus, it work on the combination of restriction digestion and PCR.



The DNA of the genome is extracted and subjected to restriction digestion by two enzymes. The cut ends on both sides are then ligated (joined) to known sequences of oligo-nucleotides. PCR is now performed for the pre-selection of a fragment of DNA which has a single specific nucleotide. By this, pre-selective amplification, pool of fragments can be reduced. In the second round of amplification by PCR, three nucleotide sequences are amplified which

further reduces the pool of DNA fragments to a manageable level (<100). Autoradiography can be used for the detection of DNA fragments. The amplified products may be 80-500 base pair in size.

Merits

1. Large number of polymorphic fragments can be produced and detected.

Demerits

1. Dominant markers
2. AFLP is tedious process, very sensitive and reproducible.

ELECTROPHORESIS

It is the latest method of cultivar identification based on protein banding and isoenzyme activity. In Greek, electrophoresis means **‘borne by electricity’**. Molecules with similar charge and size will have differential migration in solution with porous support medium in an electric field, based on the differences in net electrical charges as molecules with higher charge migrate faster than those with lower charge. Particle with smaller molecular weight migrates faster than those with higher weight. This property of molecular movement based on size and net electrical charge is utilized in electrophoresis.

Proteins and nucleic acids are electrophoresed within a matrix of gel. Mostly the gel is in the shape of thin slab, with wells for loading the sample. The gel is immersed within an electrophoresis buffer that provides ions to carry a current and some type of buffer to maintain the pH at a relatively constant value. The gel itself is composed of either **agarose or polyacrylamide**. Based on the banding pattern of protein and esterase's the varieties can be differentiated and identified.

Agarose (mix agarose powder with buffer solution, melt it by heating and pour the gel) is a polysaccharide which is non-toxic extracted from seaweed. It is typically used at a concentration of 0.5-2%. The higher the concentration stiffer the gel. They have large range of

separation but low resolving power. By varying the concentration, fragments of DNA from about 200-50000 base pairs can be separated using electrophoresis technique.

Polyacrylamide is a cross-linked polymer of acrylamide. The concentration used is between 3.5 and 20%. These gels are more annoying to prepare because oxygen inhibits the polymerization process, they must be poured between glass plates. These gel have smaller range of separation but high resolving power. In case of DNA, only less than 500 base pairs are separated.

Electrophoresis for proteins and enzymes:

Proteins and enzymes are primary products of genes specific to each variety and hence are most suited for varietal purity determination. Seeds, seedlings or mature leaves etc. of a crop plant have a specific mix of proteins which are not only crop specific but also variety specific (genotype specific). The electrophoresis in a suitable medium **separates the mixture of proteins extracted from seeds, seedlings or mature leaves into distinct bands.**

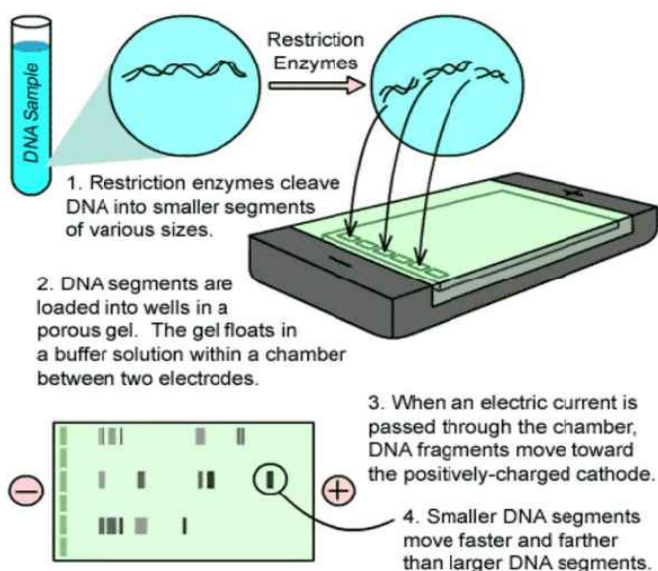
Each variety (or genotype) thus has a specific "banding pattern" on the basis of which admixtures of other varieties, differing in "banding pattern" could be detected. This is done by comparing the banding pattern of analysed sample with the standard banding pattern of that variety. The electrophoresis is now being increasingly used for determining the **genetic purity of seed samples.**

Principle

The term 'electrophoresis' refers to the migration of a charged particle under the influence of an electric field. The movement of ions takes place in a suitable medium, such as, polyacrylamide gel, which acts as a molecular sieve and cut down convection currents and diffusion, so that the separated components remain as sharp zones with maximum resolution. The separation into distinct bands is due to,

1. differences in the size of molecules (molecular weight) of various proteins. Particles with smaller molecular weights migrate faster than those with higher weights, and

2. differences in charge. The molecules with the higher charge migrate faster than those with a lower charge.



Since proteins carry a net charge at any pH other than their isoelectric point, they migrate in an electric field, the rate of which depends on the charge density (that is, the rate of charge to mass of the molecule). Proteins with higher charge density will migrate faster, thus resulting in differential rates of movement of proteins when a mixture of different proteins is subjected to an electric field. By altering the gel pore size (using polymers at different concentrations) and the charge on the protein molecule (by changing the pH of the system) a high degree of resolution can be achieved for separation of protein molecules in a mixture. A reference protein sample consisting of various proteins molecular weight is run along side as reference which is known as protein markers. Based on the protein marker as a reference point, the distance travelled by the protein is calculated as **Relative Mobility (R_m)**;

$$R_m = \frac{\text{Distance traveled by protein}}{\text{Distance traveled by tracking dye}}$$

On the basis of R_m value and thickness of the band a zymogram (electrophoretic strip of exhibiting the pattern of separated enzymes and isoenzymes after electrophoresis) is drawn to show the banding pattern.

The varieties are verified based on the banding pattern: (a) By measuring R_m of bands, (b) Total number of bands (c) Presence or absence of specific band, (d) Intensity of band and (e) Difference in banding pattern in comparison with authentic zymogram of the variety under test.

LECTURE 31 & 32

SEED STORAGE

Maintenance of seed vigour and viability in terms of germination from harvest until planting is of the utmost importance in any seed production programme. Care should be taken at every stage of processing and distribution to maintain the viability and vigour. The harvested seeds of most of the orthodox crop seeds are usually dried and stored for atleast one season until the commencement of the next growing season, except those of the recalcitrant seeds which require high moisture content for safe storage (once dried the viability will be lost. E.g. – Jack, Citrus, Coffee, Cocoa, Polyalthia, etc.,). In such recalcitrant seeds senescence starts in the mother plant itself. The dry weather alters moisture content of the seed, thereby reducing the viability. Some seeds require an after ripening process as in Pinus and Fraxinus. In most of the Agricultural crops ageing starts at physiological maturity, which is irreversible. Hence seeds become practically worthless if they fail to give adequate plant stands in addition to healthy and vigorous plants. Good storage is therefore a basic requirement in seed production.

Purpose of seed storage

The purpose of seed storage is to maintain the seed in good physical and physiological condition from the time they are harvested until the time they are planted. It is important to get adequate plant stands in addition to healthy and vigorous plants.

STAGES/SEGMENTS OF SEED STORAGE

In the broadest sense the storage period for seed begins with attainment of physiological maturity and ends with resumption of active growth of the embryonic axis, i.e., germination.

The entire storage periods can be divided into:

1	Post maturation/ Pre harvest segment	Period from physiological maturity to harvest (seed in field).
2	Bulk seed segment	Period from harvest to packaging (bulk seed in aeration drying bins, surge bins, etc.).
3	Packaged seed segment	Period from packaging to distribution (seed in Packages in warehouse).
4	Distribution /Marketing Segment	Period during distributing and marketing (packaged seed in transit and / or retailer's storehouse).
5	On-farm segment	Period from purchase to planting of seed (seed in on-farm storage).

Types of storage

The types of storage needed can be related to the time of storage expected. Broadly this can be classified into 4 types.

5. Storage of commercial truthfully labelled and certified seed.
6. Storage of carry over seeds.
7. Storage of foundation seed stocks and enforcement seed samples.
8. Storage of germplasm seeds.

a) Storage of commercial seeds

This storage of commercial seed requires the largest storage need from harvest until planting. The storage period ranges from 8-9 months. Seed must be dried to 14 per cent moisture content for starchy seed and 11 percent for oilseeds.

b) Carryover seeds

About 20-25 per cent of stored seed may have to be carried over through one season to the second planting time. The storage period may range 1-1½ year. Storage of seeds in metal bins with tight fitting lids or in a moisture proof bag will solve the problems of moisture penetration, provided the seeds are already dry enough for sealed storage.

c) Foundation stock and enforcement seed sample

It is desirable to store foundation and enforcement seeds for several years since genetic drift are minimized by reproducing foundation or stock seeds. Since the quantity of seeds involved is not large, the storage room is only a small part of the total storage area and in fact, is often a small room within a large warehouse. Relative humidity and temperature combination has to be provided for maintaining the viability. A combination of 25 per cent RH at 30°C temperature or less or a RH of about 45 per cent at 20°C or less will be ideal. The required RH can be achieved by making the room moisture proof and by using a dehumidifier.

d) Germplasm seed storage

Germplasm seeds are required to be kept for many years, perhaps very long periods. Basic requirements for such long term storage are the coldest temperature economically possible and seed moisture is in equilibrium with 20-25 per cent RH. Germplasm storage built up so far have rooms which can be maintained at 5°C to 10°C and 30 per cent RH. In addition, the stored samples are dried to perfect moisture level.

PRINCIPLES OF STORAGE

- Seed storage conditions should be dry and cool
- Effective storage pest control
- Proper sanitation in seed stores
- Before placing seeds into storage they should be dried to safe moisture limits.
- Storing of high quality seed only i.e., well cleaned treated as well as high germination and vigour

FACTORS AFFECTING SEED LONGEVITY IN STORAGE

1. Kind (or) variety of seed
2. Initial seed quality
3. Moisture content
4. Relative humidity and temperature during storage
5. Provenance
6. The activity of organisms associated with seeds in storage.

1. Kind or variety of seed

Seed storability is considerably influenced by the kind or variety of seeds. Some seeds are short lived. E.g.: Onion, Soybean and Groundnut. As a general rule starchy seeds can be stored considerably for a longer period compared to proteinaceous or oily seeds because of their hygroscopic nature.

2. Initial seed quality

Seed lots having plumpy, vigorous undamaged seeds store longer than that of deteriorated. Even seed lots having good germination at the beginning of storage period, may deteriorate at a faster rate depending upon the severity of weathering damage, mechanical injury or otherwise in the field. The low quality seeds should invariably be rejected. Even at best storage conditions, the initial quality of the seed cannot be improved (except for the dormant seed) but can only be maintained.

3. Moisture content

The most important factor influencing seed viability during storage is the moisture content and the rate of deterioration increases, as the seed moisture content increases. The drier the seed the higher will be the storage life.

Seed moisture content (%)	Storage life
11-13	½ year
10-12	1 year
9-11	2 years
8-10	4 years

It is well known that higher moisture content enhances the biological activity in the seeds and causes excessive heating, besides promoting mould and insect activities. The relationship of moisture content of seeds during post harvest stages furnished below would clearly indicate the role of moisture in the life of seeds in storage.

The importance of seed moisture content in extending the shelf life of seeds under ideal storage conditions can be well known and understood from the Harrington's thumb rule, that one per cent decrease in seed moisture content nearly doubles the storage potential of the seed. Again this rule is applicable only at a moisture range of 5-14 per cent because, moisture content below 5 per cent the physio chemical reaction may take place and at above 14 per cent fungi and insects become active. Another rule given by Harrington states that for every 5°C decrease in storage temperature, the seed life will be doubled. Again this can hold good only in the temperature range of 0-50°C. There are exceptions in this rule in a few crops like chillies, brinjal and bhendi. The safe moisture content again depends upon the period of storage, storage structures, kind and variety of seed and the packaging materials used. For cereals under open storage, seed drying upto 10 per cent moisture content appears quite satisfactory. The storage in sealed containers

during upto 4-8% moisture content depending upon the particular kind of seed may be necessary.

Use of desiccants

Desiccant like silicagel can maintain the moisture content in equilibrium with the Relative Humidity of 45%. It is kept @ 1kg/10 kg of seeds. When the silicagel turns to pink colour it should be dried at 175° in oven and then again placed in the container.

4. Relative humidity and temperature during storage

Seeds are hygroscopic. They attain rather specific and characteristic moisture content when subjected to given level of atmospheric humidity at a particular temperature (equilibrium moisture content). The equilibrium moisture content for a particular kind of seed at a given relative humidity tends to increase as temperature decreases and the deterioration starts.

Equilibrium moisture content varies among seed kinds. In general, the equilibrium moisture content of “oily” seed is lower than that of “starchy” seed at the same relative humidity and temperature. This phenomenon can be accounted for by the fact that fats and oils do not mix with water. Thus, in a seed with 50% oil content, the moisture has to be concentrated in half the seed, while in a seed containing 10% oil, the moisture is distributed throughout 90% of the seed. Thus the maintenance of moisture content of seed during storage is a function of RH and to a lesser extent of temperature. At equilibrium moisture content there is no net gain or loss in seed moisture content when seed is placed in a new environment with RH higher or lower than that of the seed, the seed will gain or lose moisture till it reaches a new equilibrium moisture content at this particular new environment.

Dry, cool conditions during storage

The general prescription for seed storage is a dry and cool environment. At this point, the question naturally arises: How dry and how cool? It is difficult to answer this question unless three factors are known:

b) kind(s) of seed to be stored; (2) desired period of storage and (3) physiological quality of the seed.

Seed of most grain crops, e.g., corn, wheat, sorghum, barley, rye, oats, rice, will maintain germination for the 8-9 months period from harvest to planting at moisture content of 12-13% and normal warehouse temperature except possibly in Southern coastal areas.

5. Provenance

The seeds harvested in different climates (or) at different times show differences in viability. Because they would have been subjected to different pre harvest conditions which will have caused different amounts of deterioration by the time, the seeds are harvested.

6. The activity of organisms associated with seeds in storage

The bacteria, fungi, mites, insects, rodents and birds may do harm to seeds in storage. The general limits of temperature and relative humidity for the multiplication of the various biological agencies infesting stored seeds are,

Organism	Temperature		Relative humidity
	Range for	Optimum	
	multiplication	range	
Insects	21-42°C	27-37°C	30-95%
Mites	8-31°C	19-31°C	60-100%
Fungi	8-80°C	20-40°C	60-100%
Microbes	8-80°C	26-28°C	91-100%

It is also interesting to note that the favourable limits of temperature and RH for germination are 16-42°C and 95-100 per cent respectively.

Sanitation in storage

There are several other recognized procedures for good seed storage that most seeds men already know. Seeds should be stored in a seed warehouse, not a fertilizer, chemicals, herbicide, or feed warehouse. Good sanitation should be a continuous practice. It will minimize storage insect infestations. If storage insects are a problem, the judicious use of insecticides and fumigants, combined with sanitation, will alleviate the problem. The best procedure is not to place insect infested lots in storage with other lots unless all the insects have been killed by fumigation or insecticide treatment.

In warehouse with concrete floors, seed bags should be stacked on wooden pallets to keep them from contact with the floor as considerable moisture can be transmitted through concrete floors. Seed warehouses should also be adequately ventilated (unless they are conditioned) and protected against rodents.

Storage Conditions

Since seed moisture content and ambient relative humidity are in equilibrium during storage, maintenance of a “safe” moisture content requires an average level of relative humidity in the storage environment no higher than that in equilibrium with the “safe” or desired moisture content. This favorable situation can be achieved in only three ways: (1) location of the storage facility in a region where relative humidity does not rise – on the average – above the critical level; (2) maintenance of the relative humidity at the desired level by packaging seed in moisture vapor proof containers; or

d) dehumidification of the storage room atmosphere to the desired level. The desired level of relative humidity for successful storage of seed depends, of course on the kind of seed, the duration of the storage period, and the temperature.

SEED PACKAGING IN RELATION TO SEED STORAGE

In reality the seed package is a small storage container. The kind of container needed is affected by several factors including :

6. The quantity of seed desired in each package
7. The protection desired
8. The cost of the package
9. The value of the seed
10. The storage conditions into which the container is to be placed and
11. The facilities for drying the seeds

Classification of packing materials or containers

1. Moisture and vapour pervious containers

These containers allow entry of water in the form of vapour and liquid. These are suited for short term storage. The seeds in these containers will attain seed equilibrium moisture with the surrounding atmosphere (eg) cloth bags, gunny bags, paper bags etc.

2. Moisture impervious but vapour pervious containers

These allow entry of water in the form of vapour and not in liquid. The seeds in the containers can't be carried over for long period in hot humid conditions (.g.) polythene bags of <300 gauge thickness and urea bags.



3. Moisture and vapour proof containers

These containers will not allow entry of moisture in the form of liquid or vapour. These are used for long term storage even in hot humid conditions if the seeds are sealed at

optimum m.c. eg. Polythylene bags of >700 gauge thickness, aluminium foil pouches, rigid plastics etc.



Certified seeds of cereals, pulses and oil seeds are normally packed either in gunny bags or cloth bags. However, paper bag, aluminium foil pouches and polyethylene bags are used for packing flower and vegetable seeds.

Seed storage in relation to seed deterioration

The Purpose of seed storage has been previously stated, *viz.*, to preserve or maintain the physiological quality of seed for the period desired through minimization of the rate of deterioration. Since seed storage is basically concerned with “control” of deteriorative processes, some knowledge of these processes is essential for successful seed storage operations.

Deteriorative changes in seed and their consequences

In our consideration of some of the characteristics of deterioration in seed, another might have been added that deterioration is characterized by change. Indeed, in our context, deterioration and change – detrimental change – are almost synonymous. For deterioration is identifiable only in terms of observable or measurable changes in the response reactions of the seed. Conversely, detrimental changes, e.g., loss of germination or vigour, are said to be the result of deterioration.

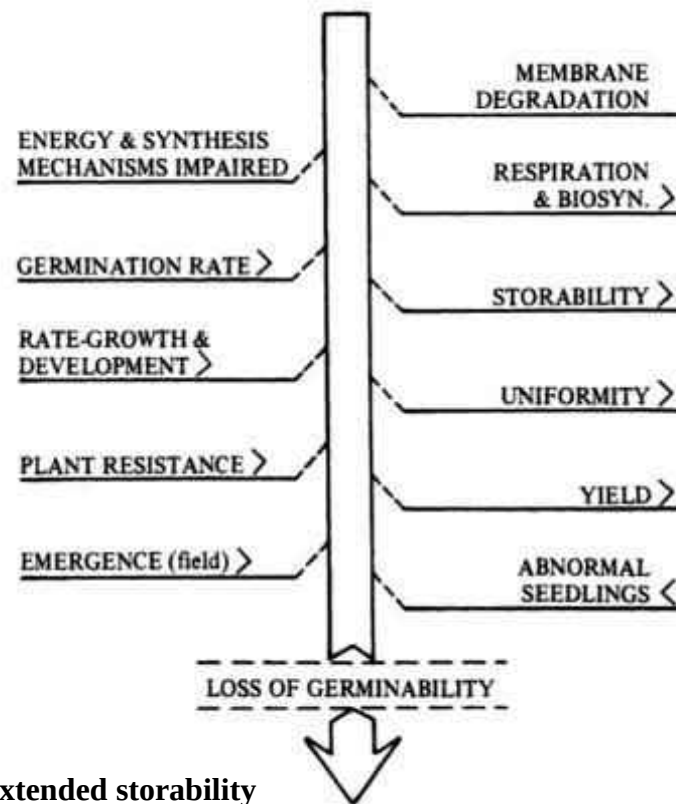
In the sequence of deteriorative changes postulated in figure 1, it can be readily seen that during deterioration, the “performance potential” of seed becomes progressively impaired (reduced) until they lose their capacity to germinate, at which time “performance potential” is zero. Since loss of the capacity to germinate is the last practically significant consequence of deterioration, the design and evaluation of storage conditions only in terms of “maintenance of germination” is not sufficient. The “lesser consequences” of deterioration must also be considered because collectively they determine the “vigour” level of the seed. And, the vigour of seed determines how well they germinate, emerge, grow, and develop in the farmer’s field.

Longevity of seed is a characteristic of the species or variety

Some kinds of seed are inherently long-lived, others are short-lived, while others have an “intermediate” life span. Differences in storability extend even down to the variety level. It has been known, for example, the certain inbred lines of corn are “poor storers” and that this characteristic is inherited.

Inherent differences in seed longevity are facts, the seeds man must accept and contend with as best he can. Among the vegetables, onion seed are notoriously short-lived, radish seed are intermediate in longevity, and watermelon seed are relatively long-lived. Soybean and peanut seed do not store well as compared to seed of wheat, corn, cotton, sorghum and rice. In some cases, seed kinds which have very similar chemical and physical properties differ substantially in longevity. Tall fescue and annual ryegrass seed are similar in structure, chemical composition, and yet, ryegrass seed store better than tall fescue seed.

Possible Sequence of changes in seed during deterioration



Seed selection for extended storability

- ☐ Store well mature seeds.
- ☐ Store normal coloured seeds
- ☐ Seeds should be free from mechanical injury
- ☐ Seeds should not have met with adverse conditions during maturation
- ☐ Seeds should be dried to optimum moisture content.
- ☐ Seeds should be treated with fungicides before storage.
- ☐ Suitable packaging materials should be used for packing.

STORAGE GODOWNS AND THEIR MAINTENANCE

Seeds undergo deterioration due to aging in storage. This is accelerated by climatic factors and external biotic factors like insects and pathogen. In addition to seed borne pathogen and storage insects, seeds are damaged by birds and rats for their feed.



Clean and hygienic godowns protect the seed from external insects and preserve the seed. Hence care should be taken in construction of godown. The points to be noted are as follows.

- ☐ Seed godown should be in a place where transport facilities are easily available.
- ☐ Seed godowns should not be constructed in areas near seashore. Since the high RH of atmospheric air accelerate the deterioration of seed.
- ☐ Seed godown should not be constructed in low lying water stagnating areas.
- ☐ Seed godown should be constructed in places where atmospheric RH is low, free circulation of air is possible, sunlight is adequate and elevated in nature.
- ☐ The ventilators should be at bottom for free air circulation.
- ☐ Ground moisture should not reach the floor.
- ☐ Should be rat proof with wire mesh
- ☐ Should not be near industries as smoke is injurious

In maintenance of seed in godown following points are to be considered.

1. Godown should be clean and dry
2. Seed bags should not be stacked directly on floor. Should be stacked on wooden ballets.
3. The height of the stack should not be more than 6-8 bags.
4. Different seed lot should be kept separately.
5. Godown should be sprayed periodically once in a week or fortnightly with Malathion 50 EC (1 : 300 Chemical : Water) @ 5 lit. sq. m⁻¹ or 0.25% Nuvan @ 1 lit. 100 m³⁻¹.
6. Altering the chemicals at weekly intervals will give better control.

7. Seed lots can be fumigated with Aluminium phosphide @ 3 gm/cu.m in air tight condition for 7 days. This can be done as prophylactic measure and on minimum infestation by insects.
8. Seed lots should be periodically (once in month) tested for seed quality.
9. Based on seed testing result, seeds can be dried under sun for the removal of moisture. It reduces insect and pathogen infestation.
10. New seed lots should be kept away from old seed lots to avoid secondary infestation of insects.
11. Seeds should be treated with combination of fungicide and insecticide (eg.) Thiram @ 2 g kg⁻¹ + carbaryl @ 200 mg kg⁻¹.
12. Frequent supervision of each and every lot is must.
13. Seed bag should be restacked once in 3 months for free aeration.
14. .Instead of gunny bags low cost interwoven polythene bags should be used to prolong the life of seed.



15. Pesticides, fungicides, fertilizers, rejects should not be stored with seed.
16. Each lot should be labeled accurately and registers for stocks should be maintained.
17. Per acre or per hectare packing (small) is preferable for easy handling and effective supervision.

STORAGE INSECT MANAGEMENT

Maintenance of store house hygiene

- A Cracks and crevices around corners have to be brushed to eliminate hiding pests. All debris should be removed.
- B Reduce the moisture content of seed to prevent insect build up (usually below 10%). Previously used bags, bins, etc. should be dried in the sun repeatedly.
- C Elimination of conditions which favour storage pests. Uniformly graded seeds should be used, broken seeds should be removed before bagging since they favour pest build up. Stitching of all torn bags, filling bags up to the brim, no loose packing.
- D Surface treatment of storehouse before storage with malathion dust 4% @ 25 g/sq m or malathion 50% EC spray @ 10 ml/lit of water and 3 lit of solution per sq. metre.
- E Good dunnage by arranging wooden planks or bamboo poles or spreading thick polythene sheets on the floor. Treatment of dunnage materials with malathion as specified, arrange the bags in crisscross pattern with a maximum of 15 bags and provide adequate space between the roof and the seed bags.

Prophylactic treatment of seeds

Application of malathion 4 per cent dust 25 g/sq metre or malathion 50 per cent EC 10 ml per litre of water and 3 litres of spray solution for 100 sq.m. The chemicals have to be sprayed on the walls and floors and the treatment has to be repeated based on the extent of flying and crawling insects.

Chemicals

Two chemicals are widely used : Phosphine and Methyl Bromide.

Phosphine : Available in a solid form (0.6 g pellets, 3 g tablets). The active ingredient is Aluminium phosphide mixed with Ammonium carbonate and Paraffin. After exposure to the atmosphere, the pellets decompose and release the active substance, hydrogen phosphide

(PH₃), which has the same specific weight as air, and is thus evenly distributed in the fumigated material or chamber. Phosphine is also able to penetrate bags, carton boxes and other containers.

It must be borne in mind that fumigation particularly repeated fumigation, may seriously reduce the vigour and viability. This is particularly true for seeds with a higher moisture content of 14 per cent. Seeds with moisture content above 14 per cent should be dried, before fumigation.

The fumigant has to be chosen and the requirement worked out on the following guidelines
Aluminium phosphide: Three tablets of 3 g each per ton of seed for cover fumigation (only selected blocks of bags)

Twenty one tablets of 3g each for 28 cubic metres, for shed fumigation (entire godown). Period of fumigation - 5 days. The major advantages of Phostoxin are that it lacks residues and does not affect flavor or germination and is easy to handle.

Methyl bromide: Above 5.6°C, methyl bromide is in the gas phase and is available in cylinders similar to those used for cooking gas. Since, it is odorless, other gases such as chloropicrin are sometimes added to facilitate detection of leaks. Because methyl bromide is 3.5 times heavier than air, care has to be taken that it is properly distributed within the goods to be fumigated (fan can be used). The recommended dosage is 20 g/m³ for 24-48 hrs.

Rodent Management in Store Houses

Provide of wire mesh to windows, ventilators, drains and leave no gaps to doors. Use rodent baits with multi dose or anticoagulant rodenticides. The bait may be prepared as follows:

Cereal flour	450 g
Any edible oil	10 g
Powdered jaggery	15 g

Anticoagulant or rodenticide such as coumarin	25 g
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Replace the consumed bait daily. If needed the single dose or acute poison bait may be prepared as follows :

Food material	97 g
Edible oil	1 g
Zinc phosphide	2 g

Before providing the poisoned zinc phosphide bait, the plain or non-poisoned bait are to be provided for two or three days to make the rats accept the bait.

LECTURE 33

Seed marketing-Structure and organization-sales promotional activities – factors affecting seed marketing and demand

SEED MARKETING

Seed marketing is one of the most vital components of seed technology. On it depends the size and scope of the seed industry. Broadly, it includes such activities as production, processing, storage, quality control and marketing of seeds. Seed marketing refers to the actual acquisition and selling of packed seeds, intermediate storage, delivery and sales promotional activities.

Seed Marketing Comprises the Following:

1. **Demand forecasts (assessment of effective demand)** - Seed supply keeps space with seed demand in terms of quantity, quality, price, place and time.

2. **Marketing structure** – refers to the establishment of effective channel for seed distribution by farmer distribution, co-operatives distribution and departments of agriculture.
3. **Arrangements for storage of seeds** – it is necessary to establish seed marketing regional in the end-use areas and to arrange food godown.
4. **Sales promotional activities** – such as advertising, publicity and public relations.
5. **Post- sales service** – more importance should be given on education of farmers, technical help.
6. **Economics of seed production and seed pricing** – seed production cost must be low. Seed producer should be allowed sufficient margin to produce quality seeds.

Types of seed production and distribution organization

1. Government departments
2. State owned corporations
3. Private enterprise

MARKETING STRUCTURE

Establishment of Effective Channel for Seed Distribution

The key to success in seed marketing is the establishment of effective channel of distribution. There are ten major channels of marketing seeds within the state. They are,

Channel I : Represents the movement of breeder seeds from Research Station (ICAR's and SAU's) to the department of Agriculture, National seed corporation and private producer to seed growers for raising the foundation seeds.

Channel II : The foundation seeds from the department of Agriculture, National seed corporation and private producer to seed growers for raising the certified seeds.

Channel III : The department of Agriculture act as a seed producer and organizes seed farms at the level of seed farm growers, produces the certifies seeds from them and moves it to

its agricultural depots after processing through which it retails the seeds to the end user-farmers. Used for paddy, pulses and oil seeds (marketed in state).

Seed grower → Agricultural department → Seed user farmer

Channel IV : The cooperatives act as a seed producers and organize the seed farms at the level of seed farm growers, procure the seeds from the growers and move through their own cooperatives retail outlets to the seed users. Used for paddy, cotton and groundnut are handled by cooperatives.

Seed grower → Cooperative seed producer → Cooperative retail outlet → Seed user farmer

Channel V : Government undertaking like National Seed Corporation organizes seed farms with private growers, procures and distributes seeds both within and outside the state.

Seed grower → National Seed Corporation → District retail outlet → Seed user farmer

Channel VI : The private seed producer handles hybrids of millets as well as both varieties and hybrids of cotton and sells to the seed user farmer through retail outlets.

Seed grower → Private Seed Producer → Retail outlet → Seed user farmer

Channel VII : National Seed Corporation produces seeds from its seed growers and sells to the department of Agriculture, which in turn sells to the user farmers through its depots.

Seed grower → National Seed Corporation → Government Agricultural Depot → Seed user farmer

Channel VIII : Cooperatives, National Seed Corporation and private seed producers procure seeds from seed growers and sell through their agencies to the wholesale retailers of other states and pass on finally to the users farmers of other state.

Seed grower —————> Cooperative, National —Seed—> Agents of other state —————> Seed user
Corporation and Private farmer

Channel IX : Cooperatives, National Seed Corporation and private seed growers procure seeds from seed growers of other states and bring to respective state and sell either through their retail outlets to the users farmers or to the government depots, which in turn sells through the agricultural depots to the user farmers.

Seed grower of —————> Cooperative, National Seed —————> Cooperative retail, National —————> Seed user
other states Corporation and Private Seed Corporation retail and farmer
Private retail

Present Status of Seed Distribution:

The types of seed distribution systems in India are:

a) Farmer to farmer distribution - This is the traditional method, whereby farmers obtain their requirements from neighbours either on cash payment or on exchange basis. No formal marketing organisation is required for this type of distribution.

b) Distribution by co-operatives - This involves procurement of seeds by cooperatives and its subsequent distribution. The distribution of seeds through cooperatives has often been encouraged by the government through subsidies and guarantees.

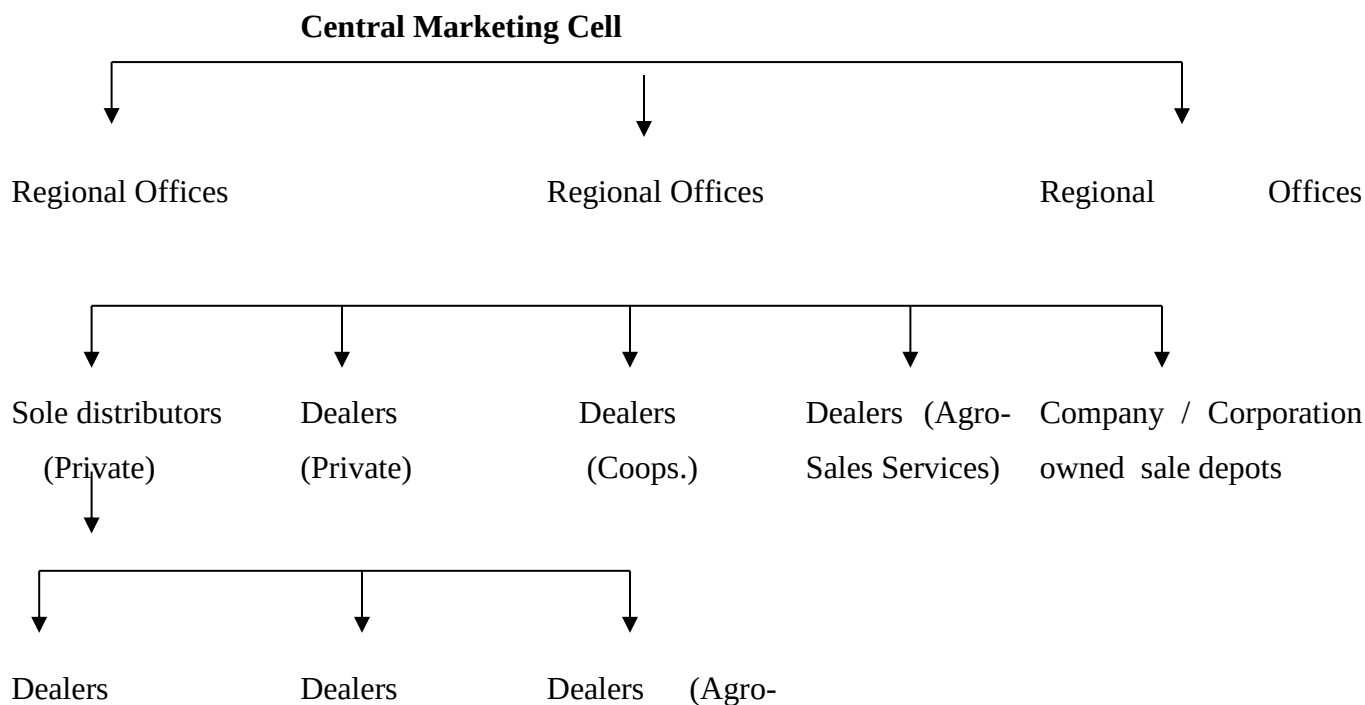
c) Distribution by departments of Agriculture - Seeds are purchased by the governments, out of the government funds, and are distributed district Agricultural Officers and Block Development officers.

d) Distribution of seeds by non-government or quasi-government agencies - seeds are distributed through a network of seed distributors and seed dealers.

Both the seed Review Team (1968) and the National Commission on Agriculture (1976) have recommended that the State government should withdraw from the seed procurement and distribution fields in a phased manner, so as to be able to concentrate on their principal function of providing extension education in the use of high quality seeds for improving productivity. Thus, emphasis is on the establishment of a seed marketing network to replace the role of state government, and to establish a system that will be adequate for the anticipated increase in seed demand. In this connection, it is considered necessary that a network of seed dealers should be established.

MARKETING ORGANISATION

There are a number of possible ways a marketing network could be organised. The simple and most efficient systems are to establish a central marketing cell and regional offices in end-use areas. The retail sale could be organised either by appointing distributors/ dealers such as private dealers, cooperatives, agro-sales service centres, etc, or by opening seed company/ corporation –owned sales points, or both as illustrated.



(Private)

(Coops.)

Sales Services)

Under such a scheme, the central marketing cell is responsible for planning, appointment of dealers/ distributors, seed movement, marketing intelligence research , pricing , promotional activities, financing and record keeping.

SEED DEMAND FORECASTING

The assessment of effective seed requirement is critical to any planned seed programme. **“Anticipation of required seed quantity in open market by understanding the behavior of consumers and other factors”**. The underlying **principle** in making demand forecasting should be that the seed supply keeps place with seed demand (both present and future) in terms of quantity, quality, price, place and time. The outcome of such approach would be planned seed production and marketing would also avoid shortage and gluts and as well as ensure stable price and profits.

In Making Demand Forecast, the following factors must be considered carefully:

- a) Total cultivated acreage, seed rate, quantity replacement period and assessment of total potential seed requirement of each important crop.
- b) Impact of extension efforts on the introduction of improved production techniques and future plans for promotion.
- c) Current acreage under high yielding varieties and amount of seed sold in last year.
- d) Cultivar preference for varieties, package, kind of packing, quality and price
- e) Number and size of competition.
- f) Kinds of publicity and sales promotion those are most effective.
- g) Climate of the area where seed is being marketed.

Assessment of potential effective seed demand of the market, based on total seed requirements is of very little values, since the demand for high quality seed normally exists for the crop area which is under good fertility and irrigated conditions. The requirements for the remaining crop area are covered by uncontrolled production material obtained from the previous crop production. Furthermore, experience shows that the varietal purity and the yield potential of

high quality seed of the self pollinating varieties can be maintained by farmers during the reproduction process, without significant deterioration for three to four generations.

Therefore, individual farmers only need to **replace seed of self –pollinating varieties every third or fourth year because they loss their purity and quality**. Thus, the demand for high quality seed of self –pollinated crops is normally not higher than 25 to 30 percent of the total requirement for areas under irrigated and high fertility conditions. However, there could be some exceptions, Eg. if the climate of the region is not suitable for retaining viability of seed from the preveious crop production. The farmers of such areas, if properly trained, can buy high quality seed each year, even of self-pollinated crops.

A rather different approach must be taken in the marketing of **hybrid seeds**, in which case **new seed is needed by the farmer each season**. Although, the critical period may be rather difficult, the subsequent planning is easier, particularly after sale statistics are seen to point in a certain direction.

**Demand forecasting in the classes of seed **

Breeder seeds – forecasting is done before 18 months

Progeny of nucleus seed



Indents form agencies (National Seed Corporation, State Seed Corporation, State Agricultural Universities)



State department of agriculture



Department of agriculture and co-operation compiled



Project co-ordinator / Project director



Breeder / producer (based on environment conditions)

Foundation seeds – forecasting is done by agencies (National Seed Corporation, State Seed Committee, State Agricultural Universities, KVK's) in advance and submitted to producer.

Certified seeds – State government, State Seed Committee, Cooperatives.

The dealers need to make periodic surveys of the market areas, to determine market potential at least one season in advance. Dealer advance orders should be treated as informational material to aid the production section in organising an effective production programme. The dealer should, however, not be held to the exact amounts of such advanced orders.

The uses of demand forecasts are many and varied. A reliable forecast is the sheet anchor of all planning in business. Long term demand indications, in terms of quality, prices and locations, help to make an investment decision, that is how much to invest, where to locate the production facility, and how to organise marketing. Intermediate-range demand forecast help to make decisions on action necessary to optimise profits by balancing production and sales. Uses of short-term demand forecast include production planning and scheduling, distribution planning and scheduling, determination of targets and quotas for dealers and salesmen, planned buying of inputs, preparation of cash flow budgets, preparation of overall budgets and profits and loss statements, modifications of prices, policies, etc.

Sales promotional activities

- **A variety of techniques designed to encourage customers to buy**

Selling out of the marketing channel

- Free samples
- Seed exchange
- Coupon offers - to encourage repeat orders or purchases of other products in the company's range.
- Price reductions
- Competitions

- Premium offers - The offer of an extra product for a promotional price can be made
- Field demonstrations and group discussions

Selling to the marketing channel

- Dealer 'loaders' - encouraging dealers to build up their stocks in sufficient time
- Point-of-sale materials - materials supplied by the seed company to the sale points to support sales of their product range. (colourful display boxes, leaflets, seed catalogues and audio-visual displays).
- Dealer competitions - afford to offer significant prizes linked to dealer sales' performance.
- Dealer training - Training dealers in the use and benefits of the product is a vital form of support.
- Credit - incentive to dealers
- Sale or return
- Promotional gifts

Factors affecting the final marketing price

1. Input cost – price paid to growers for raw seed, storage and transportation cost, distribution costs, wholesale margin, retail margin, risk costs, profit. The final market price depends on these factors.
2. Supply and demand - if the supply is short, the seeds could be sold at relatively high price and vice versa, seed prices have to be very realistic and the profit margin reduced.
3. Prices of other farm products and time trend – if due to food scarcity in a particular zone food prices are fairly high, there is no reason why seed should not be sold at comparatively high prices. High prices in such areas would also help in preventing the consumption of seed as food grains.
4. Clear –cut Policy
5. Availability of well – identified and Adapted Varieties
6. Adequate production, storage and testing facilities

7. Official Programme
8. Demand Forecast
9. Market intelligence
10. Transport and storage arrangements
11. Nature of product
12. Quality control programme
13. Publicity
14. Rewards to dealers

LECTURE 34

International seed movement - role of international organizations - WTO and OECD in seed trade

International seed movement

The following organizations are involved in international seed movement.,

1. The Organization for Economic Co-operation and Development (OECD)
2. World Trade Organization (WTO)
3. International Seed Testing Association (ISTA)
4. International Seed Trade Federation (FIS)

The Organization for Economic Co-operation and Development (OECD)

- **History** : Established in 1961
- **Headquarters** : Paris, France
- **Membership** : 36 countries
- India is also member in OECD

- Inter - governmental Organization

The Organization for Economic Co-operation and Development (OECD) **Seed Certification Schemes** is a global frameworks for the varietal certification of internationally - traded seed. The OECD schemes have made a substantial contribution to growth in the global seed trade, especially with respect to **field crops**. The United States, China, France and Brazil are the **largest seed markets in terms of value**. OECD **varietal certification standards are developed by an expert committee** consisting of the national designated authorities (NDA), researchers, industry and farmer representatives, as well as representative of other international organizations. The certification standards are based on two key criteria:

Varietal identity and varietal purity.

The OECD Seed Certification Schemes

Schemes established in 1958

- Grass and legume seed
- Cereal seed
- Crucifer and other oil or fiber species seed
- Maize and sorghum seed
- Sugar beet and fodder beet seed
- Seed of subterranean clover and similar species
- Vegetable seed

The maize and sorghum seed scheme is the largest of the seven. The schemes facilitate imports and exports of seed by the removal of technical barriers using labels recognized globally which, in effect, act as a ‘passport’ for the seed traded.

World Trade Organization (WTO)

- **Inter –governmental organization that regulates international trade**
- **Established during 1st January 1995**
- **WTO is result of ‘Uruguay round of Negotiations’**
- Membership : 164 member countries
- India is also one of the member
- **Headquarters : Geneva, Switzerland**
- The WTO is a successor to GATT (General Agreement on Tariffs and Trade)

Objectives of WTO

1. To improve the standard of living of people in the member countries.
2. To ensure full employment and broad increase in effective demand
3. To enlarge production and trade of goods
4. To increase the trade of services
5. To ensure optimum utilization of world resources
6. To protect the environment
7. To accept the concept of sustainable development

Functions of WTO

1. To implement rules and provisions related to trade policy review mechanism

2. To provide a platform to member countries to decide future strategies related to trade and tariff
3. To provide facilitates for implementation, administration and operation of multilateral and bilateral agreements of the world trade
4. To administer the rules and processes related to dispute settlement
5. To ensure the optimum use of world resources

i. International Seed Testing Association (ISTA)

- History : established in 1924
- Headquarters : zurich, Switzerland
- Membership : 225 Member Laboratories

35 Personal Members

63 Associate Members

Functions :

- To ensuring the uniformity in seed testing methods
- For global seed trade issue the international certificates
 1. Orange or Green - Seed lot Blue – Sample
 2. Orange – sampling and testing done in same country
 3. Green – Sampling in one country and testing in another country

International Seed Trade Federation

- **History : established in 1924**

Objectives and functions:

- To formulate and deliberate opinion of all concerned in the seed trade
- To carry out improvement of seed trade condition between various countries
- To find out the solution to problems in international seed trade
- To facilitate settlement of international disputes in seed trade